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ABC-PMX-II SOFTWARE COMPOSITE BEAM DESIGN



SPECIFICATIONS

« Etablissement certifié qualité ISO 9001, le CTICM assure un suivi de chaque étude dans le plus strict respect de ses procédures qualité »

Date :	Apr. 24 th , 2024	Author :	P-O. MARTIN	Total number of pages: (including this one)	200
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FOR O2-03 A

 Construire en métal, un art, notre métier	Research and Valorisation Department		
	Project : ACB-PMX-II Software		
	Document index : DRV/PMX/ST/001	Rev. :	C

Revision	Date	Authors	Subject
A	25/04/2023	P-O. MARTIN	Specifications of the software – Phase 1 Fire design and slim floors excluded
B	26/09/2023	P-O. MARTIN	Phase 2 Introduction of slim floors and fire design
C	24/04/24	P-O. MARTIN	Complements on Fire design (see § 10, and new annexes F and G). New § 7.6.8 for the check of a steel deck used as a lateral restraint in the construction phase. Modifications in blue
D			

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1. INTRODUCTION

1.1. General

- (1) ABC-PMX software aims at the calculation of composite beams according to EN 1994-1-1. Its development is co-funded by ArcelorMittal and CTICM. Two separate installation packages will be prepared with specific settings.

This document provides the technical specifications of the software. It describes:

- the parameters;
- the assumptions and methods applied for the section and global analyses;
- the calculations of loads and combinations of loads;
- the design checks applied for the verification of the beams at ultimate limit state (final stage and construction stage);
- the design checks at serviceability limit states.

It also gives a description of the main features of the User Interface of the software.

1.2. Scope of application

- (1) The software deals with composite beams obtained by the connection of the slab to the upper flange of a steel profile (standard beams) or by a steel profile embedded in the concrete slab (slim floors). It also covers the case of partially encased steel profiles.
- (2) It covers application for buildings only, according to EN 1994-1-1 (first and second generations). Applications for bridges are not covered.



Except when otherwise stated in this document, it is assumed that the same rules apply for both generations of Eurocodes.

- (3) The software does not cover the possibility to deal with specific National Annexes. The User has the possibility to set the parameters of the software to reproduce the choices of any National Annex.
- (4) Single span beams are considered. It is also possible to add cantilever spans at each support, except for slim floor beams.
- (5) Continuous beams with multiple spans are not covered in the first edition of the software. The program will nevertheless be built in order to facilitate the future integration of multi-spans continuous beams.
- (6) The software covers primary as well as secondary beams.
- (7) Large web openings are not covered by the present specifications. Slab openings are possible.
- (8) Class 4 cross-sections are not covered.

1.3. Assumptions

- (1) All beams have a uniform section along the span length. The same cross-section is considered in the cantilever spans, if any.
- (2) In presence of cantilever spans, the joints which provide the continuity at supports are assumed to be rigid and resistant enough. The software will not provide any calculations for these joints.
- (3) Torsion effects are not taken into account, even for asymmetrical configurations of slim floors. In the latter case, specific devices are assumed to be present in order to prevent torsion in the beam.

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1.4. Axes reference

- (1) For steel sections below the slab, with or without connection, the location of centroids are measured from the interface between concrete and steel, i.e. from the top fibre of the upper flange. Values are negative below the interface and positive above (**Figure 1**).

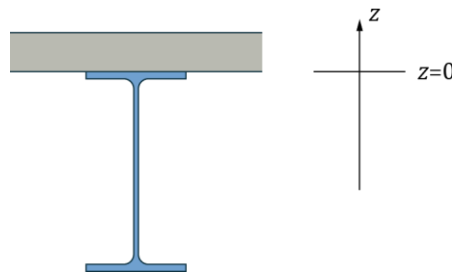


Figure 1 : Axis z for usual sections



The term “usual sections” is used here for all the cross-sections defined in § 2.2.1 (2) and (3).

- (2) For steel and composite slim floor sections, the location of centroids is measured from the bottom of the slab. Values are negative below the interface and positive above (**Figure 2**).

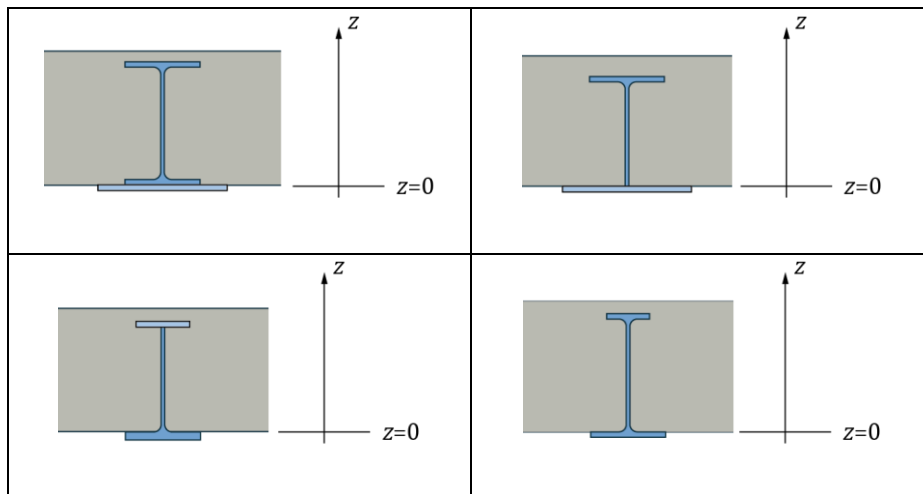


Figure 2 : Axis z for slim floor sections

1.5. Terms

- (1) The term “composite slabs” is used for a concrete slab where a steel profiled sheeting is used initially as permanent formwork and subsequently combines structurally with the hardened concrete to act as a tensile reinforcement in the finished floor.

The software does not cover the design of the slab, so that it can be also used for slabs with steel profiled sheeting which acts only as lost formwork.

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2. BASIC PARAMETERS

2.1. Geometrical description of the beam

- (1) A beam has to be defined either as an intermediate beam or as an edge beam (**Figure 3** for usual sections and **Figure 4** for slim floor sections).

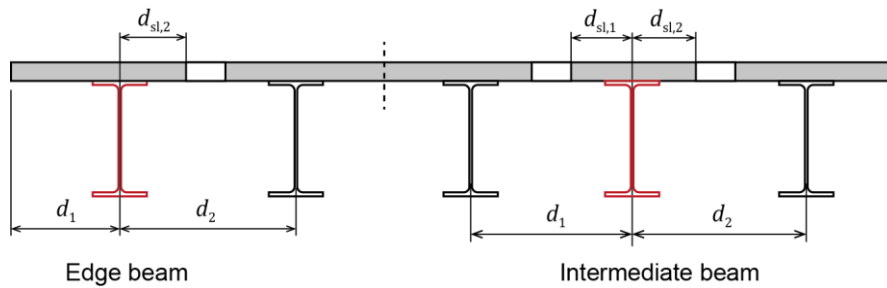


Figure 3 : Definition of edge beams and intermediate beams for usual sections

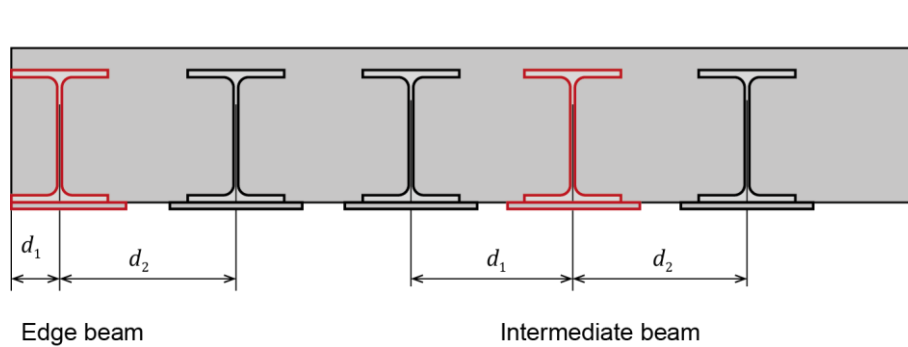


Figure 4 : Definition of edge beams and intermediate beams for slim floor sections

- (2) The geometry of the main span of the beam is defined as follows (see **Figure 3** to **Figure 5**):

- For intermediate beams:
 - L is the length of the main span
 - d_1 is the spacing of the beam to the left beam
 - d_2 is the spacing of the beam to the right beam
- For edge beams:
 - L is the length of the main span
 - d_1 is the spacing of the beam to the slab edge
 - d_2 is the spacing of the beam to the adjacent beam

L should be at least 5 m: $L \geq L_{\min} = 5$ m. It is possible to modify $L_{\min}$ in the options of the software in Expert mode.

The minimum spacing of intermediate beams is given by: $d_{\min} = \max\{1 \text{ m}; 2 \sum b_f; 2 \sum b_p\}$

For an edge beam, if d_1 is set to zero by the User, the following values will be considered in the calculations:

- for usual beams: $d_1 = b_{fs} / 2$
- for slim floor beams: $d_1 = b_s / 2$, where b_s is the section width.

- (3) For slim floors, the distance d_1 for edge beams is always assumed to be at the edge of the profile.

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- (4) For cantilever spans, the following additional parameters are required:

L_{cA} is the length of the left cantilever span

L_{cB} is the length of the right cantilever span

The length of a cantilever span should fulfil the following conditions:

$$500 \text{ mm} \leq L_{cS} \leq 0,3 L \quad (1)$$

Cantilever spans are not available for slim floor steel or composite beams.

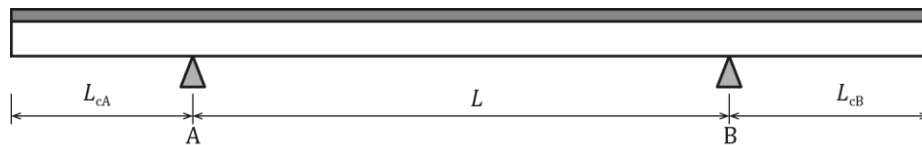


Figure 5 : Spans of the beam

- (5) In the central span only, it is possible to define the presence of slab openings. A slab opening will be defined simply by the distance d_{sl} of the opening edge to the beam axis. The presence of a slab opening will be possible only between two adjacent beams. So that, the following parameters will be defined by the User for slab openings (see **Figure 3**):

- For intermediate beams:

$d_{sl,1}$ distance of a slab opening edge to the left

$d_{sl,2}$ distance of a slab opening edge to the right

- For edge beams:

$d_{sl,2}$ distance of a slab opening edge to the side of the adjacent beam



Slab openings are not possible on the cantilever side of the slab, in the case of edge beams.

- (6) There is no slab openings for beams with slim floor sections.
- (7) At any support of the main span without adjacent cantilever span, the User should define, for beams with usual sections, if the slab is continuous at the support. See § 11.4 for the effect of the continuity of the slab at a simple support.

2.2. Steel cross-sections

2.2.1. Types of cross-sections

- (1) Following types of cross-sections, as given in (2) to (5), will be available in the software:

- (2) **Steel profiles below the slab, without connection to the slab**

The first set of sections concerns steel profiles placed below the concrete slab without a connection to the slab. The steel element may be a hot-rolled profile (SHR type in **Table 1**) or may be obtained by welding of steel plates, for either bi-symmetrical (SWB type) or mono-symmetrical sections (SWM type). The design of these beams is carried out according to EN 1993-1-1 [4].

For bi-symmetrical sections, either hot-rolled or welded, a partial encasement by concrete between the flanges will be available (see **Table 2**). For these sections, the composite action of the concrete between the flanges is taken into account and the design is carried out according to EN 1994-1-1 [7] but neglecting the contribution of the concrete slab.

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This type of cross-sections will be denoted hereafter in this document as **usual steel sections**, with or without partial concrete encasement.

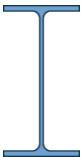
		
SHR	SWB	SWM
hot rolled section	bi-symmetrical welded section	mono-symmetrical welded section

Table 1: Types of steel section without connection to the slab

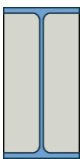
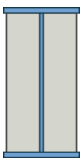
	
SHR-E	SWB-E
hot rolled section	bi-symmetrical welded section

Table 2: Types of steel section with partial concrete encasement but without connection to the slab



Even for these steel profiles, a slab is to be defined. The beams in the software are always assumed to be part of a floor system.

(3) **Steel profiles below the slab, with a connection to the slab**

The second set of sections concerns steel profiles placed below the concrete slab with a connection between them, which allows to take benefit from the composite action. The steel element may be a hot-rolled profile (CHR type in **Table 3**) or may be obtained by welding of steel plates, for either bi-symmetrical (CWB type) or mono-symmetrical sections (CWM type). The design of these beams is carried out according to EN 1994-1-1 [7].

A partial encasement by concrete between the flanges will also be available (see **Table 4**) for bi-symmetrical sections.

This type of cross-sections will be denoted hereafter in this document as **usual composite sections**, with or without partial concrete encasement.

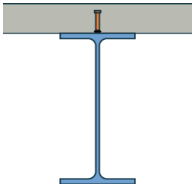
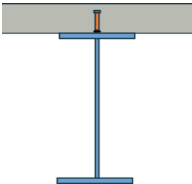
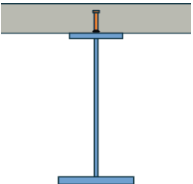
		
CHR	CWB	CWM
hot rolled section	bi-symmetrical welded section	mono-symmetrical welded section

Table 3: Types of steel section with a connection to the slab

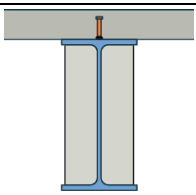
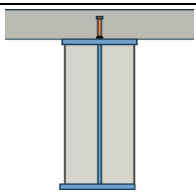
	
CHR-E	CWB-E
hot rolled section	bi-symmetrical welded section

Table 4: Types of steel section with partial concrete encasement and with a connection to the slab

(4) Slim floor steel sections (without connection to the slab)

The third set of sections deals with steel profiles encased in a concrete slab, denominated as slim floors sections. No connection is provided between the steel profile and the concrete, so that no composite action is developed. The steel elements are obtained from hot-rolled profiles only. In order to have the possibility to build this type of floors, the lower flange or plate should be larger than the upper flange. For the SFB type (see Table 5), a steel plate is welded below the lower flange of a hot rolled profile. For the IFB sections, the hot-rolled profile is first cut longitudinally, in order to create a tee section. A steel plate is then welded to the tee section. For the IFB-A sections, the steel plate constitutes the lower flange of the section whereas for the IFB-B sections, it is the upper flange. The last type of section (SAB) is obtained by cutting in the longitudinal direction the upper flange of a profile in order to reduce its width.

This type of profiles is designed according to specific methods detailed in the present specifications, based on the principles of Eurocode 3.

This type of cross-sections will be denoted hereafter in this document as (non-composite or) steel slim floor sections.

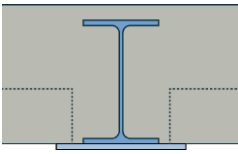
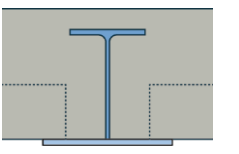
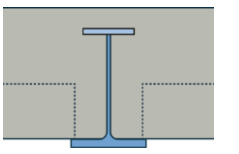
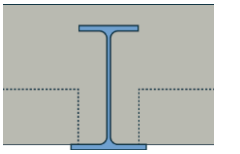
Non composite slim floors			
			
SFB	IFB – A	IFB – B	SAB

Table 5: Non-composite sections for slim-floors

(5) Slim floor composite sections (with connection to the slab)

The fourth set of sections deals with steel profiles encased in a slim concrete floor with a connection, which provides a composite action. These sections are denominated as composite slim floors sections. The same four types of steel cross-section are considered as for non-composite slim floor sections, together with three types of connectors (Table 6):

- composite dowels, by a steel reinforcement passing through the web (Group A);
- studs welded on the upper flange of the steel profile (Group B);
- studs welded on both sides of the profile’s web (Group C).

This type of cross-sections will be denoted hereafter in this document as composite slim floor sections.

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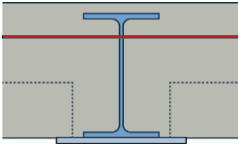
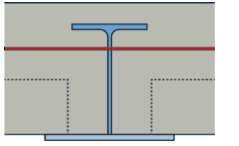
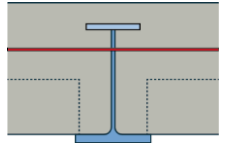
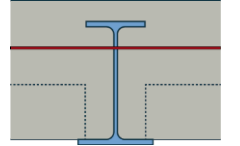
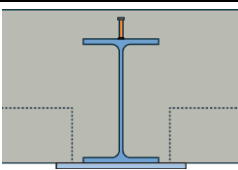
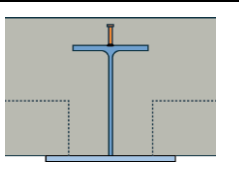
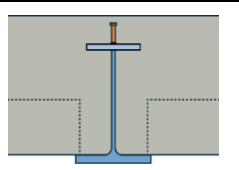
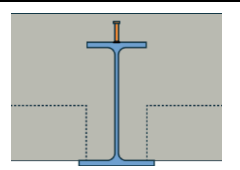
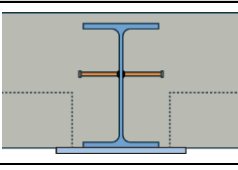
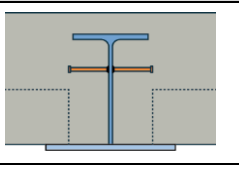
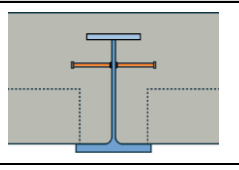
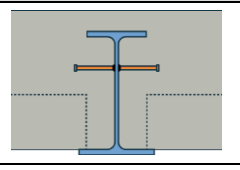
Composite slim floors				
Group A – Composite dowels				
				
CA-SFB	CA-IFB-A	CA-IFB-B	CA-SAB	
Group B – Studs welded on the upper flange				
				
CB-SFB	CB-IFB-A	CB-IFB-B	CB-SAB	
Group C – Studs welded on the web				
				
CC-SFB	CC-IFB-A	CC-IFB-B	CC-SAB	

Table 6: Composite sections for slim-floors

2.2.2. General description of a cross-section

- (1) Any steel profile can be described by the following geometrical parameters:
- h_s total depth of the section
 - h_b depth of the parent profile (when a part or all the section is obtained from a hot rolled profile)
 - b_{fs} width of the upper flange
 - t_{fs} thickness of the upper flange
 - b_{fi} width of the lower flange
 - t_{fi} thickness of the lower flange
 - r_{cs} radius of the upper fillet
 - r_{ci} radius of the lower fillet
 - h_w depth of the web (accounted between flanges)
 - t_w thickness of the web
 - d_w clear depth of the web (accounted between fillets)
 - b_p width of the plate (slim floors SFB and IFB)
 - t_p thickness of the plate (slim floors SFB and IFB)
 - a thickness of the weld throat, for welded sections. The same thickness is used for both flanges.

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- (2) The following relationships are verified by these parameters:

$$h_s = t_{fs} + h_w + t_{fi} + t_p \quad (2)$$

$$h_w = d_w + r_{cs} + r_{ci} + 2\sqrt{2}a$$

The parameters are given in the next chapter for each type of section.

2.2.3. Hot rolled profile

- (1) For a hot rolled profile, the parameters of the section are shown in **Figure 6**. In this case, the following formulas are applied:

$$b_{fs} = b_{fi}$$

$$t_{fs} = t_{fi}$$

$$r_{cs} = r_{ci}$$

$$b_p = t_p = 0$$

$$h_s = h_b$$

$$a = 0$$

- (2) In the software, hot rolled profiles are defined from the ArcelorMittal database.
Custom hot rolled sections can be defined in Expert mode.

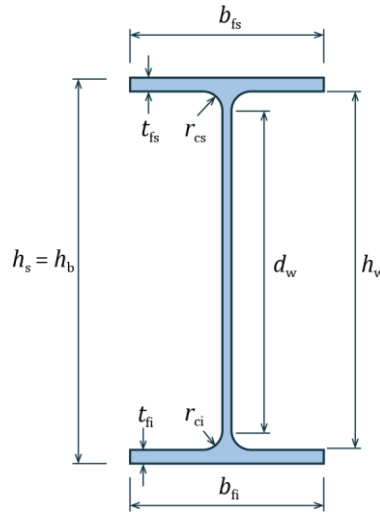


Figure 6 : Dimensions of a hot rolled profile

2.2.4. Bi-symmetrical welded section

- (1) For a bi-symmetrical welded section, the parameters of the section are shown in **Figure 7**. They fulfil the following formulas:

$$b_{fs} = b_{fi}$$

$$t_{fs} = t_{fi}$$

$$r_{cs} = r_{ci} = 0$$

$$b_p = t_p = 0$$

$$h_s = h_w + t_{fs} + t_{fi}$$

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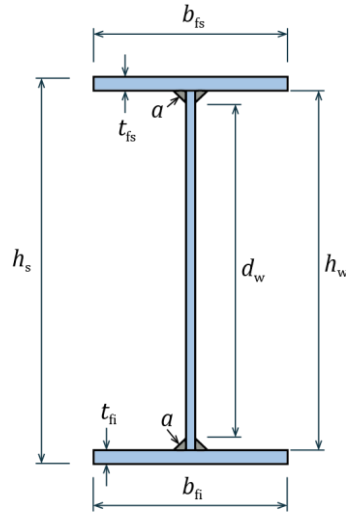


Figure 7 : Dimensions of a bi-symmetrical welded section

(2) The following scope of application will be considered for welded sections:

$$h_w / t_w \leq 1000 / 6$$

$$100 \leq h_w < 1500 \text{ mm}$$

$$100 \leq b_f < 400 \text{ mm}$$

$$3 \leq t_f \leq 100 \text{ mm and } 3 \leq t_w < 100 \text{ mm}$$

$$a \geq 3 \text{ mm}$$

2.2.5. Mono-symmetrical welded section

(1) For a mono-symmetrical welded section, the parameters of the section are shown in **Figure 8**. They fulfil the following formulas:

$$b_{fs} \neq b_{fi}$$

$$t_{fs} \neq t_{fi}$$

$$r_{cs} = r_{ci} = 0$$

$$b_p = t_p = 0$$

$$h_s = h_w + t_{fs} + t_{fi}$$

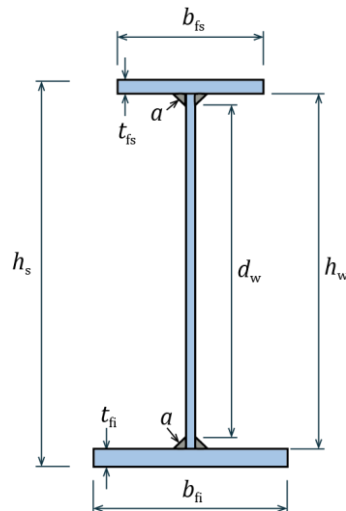


Figure 8 : Dimensions of a mono-symmetrical welded section

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- (2) The area of flanges are given by:

$$\begin{aligned} A_{fs} &= b_{fs} t_{fs} \\ A_{fi} &= b_{fi} t_{fi} \end{aligned} \quad (3)$$

For mono-symmetric sections in Normal mode, the following criterion should be fulfilled:

$$A_{fi} / A_{fs} \leq 3 \quad (4)$$

The upper limit can be modified in the options of the software, in Expert mode.

- (3) § 2.2.4 (2) applies.

2.2.6. SFB slim floor section

- (1) A SFB section comprises a hot rolled profile and a plate welded beneath the lower flange. The geometrical parameters of the section are shown in **Figure 9**. They fulfil the following formulas:

$$\begin{aligned} b_{fs} &= b_{fi} \\ t_{fs} &= t_{fi} \\ r_{cs} &= r_{ci} \\ h_s &= h_b + t_p \end{aligned}$$

- (2) In the software, hot rolled profiles are defined from the ArcelorMittal database.

Custom hot rolled sections can be defined in the Expert mode.

- (3) The following condition should be fulfilled:

$$\begin{aligned} b_p &\geq b_{fi} + 2 b_{app,min} + 2 t_{fi} && \text{(intermediate beams)} \\ b_p &\geq b_{fi} + b_{app,min} + t_{fi} && \text{(edge beams)} \\ t_p &\geq t_{p,inf,min} = 10 \text{ mm} \\ h_b &\leq h_{slim,max} \end{aligned} \quad (5)$$

where: $b_{app,min}$ is the minimum width of the slab support, with a default value of 50 mm;

$t_{p,inf,min}$ is the minimum thickness of the plate for slim floors, with a default value of 10 mm;

$h_{slim,max}$ is the maximal depth of a slim floor hot rolled profile, with a default value of 650 mm.



$b_{app,min}$, $t_{p,inf,min}$, $h_{slim,max}$ are parameters of the software. It will be possible to modify them in Expert mode.
The terms t_{fi} account for the spacing required for the flange-to-plate welds.

- (4) The embedded depth of the profile is given by:

$$h_{ec} = h_b$$

- (5) For edge beams, the configuration of the lower plate is unsymmetrical. The width b_p is accounted from the edge of the hot rolled profile. Any additional width potentially required for the welding is not accounted for in the software.

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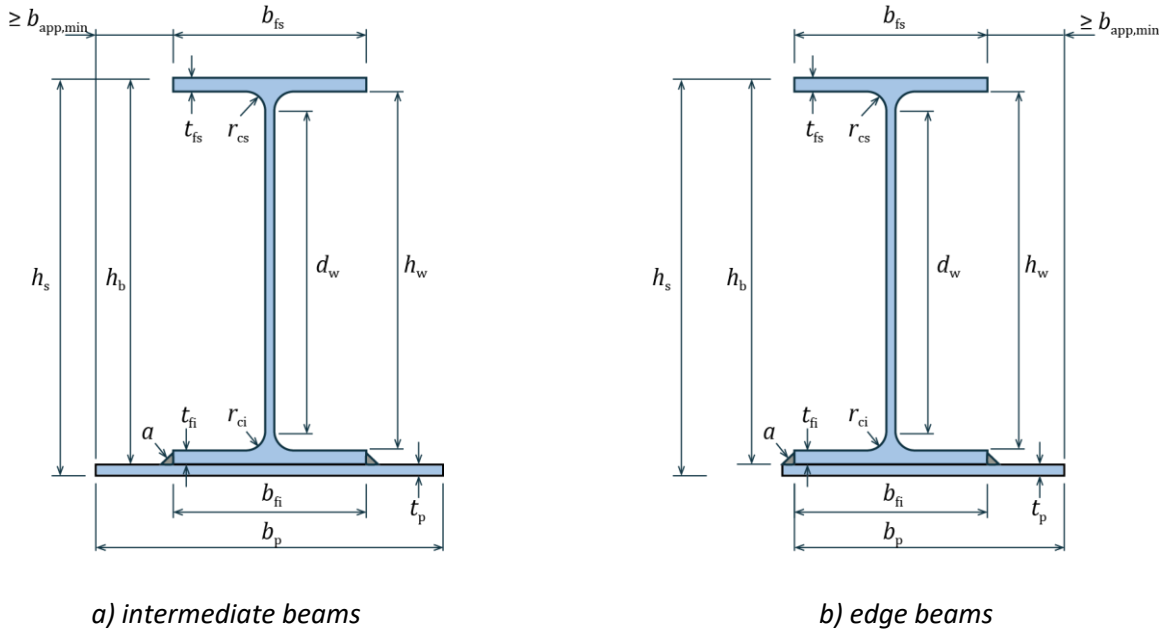


Figure 9 : Dimensions of SFB profile

2.2.7. IFB type A slim floor section

- (1) An IFB-A section comprises an upper part obtained from a hot-rolled profile that has been longitudinally cut in the web and that is welded to a lower plate. The lower plate is wider than the upper flange. The geometrical parameters of the section are shown in **Figure 10**. They fulfil the following formulas:

$$h_s = h_w + t_{fs} + t_p$$

$$r_{ci} = t_{fi} = b_{fi} = 0$$

$$h_s \leq h_b - t_{fs} - r_{cs} + t_p$$

- (2) In the software, hot rolled profiles are defined from the ArcelorMittal database. Custom hot-rolled sections may be defined in Expert mode.

- (3) The following conditions should be checked:

$$b_p \geq b_{fs} + 2 b_{app,min} \quad (\text{intermediate beams})$$

$$b_p \geq b_{fs} + b_{app,min} \quad (\text{edge beams})$$

$$t_p \geq t_{p,inf,min} = 10 \text{ mm}$$

$$h_s - t_p \leq h_{slim,max}$$

(6)

where: $b_{app,min}$, $h_{slim,max}$ and $t_{p,inf,min}$: see § 2.2.6 (3)

- (4) The embedded depth of the profile is given by:

$$h_{ec} = h_s - t_p$$

- (5) For edge beams, the configuration of the lower plate is unsymmetrical. The width b_p is accounted from the edge of the hot-rolled profile.

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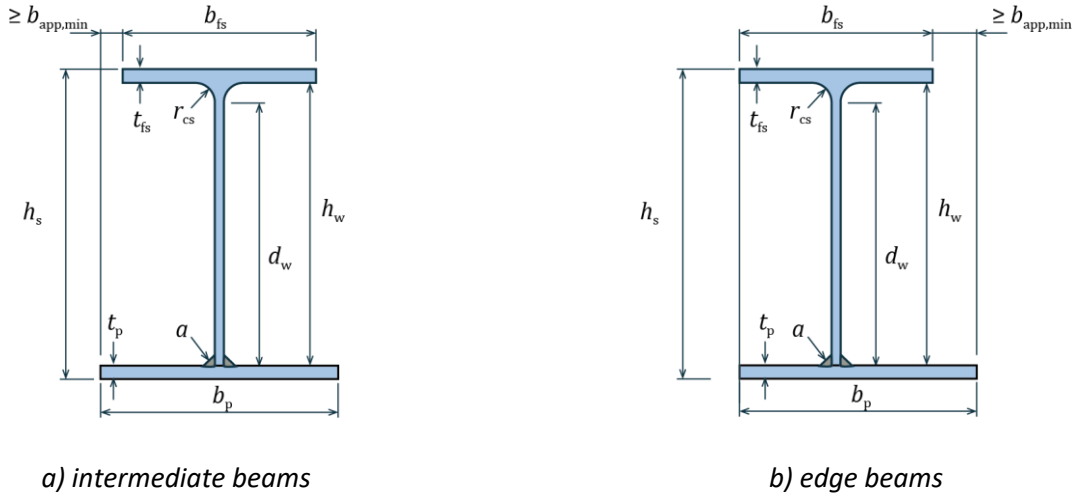


Figure 10 : Dimensions of IFB-A profile

2.2.8. IFB type B slim floor section

- (1) An IFB type B section comprises a lower part obtained from a hot-rolled profile that has been longitudinally cut in the web and that is welded to an upper plate. The lower flange is wider than the upper plate. The geometrical parameters of the section are shown in **Figure 11**. They fulfil the following formulas:

$$h_s = h_w + t_{fi} + t_p$$

$$r_{cs} = t_{fs} = b_{fs} = 0$$

$$h_s \leq h_b - t_{fs} - r_{cs} + t_p$$

- (2) In the software, hot-rolled profiles are defined from the ArcelorMittal database. Custom hot-rolled sections can be defined in Expert mode.
- (3) The following criteria should be checked:

$$b_{fi} \geq b_p + 2 b_{app,min}$$

$$b_p \geq 10 t_p$$

$$t_p \geq t_w$$

$$h_s - t_p \leq h_{slim,max}$$

(7)

where: $b_{app,min}$ and $h_{slim,max}$: see § 2.2.6 (3).

- (4) The embedded depth of the profile is given by:

$$h_{ec} = h_s - t_{fi}$$

- (5) For both edge and intermediate beams, the profile is symmetrical.

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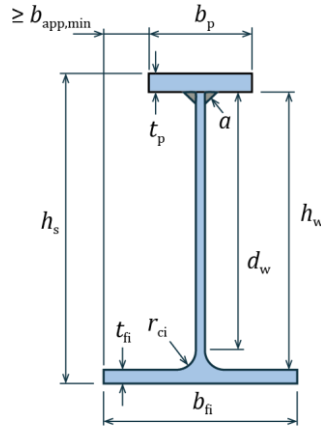


Figure 11 : Dimensions of IFB Type B profile

2.2.9. SAB slim floor section

- (1) A Steligence Asymmetric Beam (SAB) section is obtained by reducing the width of the upper flange of a hot rolled profile. The geometrical parameters of the section are shown in **Figure 12**. They fulfil the following formulas:

$$b_{fs} < b_{fi}$$

$$t_{fs} = t_{fi}$$

$$r_{cs} = r_{ci}$$

$$b_p = t_p = 0$$

$$h_s = h_b$$

- (2) In the software, hot-rolled profiles are defined from the ArcelorMittal database.

Custom hot-rolled sections can be defined in Expert mode.

- (3) The following conditions should be checked:

$$b_{fi} \geq b_{fs} + 2 b_{app,min} \quad (\text{intermediate beams})$$

$$b_{fi} \geq b_{fs} + b_{app,min} \quad (\text{edge beams})$$

$$b_{fs} \geq 10 t_{fs}$$

$$h_s - t_p \leq h_{slim,max}$$

(8)

where: $b_{app,min}$, $h_{slim,max}$: see § 2.2.6 (3)

- (4) The embedded height of the profile is given by:

$$h_{ec} = h_s - t_{fi}$$

- (5) For edge beams, the configuration of the upper flange is unsymmetrical. The width b_{fs} is accounted from the external side of the hot rolled profile (**Figure 12 b**).

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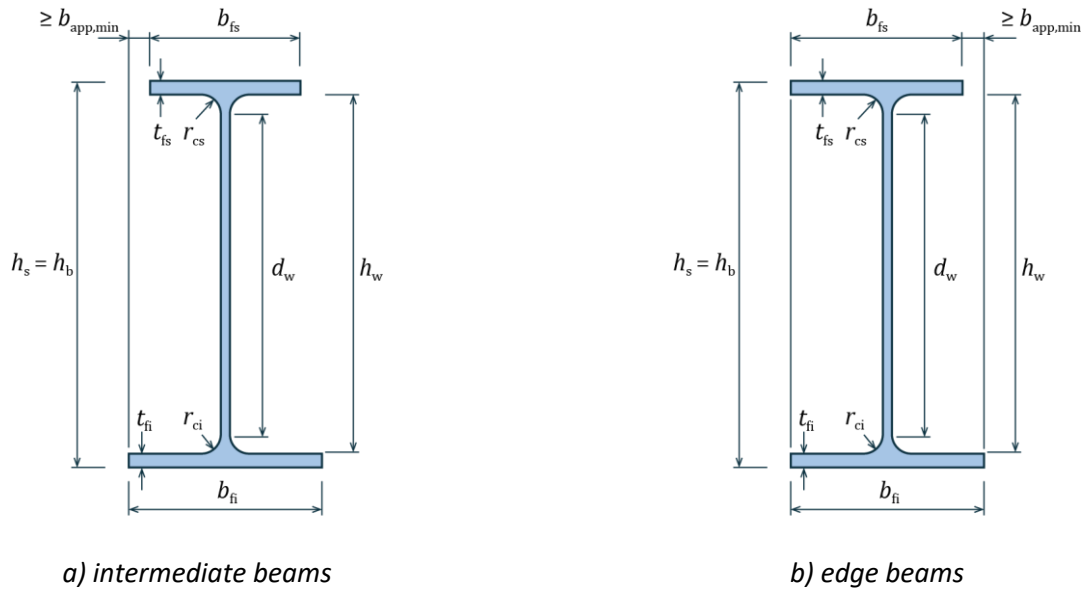


Figure 12 : Dimensions of SAB hot rolled profile

2.2.10. Additional requirements for slim floor sections

- (1) Slim floor sections are limited to the hot-rolled profiles from HE, IPE and HP series.

2.3. Concrete slab above steel profiles

2.3.1. Definition of the slab

- (1) For usual steel or composite sections, steel profiles are below the slab (cross-section from **Table 1** to **Table 4**) and three configurations of slabs are possible:
 - solid slabs;
 - solid slabs with precast elements;
 - composite slabs.

- (2) The parameters for a solid slab are as follows (**Figure 13**):

t_d total thickness of the slab;
 t_h thickness of the haunch (by default = 0).

The following requirement should be fulfilled:

$$t_{c,min} \leq t_d \quad (9)$$

where: $t_{c,min}$ is the minimum thickness of the active part of slab, with $t_{c,min} = 50$ mm.

$t_{c,min}$ is a parameter of the software. It will be possible to modify it in Expert mode.

- (3) Haunches are available only for solid slabs.

The width of the concrete haunch (b_h) will be assumed to be equal to the width of the upper flange ($b_h = b_{fs}$). For the resistance of the slab, the haunch will be neglected. For the calculation of the mass and for graphical purposes, the edges of the haunches will be assumed to have a slope defined by its angle to the vertical axis $\theta_h = 30^\circ$. The thickness of the haunch should fulfil the following criterion:

$$t_h \leq 0,5 t_d \quad (10)$$

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The value of θ_h is an option parameter of the software, that may be modified in Expert mode – see § 3.4.

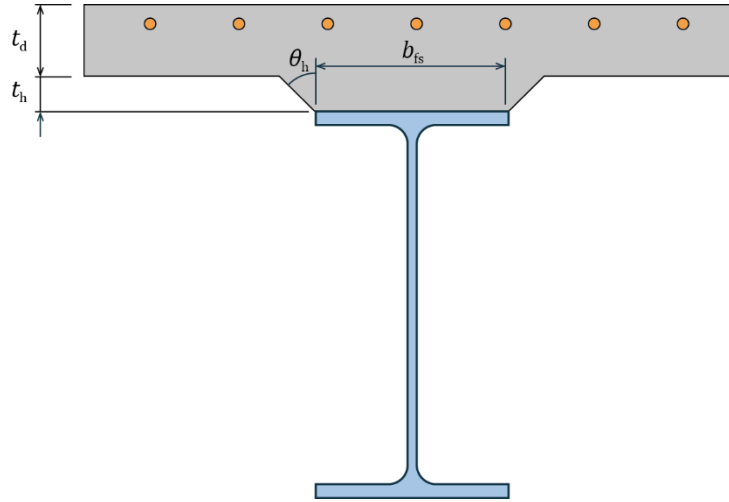


Figure 13 : Parameters for concrete solid slabs

- (4) For solid slabs with precast elements, it will not be possible to define a concrete haunch. The parameters for this configuration are as follows:

t_d total thickness of the slab;
 t_{pc} thickness of the precast elements;
 t_j thickness of the joint between precast elements that is filled with concrete.

The following requirements should be fulfilled:

$$\begin{aligned}
 t_{c,min} &\leq t_d \\
 10 \text{ mm} &\leq t_{pc} \leq t_d - 40 \text{ mm} \\
 0 &\leq t_j \leq t_{pc}
 \end{aligned}
 \tag{11}$$

- (5) A composite slab is constituted from a profiled steel sheeting, above which the concrete is poured. The parameters for this configuration of slabs are as follows:

t_d total thickness of the slab;
 orientation of the profiled steel sheeting, either longitudinal or transversal.

Additional parameters for the steel profiled sheeting should be defined according to § 2.4.

For composite slabs, the following condition should be checked:

$$t_c = t_d - h_p \geq t_{c,min} \tag{12}$$

where: t_c is the thickness of the slab above the profiled steel sheeting;

h_p is the height of the sheeting (ignoring the potential upper stiffener – see § 2.5.1);

$t_{c,min}$ see (2).

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2.3.2. Configurations of composite slabs at beam support

- (1) For composite slabs above steel profiles (steel or composite usual sections), the User will define the configuration at beam supports as given in this chapter.
- (2) The same configuration will be considered for all the beam, including the cases with multiple spans.
- (3) Composite beams with ribs perpendicular to the beam axis can have one of the three following configurations, to be defined by the User (see **Figure 14**):

- Configuration PE1:

The concrete rib and the steel sheeting are both continuous above the beam

With this configuration:

- the shear resistance of the connector is calculated with the reduction factor k_t according to § 5.7.2 (3);
- the steel profiled sheeting is taken into account for the resistance to longitudinal shear, according to § 8.5.7.

- Configuration PE2 (default configuration):

The concrete rib is continuous but the steel sheeting is not.

With this configuration:

- the shear resistance of the connector is calculated with the reduction factor k_t according to § 5.7.2 (3);
- the steel profiled sheeting is not taken into account for the resistance to longitudinal shear, see § 8.5.6;

- Configuration PE3:

The steel sheeting and the concrete rib are both interrupted at each end of the upper flange, so that a continuous parallel concrete flange is formed above the steel flange.

With this configuration:

- the shear resistance of the connector is calculated with the reduction factor k_ℓ according to § 5.7.2 (2), considering a representative width of the concrete flange equal to:

$$b_0 = b_{fs} - 2 b_{app,min}; \text{ (see § 2.2.6 (3) for the definition of } b_{app,min}\text{);}$$

- the steel profiled sheeting is not taken into account for the resistance to longitudinal shear;

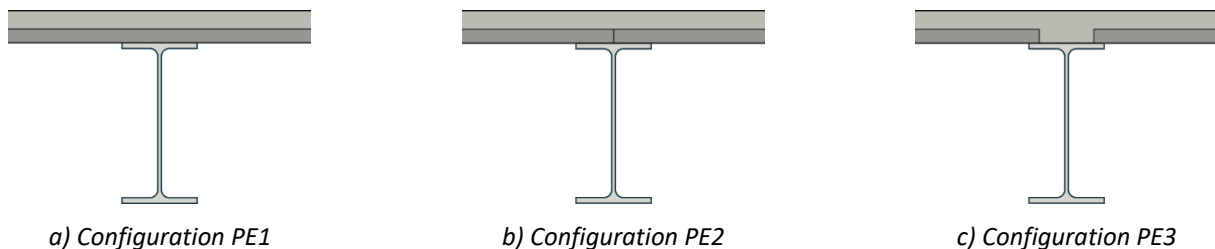


Figure 14 : Configurations of perpendicular steel sheeting across a beam

- (4) For composite beams with ribs parallel to the beam axis, the sheeting is always assumed to be continuous over the flange, so that the concrete rib above the flange has the same dimensions as other ribs (see **Figure 15**). In practice, if the sheeting is not continuous, the concrete rib above the flange has a width larger than the other ribs, which is more favorable for the shear resistance of the stud.

The shear resistance of the connector is calculated with the reduction factor k_ℓ according to § 5.7.2 (2) or § 5.7.3 (2).

- (5) **Annex A** gives the specific configuration of the composite slabs when Cofraplus 220 decking is used.

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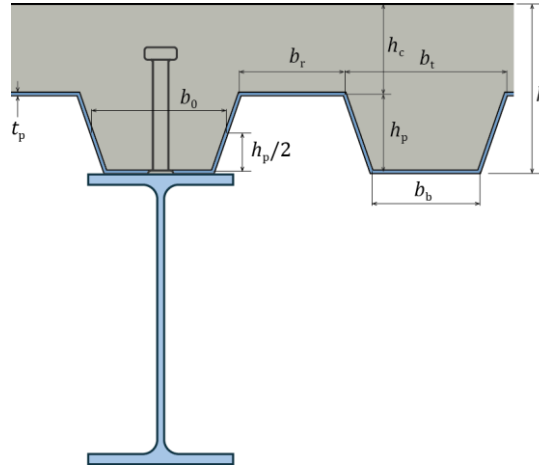


Figure 15 : Configuration of parallel steel sheeting above a beam

2.3.3. Definition of the longitudinal reinforcement in the slab

- (1) For beams with multiple spans (i.e. with cantilevers), longitudinal reinforcement of the slab should be defined by the User. By default, longitudinal reinforcements are taken into account for the stiffness and the resistance of the cross-section only when they are under tension. They may also be taken into account when in compression, see § 5.5.5 (1).
- (2) The longitudinal reinforcement consists of at least one layer of steel rebars. When the thickness of the slab is sufficient, a second layer may be defined. By default, one layer will be proposed.
- (3) A layer of steel rebars is defined by the following parameters (**Figure 16**):

e_{si} spacing of rebars axes;

d_{si} diameter of the rebars (to be selected among the following list: 6; 8, 10, 12, 14, 16, 20, 25 and 32 mm);

z_{si} location of the layer axis, measured from the upper side of the slab.

The location z_i of the rebars in the axes of the cross-section (see § 1.4) is given by:

$$z_i = t_h + t_d - z_{si}$$

These parameters should fulfil the following conditions:

$$\begin{aligned} 2 d_{si} &\leq e_{si} \leq e_{si,max} = 500 \text{ mm} \\ d_{si} &< z_{si} < t_c - d_{si} \end{aligned} \quad (13)$$

See § 5.2 for t_c ; See § 2.3.4 for the conditions on concrete cover.

- (4) For each layer, the number of rebars located in the effective width of the concrete flange, b_{eff} (see § 5.1), is calculated by:

$$n_{si} = b_{eff} / e_{si} \quad (n_{si} \text{ can be a real value in the calculations})$$

The area of steel rebars accounted for the bending moment resistance of the composite cross-section will be obtained by:

$$A_{si} = n_{si} \pi d_{si}^2 / 4 \quad (14)$$

- (5) The total area of the longitudinal reinforcement is given by:

$$A_s = \Sigma A_{si} \quad (15)$$

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The ratio of reinforcement of the slab will be assessed as follows:

$$\rho_s = \frac{A_s}{b_{\text{eff}} t_c} \quad (16)$$

See § 5.2 for t_c .

- (6) The class of reinforcing steel is to be chosen among the following list: B450, B500. In Expert mode, the User will be able to directly define the value of the yield strength f_{sk} (see § 2.10.6).

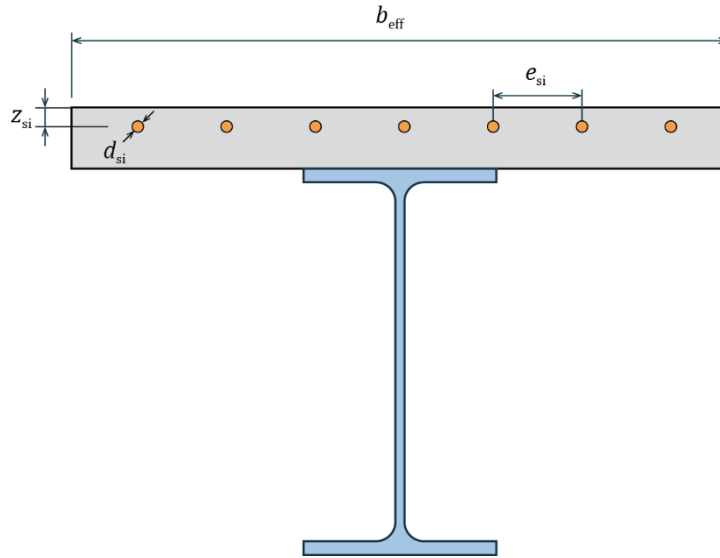


Figure 16 : Definition of a layer of steel reinforcement

2.3.4. Control of concrete cover

- (1) A control of the minimum concrete cover according to EN 1992, 4.4.1 will be performed. Only a warning message will be displayed if the conditions are not fulfilled. The control will not deal with environment conditions but only with bond requirement.
- (2) Any layer should fulfil the following conditions:

$$\frac{d_{si}}{2} + c_{\min} + \Delta c_{\text{dev}} \leq z_{si} \leq t_c - \left(\frac{d_{si}}{2} + c_{\min} + \Delta c_{\text{dev}} \right) \quad (17)$$

where: Δc_{dev} is the allowance for the deviation of rebars; $\Delta c_{\text{dev}} = 10$ mm by default, it can be modified in the software settings;

$c_{\min}$ is the minimum concrete cover, which will be calculated by:

$$c_{\min} = \max\{10 \text{ mm}; c_{\min,b}\}$$

$c_{\min,b}$ is the minimum concrete cover for bond requirement, given by: $c_{\min,b} = d_{si}$



Requirements for protection against corrosion are not covered.

2.3.5. Definition of the transverse reinforcement

- (1) For composite beams, the transverse reinforcement is automatically defined by the software, in order to resist the longitudinal shear – see § 8.5.

Transverse reinforcements are not required for the design of other configurations.

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- (2) The transverse reinforcement required to resist the transversal bending in the slab is not treated by the software.

2.3.6. Limitation of the concrete cracking

- (1) The User should define if the limitation of the concrete cracking should be verified or not (see § 11.4). In case of a positive option, the maximum crack width $w_{k,max}$ should be chosen among one of the following values: 0,2 mm, 0,3 mm and 0,4 mm.

2.4. Concrete slabs for slim floor beams

2.4.1. Definition of the slab

- (1) For slim floor section, either steel or composite, steel profiles are embedded in the concrete and four configurations of slabs are possible:

- solid slabs;
- solid slabs with prefabricated parts;
- composite slabs;
- prefabricated floors.

- (2) The parameter for a solid slab is as follows (**Figure 17**):

t_d total thickness of the slab.

It should fulfil the following requirements:

- for composite section with a stud welded on the upper flange (Group B in **Table 6**):

$$h_{ec} + h_{sc} + c_{min} + \Delta c_{dev} \leq t_d \quad (18)$$

- for all types of slim floor section, either steel or composite:

$$h_{ec} + t_{c,slim,min} \leq t_d \quad (19)$$

where: h_{ec} is the embedded depth of the steel profile (see § 2.2.6 to 2.2.9);

h_{sc} is the height of the shear stud (see § 2.5.1);

c_{min} and Δc_{dev} : see § 2.3.4;

$t_{c,slim,min}$ is the minimum thickness of concrete above the upper flange for slim floor sections. By default, $t_{c,slim,min} = 40$ mm. $t_{c,slim,min}$ is a parameter of the software that may be modified in Expert mode.

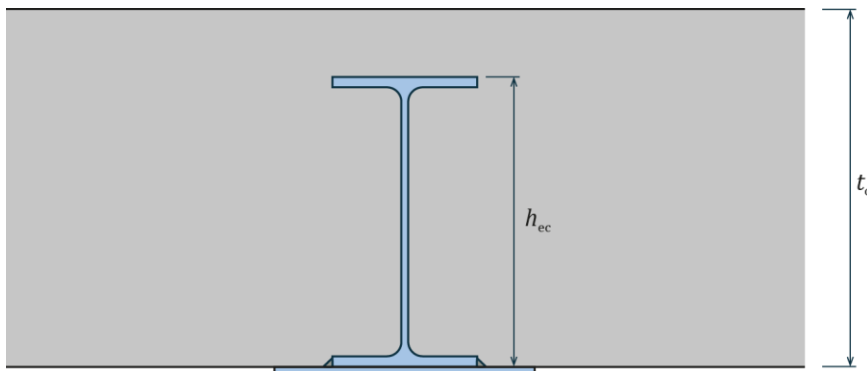


Figure 17: Slim floor section with solid slab

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(3) The parameters for a solid slab with precast elements are as follows (**Figure 18**):

- t_d total thickness of the slab;
- t_{pc} thickness of the precast elements;
- t_j thickness of the joint between precast elements that is filled with concrete.

It should fulfil the following requirement:

$$t_{pc} \leq h_{ec}$$

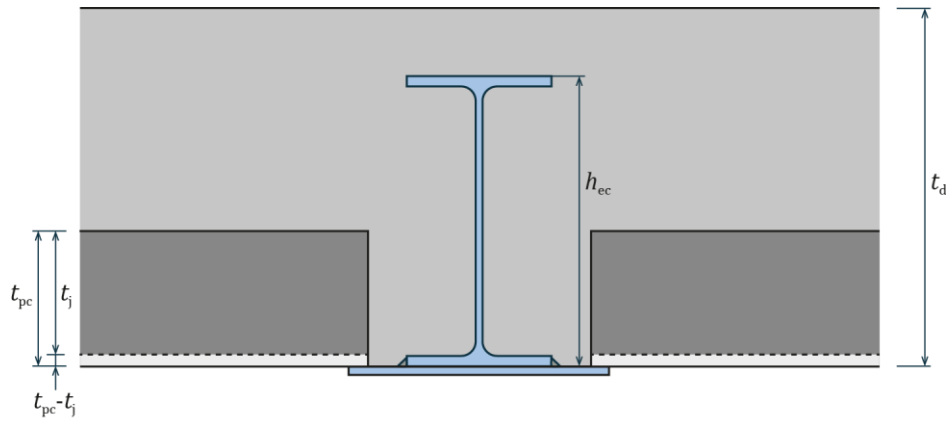


Figure 18: Slim floor section with solid slab and precast elements

(4) A composite slab is constituted by a steel profiled sheeting above which the concrete is poured. It is defined by the following parameter (**Figure 19**):

- t_d total thickness of the slab;

In slim floors, the ribs of the sheeting are always oriented perpendicularly to the beam axis.

The following requirement should be met:

$$h_p \leq h_{ec}$$

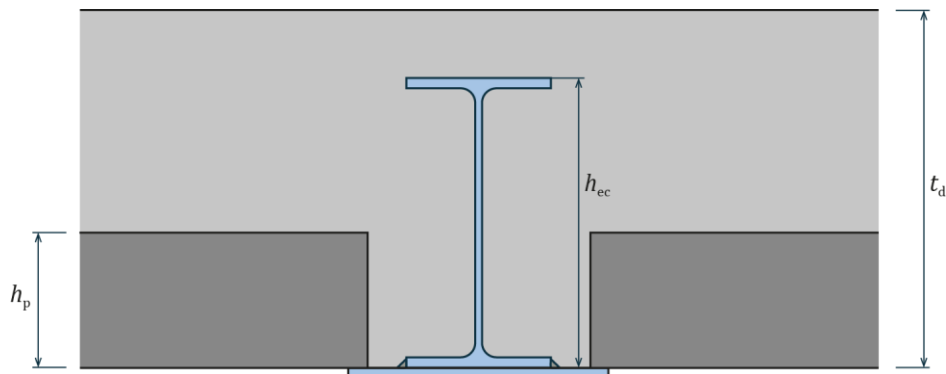


Figure 19: Slim floor section with composite slab

(5) Prefabricated floors are defined through a specific database, in which the following parameters describe the products:

- the name;
- the depth of the prefabricated elements d_p ;
- the surface weight μ_{pf} .

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The specific database will contain the Cofradal elements given in **Table 7**.

The following requirement should be met:

$$d_p \leq h_{ec}$$

Name	Depth [mm]	Surface weight [kN/m ²]
Cofradal 200 PAC	125	0,525
Cofradal 230 PAC	125	0,475
Cofradal 260 PAC	185	0,925
Cofradal 200 Prefab	200	2,4
Cofradal 230 Prefab	230	3,1
Cofradal 260 Prefab	260	2,8

Table 7: Cofradal prefabricated floor systems

2.4.2. Definition of the longitudinal reinforcement in the slab

- (1) The software will not take into account the presence of longitudinal reinforcement, even if bending moments develops at supports.

2.4.3. Definition of the transverse reinforcement

- (1) For slim floor composite beams, the transverse reinforcement is automatically defined by the software, in order to resist the longitudinal shear. This is applied only for studs welded on the upper flange.

2.4.4. Limitation of the concrete cracking

- (1) Limitation of concrete cracking is not carried out by the software for slim floors.

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2.5. Profiled steel sheeting

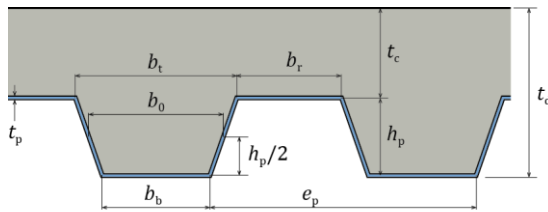
2.5.1. Geometry

- (1) The profiled steel sheeting will be defined either by access to a database or by direct definition of the parameters (custom sheeting).
- (2) The geometry of a profiled steel sheeting is defined by the following parameters (see **Figure 20**):

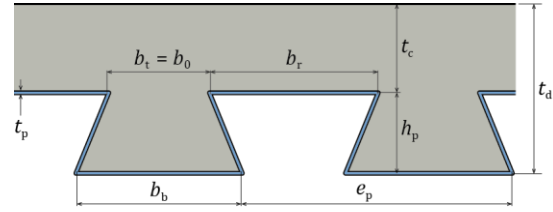
- t_p thickness of the sheet
- e_p spacing between the axes of two adjacent ribs
- h_p height of the sheet, ignoring any upper stiffener
- h_{pg} height of the sheet, including any upper stiffener
- b_b width of a rib at its bottom side
- b_t width of a rib at its upper side



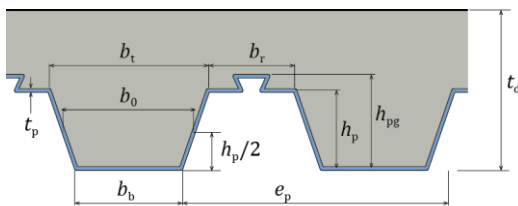
If $h_p = h_{pg}$, no upper stiffener is present.



a) Open trough



b) Re-entrant trough



c) Open trough and upper stiffener

Figure 20 : Dimensions of profiled steel sheeting

- (3) From the initial parameters in (2), additional ones are calculated as follows:

- b_r width of the sheet between two adjacent ribs: $b_r = e_p - b_t$
- b_m average width of a rib: $b_m = (b_t + b_b) / 2$
- b_0 representative width of a rib, obtained as follows:
 - for open troughs: $b_0 = b_m$ (width at mid height)
 - for re-entrant troughs: $b_0 = b_t$ (width at the top side)

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2.5.2. Other parameters

- (1) A profiled steel sheeting is also described by the following parameters:

μ_p surface mass of the sheet (in kg/m²);
 f_p yield strength of the steel of the sheet (in MPa);
 I_{pu} second moment of area per unit of width



I_{pu} is required only when the flexural stiffness of the deck will be used for the verification of the resistance of the steel profile to lateral torsional buckling in construction phase – see § 7.6.6 (2).

- (2) For usual composite cross-sections, with ribs perpendicular to the beam axis, the User should select one of the following options for the connection:

- prepunched steel sheeting (default choice);
- through deck welding of the connectors.

This choice is requested only for continuous ribs on the beam support (configuration PE1 and PE2 according to § 2.3.2 (3)).

2.6. Connection for usual composite beams

2.6.1. Welded studs

- (1) The parameters describing a welded stud are as follows (**Figure 21**):

h_{sc} height of the stud
 d diameter of the shank
 f_{usc} ultimate strength of the steel

The dimensions of a stud should fulfil the following condition:

$$h_{sc}/d \geq 3$$

- (2) All the studs used in a beam are identical.
 (3) It will be possible to define the studs from a database.
 (4) Studs should fulfil the following conditions for the minimum concrete coverage:

$$h_{sc} \leq t_d - 20 \text{ mm} \quad (20)$$

If this condition is not fulfilled, a warning message will be displayed. In any case, the condition $h_{sc} \leq t_d$ should be met.

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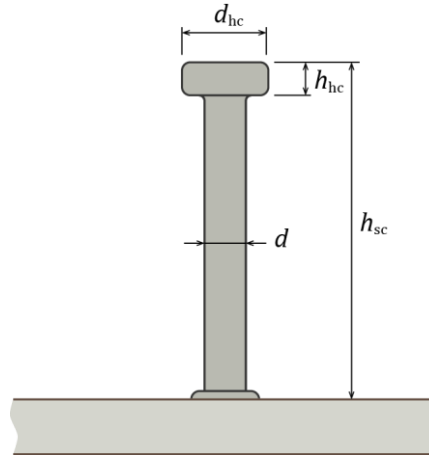


Figure 21 : Dimensions of a welded stud

2.6.2. Connection arrangement

- (1) Except in the case of an automatic process as described in § 8.4, the User will have to define the arrangement of the connection. The connection arrangement is continuous along each connection zone. 1 to 3 connection zones may be defined by the User.
 - (2) A connection arrangement on a connection zone (of index i) is defined by:
 - s_{xi} the longitudinal spacing between lines of connectors;
 - n_{ri} the number of connectors in each line.
 - (3) For composite slabs with perpendicular ribs, with configurations PE1 or PE2 according to § 2.3.2, the spacing s_{xi} is equal to a multiple of the spacing e_p between the axes of two adjacent ribs (see § 2.5.1). In all other cases, s_{xi} is to be defined by the User.
 - (4) The maximum number of connectors in each row, $n_{ri,max}$, where $n_{ri} \leq n_{ri,max}$, is defined as follows:
 - for all types of slabs, the following relation should be fulfilled:

$$(n_{ri} - 1) s_{yi,min} + d + 2 e_{D,min} \leq b_{fs}$$
 - for all composite slabs with perpendicular ribs and a PE3 arrangement at beam support, the criterion is as follows:

$$(n_{ri} - 1) s_{yi,min} + d + 2 e_{D,min} \leq b_{fs} - 2 b_{app,min}$$
 - $n_{ri,max} = 2$ for composite slabs with perpendicular ribs using PE1 ou PE2 configurations;
- where: $b_{app,min}$: see § 2.2.6 (3);
- $s_{yi,min}$ is the minimum transverse spacing between two studs, see § 2.6.3 (3);
- $e_{D,min}$ is the minimum distance of a connector to the edge of the flange, see § 2.6.3 (3).

2.6.3. Detailing of connectors

- (1) According to EN 1994-1-1, § 6.6.5.7, the following rules in (2) to (4) will be used in the software for the spacing of headed studs.
- (2) The longitudinal spacing s_{xi} between two lines of connectors should fulfil the following criteria:

$$s_{xi,min} = 5 d \leq s_{xi} \leq s_{xi,max} = \min\{800 \text{ mm} ; 6 t_d\} \quad (21)$$

where d is the diameter of the headed stud (see § 2.6.1)

t_d is the total thickness of the slab (see § 2.3.1).

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- (3) The transverse spacing s_{yi} between two adjacent connectors should fulfil the following criterion:

$$s_{yi,min} \leq s_{yi} \quad (22)$$

where: $s_{yi,min} = 2,5 d$ for solid slabs;

$s_{yi,min} = 4 d$ for composite slabs.

- (4) The distance of a connector to the edge of the flange should be less than $e_{D,min} = 20$ mm (acc. to EN 1994-1-1, 6.6.5.6 (2)).
- (5) According to EN 1994-1-1, 6.6.5.7 (5), the following criterion will be checked for usual composite beams, when the number of connectors per line, n_r , is equal or greater than 2:

$$d \leq 2,5 t_{fs} \quad (23)$$



when $n_r = 1$, it is assumed that the studs are in the axis of the web.

2.7. Connection for composite slim floors

2.7.1. Dowels

- (1) The connection by dowels is defined by the following parameters (**Figure 22**):

d_s : diameter of the rebar

d_{hs} : diameter of the holes in the web

a_{hv} : distance of the holes centres to the upper flange

s_x : longitudinal spacing of holes along the beam

L_s : length of the rebars

- (2) The following conditions should be fulfilled:

$$s_x \leq s_{x,slimmin} = 125 \text{ mm}$$

$$25 \text{ mm} \leq d_{hs} \leq 40 \text{ mm}$$

$$a_{hv} \geq r_{cs} + d_s / 2$$

(24)



$s_{x,slimmin}$ is a parameter of the software that may be modified in Expert mode.

- (3) Connection by dowels is available only for profiles where $t_w \geq t_{w,cdmin} = 7,5$ mm.



$t_{w,cdmin}$ is a parameter of the software that may be modified in Expert mode.

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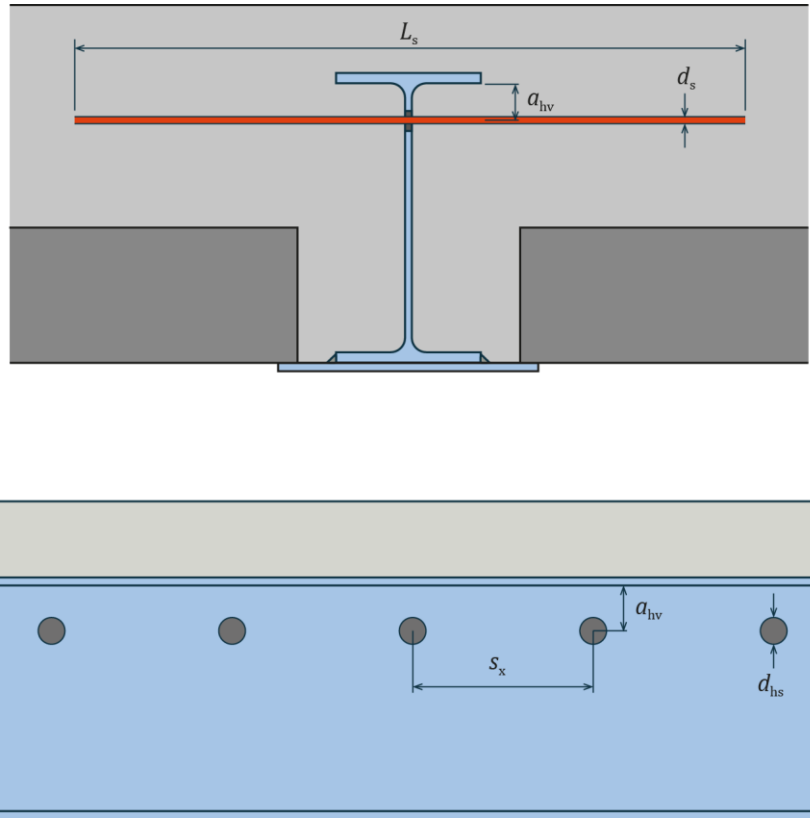


Figure 22: Composite dowel shear connection

2.7.2. Studs welded on the web

- (1) For slim floor composite beams with group C connectors (see **Table 6**), the connection is defined by:
 - the connector, with the parameters as defined in § 2.6.1;
 - the horizontal spacing s_x between two rows of connectors.
- (2) A row of studs always consists of two connectors, one on each side of the web.
- (3) The spacing s_x between rows of connectors should fulfill the following conditions, according to EN 1994-1-1, 6.6.5.6 and 6.6.5.7:

$$5 d \leq s_x \leq \min\{6 t_d; 800 \text{ mm}\} \quad (25)$$

- (4) Only one connection pattern is defined for all the beam.

2.7.3. Studs welded on the upper flange

- (1) For slim floor composite beams with group B connectors (see **Table 6**), the connection is defined as in § 2.6, considering only one connection zone.

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2.8. Compatibility of the connectors and the profiled steel sheeting

2.8.1. First generation of Eurocodes

- (1) The following criteria apply to usual composite sections.
- (2) For composite slab with transverse ribs and with the configurations PE1 or PE2 (see § 2.3.2), the following criteria should be fulfilled (see EN 1994-1-1, 6.6.4.2):
 - the height of the steel profiled sheeting should not be greater than 85 mm: $h_p \leq 85 \text{ mm}$;
 - the representative width of the rib, b_0 , is at least equal to h_p : $b_0 \geq h_p$;
 - for through deck welded studs (see § 2.5.2), the diameter of the stud should not be greater than 20 mm: $d \leq 20 \text{ mm}$;
 - for prepunched sheets, the diameter of the stud should not be greater than 22 mm: $d \leq 22 \text{ mm}$.
- (3) For all composite slabs, the height of the stud above the steel profiled sheeting should not be lower than twice the diameter of the stud (see EN 1994-1-1, 6.6.5.8 (1)): $(h_{sc} - h_p) \geq 2 d$.
- (4) For all composite slabs, the minimum width of the ribs should not be lower than 50 mm (see EN 1994-1-1, 6.6.5.8 (2)): $\min\{b_t; b_b\} \geq 50 \text{ mm}$

2.8.2. Second generation Eurocodes

- (1) § 2.8.1 applies except the following criteria.
- (2) For composite slab with transverse ribs and with the configurations PE1 or PE2 (see § 2.3.2), the following conditions should be fulfilled (see prEN 1994-1-1, 8.6.9.2):
 - the height of the steel profiled sheeting should not be greater than 105 mm: $h_p \leq 105 \text{ mm}$;
 - minimum thickness of sheeting: $t_p \geq 0,7 \text{ mm}$.

2.9. Partial concrete encasement

- (1) Partial concrete encasement of a steel profile is available for bisymmetrical steel profiles, either welded or hot rolled, i.e. for sections of type SHR-E, SWB-E, CHR-E and CWB-E (see § 2.2.2).
- (2) The partial concrete encasement of a steel profile should be defined by the parameters as described in (3) to (12).
- (3) The total width of the concrete encasement is denoted b_c and should fulfil the following criterion:

$$0,8 b_{fs} \leq b_c \leq b_{fs} = b_{fi} \quad (26)$$

- (4) The concrete of the encasement is defined by its concrete class (acc. to Table 9) and by the mass of concrete per unit of volume, ρ_{ce} (in kg/m^3 , default value of 2500 kg/m^3). The characteristic value of the compression strength of the concrete for the encasement is denoted f_{cek} (see § 2.10.5). Only normal concretes will be available for partial encasements.



Concrete for the encasement and concrete for the slab are defined separately and may be different.

- (5) The reinforcement of the encasement is defined by the stirrup, the longitudinal rebars and the concrete cover.
- (6) The diameter of the stirrups d_{est} is chosen among the following list: 6; 8, 10, 12, 14 and 16 mm. The location of the stirrup is defined by the concrete cover to the external side, c_{est} , and the concrete cover to the internal side of the flanges, c_{fst} . These concrete coverings are defined by the User.

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2.10. Material properties

2.10.1. Structural steel for hot rolled profiles

- (1) Structural steel is defined by its grade. For hot rolled profiles, the available steel grades are defined according to the ArcelorMittal database (see § 4.1). The latter includes at least the list of usual steel grades as given in **Table 8**.
- (2) In the CTICM and in the ArcelorMittal versions, all steel grades will be available.

S235	S275	S355	S420	S460
------	------	------	------	------

Table 8: Usual steel grades

- (3) In expert mode, it will be possible to replace the definition of a steel grade by the definition of its yield strength and its ultimate tensile strength.
- (4) The software will accept higher steel grades in the database. For composite sections using profiles with such high strength steels, plastic design of the beam will not be available and the elastic design will be imposed, i.e. the sections will be treated as Class 3 cross-sections.
- (5) The functions that give the nominal yield strength f_y from the steel grade and the thickness will be included in the database.
For the ArcelorMittal version, in Normal mode, only the functions from EN 10025 will be proposed.
For the CTICM version, the simplified functions from Table 3.1 of EN 1993-1-1 will also be available.
- (6) For hot-rolled profiles, one common nominal yield strength f_y will be considered uniformly for the whole profile. It will be assessed from the greatest value of the flange thicknesses and the web thickness.

$$f_{yfs} = f_{yfi} = f_{yw} = f_y \quad (32)$$

where f_{yfs} , f_{yfi} and f_{yw} are the yield strength associated to the upper flange, to the lower flange and to the web, respectively.

- (7) The design yield strength of the structural steel is obtained by:

$$f_{yd} = f_y / \gamma_{M0} \quad (33)$$

2.10.2. Structural steel for welded sections

- (1) For welded sections, steel grade will be selected among the list of steels in the steel database.
- (2) The nominal yield strength f_y will be assessed separately from the steel grade, using the thickness of the upper flange, the lower flange and the web. Associated parameters will be: f_{yfs} , f_{yfi} and f_{yw} .
- (3) For the ArcelorMittal version, in Normal mode, only the functions from EN 10025 will be proposed.
For the CTICM version, the simplified functions from Table 3.1 of EN 1993-1-1 will also be available, as long as the thickness of the plates does not exceed 80 mm.

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2.10.3. Structural steel for plates

- (1) For steel plates used in slim floor sections, the steel grade of the plate is not defined from the database but directly from the list in the steel database.
- (2) The nominal yield strength f_{ysp} will be assessed uniformly from the steel grade, using the thickness of the plate, according to EN 10025 or to Table 3 of EN 1993-1-1.
- (3) The design yield strength of the steel plate is obtained by:

$$f_{yspd} = f_{ysp} / \gamma_{M0} \quad (34)$$

2.10.4. Other structural steel properties

- (1) According to EN 1993-1-1 [4], the following common properties will be considered for all the types of structural steel:

E_a Young's modulus: $E_a = 210\,000\text{ MPa}$;
 ν_a Poisson factor: $\nu_a = 0,3$;
 G_a shear modulus: $G_a = 80\,770\text{ MPa}$;
 ρ_a density: $\rho_a = 7\,850\text{ kg/m}^3$.

- (2) The steel parameters ε_{fs} , ε_{fi} , ε_w and ε_p , for respectively the upper flange, the lower flange, the web and the steel plate, will be obtained as follows, where f_y is in MPa:

$$\varepsilon = \sqrt{235 / f_y} \quad (35)$$

2.10.5. Concrete

- (1) Concrete is defined by:
 - its type - see (2);
 - its density - see (3);
 - its class - see (4).

- (2) Two types of concrete are possible:

- Normal weight concrete (NWC)
- Light weight concrete (LWC)



Light weight concrete is not available for partial encasements.

- (3) The concrete density is denoted ρ_c . It should fulfil the following conditions:
 - normal concrete: $2\,000\text{ kg/m}^3 \leq \rho_c$
 - light concrete: $800 \leq \rho_c \leq 2\,200\text{ kg/m}^3$
- (4) The concrete is defined through its class, to be chosen among the list given in **Table 9** for NWC and in **Table 10** for LWC. When the prEN 1994-1-1 standard will be used (next generation), concrete classes of **Table 11** will also be proposed.

For slim floors composite sections, light weight concrete will not be available. Additionally, the use of C20/25 concrete class will not be possible with composite dowels connection.

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Concrete class	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
f_{ck} (MPa)	20	25	30	35	40	45	50

Table 9: Classes of normal weight concrete

Concrete class	LC20/22	LC25/28	LC30/33	LC35/38	LC40/44	LC45/50	LC50/55
f_{ck} (MPa)	20	25	30	35	40	45	50

Table 10: Classes of light weight concrete

	NWC	LWC
Concrete class	C70/85	LC60/66
f_{ck} (MPa)	70	60

Table 11: Additional classes of concrete (next generation)

- (5) The mechanical properties for concrete, given in Table 3.1 of EN 1992-1-1 [3] for the normal weight concrete or in Table 11.3.1 of EN 1992-1-1 for the light weight concrete, are defined by:

f_{ck} is the characteristic value of the compression strength of the concrete “cylinder compressive test at 28 days”. Its value is obtained from the concrete class, see **Table 9**, **Table 10** and **Table 11**.

f_{cd} is the design compressive strength of the concrete, calculated by:

$$f_{cd} = f_{ck} / \gamma_c \quad (36)$$

E_{cm} is the mean secant modulus of elasticity of the concrete for short term loading. For normal concrete, its value is obtained by the following formula:

$$E_{cm} = 22 [f_{cm} / 10]^{0.3} \text{ with } f_{cm} \text{ in MPa} \quad (37)$$

For light weight concrete LWC, the mean secant modulus of elasticity is calculated by using the following formulae:

$$E_{lcm} = E_{cm} \eta_E \quad (38)$$

where: $\eta_E = (\rho_c / 2200)^2$ and E_{cm} is obtained from Formula (37).

f_{cm} is the mean value of concrete cylinder compressive strength. For normal concrete, its value is obtained by the following formula:

$$f_{cm} = f_{ck} + 8 \text{ MPa} \quad (39)$$

ρ_c is the concrete density (in kg/m³)

f_{ctm} is the mean tensile strength of the concrete, given by:

$$f_{ctm} = 0,3 [f_{ck}]^{2/3} \quad (40)$$

For light weight concrete LWC, the mean tensile strength is calculated by using the following formula:

$$f_{lctm} = f_{ctm} \eta_1 \quad (41)$$

where: $\eta_1 = 0,4 + 0,6 (\rho_c / 2200)$ and f_{ctm} is obtained from Formula (52)).

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- (6) The development of creep in concrete depends on the relative humidity RH , which will be defined by the User, from the two predefined values of 50% (interior of a building with controlled temperature) or 80% (exterior atmosphere). Additionally, it will be possible in Expert Mode to enter a custom value, between 0% and 100%. The default value will be 50%.
- (7) The deformation for shrinkage ε_{sh} has to be defined. The default value is according to § 3.2 (7).
- (8) For sections with partial concrete encasement, a specific class for the concrete between the flanges is selected. The characteristic value of the compressive strength of this concrete is denoted f_{cek} and all previous formulas for concrete are applied.
- (9) For the next generation of Eurocodes, the properties of concrete will be obtained as follows:

f_{ck} : see (5);

f_{cd} : (according to prEN 1994-1-1, 5.1 (3)):

$$f_{cd} = \eta_{cc} k_{tc} f_{ck} / \gamma_c \quad (42)$$

where: k_{tc} is a factor considering the effect of high sustained loads and of time of loading on concrete compressive strength. See § 2.12.4;

η_{cc} is given by:

$$\eta_{cc} = (40 / f_{ck})^{1/3} \leq \eta_{cc,up} \quad \text{for NWC} \quad (43)$$

$$\eta_{cc} = (\eta_{lw,fc} 40 / f_{ck})^{1/3} \leq \eta_{cc,up} \quad \text{for LWC}$$

$\eta_{cc,up} = 0,85$ for plastic resistance of sections, $\eta_{cc,up} = 1$ otherwise.

E_{cm} :

$$E_{cm} = k_E [f_{cm}]^{1/3} \quad \text{with } f_{cm} \text{ in MPa} \quad (44)$$

f_{cm} : see (5);

f_{ctm} : see (5) if $f_{ck} \leq 50$ MPa; if $f_{ck} > 50$ MPa:

$$f_{ctm} = 1,1 [f_{ck}]^{1/3} \quad (45)$$

$k_E = 9\,500$ for concrete with quartzite aggregate. k_E as a parameter in the option of the software.

- (10) For the next generation of Eurocodes, the class of cement will be defined by the User among the following list:

class CS;

class CN;

class CR.

See § 5.3.3 for the use of this parameter.

2.10.6. Steel for reinforcements

- (1) The yield strength of rebars is denoted f_{sk} for the slab reinforcement and f_{sek} for the partial concrete encasement reinforcement.
- (2) The yield strength is 450 MPa for class B 450 and 500 MPa for class B 500.
- (3) Custom values may be defined by the User in Expert mode.
- (4) The Young's modulus of steel rebars is E_s . By default, according to EN 1994-1-1, 3.2 (2), $E_s = E_a$. See § 3.2.1 (6).

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- (5) The design yield strength of reinforcement is obtained by:

$$\begin{aligned} f_{sd} &= f_{sk}/\gamma_s \\ f_{sed} &= f_{sek}/\gamma_s \end{aligned} \quad (46)$$

2.11. Lateral restraint of the beam

2.11.1. General

- (1) Lateral restraints may be defined for usual steel beams, with or without partial **concrete** encasement. Lateral restraints may also be defined for usual composite beams, with or without partial **concrete** encasement, to deal with the construction stage only.
- (2) Lateral restraints may consist of a continuous restraint of one or both flanges. It may also be constituted by a set of cross-sections with point lateral restraints (see § 2.11.2).
- (3) For usual composite beams in construction stage, it is assumed that a specific layout of lateral restraints may be defined. An option will allow to consider the same restraint as in final stage.
For the construction stage, if the slab is composite with perpendicular ribs, it will be possible to activate the profiled steel sheeting as a contribution to the lateral restraint of the beam (see § 2.11.3).

2.11.2. Point lateral restraints

- (1) Point lateral restraints are defined as follows, for each span:
 n_{slr} number of sections with lateral restraint;
 $x_{slr,i}$ location of the cross-section with lateral restraints, from the left end of the span;
 $typ_{slr,i}$ type of restraint: both flanges, upper flange or lower flange (by default, both flanges)
- (2) By default, no intermediate lateral restraint is present.
- (3) Cross-sections at supports are always assumed to be laterally restrained, for both flanges.
- (4) No lateral restraints are assumed by default at the cross-sections with point props.

2.11.3. Lateral restraints of steel beams by the steel profiled sheeting

- (1) For a beam with usual composite cross-section, with or without partial **concrete** encasement, the User can choose to provide a restraint against lateral torsional buckling by the stabilising action of the profiled steel sheeting. This option is available only when the slab is composite and when the ribs are perpendicular to the beam axis. It is used only for the construction stage for unpropped or point propped configurations (see § 2.11.3).
- (2) When the option is activated, the following parameters should be defined by the User (**Figure 24**):
 b_p width of a single sheet
 n_t total number of sheets of the deck in the direction perpendicular to the beam;
 m number of spans crossed by an individual sheet, chosen in the list: 1, 2, 3, 4, 5;
 t_{pr} thickness of the coating applied to the sheets (see § 7.6.8 (9)).

When $n_t > 1$, the User should define the type of transition between sheets at beam support, among one of the following options (**Figure 25**):

- overlapping of sheets;

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- sheets with a gap;
- adjacent sheets without a gap.

This choice is to be done only for the configuration PE1 according to **Figure 14**. For the two other configurations, the automatic following association will be done:

- configuration PE2 => adjacent sheets without a gap;
- configuration PE3 => sheets with a gap.

Moreover, for a configuration PE1, the value $n_t = 1$ is not consistent.

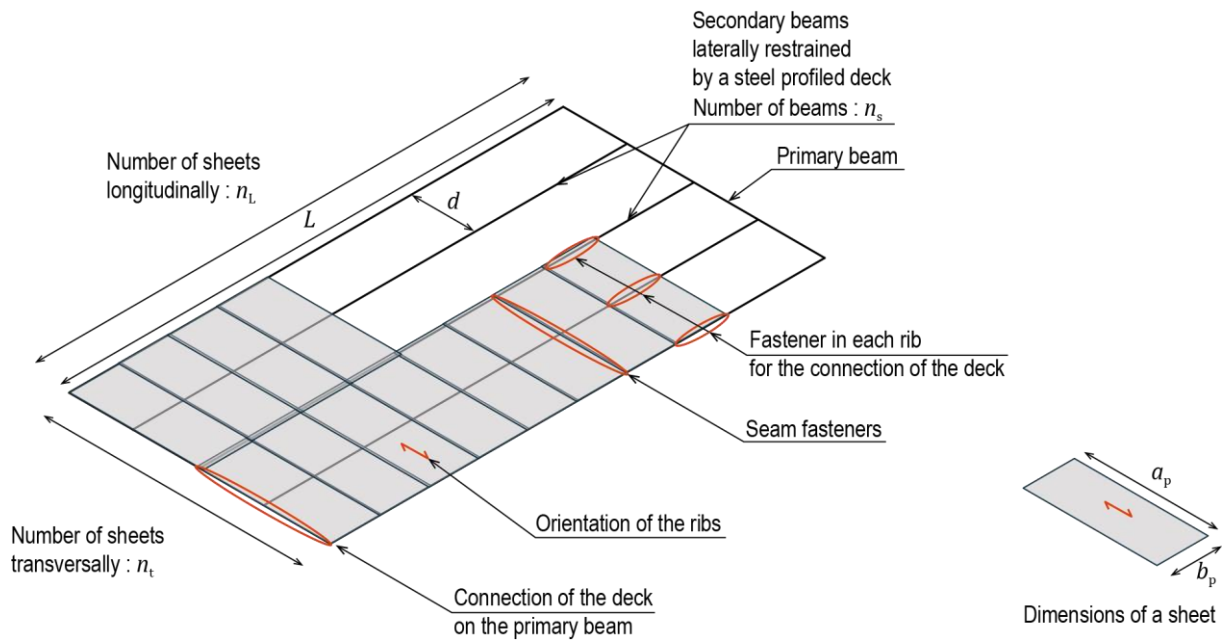


Figure 24 : Definition of the parameter of the steel decking

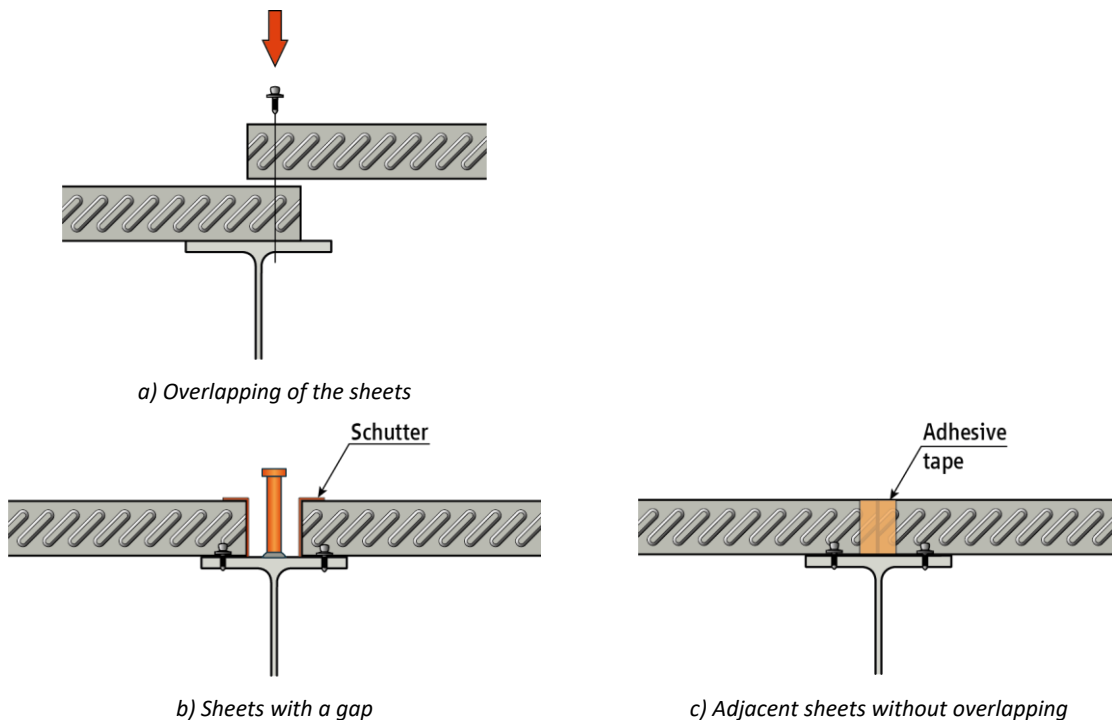


Figure 25: Transition between sheets at a beam support

- (3) The method applied to take into account the effect of the profiled steel sheeting on the lateral torsional buckling of the beam (see **Annex E**) assumes that the steel deck connects together several regularly spaced secondary beams. The constant spacing of these beams is denoted d .

The distance d will be obtained as follows:

- for edge beams (see § 2.1): $d = d_2$;
- for intermediate beams: $d = (d_1 + d_2) / 2$

If for intermediate beams $|d_1 - d_2| > 0,25 \max\{d_1 ; d_2\}$, the method is theoretically not valid. A warning message will be displayed to notify the User that the floor is outside the scope of application of the method.

- (4) Parameters for the fastening of the sheeting to the beams are as follows. The User should define if the sheets are fastened on the beams by one fastener in each rib or one fastener every 2 ribs.

e_L is the longitudinal spacing between these fasteners (**Figure 26**):

$e_L = e_p$ if there is one fastener in each rib;

$e_L = 2 e_p$ if there is one fastener every two ribs.

The type of fasteners and their diameter should be chosen according to **Table 12**.

Fastener	Type	Diameter d_{fb} (mm)	s_p (mm/kN)
Self-drilling screws	Normal washer	from 5,5 to 6,3	0,15
	Neoprene washer	from 5,5 to 6,3	0,35
Fired pins	washer+pin	from 3,7 to 4,8	0,10

Table 12: Definition of the beam-to-sheeting fasteners

- (5) Parameters for the fastening of the sheeting between adjacent sheets are as follows. The User should define e_c , the spacing of seam fasteners used for the connection between two adjacent sheets (**Figure 26**).

e_c should fulfil the following condition: $e_c \leq e_{c,max} = 500$ mm.

The type of fasteners and their diameter should be chosen according to **Table 13**.

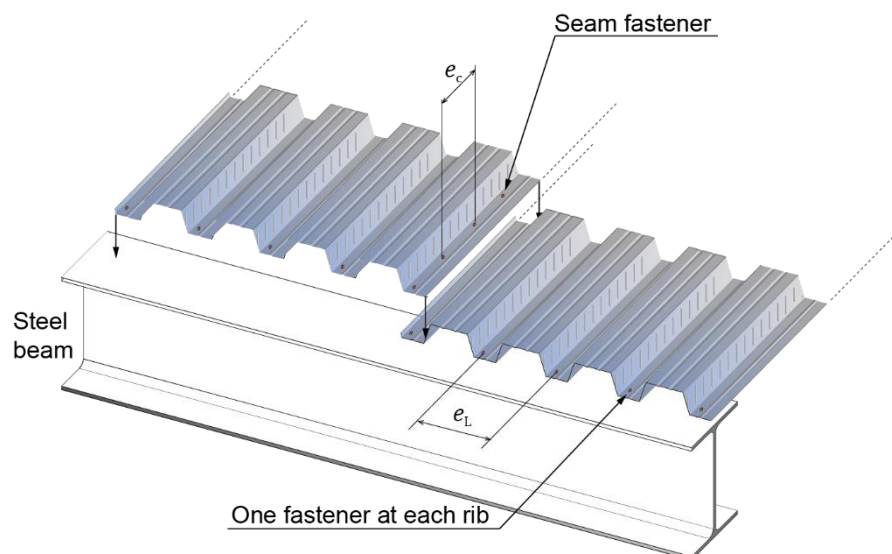


Figure 26: Sheet/girder fasteners and seam fasteners

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Fastener	Diameter d_{fs} (mm)	s_s (mm/kN)
Self-drilling screws	4,1 à 4,8	0,25
Rivet	4,8	0,30

Table 13: Definition of the seam fasteners

- (6) The following additional parameters will be calculated from the values defined by the User:
- n_L is the number of sheets in the longitudinal direction, given by: $n_L = L / b_p$;
- n_c is the number of seam fasteners along the deck span, excluding those which are fixed to the beams. It may be obtained by: $n_c = (d / e_c) - 1$;
- n_f is the number of fasteners on the beam per individual sheet, which is given by:
- $n_f = b_p / e_p$ if there is one fastener in each rib;
- $n_f = b_p / (2 e_p)$ if there is one fastener every two ribs.
- (7) As a calculation option, the User can choose to take into account only the shear stiffness of the deck or the shear stiffness and the rotation restraint of the beam.
- In the latter case, the second moment of area per unit of length, I_{pu} , should be defined.



As far as possible, it will be seen if it is possible to include this parameter in the database for steel profiled sheetings – see § 2.5.2.

2.12. Nationally Determined Parameters

2.12.1. General

- (1) For each composite beam analysed with the software, the User will have the possibility to define the nationally determined parameters considered in the calculations. The default values will be the values defined in the options of the software – see § 3.2.

2.12.2. Partial factors

- (1) The following partial factors on actions for ULS combinations will be used:
- $\gamma_{G,sup}, \gamma_{G,inf}$ for permanent actions;
- γ_Q for variable actions;
- (2) The partial factors for design resistances in normal situations are:
- γ_{M0} for the section resistance of the structural steel;
- γ_{M1} for the element resistance of the structural steel;
- γ_{M2} for the element resistance of the welds;
- γ_C for the compression resistance of the concrete;
- γ_V for the resistance of shear connectors;
- γ_s for the resistance of reinforcement steel bars;
- γ_p for the resistance of the profiled steel sheeting.
- (3) For the resistance of shear connectors, it will be possible to define the use of one common partial factor γ_V in both formulas (109) and (110) or one separate value for each formula, denoted respectively γ_{Vs} and γ_{Vc} .

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(4) In fire situations, the following partial factors on resistances are considered:

$\gamma_{M,fi}$	for the resistance of structural steel (steel beams);
$\gamma_{M,fi,a}$	for the resistance of structural steel (composite beams);
$\gamma_{M,fi,c}$	for the compression resistance of the concrete;
$\gamma_{M,fi,V}$	for the resistance of shear connectors;
$\gamma_{M,fi,s}$	for the resistance of reinforcement steel bars.

2.12.3. Combination factors

(1) For each live load case Q_1 and Q_2 , the three combination factors ψ_0 , ψ_1 and ψ_2 are to be defined.

2.12.4. Other factors

- (1) The parameter η used for the shear resistance of steel profiles will be defined. Two choices are possible: either 1 or 1,2.
- (2) The relaxation factor k_{cc} used for the resistance of welded studs (next generation of Eurocodes, see § 5.7.3, Formula (115)) is to be defined. Its default value is 1. It should verify $0 < k_{cc} \leq 1$.
- (3) η_V is used for the criterion of MV interaction in the next generation of Eurocodes, see § 7.5.1. Its default value is 0,5. It should verify: $0,3 < \eta_V \leq 0,5$.
- (4) k_{tc} is used for the design compressive strength of concrete, in the next generation of Eurocodes, see § 2.10.5 (7). Its default value is $k_{tc} = 1$ and it should verify: $0 < k_{tc} \leq 1$.

2.13. Load cases

2.13.1. Type of loading

- (1) For each beam, it will be possible to define 4 or 5 load cases, as follows:
 - one load case Q_c for the construction phase;
 - one load case G for the permanent loads;
 - one load case G_1 for the permanent loads which already act on the beam when concrete is poured;
 - one load case G_2 for the additional permanent loads which act on the beam only after the concrete has been poured;
 - two load cases Q_1 and Q_2 for the live loads.

G is defined only for usual steel sections or for fully propped composite beams. G_1 and G_2 are defined only for usual composite sections unpropped or with point proppings.

Concrete is always assumed to be poured in one phase only.
- (2) Load case Q_c will be available for beams with composite steel sections (according to (3) for usual sections and (5) for slim floors of § 2.2.1) and without continuous propping.
- (3) Load case G_1 acts according to the propping – see § 2.14.1 (2) to (4).
Load case G_2 acts in the composite phase (long term actions).
- (4) For sections with partial concrete encasement, either steel or composite, it is assumed that the concrete of the encasement is poured at least 28 days before the installation on site and that it may contribute to the stiffness and the resistance of the cross-section (if the relevant option is activated – see § 2.9 (13)).
- (5) For composite sections, both load cases Q_1 and Q_2 act in composite phase.

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- (6) For load case G_2 , which only acts in composite phase, the software will use the time $t_{0,G2}$ first loading of permanent load case, in days, by default equal to 28 d.
In Expert mode, it will be possible to define another value for $t_{0,G2}$, which then should fulfil the criterion $t_{0,G2} \geq 3 \text{ d}$. If the value is lower than 28 days, a warning message will be displayed that a reduced resistance of concrete should be considered, in addition to the software calculations.
- (7) For shrinkage load case (see § 6.7.1), the time $t_{0,SH}$ at first loading will be assumed to be equal to $t_{0,SH} = 1 \text{ d}$.
- (8) Time t of the concrete at the moment considered for the design checks will be defined for a default value associated to a life time of 50 years, i.e. $t = 18\,250 \text{ d}$.
In the parameters of the software, in Expert mode only, it will be possible to define another value.
- (9) See **Table 41** for the parameters associated to load cases.
- (10) See § 6.7.4 for the live loads arrangements.

2.13.2. Definition of the loadings

- (1) Any load case is defined by the following sets of forces, to be defined by the User (**Figure 27**):
 - a surface load Q_L ;
 - a set of point loads, defined by a location $x_{L,Fi}$ and a force F_{Li} . The location is measured from the left end of the span;
 - a set of distributed loads, defined by the locations $x_{L,1qi}$ and $x_{L,2qi}$ between which the load is applied and by the loads $q_{L,1i}$ and $q_{L,2i}$ at each end of the loaded segment.
- (2) By default, no load is applied, except for the dead load (load case G or G_1).
- (3) For the load case G or G_1 , the self-weight of the steel profile, the concrete slab, the partial concrete encasement and the steel profiled sheeting is automatically assessed (see § 6.7.2). The self-weight of reinforcements is not explicitly assessed and is assumed to be included in the weight of the concrete.

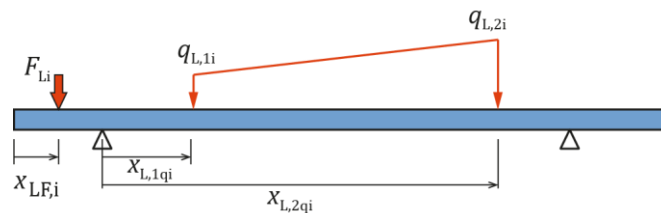


Figure 27 : Definition of loadings

2.14. Propping

2.14.1. General

- (1) For all beams having a composite cross-section (according to (3) for usual sections and (5) for slim floors of § 2.2.1), the User will define the type of propping considered for the construction stage (pouring of concrete), among the following list:
 - un-propped beam (by default);
 - fully propped beam;
 - point propping of the beam, to be defined by the User.

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- (2) For un-propped beams:
 - the G_1 loads are resisted by the steel profile alone;
 - the resistance of the beam is checked at ultimate limit state during the construction stage, considering the steel cross-section, according to § 7;
 - the resistance of the beam is checked at ultimate limit state for the final stage, considering the composite action, according to § 7.
- (3) For fully propped beams:
 - the G loads are resisted by the beam with composite effect between the concrete slab and the steel profile, considering a modular ratio for long term effects;
 - the resistance of the beam is checked at ultimate limit state for the final stage only, considering the composite action, according to § 7.
- (4) For propped beams with temporary point supports:
 - the G_1 loads are decomposed in two phases (see § 6.7.1): during a first period, for the pouring of concrete, the G_1 loads are supported by the steel profile alone ($G_{1,pp}$), considering a continuous beam with n_{pp} intermediate supports. A second period, after the removal of the propping, activates the composite action, with a modular ratio for long terms effects. The loads applied to this load case are the reactions at intermediate temporary support obtained from the first phase ($G_{1,c}$);
 - the resistance of the beam is checked at ultimate limit state during the construction stage, considering the steel cross-sections, according to § 7;
 - the resistance of the beam is checked at ultimate limit state for the final stage, considering the composite action, according to § 7.
- (5) For fully propped (see (3)) or partially propped (see (4)) beams, the time $t_{0,1c}$ when the temporary supports are removed, is assumed to be equal to 28 d. It will not be possible to change this value in Normal mode.
- (6) If $t_{0,1c} < 28$ days, an additional stage at the time t_u should be considered for the verification at ultimate limit state, considering the composite actions with reduced properties of concrete. It will not be covered by the software.
- (7) For cross-section with partial concrete encasement, § 2.13.1 (4) applies.

2.14.2. Definition of the point propping

- (1) For the main span, the User can define the number of point-props, n_{pp} , that will be uniformly distributed along the span length (Figure 28).
- (2) For cantilever spans, the User can only choose to prop the end of the span.

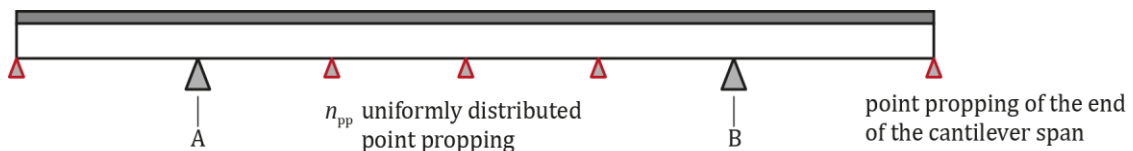


Figure 28 : Definition of the point props

- (3) For the main span, the location of the i^{th} temporary support to the left intermediate support of the beam (A) is given by:

$$x_{pi} = i L / (n_{pp} + 1)$$
- (4) An option is provided to choose the location of the temporary supports, either below the lower flange or below the lower side of the slab. The latter option is a better solution for the stability, and is set by default. No lateral torsional buckling occurs in this case.

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2.15. Combination of loads

2.15.1. Ultimate Limit States (ULS) – Persistent or transient situations

- (1) The software allows the User to define up to 6 ULS load combinations: 4 predefined combinations proposed by the program and two custom combinations defined by the User.
- (2) By default, the four predefined following ULS load combinations are automatically generated and proposed by the program (Expressions 6.10 of EN 1990 [1]):

- When the permanent loads have unfavourable effects:

$$Q_1 \text{ is the basic variable action: } \gamma_{G,\text{sup}} G + \gamma_Q Q_1 + \psi_{0,2} \gamma_Q Q_2$$

$$Q_2 \text{ is the basic variable action: } \gamma_{G,\text{sup}} G + \gamma_Q Q_2 + \psi_{0,1} \gamma_Q Q_1$$

Without variable actions, the two later resumes as follows: $\gamma_{G,\text{sup}} G$

- When the permanent loads have favourable effects:

$$Q_1 \text{ is the basic variable action: } \gamma_{G,\text{inf}} G + \gamma_Q Q_1 + \psi_{0,2} \gamma_Q Q_2$$

$$Q_2 \text{ is the basic variable action: } \gamma_{G,\text{inf}} G + \gamma_Q Q_2 + \psi_{0,1} \gamma_Q Q_1$$

Without variable actions, the two later resumes as follows: $\gamma_{G,\text{inf}} G$

For composite beams, unpropped or point propped during construction, G is replaced by $(G_1 + G_2)$ in the previous combinations.

- (3) For a composite beam at the construction stage, without continuous propping (see § 2.14), only one predefined load combination is considered:

$$\gamma_{G,\text{sup}} G_1 + \gamma_Q Q_c$$

It is also possible to add a custom combination defined by the User.

2.15.2. Serviceability Limit States (SLS)

- (1) It is possible to define up to 6 load combinations for the calculations at the Serviceability Limit States: 4 predefined combinations proposed by the program and two custom combinations defined by the User.
- (2) By default, the four predefined following SLS load combinations are proposed by the program (Expressions 6.14b of EN 1990 [1]), for characteristic combinations:

$$Q_1 \text{ is the basic variable action: } G + Q_1$$

$$G + Q_1 + \psi_{0,2} Q_2$$

$$Q_2 \text{ is the basic variable action: } G + Q_2$$

$$G + Q_2 + \psi_{0,1} Q_1$$

For composite beams, unpropped or point propped during construction, G is replaced by $(G_1 + G_2)$ in the previous combinations.

- (3) For a composite beam without continuous propping, only one predefined SLS load combination is considered at the construction stage:

$$G_1 + Q_c$$

It is also possible to add a custom combination defined by the User.

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- (4) When a control of cracking is performed for beams with cantilevers, quasi-permanent SLS combinations will be automatically considered for the calculation of the tension stresses in the reinforcements (see § 11.4.4). These combinations are defined by:

$$G + \psi_{2,1} Q_1 + \psi_{2,2} Q_2$$

2.15.3. Ultimate limit state – Fire situation

- (1) It is possible to define up to 5 combinations of loads for the limit states associated to the accidental fire situation. 3 combinations are predefined according to Equation (6.11a) of EN 1990:

$$G + \psi_{1,1} Q_1 + \psi_{2,2} Q_2$$

$$G + \psi_{2,1} Q_1 + \psi_{2,2} Q_2$$

$$G + \psi_{2,1} Q_1 + \psi_{1,2} Q_2$$

For composite beams, unpropped or point propped during construction, G is replaced by $(G_1 + G_2)$ in these combinations.

- (2) The last 2 possible combinations of loads are directly defined by the User.
- (3) By default, all 5 combinations of loads for the fire situation are deactivated. Fire calculations are carried out by the software only if at least one of the combinations of loads for the fire situation is activated.

2.16. Parameters for the serviceability limit states

2.16.1. Comfort under vibrations

- (1) The User will define the following parameters for the application of the Hivoss method (see § 11.3):
- Application of the HiVoSS method (Yes by default);
 - Selection of the combination of mass to be considered, among the following list:
 - G
 - $G + 0,1 Q$
 - $G + 0,2 Q$
 - $G + 0,3 Q$
 - $G + 0,4 Q$
 - $G + 0,5 Q$

where G means $G_1 + G_2$ when relevant and Q is to be chosen by the User as either Q_1 , Q_2 or $Q_1 + Q_2$;
 - Definition of the use of floors for the damping D_2 (by default office buildings) among the following list:
 - Critical areas,
 - Hospitals, surgeries,
 - Schools, training centres,
 - Residential buildings,
 - Office buildings,
 - Meeting rooms,
 - Senior citizen residential buildings,
 - Hotels,
 - Industrial workshops,
 - Sport facilities.
 - Presence of a ceiling under the floor (No by default). (If yes, 1% is added to D_3);
 - Presence of a swimming screed (No by default). (If yes, 1% is added to D_3).

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- Definition of the type of furniture for the floor, among the following list:
 - Traditional office with separation walls,
 - Paperless office,
 - Open plan office (by default),
 - Library,
 - Houses,
 - Schools,
 - Gymnasium.

The latter choice influences the definition of damping D_2 . It will be possible for the User to define its own value of D_2 damping.

2.16.2. Control of concrete cracking

- (1) The User can specify if a limitation of concrete cracking is to be checked.
- (2) If the option is activated, the maximum width of cracks should be chosen among the following list of values: 0,2; 0,3 and 0,4 mm (see § 11.4).

2.17. Fire design

- (1) When a fire design calculation is to be carried out, the User should define specific parameters.
- (2) For usual steel or composite sections, the steel profile can be either unprotected or protected. [It is not possible to define a fire protection for partially encased profiles.](#)
- (3) The protection of a steel profile may be obtained by:
 - spray;
 - intumescent paint;
 - boards (or panels).

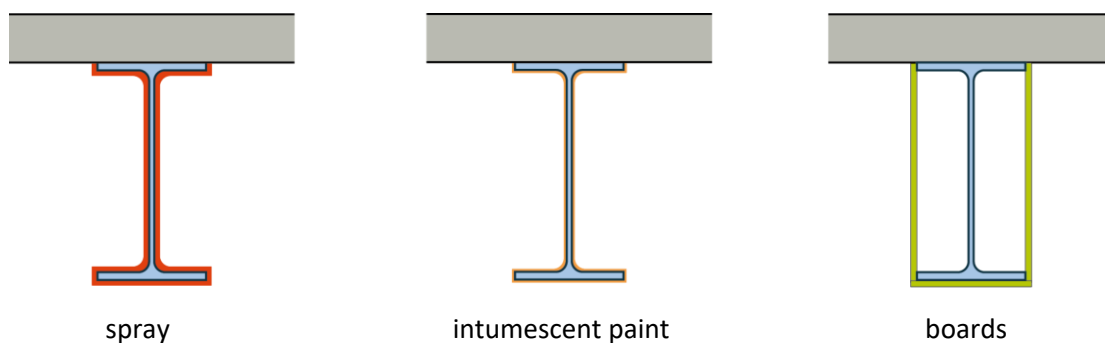


Figure 29: Type of fire protection

- (4) Only the generic properties for the insulation material are considered in the software. The thermal protection is assumed to be applied on all exposed sides of the profile.
- (5) According to the type of fire protection, the User will choose one of the predefined generic material given in **Table 14**. All properties of generic materials are assumed to be independent of the temperature.

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Type of insulation	Density ρ_p [kg/m ³]	Thermal conductivity λ_p [W/mK]	Specific heat c_p [J/kgK]
Low density spray mineral fibres vermiculite	300	0,12	1200
	350	0,12	1200
High density spray perlite with cement perlite with plaster	550	0,12	1100
	650	0,12	1100
Intumescent paint	0	(a)	0
Boards:			
vermiculite or perlite with cement	800	0,2	1200
silicate fibre	600	0,15	1200
fibro-cement	800	0,15	1200
plaster	800	0,20	1700

(a) the value of λ_p is to be defined by the User in the range of [0,005 to 0,012]

Table 14: Thermal properties of insulation material

- (6) For unprotected sections, the User may define the type of finishing as carbon steel (CS) or as hot-dip galvanised steel (HDG). **To be discussed with AM.**
- (7) The User should define if the reinforcements are made of cold worked (by default) or hot rolled steel – see § 10.9.3.
- (8) As an option, the User indicates if the concrete strength is reduced or not for temperature below 250°C.

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3. SOFTWARE SETTINGS

3.1. Software options

- (1) Following settings are available for the User in the options of the software.
- (2) General settings:
 - language of the interface;
 - language of the calculation report;
 - control of updates;
 - installation and working directories;
 - units;
 - content of the calculation sheet.

3.2. Calculation options

3.2.1. Default values for the projects

- (1) Following parameters can be chosen/defined by the User in the settings of the software for the calculation options. These choices are available in Normal mode.
- (2) For these parameters, the User will define the default values to be considered for each project of beam. A reset button will be proposed in this window, where it will be possible to reset all values to the values defined by:
 - the recommended values of Eurocodes;
 - or the French National Annexes.
- (3) Default values of partial factors – see § 2.12.2.

The default setting values of the software are as given in **Table 15** and in **Table 16**.

French NA values will be used for the CTICM version and recommended values for the ArcelorMittal one.

Partial factor	Recommended values	French NAs	Partial factor	Recommended values	French NAs
$\gamma_{G,sup}$	1,35	1,35	γ_{M2}	1,25	1,25
$\gamma_{G,inf}$	1,0	1,0	γ_c	1,5	1,5
γ_Q	1,5	1,5	γ_s	1,15	1,15
γ_{M0}	1,0	1,0	γ_v	1,25	1,25
γ_{M1}	1,0	1,0	γ_p	1,0	1,0

Table 15: Default setting values of the partial factors

Partial factor	Recommended values	French NAs
$\gamma_{M,fi}$	1,0	1,0
$\gamma_{M,fi,a}$	1,0	1,0
$\gamma_{M,fi,c}$	1,0	1,0
$\gamma_{M,fi,v}$	1,0	1,5
$\gamma_{M,fi,s}$	1,0	1,0

Table 16: Default setting values of the partial factors for fire design

- (4) Default values of combination factors – see § 2.12.3.

The default setting values of the software are as given in **Table 17**. French NA values will be used for the CTICM version and recommended values for the ArcelorMittal one. Values for offices are considered.

Comb. factor	Q_1		Q_2	
	Recommended values	French NAs	Recommended values	French NAs
ψ_0	$\psi_{0,1} = 0,7$	$\psi_{0,1} = 0,7$	$\psi_{0,2} = 0,7$	$\psi_{0,2} = 0,7$
ψ_1	$\psi_{1,1} = 0,5$	$\psi_{1,1} = 0,5$	$\psi_{1,2} = 0,5$	$\psi_{1,2} = 0,5$
ψ_2	$\psi_{2,1} = 0,3$	$\psi_{2,1} = 0,3$	$\psi_{2,1} = 0,3$	$\psi_{2,1} = 0,3$

Table 17: Default setting values of the combination factors

- (5) The default values of the other NDP parameters – see § 2.12.4- are as follows:

$\eta = 1$	Limits: $\eta = 1$ or $1,2$
$k_{cc} = 1$	Limits: $0 < k_{cc} \leq 1$
$\eta_v = 0,5$	Limits: $0,3 < \eta_v \leq 0,5$
$k_{tc} = 1$	Limits: $0 < k_{tc} \leq 1$
$k_E = 1$	Limits: $5\,000 < k_E \leq 13\,000$

- (6) The Young's modulus for the reinforcement steel may be modified (see § 2.10.6).

The default value is $E_s = E_a = 210$ GPa.

The limits are: $190 \leq E_s \leq 210$ GPa.

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- (7) The default value of ε_{sh} (deformation of concrete under shrinkage effect – see § 2.10.5 (5)) may be chosen. Default setting of the software is $\varepsilon_{sh} = 325 \times 10^{-6}$, according to Annex C of EN 1994-1-1. Limits are: $0 \leq \varepsilon_{sh} \leq 1000 \times 10^{-6}$.
- (8) The User can choose between the normal or the simplified distribution of effective width along the beam – see § 5.1.4. Default value: normal distribution.

3.2.2. Standards

- (1) The User will have the possibility to choose between the first and the second generation of Eurocodes. For the first version of the software, as long as the second generation is in its preparation process, only the first generation will be available in the Normal mode.

3.2.3. Other options

- (1) The fineness of the discretisation of the beam may be defined by the User. See **Annex C**.
- (2) δ_d is the slip of a stud under a shear force equal to P_{Rd} . By default $\delta_d = 0,5$ mm. See § 11.1 (3).

3.3. Software constants

- (1) Following additional, not mentioned before, are considered in the software:

g : Acceleration of gravity: $g = 9,81 \text{ ms}^{-2}$

σ : Boltzmann constant: $\sigma = 5,67 \cdot 10^{-8} \text{ W/m}^2\text{K}^4$

3.4. Fire calculations

- (1) Time increment for the heating of unprotected steel sections (Δt) – see § 10.7.1 (3).
Default value: $\Delta t = 5$ s
According to EN 1994-1-2, 4.3.4.2.2 (5), Δt should not be taken greater than 5 s.
- (2) Time increment for the heating of unprotected steel sections (Δt) – see § 10.7.2 (3).
Default value: $\Delta t = 10$ s
According to EN 1994-1-2, 4.3.4.2.2 (8), Δt should not be taken greater than 30 s.
- (3) Following parameters are used by the software for the fire calculations. The value of some of these parameters may be modified in Expert mode (these parameters are indicated in the list with the label Expert).
 - reference temperature: $\theta_{ref} = 20^\circ\text{C}$ (Expert);
 - maximal temperature: $\theta_{max} = 1\,200^\circ\text{C}$;
 - form factor: $\Phi = 1,0$ (Expert);
 - emissivity of steel surfaces: $\varepsilon_m = 0,7$ (Expert);
 - emissivity of fire: $\varepsilon_f = 1,0$ (Expert);
 - factor of thermal transfer by convection: $\alpha_c = 25 \text{ W/m}^2\text{K}$ (Expert);
 - correction factor for the shadow effect: $k_{sh} = 1$ (Expert);
 - coefficient of heat transfer by convection on the upper side of a slab: $\alpha_{cc} = 4 \text{ W/m}^2\text{K}$;
 - factor for the plastic resistance to axial compression of the concrete slab: $\alpha_{slab} = 1$.

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3.5. Expert mode settings

Minimum span length : $L_{\min} = 5$ m- see § 2.1 (2);

Maximum ratio of A_{fi} / A_{fs} , by default 3 - see § 2.2.5 (2);

Default value of $b_{app,min}$, see § 2.2.6 (3).

Haunches: inclination angle θ_h - see § 2.3.1 (3);

limit value of the haunch depth t_h / t_d - see § 2.3.1 (3);

Minimum depth of the active part of the slab: $t_{c,min} = 50$ mm - see § 2.3.1 (7);

Time t of the concrete at the moment considered for the design checks, default value associated to a life time of 50 years, i.e. $t = 18\,250$ d – see § 2.13.1 (8);

Time $t_{0,SH} = 1$ d for shrinkage load case – see § 2.13.1 (7);

Time $t_{0,G2} = 28$ d for G and G_2 load cases – see § 2.13.1 (6);

Time $t_{0,1c} = 28$ d for $G_{1,c}$ load case – see § 2.14.1 (5);

Deformability of studs δ_d – see § 11.1.

3.6. Specific settings for CTICM and AM versions

To be defined

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4. GENERAL PROCESS OF THE SOFTWARE

4.1. Databases

- (1) The software will use the following databases:
- hot-rolled profiles – ArcelorMittal catalogue: binary non editable file;
 - structural steels: binary non editable file;
 - profiled steel sheetings: editable ASCII file;
 - welded studs: editable ASCII file;
 - prefabricated floors for slim floors beams: editable ASCII file.

4.2. General case

Definition of the beam, the supports, the load cases, etc.

Control of parameters

For composite beams:

- calculation of modular ratios for each load case;
- calculation of effective widths of the slab (for global analysis and for section analysis)

Properties of materials

Classification of cross-sections

Section properties for global analysis

Discretisation of the beam, definition of the first critical sections

Global analysis for each load case:

- internal forces and moments;
- deflection.

Combination of loads

Section properties for section analysis

ULS verifications

Design of welds (welded sections) and of the transverse reinforcements (composite sections)

SLS checks.

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5. BASIC CALCULATIONS

5.1. Effective width of the concrete flange

5.1.1. General

- (1) The effective width of the concrete flange is calculated according to EN 1994-1-1, 5.4.1.2.
- (2) The geometric widths b_1 and b_2 of the concrete flange are given by:

- for an edge beam:

$$b_1 = d_1$$

$$b_2 = d_2 / 2$$

- for an intermediate beam:

$$b_1 = d_1 / 2$$

$$b_2 = d_2 / 2$$

See § 2.1 for the definitions of edge beams, intermediate beams, d_1 and d_2 .

- (3) If slab openings are defined, previous equations will be replaced by:

- for an edge beam:

$$b_1 = d_1$$

$$b_2 = \min\{d_{sl,2}; d_2 / 2\}$$

- for an intermediate beam:

$$b_1 = \min\{d_{sl,1}; d_1 / 2\}$$

$$b_2 = \min\{d_{sl,2}; d_2 / 2\}$$

- (4) For the calculation of the effective width of concrete flanges as given in this part, the distance b_0 between the centres of the outstand shear connectors is taken equal to 0, according to clause 5.4.1.2 (9) of EN 1994-1-1.

5.1.2. Effective width of the concrete flange in the main span

- (1) The effective width at mid-span of the main span is given by the following formula:

$$b_{eff,m} = b_{e1} + b_{e2}$$

$$b_{e1} = \min\{L_e/8; b_1\}$$

$$b_{e2} = \min\{L_e/8; b_2\}$$

(47)

where: b_1 and b_2 are the geometric widths, according to § 5.1.1;

L_e is the distance between the points of zero bending moment.

- (2) In Formula (47), the distance L_e is obtained as follows:

- For a single span beam without cantilever: $L_e = L$;
- For a beam with one cantilever span: $L_e = 0,85 L$;
- For a beam with two cantilever spans: $L_e = 0,7 L$.

- (3) At support A or B, if there is no adjacent span, the effective width of the concrete flange is given by:

$$b_{eff,s} = \beta_1 b_{e1} + \beta_2 b_{e2}$$

(48)

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where: b_{e1} and b_{e2} are defined by Formula (47);

β_1 and β_2 are defined by:

$$\beta_i = \min\{1; 0,55 + 0,025 L_e / b_{ei}\} \quad (49)$$

L_e see (2);

index s: A or B.

- (4) At support A or B, if there is an adjacent cantilever span, the effective width of the concrete flange is given by Formula (47), where L_e is calculated by:

$$L_e = 2 L_{cs} \quad (50)$$

- (5) For any cross-section in the main span located at a distance d_x of the nearest support s (A or B) lower than $L / 4$, the effective width of the concrete flange is obtained by (Figure 30):

$$b_{eff} = b_{eff,s} + 4 \frac{d_x}{L} (b_{eff,m} - b_{eff,s}) \quad (51)$$

- (5) For any cross-section in the main span located at a distance d_x of the nearest support s (A or B) greater or equal to $L / 4$, the effective width of the concrete flange is obtained by:

$$b_{eff} = b_{eff,m} \quad (52)$$

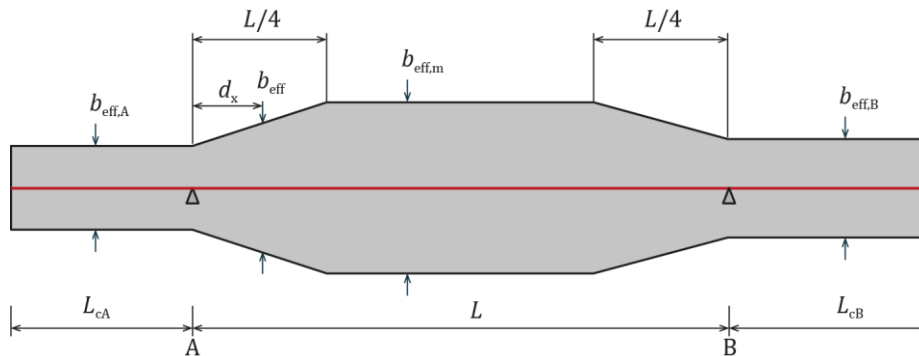


Figure 30 : Effective width of the concrete flange

5.1.3. Effective width of the concrete flange in the cantilever spans

- (1) The effective width of the concrete flange is constant along any cantilever span A or B and is given by clause (4) in § 5.1.2.

5.1.4. Simplified models for the effective width

- (1) An option will be proposed to the User in order to use a simplified distribution of the effective width along the beam, as an alternative to the model shown in Figure 30 – see § 3.2.1 (8).
- (2) If this option is activated, the effective width of the concrete flange considered for the global analysis will be assumed constant along the main span (Figure 31) and equal to $b_{eff,m}$ (according to EN 1994-1-1, 5.4.1.2 (4)).

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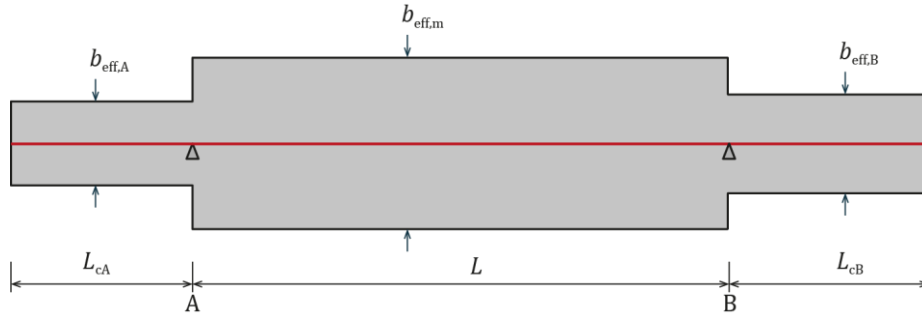


Figure 31 : Simplified effective width of the concrete flange for the global analysis

- (3) If the option is activated, the effective width of the concrete flange considered for the verification of cross-sections will be assumed to be obtained as follows (**Figure 32**) (according to EN 1994-1-1, 6.1.2 (2)).
- For any cross-section in the main span located at a distance d_x of the nearest support s (A or B) lower than $0,15 L$, the effective width of the concrete flange is equal to $b_{eff,s}$. For all other cross-sections of the main span, the effective width is equal to $b_{eff,m}$. For cantilever spans, § 5.1.3 applies.

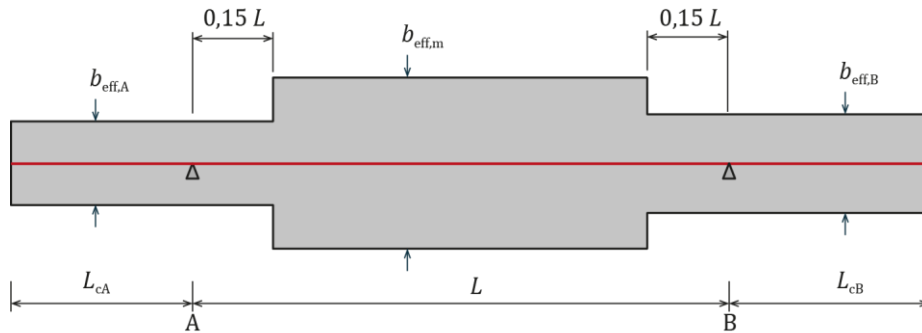


Figure 32 : Simplified effective width of the concrete flange for the verification of cross-sections

5.2. Active part of the concrete slab for composite sections

5.2.1. General

- (1) The slab participates to the resistance of the composite beams under bending moments. The participating area of the slab is defined by the effective width as given in § 5.1 and the thickness of the active part, as described here after.

5.2.2. Usual sections

- (1) The thickness t_c of the active part of the slab is obtained as follows:
- solid slabs: $t_c = t_d$
 - solid slabs with precast elements: $t_c = t_d - t_{pc} + t_j$
 - composite slabs: $t_c = t_d - h_p$ for ribs parallel to the beam axis;
 - $t_c = t_d - h_{pg}$ for ribs perpendicular to the beam axis.

where: h_p and h_{pg} : see § 2.5.1.

t_j : see § 2.3.1 (4).

5.2.3. Slim floors sections

- (1) The thickness t_c of the active part of the slab is obtained as follows:
- for solid slabs:
 - the full depth of concrete is considered: $t_c = t_d$
 - solid slabs with precast elements:
 - only the depth including the precast elements where the transverse joints have been concreted is considered: $t_c = t_d - t_{pc} + t_j$
 - composite slabs:
 - the depth above the steel sheeting is considered: $t_c = t_d - h_p$. Additionally, the part of concrete embedded in the steel section is also considered.
 - prefabricated floors:
 - the depth above the prefabricated floors is considered: $t_c = t_d - d_p$. Additionally, the part of concrete embedded in the steel section is also considered.

5.3. Modular ratio

5.3.1. Eurocode generation 1

- (1) A modular ratio is associated to each load case which is applied on the beam in a composite stage. The modular ratio is obtained with the following formula (see § 5.4.2.2 of EN 1994-1-1):

$$n_L = n_0 (1 + \psi_L \varphi_t) \quad (53)$$

where: n_0 is the modular ratio for short term loading, defined by: $n_0 = E_a / E_{cm}$

E_a is the Young's modulus of steel (see § 2.10.1)

E_{cm} is the secant modulus of concrete (see § 2.10.5)

ψ_L is the creep multiplier depending on the type of loading, which is given by **Table 18**.

Type of loading	ψ_L
Permanent loads	1,1
Shrinkage	0,55
Live loads	0

Table 18: Values of the creep multiplier

φ_t is the creep coefficient depending on the age t of concrete at the moment considered and on the age t_0 at the moment of loading (see § 2.13.1).

$$\varphi_t = \varphi(t, t_0) \quad (54)$$



A modular ratio will also be calculated for the concrete of the partial concrete encasement, if any, using the same formulas. For composite beams with partial concrete encasement, two separate values of modular ratios will thus be used, assuming the same age t and separate concrete properties and t_0 values.

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(2) For normal concrete, φ_t is obtained according to Annex B of EN 1992-1-1 [3], as follows:

$$\varphi(t, t_0) = \varphi_0 \beta_c(t, t_0) \quad (55)$$

where: φ_0 is the notional creep coefficient, calculated by:

$$\varphi_0 = \varphi_{RH} \beta(f_{cm}) \beta(t_0) \quad (56)$$

φ_{RH} is a factor which accounts for the effect of relative humidity on the notional creep coefficient, given by:

$$\begin{aligned} \varphi_{RH} &= 1 + \frac{1 - RH / 100}{0,1 \sqrt[3]{h_0}} & \text{for } f_{cm} \leq 35 \text{ MPa} \\ \varphi_{RH} &= \alpha_2 \left[1 + \alpha_1 \frac{1 - RH / 100}{0,1 \sqrt[3]{h_0}} \right] & \text{for } f_{cm} > 35 \text{ MPa} \end{aligned} \quad (57)$$

RH is the relative humidity of the ambient environment, in %, see § 2.10.5 (4);

h_0 is the notional size of the member (see § 5.3.2);

f_{cm} : see § 2.10.5;

$\beta(f_{cm})$ is a factor which takes into account the concrete strength (see (3));

$\beta(t_0)$ is a factor which takes into account the age of concrete at loading (see (4));

$\beta_c(t, t_0)$ is a coefficient which accounts for the development of creep with time (see (5));

RH is the relative humidity of the ambient atmosphere, see § 2.10.5 (4);

t_0 is the age of concrete at loading in days, see § 2.13.1;

t is the age of concrete at the moment considered, see § 2.13.1.

(3) $\beta(f_{cm})$ is obtained by:

$$\beta(f_{cm}) = \frac{16,8}{\sqrt{f_{cm}}} \quad (58)$$

(4) $\beta(t_0)$ is given by:

$$\beta(t_0) = \frac{1}{(1 + t_0^{0,20})} \quad (59)$$

(5) $\beta_c(t, t_0)$ is obtained by:

$$\beta_c(t, t_0) = \left[\frac{t - t_0}{\beta_H + t - t_0} \right]^{0,3} \quad (60)$$

where β_H is a coefficient depending on the relative humidity and on the notional size of the member, given by:

$$\beta_H = 1,5 \left[1 + \left(\frac{3 RH}{250} \right)^{18} \right] h_0 + 250 \leq 1500 \quad \text{for } f_{cm} \leq 35 \text{ MPa} \quad (61)$$

$$\beta_H = 1,5 \left[1 + \left(\frac{3 RH}{250} \right)^{18} \right] h_0 + 250 \alpha_3 \leq 1500 \alpha_3 \quad \text{for } f_{cm} > 35 \text{ MPa}$$

α_i factors are given by:

$$\alpha_1 = \left(\frac{35}{f_{cm}} \right)^{0,7} \quad \alpha_2 = \left(\frac{35}{f_{cm}} \right)^{0,2} \quad \alpha_3 = \left(\frac{35}{f_{cm}} \right)^{0,5} \quad (62)$$

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5.3.2. Notional size of a member

- (1) The general formula to assess the notional size of a member is as follows:

$$h_0 = \frac{2 A_c}{u} \quad (63)$$

where A_c is the area of the concrete member;

u is the perimeter of the member in contact with the atmosphere.

- (2) For a solid slab without haunch, with or without a prefabricated slab, h_0 will be calculated by:

$$h_0 = \frac{1}{\frac{1}{t_d} - \frac{b_{fs}}{2 b_{eff} t_d}} \quad (64)$$

- (3) For a solid slab with a haunch, h_0 will be calculated by:

$$h_0 = \frac{b_{eff} t_d + (b_f + t_h \tan \theta_h) t_h}{b_{eff} - \frac{1}{2} b_f + t_h \left(\frac{1 - \sin \theta_h}{\cos \theta_h} \right)} \quad (65)$$

- (4) For a composite slab with longitudinal ribs, h_0 will be calculated by:

$$h_0 = 2 t_c \left(1 + \frac{h_p b_m}{e_p t_c} \right) \quad (66)$$

See § 2.5.1 for the geometric parameters of the profiled steel sheeting.



For composite slabs, it is assumed that the side of concrete in contact with the steel profiled sheeting is not in contact with the atmosphere.

- (5) For a composite slab with perpendicular ribs, h_0 will be calculated by:

$$h_0 = 2 t_c \quad (67)$$

- (6) For the concrete of a partial encasement, h_0 will be calculated by:

$$h_0 = A_{ce} / h_w \quad (68)$$

where A_{ce} is the area of concrete in the two chambers of the partial encasement, as given by Formula (31).

5.3.3. Next generation

- (1) The modular ratio associated to each load case is calculated with Formula (53).

- (2) For normal concrete, φ_t is obtained according to §B.5 of prEN 1992-1-1, as follows:

$$\varphi(t, t_0) = \varphi_{bc}(t, t_0) + \varphi_{dc}(t, t_0) \quad (69)$$

where: $\varphi_{bc}(t, t_0)$ is the basic creep coefficient (see (3));

$\varphi_{dc}(t, t_0)$ is the drying creep coefficient (see (4)).

- (3) $\varphi_{bc}(t, t_0)$ is calculated by the following formula:

$$\varphi_{bc}(t, t_0) = \beta_{bc, fcm} \beta_{bc, t-t_0} \quad (70)$$

where: $\beta_{bc, fcm}$ is a function to describe the effect of concrete strength on basic creep, given by:

$$\beta_{bc, fcm} = \frac{1,8}{(f_{cm, 28})^{0,7}} \quad (71)$$

$\beta_{bc, t-t_0}$ is a function to describe the time development of basic creep, given by:

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$$\beta_{bc,t-t_0} = \ln \left(\left(\frac{30}{t_{0,adj}} + 0,035 \right)^2 (t - t_0) + 1 \right) \quad (72)$$

$t_{0,adj}$ is the adjusted age at loading in days according to (5).

(4) $\varphi_{dc}(t, t_0)$ is calculated by the following formula:

$$\varphi_{dc}(t, t_0) = \beta_{dc,fc} \beta_{dc,RH} \beta_{dc,t_0} \beta_{dc,t-t_0} \quad (73)$$

where: $\beta_{dc,fc}$ is a function to describe the effect of concrete strength on drying creep, given by:

$$\beta_{bc,fc} = \frac{412}{(f_{cm,28})^{1,4}} \quad (74)$$

$\beta_{dc,RH}$ is a function to describe the effect of relative humidity and notional size on drying creep, given by:

$$\beta_{dc,RH} = \frac{1 - RH / 100}{\sqrt[3]{0,1 h_0 / 100}} \quad (75)$$

RH is the relative humidity of the ambient atmosphere, see § 2.10.5 (4).

β_{dc,t_0} is a function to describe the effect of the adjusted concrete age at loading on drying creep, given by:

$$\beta_{dc,t_0} = \frac{1}{1 + t_{0,adj}^{0,2}} \quad (76)$$

$\beta_{dc,t-t_0}$ is a function to describe the time development of drying creep, given by:

$$\beta_{dc,t-t_0} = \left[\frac{t - t_0}{\beta_H + t - t_0} \right]^{\gamma(t_{0,adj})} \quad (77)$$

β_H is given by:

$$\beta_H = \min\{1,5 h_0 + 250 \alpha_{fc}; 1500 \alpha_{fc}\} \quad (78)$$

α_{fc} is given by:

$$\alpha_{fc} = \left(\frac{35}{f_{cm,28}} \right)^{0,5} \quad (79)$$

$\gamma(t_{0,adj})$ is defined by:

$$\gamma(t_{0,adj}) = \frac{1}{2,3 + 3,5 / \sqrt{t_{0,adj}}} \quad (80)$$

h_0 is the notional size of the concrete member, see § 5.3.2;

(5) The adjusted age at loading, $t_{0,adj}$, is calculated by:

$$t_{0,adj} = \max \left\{ 0,5; t_{0,T} \left[\frac{9}{2 + t_{0,T}^{1,2}} + 1 \right]^{\alpha_{sc}} \right\} \quad (81)$$

where: α_{sc} is a coefficient which depends on the strength development of concrete, given by:

$\alpha_{sc} = -1$ for class CS;

$\alpha_{sc} = 0$ for class CN;

$\alpha_{sc} = 1$ for class CR.

$t_{0,T}$ is the age of concrete at loading in days adjusted according to the concrete temperature. In the software, a temperature of 20°C will be assumed, so that $t_{0,T} = t_0$.

5.4. Classification of cross-sections

5.4.1. General

- (1) The software does not cover Class 4 cross-sections.
- (2) Steel cross-sections are classified according to EN 1993-1-1 Table 5.2.
- (3) Composite cross-sections, including sections with concrete encasement, are classified according to EN 1994-1-1, 5.5.1.
- (4) The classification is carried out considering pure bending only.
- (5) The final Class of a cross-section is the maximum class of the flanges and of the web.

5.4.2. Usual steel sections

- (1) The class of a flange under compression is obtained from its geometrical slenderness c / t according to **Table 19**, where:

c is the outstand width, given by: $c = (b_f - t_w) / 2 - r_c$;

t is the flange thickness t_f ;

ε is the steel parameter, either equal to ε_{fs} or to ε_{fi} see § 2.10.4 (2).

Class	Criteria
1	$c / t \leq 9 \varepsilon$
2	$c / t \leq 10 \varepsilon$
3	$c / t \leq 14 \varepsilon$
4	otherwise

Table 19: Classification of an outstand panel under pure compression

- (2) The upper flange is considered as in compression if one of the following conditions is fulfilled:

- under positive moments:
 - plastic distribution of stresses: $t_{fs} \leq -z_{pl}$
 - elastic distribution of stresses: $t_{fs} \leq -z_{el}$
- under negative moments:
 - plastic distribution of stresses: $0 \geq -z_{pl}$
 - elastic distribution of stresses: $0 \geq -z_{el}$

The lower flange is always considered in tension under positive bending moment and in compression under negative bending moment.

- (3) The class of the web under full or partial compression, is obtained from its geometrical slenderness c / t and the bending factor α or ψ , according to **Table 20**, where:

c / t is given by: $c / t = d_w / t_w$

α is the fraction of the web that is under compression considering a plastic distribution of the stresses. It is calculated as given in (4);

ψ is the ratio of stresses at edges of the web, as given in (5), considering an elastic distribution;

ε is the steel factor, equal to ε_w – see § 2.10.4 (2).

Class	Criteria
1	$\alpha > 0,5 \quad \frac{c}{t} \leq \frac{396\varepsilon}{13\alpha - 1}$
	$\alpha \leq 0,5 \quad \frac{c}{t} \leq \frac{36\varepsilon}{\alpha}$
2	$\alpha > 0,5 \quad \frac{c}{t} \leq \frac{456\varepsilon}{13\alpha - 1}$
	$\alpha \leq 0,5 \quad \frac{c}{t} \leq \frac{41,5\varepsilon}{\alpha}$
3	$\psi > -1 \quad \frac{c}{t} \leq \frac{42\varepsilon}{0,67 + 0,33\psi}$
	$\psi \leq -1 \quad c / t \leq 62\varepsilon(1 - \psi)\sqrt{-\psi}$
4	otherwise

Table 20: Classification of an internal panel under compression and bending – First generation of EN

- (4) Under positive bending moment, α is obtained as follows, from the location of the plastic neutral axis z_{pl} of the cross-section, as calculated in § 5.5.1 or 5.5.3:

- if $z_{pl} \geq -(t_{fs} + r_{cs})$: the web is fully in tension and its class is 1;
- if $z_{pl} \leq -(t_{fs} + r_{cs} + d_w)$: the web is fully under compression and $\alpha = 1$;
- otherwise :

$$\alpha = \frac{-z_{pl} - t_{fs} - r_{cs}}{d_w} \quad (82)$$



See § 1.4 for the definition of the axes.

Under negative bending moment, α is obtained as follows:

- if $z_{pl} \geq -(t_{fs} + r_{cs})$: the web is fully in compression and $\alpha = 1$;
- if $z_{pl} \leq -(t_{fs} + r_{cs} + d_w)$: the web is fully under tension and its class is 1;
- otherwise :

$$\alpha = \frac{h_s - t_{fi} - t_p - r_{ci} + z_{pl}}{d_w} \quad (83)$$

- (5) Under positive bending moment and elastic distribution of stresses, the stress ratio ψ is given by:

$$\psi = \frac{z_{el} + h_s - t_{fi} - r_{ci}}{z_{el} + t_{fs} + r_{cs}} \quad (84)$$

Under negative bending moment, it is obtained by:

$$\psi = \frac{z_{el} + t_{fs} + r_{cs}}{z_{el} + h_s - t_{fi} - r_{ci}} \quad (85)$$

- (6) For the classification of the web, the sign of bending moment is obtained as follows:

- for the global analysis, positive moments are assumed in the uncracked segments of the beam and negative moments in the cracked segments – see § 6.4;
- for the section analysis (i.e. verification at ULS or at SLS), the sign of the moment, obtained from the global analysis and the relevant combination of actions, is considered.

(7) For the second generation of Eurocodes, **Table 20** will be replaced by **Table 21**.

Class	Criteria
1	$\alpha > 0,5 \quad \frac{c}{t} \leq \frac{126 \varepsilon}{5,5 \alpha - 1}$
	$\alpha \leq 0,5 \quad \frac{c}{t} \leq \frac{36 \varepsilon}{\alpha}$
2	$\alpha > 0,5 \quad \frac{c}{t} \leq \frac{188 \varepsilon}{6,53 \alpha - 1}$
	$\alpha \leq 0,5 \quad \frac{c}{t} \leq \frac{41,5 \varepsilon}{\alpha}$
3	$\psi > -1 \quad \frac{c}{t} \leq \frac{38 \varepsilon}{0,608 + 0,343 \psi + 0,049 \psi^2}$
	$\psi \leq -1 \quad c / t > 60,5 \varepsilon (1 - \psi)$
4	otherwise

Table 21: Classification of an internal panel under compression and bending – Second generation of EN

5.4.3. Usual steel sections with partial concrete encasement

(1) The class of a flange under compression is obtained from its geometrical slenderness c / t according to **Table 22**, where:

c / t is calculated according to **§ 5.4.2 (1)**;

ε is the steel parameter, either equal to ε_{fs} or to ε_{fi} see **§ 2.10.4 (2)**.

A flange is assumed in compression when the conditions in **§ 5.4.2 (2)** are met.

Class	Criteria
1	$c / t \leq 9 \varepsilon$
2	$c / t \leq 14 \varepsilon$
3	$c / t \leq 20 \varepsilon$
4	otherwise

Table 22: Classification of an outstand panel under pure compression



According to Table 5.2 of EN 1994-1-1.

(2) The class of the web is obtained according to **§ 5.4.2 (3)**. Any Class 3 web with partial concrete encasement will be considered as Class 2.

5.4.4. Usual composite sections

(1) The class of a flange under compression is obtained according to **§ 5.4.2 (1)** and **(2)**.



For the upper flange, the positive effect of the connection according to EN 1994-1-1, 5.5.2 (1) is not taken into account in the software.

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- (2) The class of the web is obtained from § 5.4.2 (3), where the plastic and elastic neutral axes are obtained according to § 5.5.5.

5.4.5. Usual composite sections with partial concrete encasement

- (1) The class of a flange under compression is obtained according to § 5.4.3 (1).
(2) The class of the web is obtained from § 5.4.3 (2), where the plastic and elastic neutral axes are obtained according to § 5.5.6.

5.4.6. Slim floor steel or composite sections

- (1) The upper flange is considered as in compression when the following conditions are met, where $z = z_{pl}$ for a plastic distribution of stresses and $z = z_{el}$ for an elastic one:
- under positive (sagging) moments: if $z \leq h_{ec} - t_{fs}$,
 - under negative (hogging) moments: if $z \geq h_{ec}$.
- (2) The lower flange is always assumed to be under tension if positive bending moments and in compression under negative moments.
- (3) The class of a flange under compression is obtained according to **Table 22**, where c / t is given by:

$$c/t = (b_p - t_w) / (2 t_p) \quad \text{for IFB-B sections;}$$

$$c/t = [(b_f - t_w) / 2 - r_c] / t_f \quad \text{for other slim floor sections.}$$



Slim floors are always under positive bending moments, so that the flange under compression is always the upper flange. Table 22 is used as for partially encased sections, assuming that the concrete will have the same restraining effect.

For the upper flange in compression, if the concrete cover is greater than 50 mm and one sixth of the flange width, its class will not be worse than 2. These conditions may be written as follows:

$$t_d - h_{ec} \geq 50 \text{ mm}$$

$$t_d - h_{ec} \geq b / 6$$

where $b = b_p$ for IFB-B sections and $b = b_{fs}$ for other slim floor sections.

- (4) For SFB sections under negative moments, the lower plate is assumed to be under compression. Its class is the maximal class of the outstand parts and the internal part. The class of the outstand part is obtained according to **Table 19**, where $c / t = (b_p - b_{fi}) / (2 t_p)$. The class of the internal part is obtained according to **Table 23**, where $c / t = b_{fi} / t_p$

Class	Criteria
1	$c / t \leq 33 \varepsilon$
2	$c / t \leq 38 \varepsilon$
3	$c / t \leq 42 \varepsilon$
4	otherwise

Table 23: Classification of an internal panel under pure compression

- (5) The class of the web is obtained according to **Table 20** or **Table 21**, according to the standard used.

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(6) For the application of (5), α is obtained as follows, from the location of the plastic neutral axis z_{pl} of the cross-section, as calculated in § 5.5.7 or 5.5.8, for a positive bending moment:

- if $z_{pl} \geq h_{ec} - d_s$: the web is fully in tension and its class is 1;
- if $z_{pl} \leq d_i$: the web is fully under compression and $\alpha = 1$;
- otherwise :

$$\alpha = \frac{h_{ec} - d_s - z_{pl}}{d_w} \quad (86)$$

where:

- $d_s = t_f + r_c$ for SFB, IFB-A and SAB sections;
- $d_s = t_p$ for IFB-B;
- $d_i = t_f + r_c$ for SFB;
- $d_i = 0$ for IFB-A;
- $d_i = r_c$ for IFB-B and SAB sections.

(7) For the application of (5), the stress ratio ψ for a positive bending moment is given by:

$$\psi = - \frac{z_{el} - d_i}{h_{ec} - d_s - z_{el}} \quad (87)$$

(8) For an **intermediate** beam with slim floor section, a class 3 web according to (5) will be considered as class 2 if the connectors are composite dowels (group A) or welded studs on the web (group B).

5.5. Section bending properties

5.5.1. General

- (1) The properties of cross-sections will be assessed through a general method based on the discretisation of the section. This general method is described in **Annex B**.
- (2) The following chapters give the principles for the calculation of section properties. They also provide simplified formulas for limited scopes of application (for validation purposes).
- (3) For all types of cross-sections including concrete, the plastic bending moment is assessed first by assuming a full shear connection.
- (4) The procedure to assess the plastic bending moment with partial shear connection is given in the relevant chapters.
- (5) For cross-sections with partial concrete encasement, it is always assumed, for any cross-section, that the full connection of the encasement is obtained.



*This simplification may be assumed because the full connection requires very few connectors and because the influence of the partial concrete encasement on the resistance of cross-sections is not significant.
The full connection is assumed according to clause 6.3.2 (1) of EN 1994-1-1.*

5.5.2. Steel sections

- (1) The location of the elastic neutral axis of any steel cross-section is obtained with the following formula:

$$z_{el} = \frac{\sum A_i z_i}{A_a} \quad (88)$$

where: A_i is the area of any elementary part which constitutes the cross-section;

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z_i is the location of the centroid of any elementary part;

A is the area of the cross-section which is given by:

$$A_a = \Sigma A_i \quad (89)$$

The formulas for A_i and z_i are given in **Table 24**.

Elementary part	A_i	z_i	I_i
1 – Upper flange	$b_{fs} t_{fs}$	$-t_{fs} / 2$	$b_{fs} t_{fs}^3 / 12$
2 – Upper root	$r_{cs} (t_w + (2 - \pi/2) r_{cs})$	$-t_{fs} - z_2$	$t_w r_{cs}^3 / 12 + k_{cl} r_{cs}^4$
3 – Web between roots	$d_w t_w$	$-(t_{fs} + r_{cs} + d_w / 2)$	$t_w d_w^3 / 12$
4 – Lower root	$r_{ci} (t_w + (2 - \pi/2) r_{ci})$	$-(h_s - t_p - t_{fi} - z_4)$	$t_w r_{ci}^3 / 12 + k_{cl} r_{ci}^4$
5 – Lower flange	$b_{fi} t_{fi}$	$-(h_s - t_p - t_{fi} / 2)$	$b_{fi} t_{fi}^3 / 12$
6 – Lower plate (*)	$b_p t_p$	$-h_s + t_p / 2$	$b_p t_p^3 / 12$
where : $k_{cl} = 0,0151$ $k_{cG} = (10 - 3\pi) / (12 - 3\pi)$ $z_{cs} = k_{cG} r_{cs}$			
$z_2 = \frac{t_w + z_{cs} (4 - \pi)}{4 - \pi + 2 t_w / r_{cs}}$ $z_4 = \frac{t_w + z_{ci} (4 - \pi)}{4 - \pi + 2 t_w / r_{ci}}$			
(*) elementary part no 6 not considered for usual steel cross-section			

Table 24: Properties of elementary parts of a cross-section

- (2) The second moment of area of the cross-section about its strong axis $y - y$ is calculated by the following formula:

$$I_{y,a} = \Sigma I_i + \Sigma A_i (z_i - z_{el})^2 \quad (90)$$

- (3) The plastic properties of the cross-section will be assessed by considering the following assumptions:
- the area of the steel profile is stressed to its characteristic design yield strength in tension and in compression; for hot-rolled profiles, a uniform value is assumed whereas specific values are considered for each flange and the web, for welded sections;
 - the area of the lower plate (if any) is stressed to its characteristic design yield strength in tension and in compression;
 - when MV interaction is considered, the yield strength of the web (including the fillets) is reduced accordingly.

The characteristic resistance of the bending moment $M_{pl,a,Rk}$ is obtained considering the nominal yield strength of steel. The design plastic bending resistance $M_{pl,a,Rd}$ is obtained by:

$$M_{pl,a,Rd} = M_{pl,a,Rk} / \gamma_{M0} \quad (91)$$

- (4) For usual steel sections, the elastic characteristic resistance to bending moment is obtained as follows:

$$M_{el,a,Rk} = \frac{I}{\max\{z_s, z_i\}} f_y \quad (92)$$

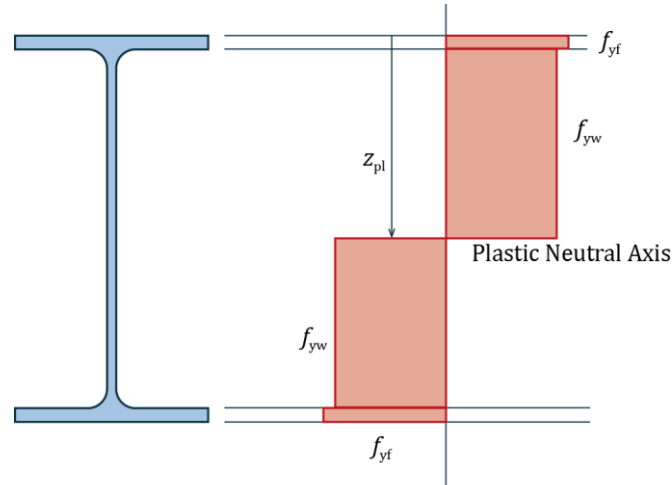
where: $z_s = |z_{el}|$

$z_i = |h_s - z_{el}|$

The design elastic bending resistance $M_{el,Rd}$ is obtained by:

$$M_{el,a,Rd} = M_{el,a,Rk} / \gamma_{M0} \quad (93)$$

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In pure bending, for hot-rolled profiles: $f_{yw} = f_{yf} = f_y$
For MV interaction, f_{yw} is calculated according § 7.4

Figure 33 : General principles for the plastic properties of a usual steel cross-section

5.5.3. Simplified methods for symmetrical steel sections

- (1) For symmetrical usual steel sections, either hot-rolled or welded (sections SHR and SWB – see **Table 1**), the location of elastic and plastic neutral axes are given by:

$$z_{el} = z_{pl} = -h_s / 2 \quad (94)$$

- (2) The area of the cross-section is given by:

$$A_a = 2 b_f t_f + h_w t_w + (4 - \pi) r_c^2 \quad (95)$$

- (3) The second moment of area is given by:

$$I_{y,a} = \frac{1}{12} [b_f h_s^3 - (b_f - t_w)(h_s - 2 t_f)^3] + 2 k_{cl} r_c^4 + (4 - \pi) \frac{r_c^2}{2} \left(\frac{h_s}{2} - t_f - k_{cG} r_c \right)^2 \quad (96)$$

- (4) The plastic bending modulus is given by:

$$W_{pl,a} = b_f t_f (h_s - t_f) + h_w^2 t_w + (4 - \pi) r_c^2 (h_s/2 - t_f - k_{cG} r_c) \quad (97)$$

where: k_{cG} see **Table 24**.

$$M_{pl,a,Rd} = W_{pl,a} f_y / \gamma_{M0}$$

5.5.4. Usual steel sections with partial concrete encasement

- (1) If the partial concrete encasement is considered in the calculations (see § 2.9 (13)), the bending properties of usual steel sections with partial concrete encasement will be assessed by an iterative procedure (**Annex B**), with the following assumptions (**Figure 34**):

- the part of concrete in tension area will be ignored;
- the longitudinal steel reinforcement in the tension area will be considered;
- the longitudinal steel reinforcement in the compression area will be considered if the associated option is activated;
- both positive and negative moments will be considered separately.

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- (2) Elastic section properties are assessed by considering the modular ratio (see § 5.2) associated to the relevant load case and to the concrete of the encasement (see § 2.9).

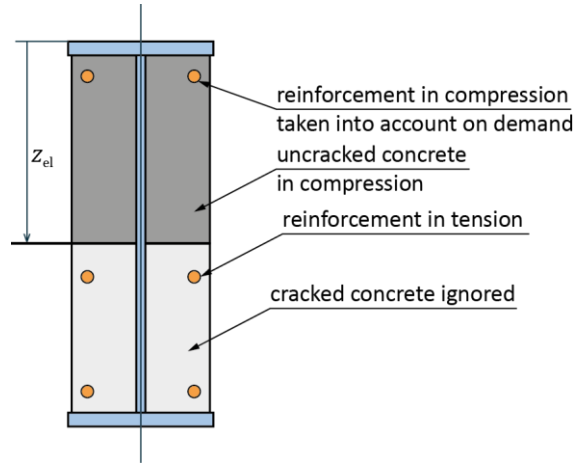


Figure 34 : Elastic bending properties of steel section with partial concrete encasement under positive bending moment

5.5.5. Usual composite sections

- (1) For the calculation of the elastic and plastic properties of usual composite cross-sections, the concrete in tension and the reinforcement in compression (by default) will be neglected. The concrete of haunches is also neglected.



The User will have the options to activate the reinforcement in compression.

If the User activates the reinforcement in compression, the following warning message will be displayed: when reinforcements in compression are accounted for in the calculation of the bending properties, the requirements of EN 1992-1-1, 9.2.1.2 (3) should be fulfilled.

If the slab contains only one layer of reinforcement, reinforcement in compression will be disregarded, whatever the option of the User is.

- (2) When a partial shear connection is used, the plastic properties are obtained considering an equivalent active part of the slab, defined as follows:

$$t_{c,\eta} = \frac{\eta N_{cx}}{0,85 b_{eff} f_{cd}} \quad \text{first generation of Eurocodes} \quad (98)$$

$$t_{c,\eta} = \frac{\eta N_{cx}}{b_{eff} f_{cd}} \quad \text{second generation of Eurocodes}$$

where: η and N_{cx} : see § 8.2.2.

5.5.6. Usual composite sections with partial concrete encasement

- (1) For the calculation of the elastic and plastic properties of usual composite sections with partial concrete encasement, § 5.5.5 applies.

5.5.7. Slim floor steel sections

- (1) The section properties of slim floor steel sections will be assessed through the iterative procedure in Annex B, where the following assumptions are considered:
- the yield strength of hot-rolled profiles is assumed to be uniform over the section and is obtained according to § 2.10.1;

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- the yield strength of the plate, if any (all slim floor sections except SAB), is assessed according to § 2.10.3;
- when the MV interaction is considered, the yield strength of the web (including fillets) is reduced accordingly;
- for the plastic properties, the area of the lower part of the sections (plate and/or lower flange), on which the slab loads are applied, are reduced to take account of the transverse bending of these parts according to one of the methods given in (2), (3) or (4).

(2) The first method to take account of the transverse bending to the lower flange is given by the ECCS guide no 83 [18], as follows. Let us consider a lower plate (either a welded plate or the lower flange), with dimensions $b \times t$ (see **Figure 35**). The plate is loaded symmetrically at both ends by a total distributed load q , with a spacing e_1 . This load is equilibrated at the centre of the plate by reactions, with a spacing e_2 .

For the calculation of the plastic bending properties, the effective area of the plate should be assessed by:

$$A_{\text{eff}} = \psi b t \quad (99)$$

where:

$$\psi = 1 - \frac{\mu^2 t 3 \sqrt{3} + \lambda \mu (2e_1 + e_2) - \lambda^2 (e_1 - e_2)}{6 \mu b}$$

$$\mu = \frac{e_1 - e_2}{t} \frac{q}{f_y}$$

$$\lambda = 1 - \sqrt{1 - \mu}$$

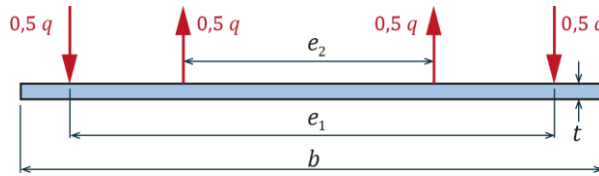


Figure 35: Transverse equilibrium of the lower part of a slim floor section

For SFB sections:

- Formula (99) is applied to the lower steel plate, with:

$$b = b_p \text{ and } t = t_p$$

$$e_1 = b_p - 2 d_{\text{app}}$$

$$e_2 = b_{\text{fi}}$$

$$f_y = f_{yp}$$

- Formula (99) is applied to the lower flange, with:

$$b = b_{\text{fi}} \text{ and } t = t_{\text{fi}}$$

$$e_1 = b_{\text{fi}}$$

$$e_2 = t_w$$

$$f_y = f_{yi}$$

For IFB-A sections, Formula (99) is applied to the lower steel plate, with:

$$b = b_p \text{ and } t = t_p$$

$$e_1 = b_p - 2 d_{\text{app}}$$

$$e_2 = t_w$$

$$f_y = f_{yp}$$

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For IFB-B sections and SAB sections, Formula **(99)** is applied to the lower flange, with:

$$b = b_{fi} \text{ and } t = t_{fi}$$

$$e_1 = b_{fi} - 2 d_{app}$$

$$e_2 = t_w$$

$$f_y = f_{yi}$$

- (3) The second method to take account of the transverse bending to the lower flange is given by the CoSFB technical specifications **[19]**. The transverse bending is taken into account by reducing the thickness of the lower flange or of the plate as follows:

$$t_{eff} = \rho_t t \quad (100)$$

where:

$$\rho_t = \frac{1}{2} (1 + \sqrt{1 - \eta_m})$$

$$\eta_m = \frac{m_{x_{bt},Ed}}{m_{x_{bt},Rd}}$$

$m_{x_{bt},Ed}$ is the transverse moment per unit length, given by:

$$m_{x_{bt},Ed} = q d_{bt}$$

$m_{x_{bt},Rd}$ is the bending resistance of the plate, given by:

$$m_{x_{bt},Rd} = 1,2 \frac{t^2 f_y}{6 \gamma_{M0}}$$



No indication is given in CoSFB software on the resistance.

The factor 1,2 in the formula here above is coming from the method of the prEN 1994-1-1 Annex I.

For SFB sections:

- Formula **(100)** is applied to the lower steel plate, with:

$$t = t_p$$

$$d_{bt} = (b_p - b_{fi}) / 2 - d_{app}$$

$$f_y = f_{yp}$$

- Formula **(100)** is applied to the lower flange, with:

$$t = t_{fi}$$

$$d_{bt} = (b_{fi} - t_w - r_{ci}) / 2 - d_{app}$$

$$f_y = f_{yi}$$

For IFB-A sections, Formula **(100)** is applied to the lower steel plate, with:

$$t = t_p$$

$$d_{bt} = (b_p - t_w) / 2 - d_{app}$$

$$f_y = f_{yp}$$

For IFB-B sections and SAB sections, Formula **(100)** is applied to the lower flange, with:

$$t = t_{fi}$$

$$d_{bt} = (b_{fi} - t_w - r_{ci}) / 2 - d_{app}$$

$$f_y = f_{yi}$$

For the calculation of q , see § 6.9.

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- (4) The third method to take account of the transverse bending to the lower flange is given in Annex I of the draft for the second generation of EN 1994-1-1 [12]. The transverse bending of the lower plate is taken into account by reducing the yield strength of that plate, as follows:

$$f_{y,eff} = \psi_y f_y \quad (101)$$

where:

$$\psi_y = \frac{\eta_m - \sqrt{\eta_m^2 - 16 \eta_m + 16}}{2 (\eta_m - 2)}$$

$$\eta_m = \frac{m_{ybt,Ed}}{m_{ybt,Rd}}$$

$m_{ybt,Ed}$ is the transverse moment per unit length, given by:

$$m_{ybt,Ed} = q d_{bt}$$

$m_{ybt,Rd}$ is the bending resistance of the plate, given by:

$$m_{ybt,Rd} = 1,2 \frac{t^2 f_y}{6 \gamma_{M0}}$$

η_m may be directly obtained from:

$$\eta_m = 5 \frac{q d_{bt} \gamma_{M0}}{t^2 f_y}$$

For SFB sections:

- Formula (101) is applied to the lower steel plate, with:

$$t = t_p$$

$$d_{bt} = (b_p - b_{fi}) / 2 - d_{app}$$

$$f_y = f_{yp}$$

- Formula (101) is applied to the lower flange, with:

$$t = t_{fi}$$

$$d_{bt} = (b_{fi} - t_w - r_{ci}) / 2 - d_{app}$$

$$f_y = f_{yi}$$

For IFB-A sections, Formula (101) is applied to the lower steel plate, with:

$$t = t_p$$

$$d_{bt} = (b_p - t_w) / 2 - d_{app}$$

$$f_y = f_{yp}$$

For IFB-B sections and SAB sections, Formula (101) is applied to the lower flange, with:

$$t = t_{fi}$$

$$d_{bt} = (b_{fi} - t_w - r_{ci}) / 2 - d_{app}$$

$$f_y = f_{yi}$$

- (6) In (2), (3) or (4), q is calculated according to § 6.9. if the value of q_i is not the same on both sides of the supporting plate, the maximal value will be assumed (i.e. $q = 2 \max\{q_1; q_2\}$).
- (7) In (2), $d_{app} = 40$ mm (according to [18]). In (3) and (4), d_{app} is calculated according to § 6.9 (3).

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5.5.8. Slim floor composite sections

- (1) The section properties of slim floor composite sections will be assessed through the iterative procedure in **Annex B**, where the following assumptions are considered:
- the yield strength of hot rolled profiles is assumed to be uniform over the section and is obtained according to § 2.10.1;
 - the yield strength of the plate, if any (all slim floors sections but SAB), is assessed according to § 2.10.3;
 - when the MV interaction is considered, the yield strength of the web (including fillets) is reduced accordingly;
 - for the plastic properties, the area of the lower part of the sections (plate and/or lower flange) on which the slab loads are applied are reduced to take account of the transverse bending of these parts according to one of the methods given in § 5.5.7 (2), (3) or (4);
 - the slab participates to the resistance of the composite section under bending moment. The participating part of the slab is defined according to § 5.2.
- (2) For edge beams, the effect of non-symmetry about the z axis is disregarded. Sections properties are calculated as if the section is symmetric, i.e. the main axes remain the natural axes.



This assumption is valid in so far as the slab laterally prevents horizontal bending, so that the bending is imposed in the vertical plane of the web.

5.5.9. Reduction of the plastic resistance for composite section with deep plastic neutral axis

- (1) For usual composite cross-sections, with or without partial concrete encasement, and with a steel profile of grade S420 or S460, the plastic bending resistance for positive bending moment should be reduced as follows:

$$M'_{pl,Rd} = \beta M_{pl,Rd} \quad (102)$$

where: $M_{pl,Rd}$ is the plastic bending resistance of the cross-section;

β is the reduction factor, as given in (2).



According to EN 1994-1-1, 6.2.1.2 (2).

For sections where the value of f_y is directly imposed by the User, the procedure is applied when $f_y > 355$ MPa.

- (2) The reduction factor β is given by:

$$\beta = 1 \quad \text{if } z_{plc} / h_t \leq 0,15 \quad (103)$$

$$\beta = 1 - \frac{0,15}{0,25} \left(\frac{z_{plc}}{h_t} - 0,15 \right) \quad \text{if } 0,15 < z_{plc} / h_t \leq 0,40$$

where: h_t is the total depth of the composite section, given by: $h_t = h_s + t_h + t_d$;

z_{plc} is the distance between the plastic neutral axis and the top fiber of the cross-section, obtained

by: $z_{plc} = t_h + t_d - z_{pl}$.

- (3) If $z_{plc} > 0,4 h_t$, the plastic design of sections is not possible. In this case, Class 3 design criteria will be used.
- (4) When the next generation of Eurocodes will be applied, the reduction factor will be used whatever the steel grade is, according to (5).
- (5) The reduction factor β for the next generation of Eurocodes is given by:

$$\beta = 1 \quad \text{if } z_{plc} / h_t \leq \alpha_1 \quad (104)$$

$$\beta = \beta_1 - \frac{\beta_2 - \beta_1}{\alpha_2 - \alpha_1} \left(\frac{z_{plc}}{h_t} - \alpha_1 \right) \quad \text{if } \alpha_1 < z_{plc} / h_t \leq \alpha_2$$

where: h_t and z_{plc} are given in (2);

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α_1 , α_2 , β_1 and β_2 are given in **Table 25**.

Steel grade	α_1	α_2	β_1	β_2
S235	0,2	0,7	1	0,93
S275	0,2	0,6		0,93
S355	0,2	0,5		0,9
S420/S460	0,15	0,4		0,9

Table 25: Parameters for the reduction factor β – Eurocodes gen. 2

- (6) If $z_{plc} > \alpha_2 h_t$, the plastic design of sections is not possible. In this case, Class 3 design criterion will be used.
- (7) **(1)** to **(6)** also apply to slim floor composite sections.

5.6. Shear areas of sections

5.6.1. Usual steel and composite sections

- (1) The shear area of a hot rolled profile is given by (acc. to EN 1993-1-1, 6.2.6 (3)):

$$A_v = \max\{A - 2 b_f t_f + (t_w + 2r_c) t_f; \eta h_w t_w\} \quad (105)$$

where: A is given by formula **(95)**;

$$b_f = b_{fs} = b_{fi}$$

$$t_f = t_{fs} = t_{fi}$$

$$r_c = r_{cs} = r_{ci}$$

η : see § 2.12.4.

- (2) The shear area of welded section is given by (idem):

$$A_v = \eta h_w t_w \quad (106)$$

- (3) The same shear areas are used for steel and for composite sections.

5.6.2. Steel and composite slim floor sections

- (1) The shear area of a SFB or SAB slim floor section is given by Formula (105).

- (2) The shear area of an IFB (either A or B) slim floor section is given by:

$$A_v = h_w t_w + (4 - \pi) \frac{r_c^2}{2} + \frac{t_f}{2} (t_w + 2 r_c) \quad (107)$$

5.7. Shear resistance of welded studs

5.7.1. General

- (1) The shear resistance of welded studs is calculated according to the type of sections and the configuration of slab (see **Table 26**).

Type of sections and configuration of slab	Formulas of P_{Rd}
Usual composite section	
Solid slabs	see § 5.7.2 (1) or § 5.7.3 (1)
Composite slab with longitudinal ribs	see § 5.7.2 (2) or § 5.7.3 (2) with reduction factor k_ℓ
Composite slab with transverse ribs and a Cofraplus 220 sheeting	formulas for solid slabs in § 5.7.2 (1) or § 5.7.3 (1)
Composite slabs with transverse ribs and common sheetings	
continuous concrete rib ⁽¹⁾	see § 5.7.2 (3) or § 5.7.3 (3) with reduction factor k_t § 5.7.3 (3) for the second generation of EN, when § 5.7.3 (4) does not apply
discontinuous concrete rib ⁽²⁾	see § 5.7.2 (2) or § 5.7.3 (2) with reduction factor k_ℓ
Composite slim floors – all types of slabs	
Composite dowels ⁽³⁾	see § 5.8
Studs welded on the upper flange ⁽⁴⁾	formulas for solid slabs in § 5.7.2 (1) or § 5.7.3 (1)
Studs welded on the web ⁽⁵⁾	formulas for solid slabs in § 5.7.2 (1) or § 5.7.3 (1)

Table 26: Calculation of the shear resistance of studs

5.7.2. First generation of Eurocodes

- (1) The shear resistance P_{Rd} of a welded stud in a solid slab is obtained with the following formula, according to § 6.6.3.1 of EN 1994-1-1 [7]:

$$P_{Rd} = \min\{P_{Rd,s}; P_{Rd,c}\} \quad (108)$$

where: $P_{Rd,s}$ is the shear resistance associated with the failure of the shank of the stud, given by:

$$P_{Rd,s} = 0,8 \frac{f_{u,sc} \pi d^2}{4 \gamma_{Vs}} \quad (109)$$

$P_{Rd,c}$ is the shear resistance associated with the failure of the concrete, given by:

$$P_{Rd,c} = 0,29 \frac{\alpha d^2 \sqrt{f_{ck} E_{cm}}}{\gamma_{Vc}} \quad (110)$$

d diameter of the shank (see § 2.6.1);

$f_{u,sc}$ ultimate tensile strength of the steel of the stud (see § 2.6.1);

f_{ck} characteristic value of the cylinder compressive strength of concrete at 28 days (see § 2.10.5);

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E_{cm} secant modulus of elasticity of concrete (see § 2.10.5);

α parameter defined as follows:

$$\alpha = 1 \quad \text{for } h_{sc} / d > 4$$

$$\alpha = 0,2 \left(\frac{h_{sc}}{d} + 1 \right) \quad \text{for } 3 \leq h_{sc} / d \leq 4$$



Formula (108) is applied to welded studs in solid slabs with or without haunch and with or without prefabricated part.

- (2) The shear resistance P_{Rd} of a welded stud in a composite slab with ribs parallel to the beam axis (**Figure 15**) is obtained by the following formula, according to § 6.6.4.1 of EN 1994-1-1:

$$P_{Rd} = k_{\ell} \min\{P_{Rd,s}; P_{Rd,c}\} \quad (111)$$

where: $P_{Rd,s}$ and $P_{Rd,c}$ are as defined by Formulas (109) and (110) respectively;

k_{ℓ} is a reduction factor to take account of the effect of the longitudinal rib, given by:

$$k_{\ell} = 0,6 \frac{b_0}{h_p} \left(\frac{h_{sc}}{h_p} - 1 \right) \quad \text{but } k_{\ell} \leq 1,0 \quad (112)$$

h_p is the height of the steel profiled decking (§ 2.5.1);

h_{sc} is the height of the stud;

b_0 is the representative width of a rib (see § 2.5.1 (4)).

(2) is also applied for composite slabs with perpendicular ribs, when the configuration at beam support is PE3 – see § 2.3.2. In this case, $b_0 = b_{fs} - 2 b_{app,min}$.

- (3) The shear resistance P_{Rd} of a welded stud in a composite slab with ribs perpendicular to the beam axis, with PE1 or PE2 configurations – see § 2.3.2, is obtained by the following formula, according to § 6.6.5.1 of EN 1994-1-1:

$$P_{Rd} = k_t \min\{P_{Rd,s}; P_{Rd,c}\} \quad (113)$$

where: $P_{Rd,s}$ and $P_{Rd,c}$ are as defined by Formulas (109) and (110) respectively;

k_t is a reduction factor to take account of the effect of the longitudinal rib, given by:

$$k_t = \frac{0,7}{\sqrt{n_r}} \frac{b_0}{h_p} \left(\frac{h_{sc}}{h_p} - 1 \right) \quad \text{but } k_t \leq k_{t,max} \quad (114)$$

h_p is the height of the steel profiled decking (§ 2.5.1);

h_{sc} is the height of the stud;

n_r is the number of studs in the through, which should not be greater than 2;

$k_{t,max}$ is the maximal value of the reduction factor k_t , which is given by **Table 27**.

Number of studs per rib	t (mm) Thickness of the steel sheeting	Through deck welding $d \leq 20$ mm	Prepunched steel sheeting $d = 19$ or 22 mm
$n_r = 1$	$\leq 1,0$	0,85	0,75
	$> 1,0$	1,0	0,75
$n_r = 2$	$\leq 1,0$	0,70	0,60
	$> 1,0$	0,80	0,60

Table 27: Upper limits of the reduction factor k_t

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(4) The scope of application for Formula **(113)** is as follows:

- through deck welding: the diameter of the shank is not greater than 20 mm: $d \leq 20$ mm
- prepunched sheeting: $19 \text{ mm} \leq d \leq 22 \text{ mm}$
- the height of the sheeting is lower than 85 mm: $h_p \leq 85$ mm
- the width b_0 of the rib should be at least equal to h_p : $b_0 \geq h_p$
- the number of studs in the rib is not greater than 2: $n_r \leq 2$

Other configurations are out of the scope of the software.

5.7.3. Second generation of EN

(1) The shear resistance of a welded stud in a solid slab is obtained from Formula **(108)**, where $P_{Rd,s}$ is obtained from Formula **(109)** and $P_{Rd,c}$ from the following Formula **(115)**:

$$P_{Rd,c} = 0,29 \frac{k_{cc} d^2 \sqrt{f_{ck} E_{cm}}}{\gamma_{Vc}} \quad (115)$$

where: k_{cc} is a relaxation factor – see § 2.12.4.

(2) The shear resistance P_{Rd} of a welded stud in a composite slab with ribs parallel to the beam axis is obtained by the Formulas **(111)**, where $P_{Rd,c}$ is obtained according to **(1)**. **(2)** is also applied for composite slabs with perpendicular ribs, when the configuration at beam support is PE3 – see § 2.3.2.

(3) The shear resistance P_{Rd} of a welded stud in a composite slab with continuous ribs perpendicular to the beam axis, with PE1 or PE2 configurations – see § 2.3.2, is obtained by the Formulas **(113)** and **(114)**, where $P_{Rd,c}$ is obtained according to **(1)**, when **(4)** applies. Otherwise, P_{Rd} is obtained according to **(5)**.

(4) The scope of application for Formula **(114)** is defined by:

- the height of the sheeting is lower than 105 mm: $h_p \leq 105$ mm
- the thickness of the sheeting is at least equal to 0,7 mm: $t_p \geq 0,7$ mm
- through deck welding: the diameter of the shank is not greater than 20 mm: $d \leq 20$ mm
- prepunched sheeting: $19 \text{ mm} \leq d \leq 22 \text{ mm}$
- re-entrant trough profile: the embedment depth of the stud h_A should be at least equal to 2 times its diameter: $h_A \geq 2 d$
- open trough profile: the embedment depth of the stud h_A should be at least equal to 2,7 times its diameter: $h_A \geq 2,7 d$
- open trough profile: assuming that the stud is located at the centre of the rib, the width b_0 of the rib should be at least 120 mm: $b_0 \geq 120$ mm
- the number of studs in the rib is not greater than 2: $n_r \leq 2$

where the embedment height of the stud is given by:

$$h_A = h_{sc} - h_p \quad (116)$$

(5) For composite slabs with ribs perpendicular to the beam and open trough profile, where the previous conditions are not met, the shear resistance of the stud is obtained by the Formula **(117)**, provided that the following conditions are met:

- the embedment depth of the stud h_A is at least equal to 2 times its diameter: $h_A \geq 2 d$
- the number of studs in the rib is not greater than 2: $n_r \leq 2$

$$P_{Rd} = \min\{P_{Rd,s}; P_{Rd,c}\} \quad (117)$$

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where: $P_{Rd,s}$ is the shear resistance associated with the failure of the steel shank of the stud, given by:

$$P_{Rd,s} = 0,58 \frac{f_{u,sc} \pi d^2}{4 \gamma_{Vs}} \quad (118)$$

$P_{Rd,c}$ is the shear resistance associated with the failure of the concrete, given by:

$$P_{Rd,c} = \frac{k_{cc} C_2 k_u}{\gamma_{Vc}} \left[\frac{f_{ctk,0,05} W_{sc}}{h_p n_r} + \frac{n_y M_{pl,sc}}{d_p} \right] \quad (119)$$

d_p is given by:

$$d_p = 0,82 h_p - d/2$$

C_2 is given by:

$$C_2 = 1,85 \frac{h_p}{b_0} \quad \text{but: } 1,0 \leq C_2 \leq 1,35$$

W_{sc} is given by:

$$W_{sc} = \frac{b_t^2}{6} (2,4 h_{sc} + (n_r - 1) s_y)$$

$M_{pl,sc}$ is given by:

$$M_{pl,sc} = \frac{1}{6} f_{u,sc} d^3$$

n_y is given by:

$$n_y = 2 \quad \text{for } n_r = 1$$

$$n_y = 1 + \frac{h_A - 2 d}{0,52 d} \quad \text{but: } n_y \leq 2 \quad \text{for } n_r = 2$$

k_u is a correction factor, which is taken equal to 1,25 for through deck welding studs with $t_p \geq 1$ mm, and which is taken equal to 1 otherwise.

s_y is the transverse spacing of studs (in the case where $n_r = 2$), which is obtained in the software as the minimum spacing between studs, i.e.:

$$s_y = 4 d$$

(6) Other configurations than those mentioned in (4) and (5) are outside the scope of the software.

5.8. Shear resistance of composite dowels

5.8.1. General

(1) Composite dowels may be used for the connection of slim floors composite sections.

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5.8.2. According to German Zulassung

- (1) The characteristic resistance of each rebar of the composite dowels connection is obtained according to **Table 28**.

Thickness of the web	Concrete class			
	C25/30	C30/37	C35/45	C40/50 to C55/67
$7,5 \text{ mm} \leq t_w \leq 15,5 \text{ mm}$	117	125	135	122
$t_w \geq 15,5 \text{ mm}$	148	157	166	122

Table 28: Characteristic values of the shear connection P_{Rk} per hole in [kN]

- (2) The design value is obtained by: $P_{Rd} = P_{Rk} / \gamma_v$.

5.8.3. According to Annex I of prEN 1994-1-1

- (1) The design resistance of each rebar of the composite dowels connection is obtained according to the following Formula, according to I.3.3 of prEN 1994-1-1:

$$P_{Rd} = \frac{\pi d_s^2 f_{sk}}{\gamma_v 4 \sqrt{3}} \quad (120)$$

5.9. Calculation of elastic stresses – Usual Sections

5.9.1. Principles

- (1) For each cross-section, elastic stresses will be assessed at predefined stress points.
- (2) Under the action of a bending moment M_{Ed} , the normal stresses at a stress point i of the structural steel profile are obtained by:

$$\sigma_i = M_{Ed} (z_i - z_{el}) / I_y \quad (121)$$

where: I_y is the second moment of area of the cross-section;

z_{el} is the location of the neutral axis;

z_i is the location of the point i .

For composite sections, properties of the transformed cross-section with the relevant modular ratios, are used.

- (3) For composite cross-section under the action of a bending moment M_{Ed} , the normal stresses at a stress point i of the steel reinforcement are obtained by:

$$\sigma_i = M_{Ed} \frac{(z_i - z_{el})}{n_s I_{y,h}} \quad (122)$$

where n_s is the modular ratio for the reinforcement, given by $n_s = E_a / E_s$;

$I_{y,h}$ is the second moment of area of the transformed cross-section.

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- (4) For composite cross-section under the action of a bending moment M_{Ed} , the normal stresses at a stress point i in concrete is obtained by:

$$\sigma_i = M_{Ed} \frac{(z_i - z_{el})}{n_c I_{y,h}} \quad (123)$$

where n_c is the modular ratio for concrete (either partial concrete encasement or slab), obtained considering the load case;

$I_{y,h}$ is the second moment of area of the transformed cross-section.

- (5) Under the action of a shear force V_{Ed} , the shear stresses at the stress points are obtained by:

$$\tau_i = \frac{V_{Ed} S_i}{t_w I_y} \quad \text{at stress points in the web of the structural steel profile} \quad (124)$$

$$\tau_i = 0 \quad \text{at other stress points}$$

where: S_i is the first moment of area associated to the stress point;

t_w is the thickness of the web;

I_y is the second moment of area of the cross-section.

For composite sections, S_i and I_y are obtained considering the transformed cross-section with the relevant modular ratios.

- (6) For composite sections, elastic stresses are calculated separately for each load case, taking into account the relevant modular ratio and the associated properties of the transformed cross-section.
- (7) For composite sections, stresses in concrete and in reinforcement are calculated for each load case considering the relevant properties under positive bending moments, both in tension and in compression. After combinations, either at serviceability limit state or ultimate limit state, the following process are applied:
- if the stresses in the concrete slab are in compression, the bending moment hypothesis is kept and the calculations are unchanged;
 - if the stresses in the concrete slab are in tension, the bending moment hypothesis is not correct, and the calculations are carried out again considering negative bending moments.

5.9.2. Usual steel sections

- (1) For usual steel cross-sections without partial concrete encasement, the following 5 stress points are defined in the vertical axis of the section (**Figure 36**):
- at the external (point ①) and internal (②) sides of the upper flange;
 - at the centroid of the section (③);
 - at the external (point ⑤) and internal (④) sides of the lower flange.

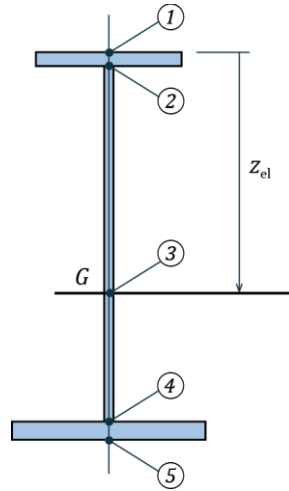


Figure 36 : Calculation of elastic stresses for usual steel sections

- (2) The location of the five stress points are given in **Table 29**.

i	z_i
①	$z_1 = 0$
②	$z_2 = -t_{fs}$
③	$z_3 = z_{ela}$
④	$z_4 = -h_s + t_{fi}$
⑤	$z_5 = -h_s$

Table 29: Location of the stress points of usual steel sections

- (3) Under the action of an axial force N_{Ed} , the normal stresses at these 5 points are obtained by:

$$\sigma_i = N_{Ed} / A_a \quad (125)$$

where: A_a is the area of the cross-section.

- (4) Under the action of a bending moment M_{Ed} , the normal stresses at the 5 stress points are obtained with Formula (121), where I_y is the second moment of area of the section, $I_{y,a}$ and the location z_i are as given in **Table 29**.
- (5) Under the action of a shear force V_{Ed} , the shear stresses are obtained by Formula (124) for the stress point in the web ② to ④, where S_i is the first moment of area as given in **Table 30**.

i	S_i
②	$S_2 = b_{fs} t_{fs} (z_{el} - t_{fs} / 2)$
③	$S_3 = b_{fs} t_{fs} (z_{el} - t_{fs} / 2) + (z_{el} - t_{fs})^2 t_w / 2$
④	$S_4 = b_{fi} t_{fi} (z_{el} - (h_s - t_{fi} / 2))$

Table 30: First moments of areas for the calculation of shear stresses

5.9.3. Usual steel sections with partial concrete encasement

- (1) For usual steel cross-sections with partial concrete encasement, the stress points comprise the five same points as in § 5.9.2 (1), plus the following points in the concrete and the reinforcements in the encasement (Figure 37):
- at the upper (point ⑥) and lower (⑦) sides of the concrete encasement;
 - at the first, second and third layers of reinforcements, if any (points ⑧, ⑨ and ⑩ respectively).

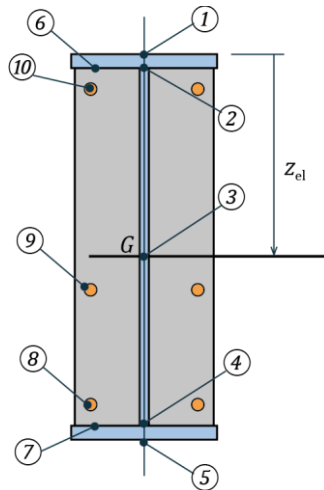


Figure 37 : Calculation of elastic stresses for usual steel sections with partial concrete encasement

- (2) The location of the five stress points in the steel profile are given in Table 29. The location of the other stress points are given in Table 31.

i	z_i
⑥	$z_6 = -t_{fs}$
⑦	$z_7 = -h_s + t_{fi}$
⑧	$z_8 = z_{es1}$
⑨	$z_9 = z_{es2}$
⑩	$z_{10} = z_{es3}$

Table 31: Location of the points for stress calculations

- (3) Under the action of a bending moment M_{Ed} , the normal stresses are obtained:
- by Formula (121), for the stress points ① to ⑤ (structural steel);
 - by Formula (122), for the stress points ⑧ to ⑩ (steel reinforcement);
 - by Formula (123), for the stress points ⑥ and ⑦ (concrete), with the modular ratio n_{ec} for the concrete of the encasement and the associated load case (see § 5.3);
- where, in these formulas, $I_{y,h}$ is the second moment of area of the transformed section, and the location z_i is given in (2).
- (4) Under the action of a shear force V_{Ed} , the same formula as in § 5.9.2 (5) applies. No shear stress is considered neither in concrete nor in steel reinforcement.

5.9.4. Usual composite sections

- (1) For usual composite cross-sections without partial concrete encasement, the 8 stress points are defined as follows (**Figure 38**):
- at the external (point ①) and internal (②) sides of the upper flange;
 - at the centroid of the steel section (③);
 - at the external (point ⑤) and internal (④) sides of the lower flange.
 - at the upper side of the concrete face (point ⑪);
 - at the first and second layers of longitudinal reinforcement of the slab, if any (points ⑫ and ⑬).

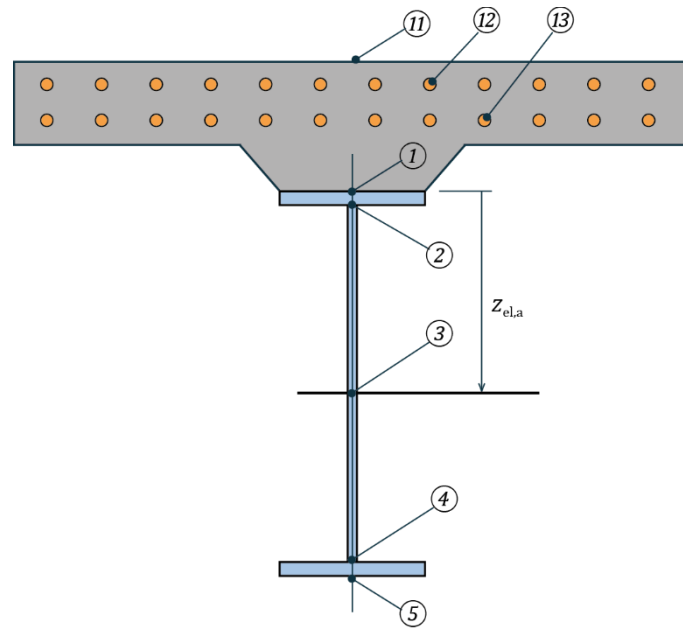


Figure 38 : Calculation of elastic stresses for usual composite sections

- (2) The location of the five stress points in the steel profile are given in **Table 29**. The location of the other stress points are given in **Table 32**.

i	z_i
⑪	$z_{11} = t_d + t_h$
⑫	$z_{12} = z_{s1}$
⑬	$z_{13} = z_{s2}$

Table 32: Location of the stress points in the slab of usual composite sections

- (3) Under the action of a bending moment M_{Ed} , the normal stresses are obtained:
- by Formula (121), for the stress points ① to ⑤ (structural steel);
 - by Formula (122), for the stress points ⑫ and ⑬ (steel reinforcement);
 - by Formula (123), for the stress point ⑪ (concrete), with the modular ratio n_c for the concrete of the slab and the associated load case (see § 5.3);

where, in these formulas, $I_{y,h}$ is the second moment of area of the transformed section, and the location z_i is given in (2).

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- (4) Under the action of a shear force V_{Ed} , the same formula as in § 5.9.2 (5) applies. No shear stress is considered neither in concrete nor in steel reinforcement.
- (5) Under the axial force N_{sh} due to the shrinkage of concrete, the normal stresses in the stress points are obtained by:

$$\sigma_i = N_{sh} / A_h \quad \text{for points ① to ⑤ (structural steel);} \quad (126)$$

$$\sigma_i = \frac{N_{sh}}{n_s A_h} \quad \text{for the stress points ⑫ and ⑬ (steel reinforcement);} \quad (127)$$

$$\sigma_i = N_{sh} \left(\frac{1}{n_c A_h} - \frac{1}{A_c} \right) \quad \text{for the stress point ⑪ (concrete)} \quad (128)$$

where: n_c is the modular ratio for the concrete of the slab and the associated load case (see § 5.3);

A_h is the area of the transformed section;

A_c is the area of the slab;

N_{sh} is the axial force in the slab due to shrinkage, defined by § 6.7.3 (4).

5.9.5. Usual composite section with partial concrete encasement

- (1) For usual composite cross-sections with partial concrete encasement, the 13 stress points are defined as follows (Figure 39):
- at the external (point ①) and internal (②) sides of the upper flange;
 - at the centroid of the steel section (③);
 - at the external (point ⑤) and internal (④) sides of the lower flange.
 - at the upper (point ⑥) and lower (⑦) sides of the **concrete** encasement;
 - at the first, second and third layers of reinforcements, if any (points ⑧, ⑨ and ⑩ respectively);
 - at the upper side of the concrete face (point ⑪);
 - at the first and second layers of longitudinal reinforcement of the slab, if any (points ⑫ and ⑬).

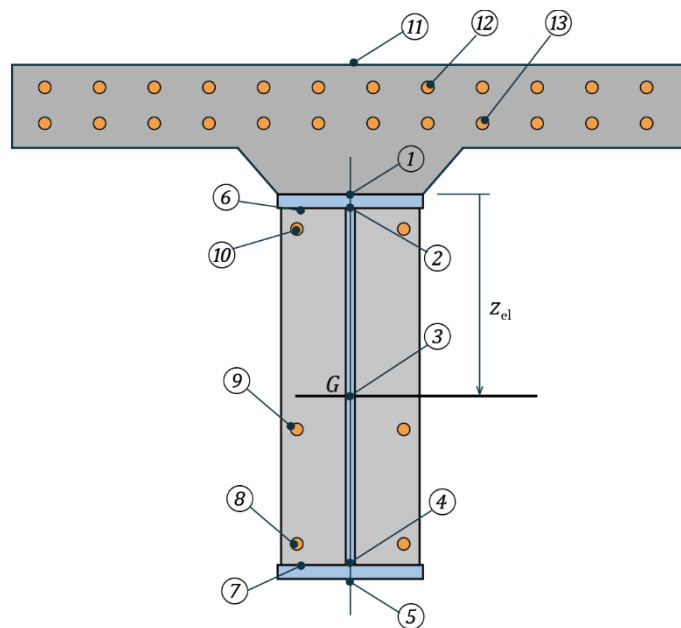


Figure 39 : Calculation of elastic stresses for usual composite sections with partial **concrete** encasement

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(2) The location of the five stress points in the steel profile (① to ⑤) are given in **Table 29**. The location of the 5 stress points of the encasement (⑥ to ⑩) are given in **Table 31**. The location of the other stress points for the slab (⑪ to ⑬) are given in **Table 32**.

(3) Under the action of a bending moment M_{Ed} , the normal stresses are obtained:

- by Formula (121), for the stress points ① to ⑤ (structural steel);
- by Formula (122), for the stress points ⑧ to ⑩, ⑫ and ⑬ (steel reinforcement);
- by Formula (123), for the stress point ⑪ (concrete), with the modular ratio n_c for the concrete of the slab and the associated load case (see § 5.3);
- by Formula (123), for the stress points ⑥ and ⑦ (concrete), with the modular ratio n_{ec} for the concrete of the encasement and the associated load case (see § 5.3).

where, in these formulas, $I_{y,h}$ is the second moment of area of the transformed section, and the location z_i is given in (2).

(4) Under the action of a shear force V_{Ed} , the same formula as in § 5.9.2 (5) applies. No shear stress is considered neither in concrete nor in steel reinforcement.

(5) Under the axial force N_{sh} due to the shrinkage of the slab concrete (defined by § 6.7.3 (4)), the normal stresses in the stress points are obtained by:

Formula (126) for points ① to ⑤ (structural steel);

Formula (127) for the stress points ⑫ and ⑬ (steel reinforcement);

Formula (128) for the stress point ⑪ (concrete of the slab);

$$\sigma_i = \frac{N_{sh}}{n_{ce} A_h} \quad \text{for the stress points ⑥ and ⑦ (concrete of the encasement)} \quad (129)$$

where: n_{ce} is the modular ratio for the concrete of the encasement (see § 5.3);

A_h is the area of the transformed section;

A_c is the area of the slab;

N_{sh} is the axial force in the slab due to shrinkage,.

(6) Under the axial force N_{she} due to the shrinkage of the encasement concrete (defined by § 6.7.3 (5)), the normal stresses in the stress points are obtained by:

$\sigma_i = N_{she} / A_h$ for points ① to ⑤ (structural steel);

$\sigma_i = \frac{N_{she}}{n_s A_h}$ for the stress points ⑫ and ⑬ (steel reinforcement);

$\sigma_i = \frac{N_{she}}{n_c A_h}$ for the stress point ⑪ (concrete of the slab);

$$\sigma_i = N_{she} \left(\frac{1}{n_{ce} A_h} - \frac{1}{A_{ce}} \right) \quad \text{for the stress points ⑥ and ⑦ (concrete of the encasement)} \quad (130)$$

5.9.6. Equivalent stresses in the steel profile

(1) For the points in the structural steel section, the equivalent von Mises stresses, combining the effect of normal and shear stresses, are obtained as follows:

$$\sigma_{eq,i} = \sqrt{\sigma_i^2 + 3 \tau_i^2} \quad (131)$$

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5.9.7. Effect of concrete cracking

- (1) In composite sections where the concrete is assumed to be cracked (i.e. $M_{Ed} < 0$), stresses in the slab reinforcements increase due to the effects of tension stiffening of concrete between cracks. The additional stresses due to this effect are given by (according to EN 1994-1-1, 7.4.3 (3)):

$$\Delta\sigma_s = \frac{0,4 f_{ctm}}{\alpha_{st} \rho_s} \quad (132)$$

where: f_{ctm} is the mean tensile stress of concrete, see § 2.10.5 (3);

ρ_s is the ratio of reinforcement, see § 2.3.3 (5);

α_{st} is obtained by:

$$\alpha_{st} = \frac{A I}{A_a I_a} \quad (133)$$

A and I are the area and the second moment of area, respectively, of the transformed composite section, calculated considering negative bending moment, according to § 5.5.5 or § 5.5.6;

A_a and I_a are the area and the second moment of area, respectively, of the steel section.

5.10. Calculation of elastic stresses – Slim floor sections

5.10.1. SFB steel sections

- (1) In the vertical axis of a SFB steel section, the following 14 stress points are considered for the calculation of elastic stresses (**Figure 40**):

- at the external (point ①) and internal (②) sides of the upper flange;
- at the centroid of the section (③);
- at the external (point ⑤) and internal (④) sides of the lower flange;
- at the external side (point ⑥) of the supporting plate;
- the eight points ⑦ to ⑭ are considered for the combined effect of the local and global bending of the supporting plate and the lower flange.

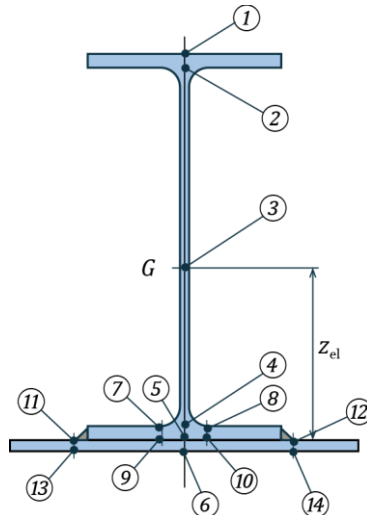


Figure 40 : Calculation of elastic stresses in SFB steel sections

- (2) The location of the stress points is defined in **Table 33**.

i	z_i
①	$z_1 = h_b$
②	$z_2 = h_b - t_{fs}$
③	$z_3 = z_{ela}$
④, ⑦, ⑧	$z_4 = t_{fi}$
⑤, ⑨, ⑩, ⑪, ⑫	$z_5 = 0$
⑥, ⑬, ⑭	$z_6 = -t_p$

Table 33: Location of the stress points in SFB steel sections

- (3) Under the action of a global axial force N_{Ed} , the normal stresses in these 14 points are obtained with Formula (125).
- (4) Under the action of a global bending moment M_{Ed} , the normal stresses in these 14 points are obtained with Formula (121), where I_y is the second moment of area of the section $I_{y,a}$ and the location z_i of the points are given in **Table 33**.
- (5) Under the action of a global shear force V_{Ed} , the shear stresses are obtained by Formula (124) for the stress point in the web ② to ④, where S_i is the first moment of area as given in **Table 34**.

i	S_i
②	$S_2 = b_{fs} t_{fs} (z_{el} - (h_b - t_{fs} / 2))$
③	$S_3 = b_{fs} t_{fs} (z_{el} - (h_b - t_{fs} / 2)) + (z_{el} - t_{fs})^2 t_w / 2$
④	$S_4 = b_{fi} t_{fi} (z_{el} - t_{fi} / 2)$

Table 34: First moments of areas for the calculation of shear stresses

- (6) Under the action of a uniformly distributed load q_1 applied on the left edge of the lower plate, bending stresses and shear stresses are calculated in points ⑦, ⑨, ⑪ and ⑬, as given in **Table 35**.

i	σ_{yi}	τ_{yi}
⑦ ⑧	$\sigma_{yi} = +6 q_i d_f / t_f^2$	$\tau_{yi} = 3 q_i / (2 t_f)$
⑨ ⑩	$\sigma_{yi} = -6 q_i d_f / t_f^2$	$\tau_{yi} = 3 q_i / (2 t_f)$
⑪ ⑫	$\sigma_{yi} = +6 q_{wi} d_p / t_p^2$	$\tau_{yi} = 3 q_{wi} / (2 t_p)$
⑬ ⑭	$\sigma_{yi} = +6 q_{wi} d_p / t_p^2$	$\tau_{yi} = 3 q_{wi} / (2 t_p)$
where:	$d_f = b_p / 2 - d_{app} - r_c - t_w / 2$ $d_p = b_p / 2 - d_{app} - a_{sp} \sqrt{2} / 2 - b_f / 2$ $q_{w1} = 0,5 [q_1 (e_1 + e_{2p}) + q_2 (e_1 - e_{2p})] / e_{2p}$ $q_{w2} = q_1 + q_2 - q_{w1}$	

Table 35: Bending and shear local stresses in the supporting plates of SFB steel sections

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By a safe sided simplification, it is assumed that the maximal shear stress in the supporting plate is obtained at the external fibres of the plates.

- (7) Under the action of a uniformly distributed load q_2 applied on the right edge of the lower plate, bending stresses and shear stresses are calculated in points (8), (10), (12) and (14), as given in **Table 35**.

5.10.2. SFB composite sections

- (1) For SFB composite section, elastic stresses are calculated in the steel profile at the same 14 points as defined in § 5.10.1 (1), plus the two following stress points (15) and (16), respectively at the upper and lower side for the slab.

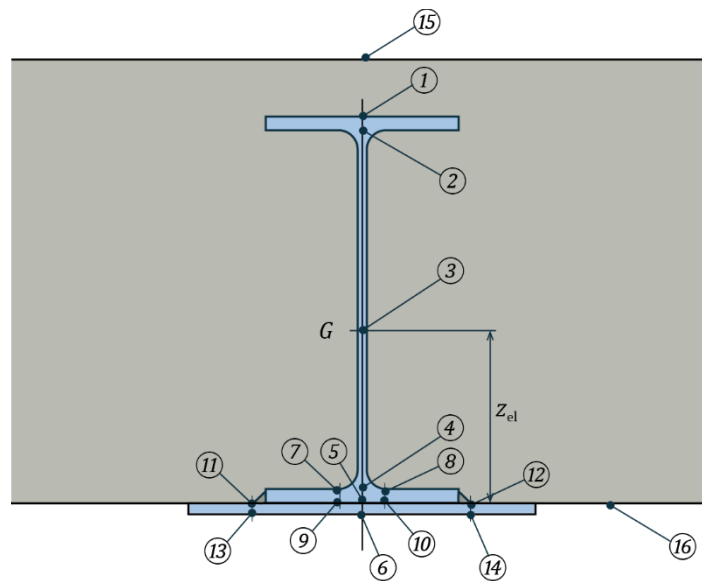


Figure 41 : Calculation of elastic stresses in SFB composite sections

- (2) The location of the stress points in the structural steel are given in **Table 33**. For stress points in concrete:
- $$z_{15} = t_d$$
- $$z_{16} = 0$$
- (3) Under the action of a global bending moment M_{Ed} , the normal stresses in these 14 points (1) to (14) of the steel profile are obtained with Formula (121), where I_y is the second moment of area of the transformed section $I_{y,h}$ and the location z_i of the points are given in (2). In the 2 stress points (15) and (16) of the concrete slab, normal stresses are obtained by Formula (122), where n is the modular ratio for the concrete of the slab and the associated load case.
- (4) § 5.10.1 (5) applies for the calculation of the shear stresses under a global shear force V_{Ed} .
- (5) § 5.10.1 (6) and (7) applies for the calculation of the local stresses in the lower flanges.

5.10.3. IFB-A steel sections

- (1) In the vertical axis of a IFB-A steel section, the following 5 stress points are considered for the calculation of elastic stresses (**Figure 42**):
- at the external (point (1)) and internal ((2)) sides of the upper flange;
 - at the centroid of the section ((3));
 - at the external (point (5)) and internal ((4)) sides of the lower plate.

Additionally, the four points ⑥ to ⑨ are considered for the local bending of the supporting plate.

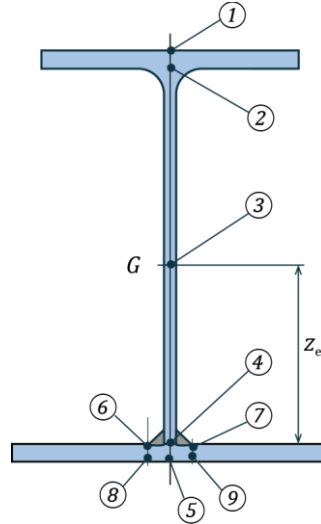


Figure 42 : Calculation of elastic stresses in IFB-A steel sections

- (2) The location of the stress points are defined in **Table 36**.

i	z_i
①	$z_1 = h_b$
②	$z_2 = h_b - t_{fs}$
③	$z_3 = z_{ela}$
④, ⑥, ⑦	$z_4 = 0$
⑤, ⑧, ⑨	$z = -t_p$

Table 36: Location of the stress points in IFB-A steel sections

- (3) Under the action of a global axial force N_{Ed} , the normal stresses in these 9 stress points are obtained with Formula **(125)**.
- (4) Under the action of a global bending moment M_{Ed} , the normal stresses in the 9 stress points are obtained with Formula **(121)**, where I_y is the second moment of area of the steel section, $I_{y,a}$ and the location z_i of the points are given in **Table 36**.
- (5) Under the action of a global shear force V_{Ed} , the shear stresses are obtained by Formula **(124)** for the stress points in the web ② to ④, where S_i is the first moment of area as given in **Table 34** for points ② and ③ and as follows for point ④:

$$S_4 = b_p t_p (z_{el} + t_p / 2) \quad \text{for point ④}$$

- (6) Under the action of a uniformly distributed load q_1 applied on the left edge of the lower plate, bending stresses and shear stresses are calculated in points ⑥ and ⑧, as given in **Table 37**.

i	σ_{yi}	τ_{yi}
⑥ ⑦	$\sigma_{yi} = +6 q_i d_p / t_p^2$	$\tau_{yi} = 3 q_i / (2 t_p)$
⑧ ⑨	$\sigma_{yi} = -6 q_i d_p / t_p^2$	$\tau_{yi} = 3 q_i / (2 t_p)$
where:	$d_p = b_p / 2 - d_{app} - a_{sp} \sqrt{2} / 2 - t_w / 2$	

Table 37: Bending and shear local stresses in the supporting plates of IFB-A steel sections

- (7) Under the action of a uniformly distributed load q_2 applied on the right edge of the lower plate, bending stresses and shear stresses are calculated in points ⑦ and ⑨, as given in **Table 37**.

5.10.4. IFB-A composite sections

- (1) For IFB-A composite section, elastic stresses are calculated in the steel profile at the same 9 stress points as defined in § 5.10.3 (1), plus the two following 2 stress points ⑩ and ⑪, respectively at the upper and lower side for the slab.

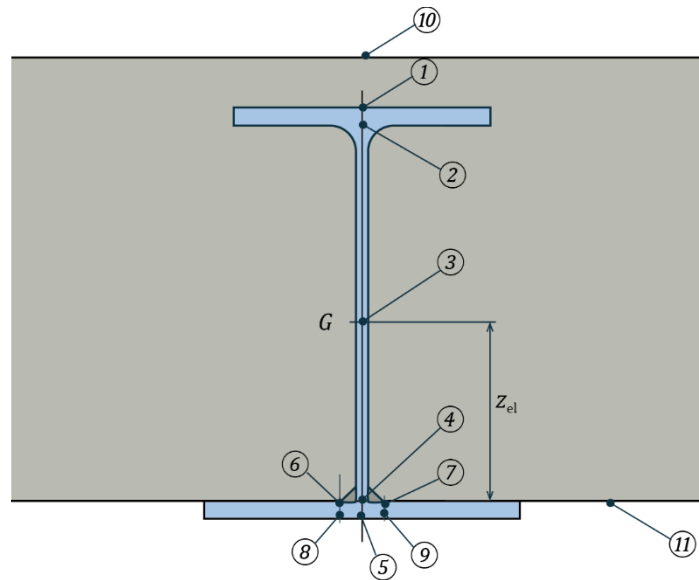


Figure 43 : Calculation of elastic stresses in IFB-A composite sections

- (2) The location of the stress points in the structural steel are given in **Table 36**. For stress points in concrete:
- $$z_{10} = t_d$$
- $$z_{11} = 0$$
- (3) Under the action of a global bending moment M_{Ed} , the normal stresses are obtained with Formula (121) for the stress points in structural steel and with Formula (122), for stress points in concrete.
- (4) § 5.10.3 (5) applies for the calculation of the shear stresses under a global shear force V_{Ed} .
- (5) § 5.10.3 (6) and (7) applies for the calculation of the local stresses in the lower flanges.

5.10.5. IFB-B steel sections

- (1) In the vertical axis of a IFB-B steel section, the following 5 stress points are considered for the calculation of elastic stresses (**Figure 44**):
- at the external (point ①) and internal (②) sides of the upper plate;
 - at the centroid of the section (③);
 - at the external (point ⑤) and internal (④) sides of the lower flange.
- Additionally, the four points ⑥ to ⑨ are considered for the local bending of the supporting lower flange.

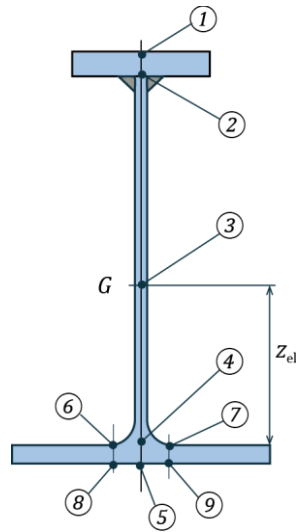


Figure 44 : Calculation of elastic stresses IFB-B steel sections

- (2) The location of the stress points are defined in **Table 38**.

i	z_i
①	$z_1 = h_b$
②	$z_2 = h_b - t_p$
③	$z_3 = z_{ela}$
④, ⑥, ⑦	$z_4 = 0$
⑤, ⑧, ⑨	$z = -t_{fi}$

Table 38: Location of the stress points in IFB-B steel sections

- (3) Under the action of a global axial force N_{Ed} , the normal stresses in these 9 stress points are obtained with Formula (125).
- (4) Under the action of a global bending moment M_{Ed} , the normal stresses in the 9 stress points are obtained with Formula (121), where I_y is the second moment of area of the steel section, $I_{y,a}$ and the location z_i of the points are given in **Table 38**.

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- (5) Under the action of a global shear force V_{Ed} , the shear stresses are obtained by Formula (124) for the stress point in the web ② to ④, where S_i is the first moment of area as given in Table 34 for point ③ and as follows for points ② and ④:

$$S_2 = b_p t_p (z_{el} - (h_b - t_p / 2)) \quad \text{for point ②}$$

$$S_4 = b_{fi} t_{fi} (z_{el} + t_{fi} / 2) \quad \text{for point ④}$$

- (6) Under the action of a uniformly distributed load q_1 applied on the left edge of the lower plate, bending stresses and shear stresses are calculated in points ⑥ and ⑧, as given in Table 39.

i	σ_{yi}	τ_{yi}
⑥ ⑦	$\sigma_{yi} = +6 q_i d_f / t_f^2$	$\tau_{yi} = 3 q_i / (2 t_f)$
⑧ ⑨	$\sigma_{yi} = -6 q_i d_f / t_f^2$	$\tau_{yi} = 3 q_i / (2 t_f)$
where:	$d_f = b_f / 2 - d_{app} - r_c - t_w / 2$	

Table 39: Bending and shear local stresses in the supporting plates of IFB-B steel sections

- (7) Under the action of a uniformly distributed load q_2 applied on the right edge of the lower plate, bending stresses and shear stresses are calculated in points ⑦ and ⑨, as given in Table 39.

5.10.6. IFB-B composite sections

- (1) For IFB-B composite section, elastic stresses are calculated in the steel profile at the same 9 stress points as defined in § 5.10.5 (1), plus the two following stress points ⑩ and ⑪, respectively at the upper and lower side for the slab.

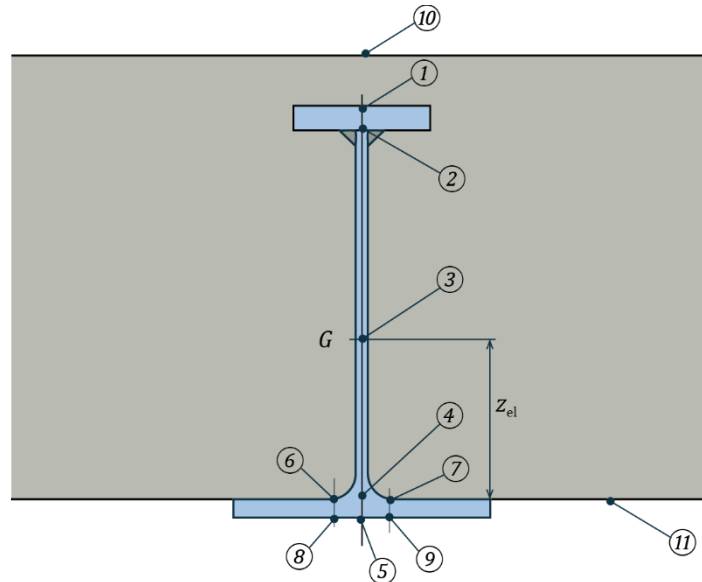


Figure 45 : Calculation of elastic stresses IFB-B composite sections

- (2) The location of the stress points in the structural steel are given in Table 38. For stress points in concrete:

$$z_{10} = t_d$$

$$z_{11} = 0$$

- (3) Under the action of a global bending moment M_{Ed} , the normal stresses are obtained with Formula **(121)** for the stress points in structural steel and with Formula **(122)**, for the stress points in concrete.
- (4) § 5.10.5 (5) applies for the calculation of the shear stresses under a global shear force V_{Ed} .
- (5) § 5.10.5 (6) and (7) applies for the calculation of the local stresses in the lower flanges.

5.10.7. SAB steel sections

- (1) In the vertical axis of a IFB-B steel section, the following 5 stress points are considered for the calculation of elastic stresses (**Figure 46**):
- at the external (point ①) and internal (②) sides of the upper flange;
 - at the centroid of the section (③);
 - at the external (point ⑤) and internal (④) sides of the lower flange.
- Additionally, the four points ⑥ to ⑨ are considered for the local bending of the supporting lower flange.

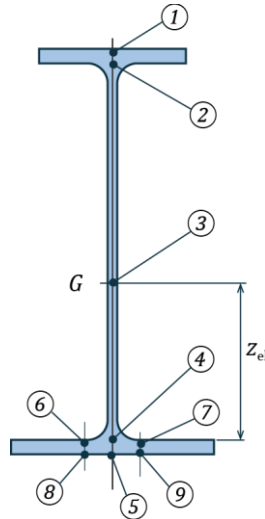


Figure 46 : Calculation of elastic stresses SAB steel sections

- (2) The location of the stress points are defined in **Table 40**.

i	z_i
①	$z_1 = h_b$
②	$z_2 = h_b - t_{fs}$
③	$z_3 = z_{ela}$
④, ⑥, ⑦	$z_4 = 0$
⑤, ⑧, ⑨	$z = -t_{fi}$

Table 40: Location of the points for stress calculations in SAB steel sections

- (3) Under the action of a global axial force N_{Ed} , the normal stresses in these 9 stress points are obtained with Formula **(125)**.
- (4) Under the action of a global bending moment M_{Ed} , the normal stresses in the 9 stress points are obtained with Formula **(121)**, where I_y is the second moment of area of the steel section $I_{y,a}$ and the location z_i of the points are given in **Table 40**.

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- (5) Under the action of a global shear force V_{Ed} , the shear stresses are obtained by Formula (124) for the stress point in the web ② to ④, where S_i is the first moment of area as given in Table 34 for points ② and ③ and as follows for point ④:

$$S_4 = b_p t_p (z_{el} + t_{fi} / 2) \quad \text{for point ④}$$

- (6) Under the action of a uniformly distributed load q_1 applied on the left edge of the lower plate, bending stresses and shear stresses are calculated in points ⑥ and ⑧, as given in Table 39.
- (7) Under the action of a uniformly distributed load q_2 applied on the right edge of the lower plate, bending stresses and shear stresses are calculated in points ⑦ and ⑨, as given in Table 39.

5.10.8. SAB composite sections

- (1) For SAB composite section, elastic stresses are calculated in the steel profile at the same 9 stress points as defined in § 5.10.7 (1), plus the two following stress points ⑩ and ⑪, respectively at the upper and lower side for the slab.

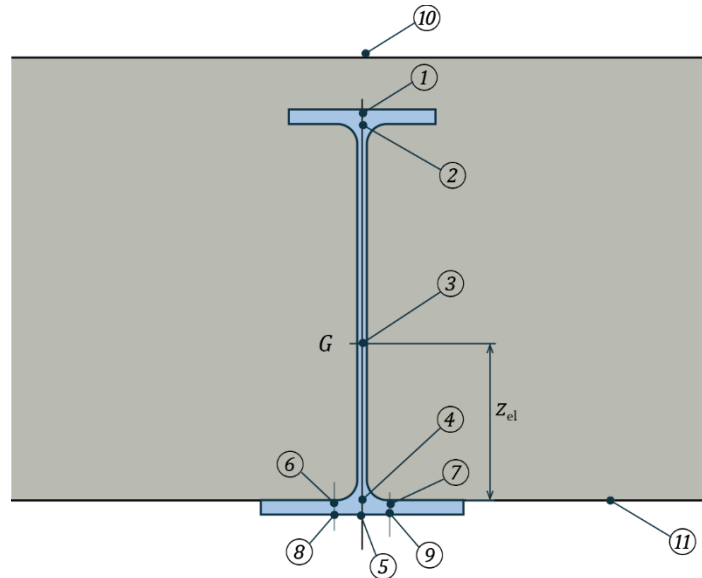


Figure 47 : Calculation of elastic stresses SAB composite sections

- (2) The location of the stress points in the structural steel are given in Table 40. For stress points in concrete:

$$z_{10} = t_d$$

$$z_{11} = 0$$

- (3) Under the action of a global bending moment M_{Ed} , the normal stresses are obtained with Formula (121) for the stress points in structural steel and with Formula (122), for the stress points in concrete.
- (4) § 5.10.7 (5) applies for the calculation of the shear stresses under a global shear force V_{Ed} .
- (5) § 5.10.7 (6) and (7) applies for the calculation of the local stresses in the lower flanges.

5.10.9. Equivalent stresses in the steel profile

- (1) For the points in the structural steel section, including the steel plate if any, the equivalent von Mises stresses are obtained as follows:

$$\sigma_{eq,i} = \sqrt{\sigma_{xi}^2 + \sigma_{yi}^2 - \sigma_{xi}\sigma_{yi} + 3\tau_i^2} \quad (134)$$

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6. GLOBAL ANALYSIS

6.1. General

- (1) Global analysis is carried out with a Finite Element module, using elastic beam elements with at 2 degrees of freedom at each node (vertical displacement and rotation).
- (2) All load cases are assessed considering the state of concrete (if any) and the associated modular ratio.
- (3) Deflections are calculated under elastic assumptions. They take into account the loading sequence.

6.2. Design sections

- (1) Shear forces, bending moments and deflections are calculated at design sections along the beam. Design checks are also performed at each of these design sections.
- (2) The design sections correspond to the discretisation of the beam used in the Finite Elements analysis (**Annex C**).

6.3. Critical sections

- (1) Critical sections are defined as follows, according to clause 6.1.1 (4) of EN 1994-1-1:
 - sections where the bending moment is maximum;
 - at supports;
 - at point load locations;
 - at point propping.



*Sections where the bending moment is maximum are not known when design sections are established first.
The vertical shear force is always assessed considering both LHS and RHS sides of the cross-sections.*

- (2) In the calculation sheet, it will be possible to display the results (internal forces and ULS verifications) only at critical sections or at all design sections. When the latter option will be used, the critical sections will be specifically identified.
- (3) In the software, ULS verifications will be carried out at each design section.

6.4. Cracked and uncracked segments of the slab

- (1) This chapter applies to usual composite sections.
- (1) For a beam with a single span, the full length of the beam is assumed to be uncracked.
- (2) For a beam with a main span and one or two cantilever spans, the segment of the beam where the concrete is cracked may be obtained by a simplified method as follows. The full length of the cantilever spans is assumed to be with cracked concrete. For the main span, it is assumed that 15% of the span on each side of each internal support (i.e. with a cantilever span) are assumed to be cracked (**Figure 48**).

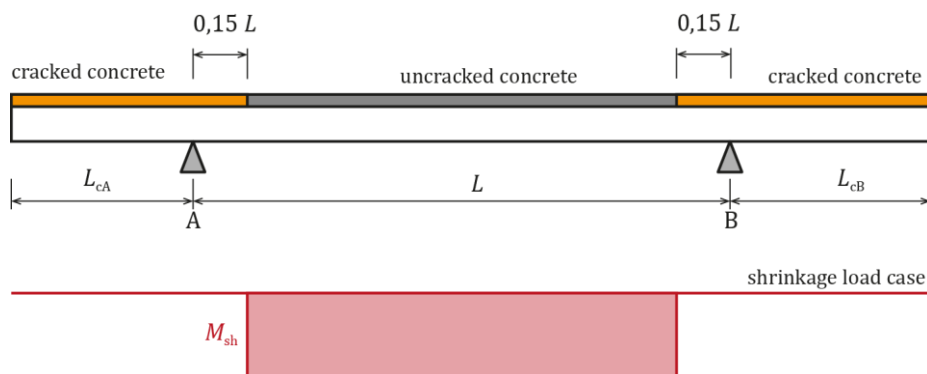


Figure 48 : Simplified determination of the cracked and uncracked concrete segments of the slab

6.5. Cracked parts of the encasement concrete

- (1) This chapter applies to usual section with partial **concrete** encasement.
- (2) The properties of the elements will be assessed according to § 5.5.4 or 5.5.6, considering a negative bending moment in the section where the slab would be cracked and positive bending moment where the slab would be uncracked.

6.6. Load cases

- (1) The load cases considered in the global analysis are described in **Table 41**.

Load case	Phase	ψ_L	t_0	n_L
Q_c (except for fully propped beams)	Steel	/	/	/
G for steel beams	Steel	/	/	/
G for fully propped composite beams	Composite	1,1	$t_{0,G1}$	long term
G_1 for unpropped composite beams	Steel	/	/	/
G_{1pp} for point propped comp. beams	Steel	/	/	/
G_{1c} for point propped comp. beams	Composite	1,1	$t_{0,G1c}$	long term
G_2 (unpropped or point propped)	Composite	1,1	$t_{0,G2}$	long term
Q_1	Steel or Composite	0	/	short term
Q_2	Steel or Composite	0	/	short term
SH shrinkage (composite beams only)	Composite	0,55	$t_{0,SH}$	long term
EM Eigen mode for vibration	Steel or Composite	0	/	short term

Table 41: Load cases considered in the global analysis

Shrinkage load case is assessed only for beams with usual composite sections, if an elastic design is carried out or if required for the calculation of deflections.

Load cases G_{1pp} and G_{1c} are detailed in § 6.7.1.

- (2) The global analysis is carried out for each load case by Finite Element Analysis.

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- (3) For each load case (except the eigen mode), the calculation sheet will give:
- the internal bending moment and shear force at each design/critical section;
 - the reaction at supports;
 - the maximal deflection of each span.

6.7. Specific load cases

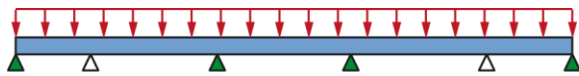
6.7.1. Permanent load case G1

- (1) Load case G_1 is assessed for composite beams, either unpropped or with point proppings. It includes the self-weight loads for concrete, profiled sheeting and steel, as described in § 6.7.2.
- (2) For beams with point propping (see § 2.14), the load case G_1 will be decomposed as follows (Figure 49):

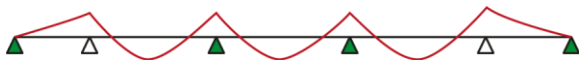
$$G_1 = G_{1,pp} + G_{1,c}$$

where: $G_{1,pp}$ corresponds to all loads applied on the beam and resisted by steel only (no composite action), considering the temporary intermediate supports;

$G_{1,c}$ is the load case corresponding to the state of the beam after the point propping has been removed. The reaction forces at temporary supports obtained from $G_{1,pp}$ are introduced downwards on the beam, which then acts compositely.



Application of the loads G_1 on the steel beam with point proppings



Global analysis for load case $G_{1,pp}$
Steel properties of the cross-sections
=> Reactions at proppings



Application of the reaction forces on the composite beam without proppings



Global analysis for load case $G_{1,c}$
Transformed composite properties of the cross-sections

Figure 49: Specific load cases for beams with point propping

6.7.2. Self-weight

- (1) The “self-weight” loads, associated to the load case G or G_1 , will automatically be assessed by the software, as a uniformly distributed load along the beam.
- (2) The distributed load for the self-weight of the beam is obtained by:

$$q_{sw} = g (\rho_a A_a + \rho_c A_c + \mu_p d_c + \rho_{ec} A_{ec}) \quad (135)$$

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where: g is the acceleration of gravity (see § 3.3);

ρ_a is the density of steel (see § 2.10.1);

ρ_c is the density of the slab concrete (see § 2.10.5);

ρ_{ec} is the density of the encasement concrete, if any (see § 2.10.5);

A_a is the area of the steel profile;

A_c is the area of the concrete slab – see (3);

A_{ec} is the area of the concrete in partial encasement – see (4);

μ_p is the surface mass of the steel profile sheeting that is supported by the beam (see § 2.5.2). $\mu_p = 0$ for solid slabs;

d_c is the width of the slab, supported by the beam: $d_c = d_1 + d_2$

d_1 and d_2 are the influence widths of the slab, obtained by (see § 2.1):

$d_1 = b_1$ for an edge beam;

$d_1 = b_1 / 2$ for an intermediate beam;

$d_2 = b_2 / 2$

(3) A_c is calculated as follows:

- for solid slabs (with or without prefabricated parts):

$$A_c = d_c t_d + t_h (b_{fs} + t_h \tan(\theta_h) / 2)$$

- for composite slabs:

$$A_c = d_c t_c \left(1 + \frac{b_m h_p}{e_p t_c} \right)$$

where $t_c = t_d - h_p$

(4) A_{ec} is equal to 0 for usual cross-sections without partial concrete encasements. For usual cross-sections with partial concrete encasement, A_{ec} is given by § 2.9 (14).

(5) For slim floors sections with prefabricated floor elements (see § 2.4 (5)), an additional distributed load equal to $g \mu_{pf} d_c$ is added, where μ_{pf} is the surface weight of the prefabricated elements and d_c is defined in (2).

6.7.3. Shrinkage load case

- The shrinkage load case is automatically assessed for the calculation of the deflection and for the calculation of elastic stresses, when elastic design is used.
- The shrinkage load case is used only for composite sections.
- The shrinkage load case consists of a normal force N_{sh} and a bending moment M_{sh} (Figure 48) applied to the uncracked segment of the main span (see § 6.4).
- For usual composite section without partial concrete encasement, the applied shrinkage bending moment, M_{sh} , due to the shrinkage in the slab, is obtained as follows:

$$M_{sh} = N_{sh} z_{sh} \quad (136)$$

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where: N_{sh} is the normal tension force in the slab due to shrinkage, given by:

$$N_{sh} = \frac{E_a}{n_{sh}} b_{eff} t_c \varepsilon_{sh} \quad (137)$$

b_{eff} is the effective width of the slab at mid-span –see § 5.1;

t_c is the thickness of active part of the slab – see § 5.2;

E_a is the Young's modulus of structural steel – see § 2.10.1;

n_{sh} is the modular ratio for the shrinkage load case – see § 5.2;

ε_{sh} is the shrinkage deformation – see § 2.10.5 (5);

z_{sh} is the distance between the centroid of the active part of the slab and the centroid of the composite cross-section, considering n_{sh} as the modular ratio.

- (5) For usual composite section with partial concrete encasement, the applied shrinkage bending moment, M_{sh} , due to the shrinkage in the slab and in the encasement, is obtained as follows:

$$M_{sh} = N_{sh} z_{sh} - N_{she} z_{she} \quad (138)$$

where: N_{sh} and z_{sh} , are given by (3).

N_{she} is the normal tension force in the slab due to the shrinkage of the encasement, given by:

$$N_{she} = \frac{E_a}{n_{she}} A_{ce} \varepsilon_{sh} \quad (139)$$

n_{she} is the modular ratio for the concrete of the encasement assuming a shrinkage load case – see § 5.2;

A_{ce} is the area of concrete in the partial encasement – see § 2.9;

z_{she} is the distance between the centroid of the steel section and the centroid of the transformed composite cross-section.

6.7.4. Live load arrangements

- (1) When the beam comprises one or two cantilever spans, the live loads Q_1 and Q_2 are split in three separate load cases associated to three loading arrangements as given in **Table 42**.

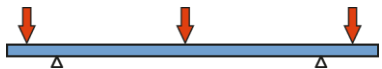
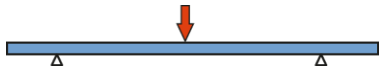
Arrangement	Q_1	Q_2	Description	Figure
#1	$Q_{1\#1}$	$Q_{2\#1}$	All spans are loaded	
#2	$Q_{1\#2}$	$Q_{2\#2}$	Only the main span is loaded	
#3	$Q_{1\#3}$	$Q_{2\#3}$	Only the cantilever spans are loaded	

Table 42: Live load arrangements

- (2) The same principle of live load arrangement will be considered for load case Q_c with point propping, for the verification of the resistance to lateral torsional buckling.

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6.8. Calculation of internal forces, moments and deflections

- (1) For each load case, the software will assess:
- the internal shear force and bending moment at each design section;
 - the deflection at each design section.
- This calculation will be carried out through Finite Element analysis (**Annex C**).

6.9. Transverse bending of the lower flange of slim floor sections

- (1) The local loads on the lower plate (or flange) of a slim floor section are calculated independently on each side of the beam. For each load case, it is obtained by the following Formula (see **Figure 50**):

$$q_i = (b_i - b_{fs}) Q_L + \sum_j \frac{(q_{L,1j} + q_{L,2j})(x_{L,2j} - x_{L,1j})}{2L} \frac{b_i - b_{fs}}{b} \quad (140)$$

where: Q_L is the surface load of the load case (see § 2.13.2);
 $q_{L,1j}$ and $q_{L,2j}$ are the values at each end of the distributed load no j ;
 $x_{L,1j}$ and $x_{L,2j}$ are the location of each end of the distributed load no j ;
 L is the span length
 b_1 and b_2 are the geometric widths as defined in § 5.1.1.



Point loads, if any, are assumed to be located directly on the steel profile.
Distributed loads are assumed as uniformly distributed along the full length of each span.

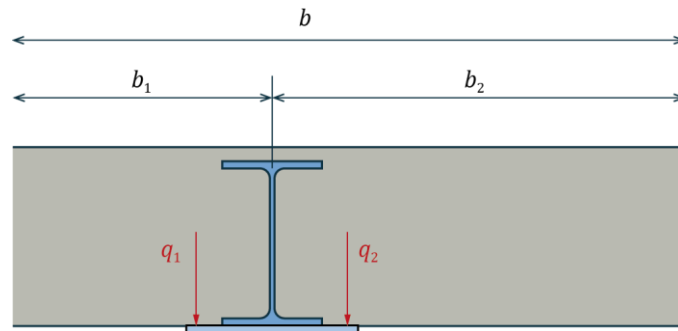


Figure 50: Transverse load on the lower flange or plate of slim floor sections

- (2) The local load is assumed to be the resultant of a contact pressure on the supporting plate. The distribution of this contact pressure between the slab and the supporting plate is assumed to be triangular, with a value of 0 at the edge of the plate.
- The distance of the local load q to the edge of the beam is denoted d_{app} .

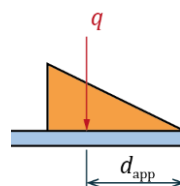


Figure 51: Local contact pressure

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(3) For slim floor with plain slabs without prefabricated parts, d_{app} is obtained as follows:

- SFB profile $d_{app} = (b_p - b_{fi}) / 3$
- IFB-A profile $d_{app} = (b_p - b_{fs}) / 3$
- IFB-B profile $d_{app} = (b_{fi} - b_p) / 3$
- SAB profile $d_{app} = (b_{fi} - b_{fs}) / 3$

For all other configurations of slim floor sections, d_{app} is obtained as follows:

$$d_{app} = \frac{2}{3} b_{app,min} \quad (141)$$

where $b_{app,min}$ is defined in § 2.2.6 (3).

(4) The local bending moment in the supporting plate is given by the following Formula (by unit length):

$$m_{x,Ed} = q_i (e_2 - e_1) / 2 \quad (142)$$

where q_i is the local load on the plate as defined in (1);

e_1 and e_2 are as defined in **Table 43** – see **Figure 35**.

Type of section	e_1	e_2	t_s	f_s
SFB plate	$b_p - 2 d_{app}$	b_f	t_p	f_{yp}
SFB lower flange	b_f	t_w	t_f	f_{yf}
IFB-A	$b_p - 2 d_{app}$	t_w	t_p	f_{yp}
IFB-B	$b_{fi} - 2 d_{app}$	t_w	t_f	f_{yf}
SAB	$b_{fi} - 2 d_{app}$	$t_w + 2 r_c$	t_f	f_{yf}

Table 43: Parameters of the supporting plates of slim floor sections

(5) The local bending moment in the supporting lower flange of a SFB section is given by the following Formula (by unit length):

$$m_{x,Ed} = q_i (e_1 - t_w - 2 r_c) / 2 \quad (143)$$

(6) For IFB and SAB section, the deflection at the end of the supporting plate due to the local bending is given by:

$$f_{0,bt} = 2 \frac{q_i (e_2 - e_1)^3}{E t_s^3} \quad (144)$$

where t_s is the thickness of the supporting plate, as given in **Table 43**.

(7) For SFB section, the deflection at the end of the supporting plate due to the local bending is given by:

$$f_{0,bt} = 4 \frac{q_i}{E} \left(\frac{b_1^2 (b_1 + 3 b_2 / 2)}{b_f^3 + t_p^3} + \frac{b_2^3}{t_p^3} \right) \quad (145)$$

where: $b_1 = (b_f - t_w) / 2 - r_c$

$b_2 = (b_p - 2 d_{app} - b_f) / 2$

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7. VERIFICATIONS OF BEAMS AT ULTIMATE LIMIT STATES

7.1. General

- (1) Chapter 7 applies to beams with steel or composite sections, either usual or slim floor, either with or without partial concrete encasement. It also applies to beams with usual composite sections in construction stage.
- (2) For each design section, following checks are performed by the software:
 - resistance of design sections to bending moment – see § 7.2;
 - resistance of design sections to shear force – see § 7.3;
 - resistance to MV interaction – see § 7.4 or 7.5;
 - resistance to lateral torsional buckling, when relevant – see § 7.6.
- (3) Sections are classified before the verifications –see § 5.4. Class 4 cross-sections are out of the scope of this document.
- (4) For continuous beams with usual composite sections, requirements for the reinforcement of the slab at supports apply if a plastic design is used, see § 7.7.
- (5) For welded profiles, a design of the weld is given in § 7.9.
- (6) For slim floor section, the resistance to local bending of the supporting plate is checked according to the rules given in § 7.8 and design of welds is given in § 7.10.
- (7) Resistance to transverse forces is not checked by the software. For concentrated loads, including reactions at supports, either in the construction phase or in the final composite one, the User should review the detailing of the beam and check the local stability of the web if required.

7.2. Verification of the resistance to bending moment

7.2.1. Usual steel sections

- (1) For each design section, the following criterion will be checked:

$$\Gamma_M = \frac{M_{Ed}}{M_{Rd}} \leq 1 \quad (146)$$

where: M_{Ed} is the design bending moment in the cross-section, obtained from the global analysis considering the relevant ULS combination of loads;

M_{Rd} is the design resistance of the cross-section to bending moment. According to the type and the class of cross-section, M_{Rd} is obtained according to the references mentioned in (2) to (4).

- (2) For Class 1 or 2 cross-sections, M_{Rd} is equal to the plastic bending resistance $M_{pl,a,Rd}$, which is obtained according to § 5.5.1 (4) or 5.5.3 (4).
- (3) For Class 3 sections, M_{Rd} is equal to the elastic bending resistance $M_{el,a,Rd}$, which is obtained according to § 5.5.1 (5) or 5.5.3 (5).
- (4) If elastic design is imposed, clause (3) applies whatever the class of the cross-section is.

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7.2.2. Usual steel sections with partial concrete encasement

- (1) For each design section of class 1 or 2, criterion **(146)** is used, where M_{Rd} is equal to the plastic resistance of the cross-section, obtained according to § 5.5.
- (2) For each cross-section of class 3, following criteria are applied:

$$\Gamma_{Ma} = \frac{|\sigma_i|}{f_{yd}} \quad \text{for the structural steel, points ① and ⑤}; \quad (147)$$

$$\Gamma_{Mce} = \frac{|\sigma_i|}{f_{ced}} \quad \text{for the encasement concrete, points ⑥ and ⑦};$$

$$\Gamma_{Mse} = \frac{|\sigma_i|}{f_{sed}} \quad \text{for the reinforcement rebars, points ⑧ to ⑩};$$

$$\Gamma_M = \max\{\Gamma_{Ma}; \Gamma_{Mce}; \Gamma_{Mse}\} \leq 1$$

where the normal stresses σ_i are calculated according to § 5.9.3;

f_{yd} is the design yield strength of the structural steel, see Table 44 and (3);

f_{ced} is the design compressive strength of the encasement concrete, obtained from § 2.10.5 considering the characteristic strength f_{cek} ;

f_{sed} is the design yield strength of the reinforcing steel in concrete encasement, see § 2.10.6.

- (3) For the criterion Γ_{Ma} , f_{yd} is obtained as given in Table 44.

i	f_{yd}
①	f_{yfs} / γ_{M0}
②	$\min\{f_{yfs}; f_{yw}\} / \gamma_{M0}$
③	f_{yw} / γ_{M0}
④	$\min\{f_{yfi}; f_{yw}\} / \gamma_{M0}$
⑤	f_{yfi} / γ_{M0}

Table 44: Design yield strengths in steel sections

- (4) If elastic design is imposed, clause (2) applies whatever the class of the cross-section is.

7.2.3. Usual composite sections

- (1) For each design section of class 1 or 2, criterion **(146)** applies, where M_{Rd} is equal to the plastic resistance of the composite cross-section, obtained according to § 5.5.5, taking into account the actual degree of shear connection η of the cross-section. η is obtained according to § 8.2.2 (1), where ΣP_{Rdi} is the sum of the resistance of all studs located between the cross-section and the support (the lowest value for the two supports being taken into account). For support with an adjacent span, ΣP_{Rdi} is replaced by $\Sigma P_{Rdi} - A_s f_{sd}$, see § 8.2.2 (2).

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- (2) For each cross-section of class 3, the following criteria are applied:

$$\Gamma_{Ma} = \frac{|\sigma_i|}{f_{yd}} \quad \text{for the structural steel, points ① and ⑤;} \quad (148)$$

$$\Gamma_{Mc} = \frac{|\sigma_i|}{f_{cd}} \quad \text{for the concrete of the slab, point ⑪;} \quad (149)$$

$$\Gamma_{Ms} = \frac{|\sigma_i|}{f_{sd}} \quad \text{for the reinforcement rebars, points ⑫ and ⑬; (*)}$$

$$\Gamma_M = \max\{\Gamma_{Ma}; \Gamma_{Mc}; \Gamma_{Ms}\} \leq 1$$

where the normal stresses σ_i are calculated according to § 5.9.3;

f_{yd} is the design yield strength of the structural steel, see § 7.2.2 (3);

f_{cd} is the design compressive strength of the concrete, obtained from § 2.10.5 considering the characteristic strength f_{ck} ;

f_{sd} is the design yield strength of the slab reinforcement, see § 2.10.6.

- (3) The criterion Γ_{Ms} is applied for reinforcement in tension. It is also applied to reinforcement in compression if the associated option has been activated.
- (4) If elastic design is imposed, clause (2) applies whatever the class of the cross-section is.

7.2.4. Usual composite section with partial concrete encasement

- (1) For each design section of class 1 or 2, criterion (146) where M_{Rd} is equal to the plastic resistance of the cross-section, obtained according to § 5.5.6, taking into account the degree of connection as described in § 7.2.3 (1).
- (2) For each cross-section of class 3, the following criteria are applied:

$$\Gamma_{Ma} = \frac{|\sigma_i|}{f_{yd}} \quad \text{for the structural steel, points ① and ⑤;} \quad (149)$$

$$\Gamma_{Mce} = \frac{|\sigma_i|}{f_{ced}} \quad \text{for the encasement concrete, points ⑥ and ⑦;} \quad (150)$$

$$\Gamma_{Mse} = \frac{|\sigma_i|}{f_{sed}} \quad \text{for the reinforcement rebars, points ⑧ to ⑩;} \quad (151)$$

$$\Gamma_{Mc} = \frac{|\sigma_i|}{f_{cd}} \quad \text{for the concrete of the slab, point ⑪;} \quad (152)$$

$$\Gamma_{Ms} = \frac{|\sigma_i|}{f_{sd}} \quad \text{for the reinforcement rebars, points ⑫ and ⑬;} \quad (153)$$

$$\Gamma_M = \max\{\Gamma_{Ma}; \Gamma_{Mc}; \Gamma_{Ms}; \Gamma_{Mce}; \Gamma_{Mse}\} \leq 1$$

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where the normal stresses σ_i are calculated according to § 5.9.3;

f_{yd} is the design yield strength of the structural steel, see § 7.2.2 (3);

f_{cd} is the design compressive strength of the concrete, obtained from § 2.10.5 considering the characteristic strength f_{ck} ;

f_{sd} is the design yield strength of the slab reinforcement, see § 2.10.6;

f_{ced} is the design compressive strength of the encasement concrete, obtained from § 2.10.5 considering the characteristic strength f_{cek} ;

f_{sed} is the design yield strength of the reinforcing steel in concrete encasement, see § 2.10.6.

- (3) If elastic design is imposed, clause (2) applies whatever the class of the cross-section is.

7.2.5. Slim floor steel sections

- (1) For each design section of class 1 or 2, criterion (146) is used, where M_{Rd} is equal to the plastic resistance of the cross-section, obtained according to § 5.5.7.

- (2) For each cross-section of class 3, the following criterion is applied:

$$\Gamma_{Ma} = \max \left\{ \frac{|\sigma_i|}{f_{yd}} \right\} \quad (150)$$

where the equivalent stresses σ_i , calculated according to § 5.10.9, taking into account the global bending moment M_{Ed} , the local bending on the supporting plate and neglecting the shear stresses;

f_{yd} is the design yield strength of the structural steel, given by $f_{yd} = f_{yi} / \gamma_{M0}$;

f_{yi} is the nominal yield strength of the structural steel, as defined for each point in Table 45.

	f_{yi}						
i	SFB	IFB-A	IFB-B	SAB			
①	f_y	f_y	f_{yp}	f_y			
②			$\min\{f_y; f_{yp}\}$				
③			f_y				
④		$\min\{f_y; f_{yp}\}$					
⑤	$\min\{f_y; f_{yp}\}$	f_y					
⑥	f_{yp}						
⑦	f_y						
⑧							
⑨	$\min\{f_y; f_{yp}\}$						
⑩							
⑪ à ⑭	f_{yp}						

Table 45: Design yield strengths in steel slim floor sections

- (3) If elastic design is imposed, clause (2) applies whatever the class of the cross-section is.

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7.2.6. Slim floor composite sections

- (1) For each design section of class 1 or 2, criterion **(146)** is used, where M_{Rd} is equal to the plastic resistance of the cross-section, obtained according to § 5.5.8.
- (2) For each cross-section of class 3, the following criteria are applied:

$$\Gamma_{Ma} = \frac{|\sigma_i|}{f_{yd}} \quad \text{for stress points in structural steel;}$$

$$\Gamma_{Mc} = \frac{|\sigma_i|}{f_{cd}} \quad \text{for stress points in the concrete of the slab;}$$

$$\Gamma_M = \max\{\Gamma_{Ma}; \Gamma_{Mc}\} \leq 1$$

where the normal stresses σ_i are calculated according to § 5.10;

f_{yd} is the design yield strength of the structural steel, see § 7.2.5 (2);

f_{cd} is the design compressive strength of the concrete, obtained from § 2.10.5 considering the characteristic strength f_{ck} .

- (3) If elastic design is imposed, clause (2) applies whatever the class of the cross-section is.

7.3. Verification of the resistance to shear force

7.3.1. General criterion

- (1) Except when an elastic design is imposed, the following criterion will be checked for each design section:

$$\Gamma_V = \frac{V_{Ed}}{V_{Rd}} \leq 1 \quad (151)$$

where: V_{Ed} is the design shear force in the cross-section, obtained from the global analysis considering the relevant ULS combination of loads;

V_{Rd} is the design resistance of the cross-section to shear force, as given in § 7.3.2 or 7.3.3.

- (2) When an elastic design is imposed, the following criterion will be applied, for all sections:

$$\Gamma_V = \frac{\tau_{Ed}}{f_{yw} / (\sqrt{3} \gamma_{M0})} \quad (152)$$

where: τ_{Ed} is the maximal shear stress at points ②, ③ and ④, obtained according to § 5.9.

7.3.2. Usual cross-sections without partial concrete encasement

- (1) For steel or composite sections without partial concrete encasement, V_{Rd} is calculated as follows.
- (2) Webs are considered as compact when the following criterion is fulfilled (acc. to EN 1993-1-5, 5.1 (2)):

$$\frac{h_w}{t_w} \leq 72 \frac{\varepsilon_w}{\eta} \quad (153)$$

where: η see § 2.12.4 (1).

ε_w is the steel factor see § 2.10.4 (2).

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- (3) For compact webs, V_{Rd} is equal to the plastic resistance and is obtained by:

$$V_{Rd} = V_{pl,Rd} = \frac{A_v f_{yw}}{\sqrt{3} \gamma_{M0}} \quad (154)$$

where: A_v is the shear area – see § 5.6.

- (4) For slender webs, V_{Rd} is equal to the resistance to shear buckling and is obtained by:

$$V_{Rd} = V_{b,Rd} = \chi_w \frac{h_w t_w f_{yw}}{\sqrt{3} \gamma_{M1}} \quad (155)$$

where: χ_w is the reduction factor for shear buckling, which is given in **Table 46**.

$\bar{\lambda}_w$ is the reduced slenderness for shear buckling, which is calculated by:

$$\bar{\lambda}_w = \frac{h_w}{t_w} \frac{1}{37,4 \varepsilon_w \sqrt{k_\tau}} \quad (156)$$

$$k_\tau = 5,34$$



It is assumed that the beam has no intermediate transverse stiffener and that it has non rigid post at supports.

$\bar{\lambda}_w$	Main span with 2 adjacent cantilever spans Cantilever spans	Main span without adjacent cantilevers spans or with one cantilever span
$\bar{\lambda}_w \leq 0,83 / \eta$	η	η
$0,83 / \eta < \bar{\lambda}_w \leq 1,08$	$0,83 / \bar{\lambda}_w$	$0,83 / \bar{\lambda}_w$
$\bar{\lambda}_w > 1,08$	$1,37 / (0,7 + \bar{\lambda}_w)$	$0,83 / \bar{\lambda}_w$

Table 46: Reduction factor for shear buckling

7.3.3. Usual cross-sections with partial concrete encasement

- (1) For steel or composite sections with partial concrete encasement, V_{Rd} is calculated as follows.
(2) Webs are considered as compact when the following criterion is fulfilled (acc. to EN 1994-1-1, 6.3.1 (2)):

$$\frac{d_w}{t_w} \leq 124 \varepsilon_w \quad (157)$$

- (3) For compact webs, V_{Rd} is equal to the plastic resistance and is obtained by Formula (154) in 7.3.2 (3).



The contribution of the concrete of the partial concrete encasement is neglected. The shear force is assumed to be resisted by the steel section alone.

- (4) Slender webs are outside the scope of application of the software.

7.3.4. Slim floor sections

- (1) For all slim floor sections, either steel or composite, § 7.3.1 applies, where the shear buckling is not accounted for.

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7.4. Verification of the resistance to MV interaction – First generation of EN

7.4.1. General

- (1) Tree MV criteria are possible according to the situation:
 - the plastic MV interaction criterion (see § 7.4.2);
 - the EN 1993-1-5 MV interaction criterion (see § 7.4.3);
 - the elastic MV interaction criterion (see § 7.4.4).
- (2) The plastic criterion is applied to all steel sections with compact webs, whatever their class is, except if an elastic calculation with Von-Mises criteria is used.
- (3) The EN 1993-1-5 criterion is used for:
 - steel or composite sections with slender webs;
 - composite sections with Class 3 cross-sections.
- (4) The elastic criterion is used when an elastic design with Von-Mises principle is used.

7.4.2. General criterion for usual sections with compact webs

- (1) The definition of compact webs is given in § 7.3.2 (2) for cross-sections without partial concrete encasement and in § 7.3.3 (2) for cross-sections with partial concrete encasement.
- (2) For each design section with a compact web, the following criterion will be checked for the MV interaction:

$$\Gamma_{MV} = \frac{M_{Ed}}{M_{V,Rd}} \leq 1 \quad (158)$$

where: M_{Ed} is the design bending moment in the cross-section, obtained from the global analysis considering the relevant ULS combination of loads;

$M_{V,Rd}$ is the plastic design resistance of the cross-section to bending moment, taking account of the presence of the shear force.



$M_{V,Rd}$ is the plastic shear resistance of the cross-section whatever its class is.

- (3) $M_{V,Rd}$ is obtained to § 5.5, by considering a reduced yield strength for the web, given by:

$$\begin{aligned} f_{y,w} &= (1 - \rho) f_y & \text{when } V_{Ed} > 0,5 V_{pl,Rd} \\ f_{y,w} &= f_y & \text{when } V_{Ed} \leq 0,5 V_{pl,Rd} \end{aligned} \quad (159)$$

where: $V_{pl,Rd}$ is the plastic shear resistance of the web, given by § 7.3.2 (3);

$$\rho = \left(2 \frac{V_{Ed}}{V_{pl,Rd}} - 1 \right)^2$$

For usual composite section, $M_{V,Rd}$ is assessed taking into account the actual degree of shear connection, as given in § 7.2.3 (1) or 7.2.4 (1).

7.4.3. General criterion for usual section with slender webs

- (1) For slender webs of sections without partial concrete encasement, the following criterion will be used, according to EN 1993-1-5, § 7.1:

$$\Gamma_{MV} = \frac{M_{Ed}}{M_{pl,Rd}} + \left(1 - \frac{M_{f,Rd}}{M_{pl,Rd}} \right) \left(2 \frac{V_{Ed}}{V_{b,Rd}} - 1 \right)^2 \leq 1 \quad \begin{aligned} &\text{if } M_{Ed} > M_{f,Rd} \\ &\text{and if } V_{Ed} > 0,5 V_{b,Rd} \end{aligned} \quad (160)$$

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$$\Gamma_{MV} = \frac{M_{Ed}}{M_{pl,Rd}} \leq 1 \quad \begin{array}{l} \text{if } M_{Ed} \leq M_{f,Rd} \\ \text{or if } V_{Ed} \leq 0,5 V_{b,Rd} \end{array}$$

where: $M_{pl,Rd}$ is the plastic resistance of the composite gross cross-section, whatever its class is;

$M_{f,Rd}$ is the plastic resistance of the composite cross-section neglecting the contribution of the web;

$V_{b,Rd}$ is the resistance to shear buckling, according to § 7.3.2 (4).



As Class 4 cross-sections are outside the scope of the software, the gross area of the flanges is always fully effective.

In the software, following limitation will be applied: $\Gamma_{MV} \leq \Gamma_M$.

7.4.4. Elastic design

- (1) When an elastic design is imposed, the following criterion will be applied to cross-sections with compact webs:

$$\Gamma_{MV} = \frac{\sigma_{eq,Ed}}{f_{yw} / \gamma_{M0}} \leq 1 \quad (161)$$

where: $\sigma_{eq,Ed}$ is the maximal equivalent von Mises stress at all points of structural steel, obtained according to § 5.9.6 for usual sections or to § 5.10.9 for slim floor sections.

- (2) When an elastic design is imposed to usual cross-sections with slender webs, § 7.4.3 will be applied.



There is no elastic design for slender webs, which would take into account the shear buckling.

7.5. Verification of the resistance to MV interaction – Second generation of EN

7.5.1. General criterion

- (1) For usual sections, the definition of compact webs is given in § 7.3.2 (2) for cross-section without partial concrete encasement and in § 7.3.3 (2) for cross-sections with partial concrete encasement. For all slim floor sections, the web is assumed as compact.
- (2) For each design section with a compact web, the following criterion will be checked for the MV interaction:

$$\Gamma_{MV} = \frac{M_{Ed}}{M_{V,Rd}} \leq 1 \quad (162)$$

where: M_{Ed} is the design bending moment in the cross-section, obtained from the global analysis considering the relevant ULS combination of loads;

$M_{V,Rd}$ is the plastic design resistance of the cross-section to bending moment, taking into account the presence of the shear force.



$M_{V,Rd}$ is the plastic shear resistance of the cross-section whatever its class is.

In the software, the following limitation will be applied: $\Gamma_{MV} \leq \Gamma_M$

- (3) $M_{V,Rd}$ is obtained according to § 5.5, by considering a reduced yield strength for the web, given by:

$$\begin{array}{ll} f_{y,w} = (1 - \rho) f_y & \text{when } V_{Ed} > \eta_V V_{pl,Rd} \\ f_{y,w} = f_y & \text{when } V_{Ed} \leq \eta_V V_{pl,Rd} \end{array} \quad (163)$$

where : $V_{pl,Rd}$ is the plastic shear resistance of the web, given by § 7.3.2 (3);

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$$\rho = \left(\frac{1}{\eta_V} \frac{V_{Ed}}{V_{pl,Rd}} - 1 \right)^2$$

η_V : see § 2.12.4 (3).



The formula of ρ has been adapted in order to keep consistency.

For usual composite section, $M_{V,Rd}$ is assessed taking into account the actual degree of shear connection, as given in § 7.2.3 (1) or 7.2.4 (1).

7.5.2. General criterion for slender webs

- (1) For slender webs of sections without partial concrete encasement, the following criterion will be used, according to prEN 1993-1-5, § 9.1:

$$\Gamma_{MV} = \eta_1 + \left(1 - \frac{M_{f,Rk}}{M_{pl,Rk}} \right) \left(2 \frac{V_{Ed}}{V_{b,Rd}} - 1 \right)^\mu \leq 1 \quad \begin{array}{l} \text{if } M_{Ed} > M_{f,Rk} \\ \text{and if } V_{Ed} > 0,5 V_{b,Rd} \end{array} \quad (164)$$

$$\Gamma_{MV} = \frac{M_{Ed}}{M_{pl,Rd}} \leq 1 \quad \begin{array}{l} \text{if } M_{Ed} \leq M_{f,Rk} \\ \text{or if } V_{Ed} \leq 0,5 V_{b,Rd} \end{array}$$

where: $M_{pl,Rk}$ is the plastic resistance of the composite gross cross-section, whatever its class is, obtained with the nominal and characteristic stress limits;

$M_{f,Rk}$ is the characteristic plastic resistance of the composite cross-section neglecting the contribution of the web;

$V_{b,Rd}$ is the resistance to shear buckling, according to § 7.3.2 (4);

$$\mu = \left(\frac{M_{f,Rk}}{M_{pl,Rk}} + 0,2 \right)^{15} + 1$$

η_1 : see (2).



As Class 4 cross-sections are outside the scope of the software, the gross area of the flanges is always fully effective.

In the software, following limitation will be applied: $\Gamma_{MV} \leq \Gamma_M$.

- (2) The factor η_1 is obtained by: $\eta_1 = \Gamma_M$, where Γ_M is the bending resistance ratio, as given in § 7.2.

7.5.3. Elastic design

- (1) For compact webs, § 7.4.4 (1) applies. For slender webs, § 7.5.2 applies.

7.6. Resistance to lateral torsional buckling

7.6.1. General

- (1) Resistance to lateral torsional buckling is not checked if both flanges are continuously restrained (see § 2.11.1).
- (2) Slim floor beams are assumed to be not prone to lateral torsional buckling. The LTB resistance is not checked, including for the construction phase. In the latter case, it is assumed that required details are adopted in order to restrain the lateral buckling of the upper flange.

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7.6.2. Beams with usual steel sections

(1) The beam is divided in segments. Segments are delimited by:

- the ends of the beam (for cantilever spans);
- the supports;
- the cross-sections with both flanges laterally restrained.

In the simple case of a simply supported beam with no lateral intermediate restraints, only one segment is considered for the full beam.

(2) The resistance of each segment i to lateral torsional buckling is checked through the following criterion:

$$\Gamma_{LTi} = \frac{M_{Ed}}{M_{b,Rd}} \leq 1 \quad (165)$$

where: M_{Ed} is the maximum bending moment along the segment;

$M_{b,Rd}$ is the buckling resistance, as given in (3).

(3) The buckling resistance $M_{b,Rd}$ is obtained by the following equation, according to EN 1993-1-1, 6.3.2.1:

$$M_{b,Rd} = \chi_{LT} M_{Rk} / \gamma_{M1} \quad (166)$$

where: M_{Rk} is the characteristic bending resistance (i.e. obtained by neglecting the partial factors) – see § 5.5.2;

γ_{M1} : see § 2.12.2;

χ_{LT} is the reduction factor for lateral torsional buckling.

(4) χ_{LT} is obtained by:

$$\chi_{LT} = \frac{1}{\Phi_{LT} + \sqrt{\Phi_{LT}^2 - \bar{\lambda}_{LT}^2}} \leq 1 \quad (167)$$

where: $\Phi_{LT} = 0,5 \left[1 + \alpha_{LT} (\bar{\lambda}_{LT} - 0,2) + \bar{\lambda}_{LT}^2 \right]$

α_{LT} is the imperfection factor;

$\bar{\lambda}_{LT}$ is the relative slenderness of the beam, given by:

$$\bar{\lambda}_{LT} = \sqrt{\frac{M_{Rk}}{M_{cr}}} \quad (168)$$

M_{cr} is the critical moment for lateral torsional buckling.

(5) α_{LT} is given by Table 47 as a function of the buckling curve, the latter being given in Table 48 from the geometry of the cross-section.

Buckling curve	a	b	c	d
Imperfection factor α_{LT}	0,21	0,34	0,49	0,76

Table 47: Imperfection factor

Cross-sections	Limits	Buckling curve
Hot rolled profiles	$h_s / b_f \leq 2$	<i>a</i>
	$h_s / b_f > 2$	<i>b</i>
Welded sections	$h_s / b_f \leq 2$	<i>c</i>
	$h_s / b_f > 2$	<i>d</i>

Table 48: Reduction curves



For welded sections, b_f is taken as the width of upper flange if $M_{Ed} > 0$ and as the width of the lower flange otherwise.

- (6) The critical moment M_{cr} is obtained by:

$$M_{cr} = \alpha_{cr} M_{Ed} \quad (169)$$

where: M_{Ed} : see (2);

α_{cr} is the critical amplification factor, obtained by use of the LTBeamN module – see **Annex D**, considering the full beam.

7.6.3. Beams with usual steel sections and partial concrete encasement

- (1) For beams with usual steel sections and partial concrete encasement, § 7.6.2 applies, where M_{Rk} is obtained according to § 5.5.4 and M_{cr} according to (2).
- (2) The critical moment M_{cr} is calculated according to § 7.6.2 (6), where α_{cr} is calculated using the LTBeamN module (**Annex D**), with the following section properties (see [17]):

- second moment of area about the weak axis:

$$I_z = I_{z,a} + \frac{h_w b_c^3}{12 n_0}$$

- torsion inertia:

$$I_t = I_{t,a} + 0,1 I_{t,c} \frac{G_c}{G_a}$$

where: $G_c = 0,3 E_a / n_x$

$$I_{t,c} = \frac{1}{3} \left(1 - 0,63 \frac{b_c}{h_w} \right) h_w b_c^3$$

n_x is the modular ratio to be considered. By default, in the software, $n_x = n_0$ (according to [17]). It will be possible in the settings of the software to skip to the long term modular ratio (according to clause 6.4.2 (9) of EN 1994-1-1).

- warping factor:

$$I_w = I_{w,a}$$

where: n_0 is the modular ratio for short term actions, see § 5.3.

$I_{z,a}$, $I_{t,a}$, and $I_{w,a}$ are the section properties of the steel section.

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7.6.4. Beams with usual composite sections – composite stage – First generation of EN

- (1) For single span beams, without negative bending moment, it is assumed that the slab is a part of a floor which prevents its lateral instability, so that the beam is not prone to lateral torsional buckling.
- (2) For beams with cantilever spans, negative bending moments are formed near the supports, which leads to lower flanges being compressed. The beams should thus be checked against LT buckling, as described below, according to the method given by EN 1993-1-1, 6.3.2.4.
- (3) The beam is separated in segments delimited by:
 - the ends of the beam (for cantilever spans);
 - the supports.

- (4) The resistance of each segment i to lateral torsional buckling is checked through the following criterion:

$$\Gamma_{LTi} = \frac{M_{Ed}}{M_{b,Rd}} \leq 1 \quad (170)$$

where: M_{Ed} is the maximum negative bending moment along the segment;

$M_{b,Rd}$ is the buckling resistance, as given in (5).

- (5) The buckling resistance $M_{b,Rd}$ is obtained by the following equation, according to EN 1993-1-1, 6.3.2.4 (2):

$$M_{b,Rd} = k_{f,B} \chi_{LT} M_{el,Rk} / \gamma_{M1} \text{ and } M_{b,Rd} \leq M_{el,Rd} \quad (171)$$

where: $M_{el,Rd}$ is the design elastic bending resistance obtained according to § 5.5 where the partial factor for steel γ_{M0} is replaced by γ_{M1} ;

γ_{M1} : see § 2.12.2;

χ_{LT} is the reduction factor for lateral torsional buckling, calculated according to § 7.6.2 (4), considering the reduced slenderness $\bar{\lambda}_f$ given in (6) and the imperfection factors as given in (7);

$k_{f,B}$ is a modification factor, where: $k_{f,B} = 1$

- (6) The reduced slenderness $\bar{\lambda}_f$ is obtained by:

$$\bar{\lambda}_f = \frac{k_C L_c}{i_{f,z} \lambda_1} \quad (172)$$

where: $k_C L_c$ is the buckling length of the segment, see (9);

$$\lambda_1 = \pi \sqrt{E / f_y} = 93,9 \varepsilon$$

$i_{f,z}$ is the radius of gyration of the cross-section constituted by the lower flange, the lower root fillet if any and 1/3 of the compressed part of the web, see (8).

- (7) α_{LT} is given by Table 47 as a function of the buckling curve, chosen as follows:

- curve d for welded sections;
- curve c for hot rolled sections.

- (8) $i_{f,z}$ is given by:

$$i_{f,z} = \sqrt{\frac{I_{fz}}{A_{fz}}} \quad (173)$$

where:

$$I_{fz} = \frac{b_{fi}^3 t_{fi}}{12} + \frac{t_w^3 h_{wc}}{12} + (4 - \pi) \frac{r_{ci}^2}{2} \left(\frac{t_w}{2} + k_{cG} r_{ci} \right)^2$$

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$$A_{fz} = b_{fi} t_{fi} + h_{wc} t_w + (4 - \pi) \frac{r_{ci}^2}{2}$$

h_{wc} is the 1/3 of the compressed part of the web, which is given by:

$$h_{wc} = (h_s - t_{fi} - z_{na}) / 3$$

z_{na} is the location of the neutral axis, obtained as follows: $z_{na} = z_{el}$;

z_{el} is the location of the elastic neutral axil of the composite section under negative bending moment;

k_{CG} : see **Table 24**.

- (9) The buckling length $k_C L_c$ will be obtained as follows:

$$k_C L_c = \pi \sqrt{\frac{EI}{\alpha_{cr} N_{Ed}}} \quad (174)$$

where: EI is the flexural stiffness of the composite section under negative bending moments. For composite sections with partial concrete encasement, I will be assessed considering the modular ratio n_x as defined in § 7.6.3 (2).

N_{Ed} is the maximum axial force in the lower flange along the segment, obtained by:

$$N_{Ed} = \sigma_i b_{fi} t_{fi} \quad (175)$$

σ_i is the elastic stress at mid-thickness of the lower flange under the relevant combination of load, obtained according to § 5.9., where $z_5 = -h_s + t_{fi} / 2$;

α_{cr} is the critical amplification factor, calculated with the LTBeamN module, considering the column constituted by the lower flange loaded by the axial forces obtained with Formula (197).

7.6.5. Beams with usual composite sections – composite stage – Second generation of EN

- (1) For single span beams, without negative bending moment, it is assumed that the slab is a part of a floor which prevents its lateral instability, so that the beam is not prone to lateral torsional buckling.
- (2) For beams with cantilever spans, negative bending moments are formed near the supports, which leads to lower flanges being compressed. The beams should thus be checked against LT buckling, as described below, according to the method given by FprEN 1993-1-1, 8.3.2.4.
- (3) The beam is separated in segments delimited by:
 - the ends of the beam (for cantilever spans);
 - the supports.
- (4) The resistance of each segment i to lateral torsional buckling is checked through the following criterion:

$$\Gamma_{LTi} = \frac{M_{Ed}}{M_{b,Rd}} \leq 1 \quad (176)$$

where: M_{Ed} is the maximum negative bending moment along the segment;

$M_{b,Rd}$ is the buckling resistance, as given in (5).

- (5) The buckling resistance $M_{b,Rd}$ is obtained by the following equation, according to EN 1993-1-1, 6.3.2.4 (2):

$$M_{b,Rd} = \chi_{c,z} M_{el,Rk} / \gamma_{M1} \quad (177)$$

where: γ_{M1} : see § 2.12.2;

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$\chi_{c,z}$ is the reduction factor for lateral torsional buckling, calculated according to § 7.6.2 (4), considering the reduced slenderness $\bar{\lambda}_{c,z,mod}$ given in (6) and the imperfection factors as given in (7).

(6) The reduced slenderness $\bar{\lambda}_{c,z,mod}$ is obtained by:

$$\bar{\lambda}_{c,z,mod} = k_c \beta_c \bar{\lambda}_{c,z} \quad (178)$$

where: $k_c L_c$ is the buckling length of the segment, see § 7.6.4 (9);

$$\bar{\lambda}_{c,z} = \sqrt{\frac{A_{ac} f_{yi}}{N_{cr,c,z}}}$$

A_{ac} is obtained by:

$$A_{ac} = b_{fi} t_{fi} + \frac{1}{3} h_{wc} t_w$$

h_{wc} is the height of the web that is under compression, given by:

$$h_{wc} = \min\{h_w; h_a - t_{fi} + z_{el}\}$$

z_{el} is the location of the elastic neutral axil of the composite section under negative bending moment;

$N_{cr,c,z}$ is the elastic critical axial force, see (8);

β_c is given by:

$$\beta_c = \sqrt{\frac{0,06 + \frac{h_a}{t_{f,max}}}{\bar{\lambda}_{c,z} + \frac{t_{f,max}}{t_{f,inx}}}}$$

but: $\beta_c \leq 1$.

$$t_{f,max} = \max\{t_{fi}; t_{fs}\};$$

$$t_{f,min} = \min\{t_{fi}; t_{fs}\}.$$

(7) The imperfection factor α_{LT} is given by **Table 47** as a function of the buckling curve, chosen as follows:

- curve *d* for welded sections;
- curve *c* for hot rolled sections.

(8) $N_{cr,c,z}$ is given by:

$$N_{cr,c,z} = \alpha_{cr} N_{Ed} \quad (179)$$

where: α_{cr} : see (6);

$$N_{Ed} = \frac{M_{Ed}}{I_y} \left[(h_a - t_{fi} + z_{el}) b_{fi} t_{fi} + \left(h_a - t_{fi} - \frac{h_{wc}}{2} + z_{el} \right) h_{wc} t_w \right]$$

z_{el} and I_y are the location of the elastic neutral axis and the second moment or area of the composite section under negative bending moment.

7.6.6. Beams with usual composite sections – construction stage

(1) For beams with usual composite sections, if point lateral restraints have been defined for the construction stage, § 7.6.1 (cross-section without partial concrete encasement) or § 7.6.3 (cross-section without partial concrete encasement) applies.

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- (2) If a restraint by the steel profiled sheeting is considered, the same chapters apply, but where the critical moment M_{cr} is calculated taking into account the presence of the sheeting – see **Annex E**. In this case, the profiled steel sheeting and its connection are checked according to § 7.6.8.
- (3) For beams with point propping, specific load arrangements will be considered for the load case Q_c – see § 6.7.4.

7.6.7. Beams with usual composite sections and partial concrete encasement

- (1) For a usual composite beam with partial concrete encasement in the final stage, it is assumed that no lateral torsional buckling may occur, event in negative bending segments of the beam. In this case, no check is carried out.

7.6.8. Verification of the profiled steel sheeting used as lateral restraint in the construction phase

- (1) This chapter describes the verifications carried out by the software at the ultimate limit states for the steel profiled sheeting used as a lateral restraint against lateral torsional buckling in the construction phase (see § 7.6.6). It also gives the verifications performed for the fastening of the steel sheeting.
- (2) The maximum shear force in the shear panel for the stabilisation of a beam is given by the following formula (according to Annex B of the ECCS Recommendations [14]):

$$V_{m,Ed} = \frac{\pi}{\mu_{cr} - 1} S_{act} \frac{e_0}{L} \quad (180)$$



Attention should be paid to the error in the formula of the ECCS Recommendations. Formula (179) here above takes into account the adequate correction – cf. [21].

where L is the span length of the beams (**Figure 24**);
 S_{act} is the shear stiffness of the deck – see § E.3;
 e_0 is the equivalent horizontal imperfection of the flange restrained by the deck, which may be obtained by:

$$e_0 = \alpha_{LT} (\bar{\lambda}_{LT} - 0,2) \frac{W_{el}}{A} \quad (181)$$

W_{el} is the elastic bending modulus of the steel profile cross-section;

A is the area of the steel profile cross-section;

μ_{cr} is minimum amplifier to reach the elastic critical buckling moment, defined by:

$$M_{cr} = \mu_{cr} M_{Ed}$$

M_{Ed} is the design bending moment in the beam;

M_{cr} is the elastic critical moment for lateral torsional buckling (see § E.2 (1));

$\bar{\lambda}_{LT}$ is the reduced slenderness for lateral torsional buckling, defined by:

$$\bar{\lambda}_{LT} = \sqrt{\frac{M_{Rk}}{M_{cr}}}$$

M_{Rk} is the characteristic resistance in bending of the cross-section, calculated as follows:

$$M_{Rk} = W_{pl} f_y \text{ for Class 1 or 2 cross-sections;}$$

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$M_{Rk} = W_{el} f_y$ for Class 3 cross-sections.

W_{pl} is the plastic bending modulus of the profile cross-section;

W_{el} is the elastic bending modulus of the profile cross-section;

f_y is the yield strength of steel;

α_{LT} is the imperfection factor, determined according to **Table 47** from the buckling curve. The choice of the latter is done according to **Table 48** from the geometry of the cross-section.



e_0 is an equivalent imperfection, which takes into account the geometric imperfection of the beam and other destabilising phenomena, such as initial stresses. Formula (179) for the calculation of the shear force of the steel deck is conservatively based on this equivalent imperfection e_0 .

- (3) The resistance of the horizontal shear panel under the action of the stabilising forces is checked through the following criterion:

$$\frac{V_{m,Ed}}{V_{m,Rd}} \leq 1 \quad (182)$$

where $V_{m,Rd}$ is the shear resistance of the panel, which is given by the following Formula:

$$V_{m,Rd} = n_c F_{s,Rd} + \frac{\beta_1}{\beta_3} F_{p,Rd} \quad (183)$$

$F_{p,Rd}$ is the design strength of an individual sheet fastener (cf. (4) ou (5));

$F_{s,Rd}$ is the design strength of an individual seam fastener (cf. (6));

n_c is the number of seam fasteners per side lap (excluding those which pass through both sheets and the supporting beam);

β_1 is a coefficient whose value is given by **Table 64** in **Annex E** from n_f ;

n_f is the number of sheet-to-beam fasteners per sheet width;

β_3 is a factor which is equal to 1 when seam fasteners are in the troughs. On the other hand, it is given by the following formula when seam fasteners are in the crests:

$$\beta_3 = \frac{n_f - 1}{n_f} \quad (184)$$

- (4) For the deck-to-beam fastening, the resistance of a self-drilling screw is given by the following formula (according to the bearing resistance defined by Table 8.2 in EN 1993-1-3):

$$F_{p,Rd} = \alpha d_n t_p f_{up} / \gamma_{M2} \quad (185)$$

where d_{fb} is the nominal diameter of the screw (see **Table 12** in § 2.11.3);

t_p is the design thickness of the fastened sheet;

f_{up} is the ultimate strength of the sheet;

γ_{M2} is the partial factor for the resistance of fasteners; its value in France is $\gamma_{M2} = 1,25$;

α is a factor defined as follows:

- if $t_{fs} = t_p$: $\alpha = \min\{2,1; 3,2 \sqrt{t_p / d_{fb}}\}$
- if $t_{fs} \geq 2,5 t_p$ and $t_p < 1,0$ mm: $\alpha = \min\{2,1; 3,2 \sqrt{t_p / d_{fb}}\}$
- if $t_{fs} \geq 2,5 t_p$ and $t_p \geq 1,0$ mm : $\alpha = 2,1$
- if $t_p < t_{fs} < 2,5 t_p$: α is obtained by linear interpolation

t_{fs} is the thickness of the beam upper flange where the screw is fastened.

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The range of validity of Formula **(185)** is defined by:

$$e_1 \geq 3 d_{fb}$$

$$e_L \geq 3 d_{fb}$$

$$3,0 \text{ mm} \leq d_{fb} \leq 8 \text{ mm}$$

where e_1 is the spacing of the centre of the hole to the closer edge of the deck.

- (5) When fired pins are used, the following formula is used to assess $F_{p,Rd}$ (according to Table 8.3 in EN 1993-1-3):

$$F_{p,Rd} = 3,2 d_{fb} t_p f_{up} / \gamma_{M2} \quad (186)$$

where d_{fb} is the nominal diameter of the pin (see **Table 12** in § 2.11.3);

The range of validity of Formula **(186)** is as follows:

$$e_1 \geq 4,5 d_{fb}$$

$$e_L \geq 4,5 d_{fb}$$

$$3,7 \text{ mm} \leq d_{fb} \leq 6 \text{ mm}$$

$$f_{up} \leq 550 \text{ MPa.}$$

- (6) The resistance of a self-drilling screw used for the seaming of two adjacent steel sheets with the same thickness, is obtained as follows, according to Table 8.2 in EN 1993-1-3 (this criterion is associated to the bearing resistance of the sheeting):

$$F_{s,Rd} = \alpha d_{fs} t_p f_{up} / \gamma_{M2} \quad (187)$$

where d_{fs} is the nominal diameter of the screw (see **Table 13** in § 2.11.3);

t_p is the design thickness of the fastened sheet;

f_{up} is the ultimate strength of the sheet;

γ_{M2} is the partial factor for the resistance of fasteners; its value in France is $\gamma_{M2} = 1,25$;

α is a factor defined by:

$$\alpha = \min \left\{ 2,1; 3,2 \sqrt{t_p / d_n} \right\}$$

- (7) Formula **(184)** for the calculation of $V_{m,Rd}$ assumes that the ultimate limit state of the steel deck is obtained by bearing of fasteners. It thus requires to prevent the other failure modes, which are considered as non permissible modes. According to Annex B of ECCS Recommendations **[14]**, the three following criteria are checked by the software, as given in **(8)** to **(10)**.

- (8) The sheet-to-beam fastener resistance is checked through the following criterion:

$$0,6 \frac{d}{e_L} F_{p,Rd} \geq V_{m,Rd} \quad (188)$$

where d is the span length of the steel sheets (equal to spacing between secondary beams – see **Figure 24**) ;

e_L is the spacing between sheet-to-beam fasteners (see **Figure 26**);

- (9) The resistance of the sheeting to end failure is checked through the following criterion:

$$F_{p,end,Rd} \geq V_{m,Rd} \quad (189)$$

where $F_{p,end,Rd}$ is the resistance of the steel profiled sheets at the deck end, given by;

$$F_{p,end,Rd} = k_n d \frac{t_p^{1,5} f_{yp}}{e_p^{0,5} \gamma_p} \quad (190)$$

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- e_p the rib spacing of the sheets (§ 2.5.1);
 f_{yp} is the yield strength of the sheets;
 t_p is the design thickness of the sheets, obtained by: $t_p = t_{nom} - t_{pr}$;
 t_{nom} is the nominal thickness of the sheets;
 t_{pr} is the thickness of the coating applied to the sheets (§ 2.11.3);
 k_n is a factor obtained as follows:
 $k_n = 0,9$ when a fastener is used in every rib at the end of the deck,
 $k_n = 0,3$ when a fastener is used in every two ribs at the end of the deck;
 γ_p is the partial factor for the resistance of the steel deck (see § 2.12.2).

- (10) The resistance to shear buckling of the sheeting is checked through the following criterion:

$$V_{b,Rd} \frac{d}{L} \geq V_{m,Rd} \quad (191)$$

where $V_{b,Rd}$ is the resistance of the steel deck to shear buckling;
 d and L are as given in **Figure 24**.

- (11) When the criterion of formula (192) is satisfied, the shear buckling resistance of the steel deck only depends on the global shear buckling.

$$\frac{b_r}{t_p} \leq 86,7 \varepsilon \quad \text{with:} \quad \varepsilon = \sqrt{\frac{235}{f_{yp}}} \quad (192)$$

In this case, the shear buckling resistance is obtained by:

$$V_{b,Rd} = V_{bg,Rd} \quad (193)$$

When the criterion of formula (192) is not satisfied, the effect of local buckling is taken into account, with the following formula:

$$V_{b,Rd} = V_{bred,Rd} = \frac{V_{bg,Rd} V_{bl,Rd}}{V_{bg,Rd} + V_{bl,Rd}} \quad (194)$$

where $V_{bg,Rd}$ is the resistance of the steel deck to global shear buckling (see (12));
 $V_{bl,Rd}$ is the resistance of the steel deck to local shear buckling (see (13));
 b_r and t_p – see **Figure 52**.

- (12) The resistance to global shear resistance of the steel deck is obtained by:

$$V_{bg,Rd} = k_{ng} \frac{L}{d^2} D_x^{0,25} D_y^{0,75} \quad (195)$$

where: k_{ng} is a factor taking into account the influence of the fastening, obtained as follows:

- $k_{ng} = 28,8$ when a fastener is used in every rib at the end of the deck;
 $k_{ng} = 14,4$ when a fastener is used in every two ribs at the end of the deck;

D_x and D_y are the stiffnesses of the deck obtained by:

$$D_x = \frac{E t_p^3 e_p}{12 (1 - \nu^2) u}$$

$$D_y = \frac{E I_p}{e_p}$$

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u is the perimeter length of a complete single corrugation of pitch e_p , which may be calculated by (see **Figure 52** for the geometrical parameters) :

$$u = b_b + b_r + \sqrt{4 h_p^2 + (b_t - b_b)^2}$$

I_p is the second moment of area of a single corrugation about its neutral axis;

E is the Young's modulus of steel.

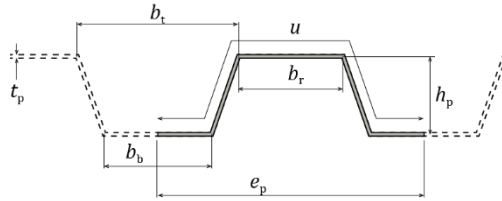


Figure 52: Perimeter length of a single corrugation

(13) The resistance to local shear resistance of the steel deck is obtained by:

$$V_{bl,Rd} = 4,83 E_a \left(\frac{t_p}{b_r} \right)^2 L t_p \quad (196)$$

where : b_r and t_p – see **Figure 52**;

L - see **Figure 24.**;

E_a - see § 2.10.4.

7.7. Reinforcement of usual composite cross-section under negative bending moment

- (1) For a composite beam with usual composite cross-sections, with or without partial concrete encasement, the longitudinal reinforcement of the slab should fulfil the criterion given in (2), when negative bending moments occur in the beam and a plastic resistance is used for the verification of cross-sections.
- (2) For the cases specified in (1), the reinforcement ratio ρ_s should fulfil the following criterion:

$$\rho_s = \frac{A_s}{A_c} \geq \rho_{s,min} = \delta \frac{f_y}{235} \frac{f_{ctm}}{f_{sk}} \sqrt{k_c} \quad (197)$$

where: A_s is the area of the longitudinal reinforcement – see § 2.3.3 (5);

A_c is the effective area of the concrete flange, given by: $A_c = b_{eff} t_c$;

b_{eff} : see § 5.1;

t_c is thickness of the active part of the concrete flange, see § 5.2;

f_y is the nominal yield strength of the structural steel, see § 2.10.1;

f_{sk} is the characteristic yield strength of the reinforcement, see § 2.10.6;

f_{ctm} is the mean tensile strength of the concrete, see § 2.10.5;

δ is equal to 1 for Class 2 cross-sections, and to 1,1 for Class 1 sections;

k_c is a factor which takes into account the stress distribution within the section immediately prior to cracking. It is given by:

$$k_c = \min \left\{ 1; 0,3 + \frac{2 z_0}{2 z_0 + t_c} \right\} \quad (198)$$

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z_0 is the vertical distance between the centroids of the uncracked concrete flange and the uncracked composite section, calculated using the short term modular ratio, n_0 – see § 5.3. It is obtained by:

$$z_0 = t_h + t_d - \frac{t_c}{2} - z_{elh} \quad (199)$$

z_{elh} is the location of the elastic neutral axis of the transformed composite cross-section, calculated according to § 5.5.

7.8. Resistance checks for the lower flange/plate of slim floor sections

7.8.1. Class 1 or 2 sections

- (1) Method in the current ABC software:

The resistance of the lower supporting plate of a slim floor section is checked through the following criterion:

$$\Gamma_q = \frac{q}{q_{Rd}} \leq 1 \quad (200)$$

$$q_{Rd} = \frac{2(e_1 - e_2)}{3} \frac{f_{ys}}{(1 + \xi) \gamma_{M0}} \left(\sqrt{1 + \frac{3 t_s^2}{(e_1 - e_2)^2}} - 1 \right)$$

where: $q = q_1 + q_2$

$$\xi = \frac{\max\{q_1; q_2\} - \min\{q_1; q_2\}}{q}$$

The criterion is checked for each supporting plate of the cross-section, using parameters of **Table 43**.

- (2) Method according to CosFB software and Annex I of prEN 1994-1-1:

The resistance of the lower supporting plate of a slim floor section is checked through the following criterion:

$$\Gamma_q = \frac{m_{x_{bt},Ed}}{m_{x_{bt},Rd}} \leq 1 \quad (201)$$

where: $m_{x_{bt},Ed}$ is the transverse moment per unit length, given by:

$$m_{x_{bt},Ed} = q d_{bt}$$

$m_{x_{bt},Rd}$ is the bending resistance of the plate, given by:

$$m_{x_{bt},Rd} = 1,2 \frac{t^2 f_y}{6 \gamma_{M0}}$$

q and d_{bt} : see § 5.5.7 (3). (See comment in § 5.5.7 (3) for the factor 1,2 in the formula).

7.8.2. Class 3 sections

- (1) The resistance of the lower supporting plate of a slim floor section is checked through the following criterion:

$$\Gamma_{\sigma q} = \frac{\sigma_{yi}}{f_d} \leq 1 \quad (202)$$

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$$\Gamma_{\tau q} = \frac{\tau_{yi}}{\sqrt{3} f_d} \leq 1$$

where: σ_{yi} is the local transverse bending stress in the supporting plate;

τ_{yi} is the local shear stress;

σ_{yi} and τ_{yi} are as given in § 5.10;

f_d is the design yield strength of the supporting plate:

$f_d = f_y / \gamma_{M0}$ for lower flanges;

$f_d = f_p / \gamma_{M0}$ for lower plates.

7.9. Design of welds for welded sections

- (1) For usual steel or composite sections with welded sections, the thickness of weld throats will be designed to fulfil the following criterion (see [15]), assuming a continuous fillet weld on both sides of the web:

$$a_w \geq \frac{g_{z,ELU,Ed}}{2 f_{vw,d}} = \frac{\sqrt{3}}{2} \frac{g_{z,ELU,Ed}}{f_u} \beta_w \gamma_{M2} \quad (203)$$

where: a_w is the thickness of throats for the welds between the flanges and the web;

$g_{z,ELU,Ed}$ is the cumulated shear flow in the welds for the relevant ULS combination of load;

f_u is the nominal ultimate tensile strength of the weaker part joined; For hot rolled profiles obtained from the ArcelorMittal database, f_u will be read from this base. For other cases, f_u will be obtained from **Table 50**;

β_w is a correlation factor given in **Table 49**;

γ_{M2} see § 2.12.2.

Steel grades	S235	S275	S355	S420, S460, higher steel grades
β_w	0,8	0,85	0,9	1,0

Table 49: Correlation factor

Steel grades	S235	S275	S355	S420	S460 and higher steel grades
f_u [MPa]	360	370	470	540	570

Table 50: Ultimate limit strength

- (2) The shear flow in welds for a section in a non-composite stage is given by:

- For the upper flange:

$$g_{zs,Ed} = \frac{V_{Ed} b_{fs} t_{fs} |z_{fs} - z_{ela}|}{I_{y,a}} \quad (204)$$

- For the lower flange:

$$g_{zi,Ed} = \frac{V_{Ed} b_{fi} t_{fi} |z_{fi} - z_{ela}|}{I_{y,a}} \quad (205)$$

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where: z_{ela} is the location of the elastic neutral axis;

$I_{y,a}$ is the second moment of area of the steel cross-section;

z_{fs} and z_{fi} are the locations of the centroids of the upper and lower flanges respectively, see **Table 24**;

(3) The shear flow in welds for a section in a composite stage is given by:

- For the upper flange, if the centroid is outside the slab ($z_{elh} < t_h$)

$$g_{zs,Ed} = \frac{V_{Ed} \left(b_{fs} t_{fs} |z_{fs} - z_{el,h}| + \frac{b_{eff} t_c}{n} |t_h + t_d - t_c / 2 - z_{elh}| \right)}{I_{y,h}} \quad (206)$$

- For the upper flange, if the centroid is in the slab ($z_{elh} \geq t_h$)

$$g_{zs,Ed} = \frac{V_{Ed} \left(b_{fs} t_{fs} |z_{fs} - z_{el,h}| + \frac{b_{eff} z_{elh}^2}{2 n} \right)}{I_{y,h}} \quad (207)$$

- For the lower flange:

$$g_{zi,Ed} = \frac{V_{Ed} b_{fi} t_{fi} |z_{fi} - z_{elh}|}{I_{y,h}} \quad (208)$$

where: z_{elh} is the location of the elastic neutral axis of the composite transformed cross-section;

$I_{y,h}$ is the second moment of area of this section.

(4) Following additional criterion will be applied for the automatic design of welds:

$$\max\{3 ; \sqrt{t_{\max}} - 0,5 \text{ mm}\} \leq a_w \leq 0,7 t_{\min} \quad (209)$$

where: $t_{\max} = t_{fs}$

$t_{\min} = t_w$

7.10. Design of welds for slim floor sections

(1) The welds of the supporting plate for SFB and IFB-A slim floor sections are designed by the software.

(2) The thickness of the weld throat should fulfil the following criterion:

$$\max\{3 ; \sqrt{t_{\max}} - 0,5 \text{ mm}\} \leq a_{sp} \leq a_u$$

$$a_u = \frac{\beta_w \gamma_{M2}}{f_u} \sqrt{2 q_{Ed}^2 + \frac{3}{4} q_{L,Ed}^2} \quad (210)$$

where: $q_{Ed} = \max\{q_{w1}; q_{w2}\}$;

$$q_{w1} = 0,5 [q_1(e_1 + e_{2p}) + q_2(e_1 - e_{2p})] / e_{2p}$$

$$q_{w2} = q_1 + q_2 - q_{w1}$$

$$e_{2p} = e_2 + \sqrt{2} a_{ws}$$

q_1 and q_2 are obtained according to § 6.9;

e_1 and e_2 are obtained from **Table 43** for the SFB plate and the IFB-A plate;

$t_{\max}$ is the maximal thickness of the two welded plates, given by:

$$t_{\max} = \max\{t_p; t_f\} \text{ for SFB sections}$$

$$t_{\max} = \max\{t_p; t_w\} \text{ for IFB-A sections}$$

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$q_{L,Ed}$ is the longitudinal shear flow in the fillet weld as given in **(2)** and **(3)**.

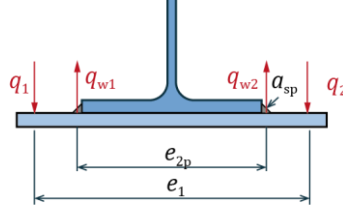


Figure 53 : Design of fillet welds

- (2) The longitudinal shear flow in the fillet weld for steel slim floor sections is given by:

$$q_{L,Ed} = \frac{V_{Ed} A_{sp} z_{sp}}{I_y} \quad (211)$$

where: V_{Ed} is the global shear force in the cross-section, obtained as the maximal shear force along the span;

A_{sp} is the area of the supporting plate, given by: $A_{sp} = b_p t_p$

z_{sp} is the distance between the centroid of the plate and the centroid of the cross-section, given by: $z_{sp} = z_{ANE} + t_p / 2$;

z_{ANE} is the location of the elastic neutral axis;

I_y is the second moment of area of the cross-section.

- (3) The longitudinal shear flow in the fillet weld for steel slim floor sections is given by:

$$q_{L,Ed} = \sum_i \frac{V_{Ed,i} A_{sp} z_{sp,i}}{I_{hy,i}} \quad (212)$$

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8. DESIGN OF THE CONNECTION BETWEEN THE SLAB AND THE STEEL PROFILE

8.1. Introduction

- (1) Chapter 0 applies to the connection between the steel profile and the slab, for beams with usual composite section, with or without partial concrete encasement.



See § 9 for the design of the connection of the partial concrete encasement.

- (2) For the connection, following types of design will be available in the software:
- plastic design, according to § 8.2, applied when a plastic design of cross-sections is performed, i.e. all cross-sections are of Class 1 or 2;
 - elastic design, according to § 8.3, applied when an elastic design of cross-sections is carried out, either because at least one section is of Class 3 or because the elastic design option has been selected.
- (3) When plastic design is applied, full or partial connection will be possible for the positive bending moment segments of the beam. In negative moment segments, only full connection is considered.
- (4) The software will propose, in a limited scope of application, the automatic design of the connection – see § 8.4.
- (5) The software will automatically design the transverse reinforcement required to resist the longitudinal shear in the connection – see § 8.5.

8.2. Plastic design of the connection

8.2.1. General

- (1) This chapter applies only to beams with usual composite section, with or without partial concrete encasement, where all the cross-sections are Class 1 or 2 (see § 5.4) and the elastic design option is not selected.

8.2.2. Degree of shear connection

- (1) For usual composite cross-section without partial concrete encasement, the degree of shear connection of the main span in positive bending is obtained as follows:

$$\eta = \frac{\sum P_{Rdi}}{N_{cx}} \quad (213)$$

where: N_{cx} is the maximal longitudinal shear force in the connection, given by:

$$N_{cx} = \min\{N_{c,Rd}; N_{a,Rd}\} \quad (214)$$

$N_{c,Rd}$ is the resistance to compression of the slab, given by:

$$N_{c,Rd} = 0,85 b_{eff} t_c f_{cd} \quad (\text{first generation of EN}) \quad (215)$$

$$N_{c,Rd} = b_{eff} t_c f_{cd} \quad (\text{second generation of EN})$$

$N_{a,Rd}$ is the axial resistance of the steel profile, given by:

- for hot rolled profiles:

$$N_{a,Rd} = A_a f_{yd} \quad (216)$$

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- for welded profiles:

$$N_{a,Rd} = (b_{fs} t_{fs} f_{yfs} + h_w t_w f_{yw} + b_{fi} t_{fi} f_{yfi}) / \gamma_{M0} \quad (217)$$

ΣP_{Rdi} is the sum of the resistance of all the studs located between the point of 0 moment in the bending moment diagram and the cross-section with the maximal positive bending moment.

- (2) As a simplification for beams with adjacent spans, ΣP_{Rdi} in Formula **(213)** will be replaced by:

$$\Sigma P_{Rdi} - A_s f_{sd} \quad (218)$$

where: A_s is the area of the longitudinal reinforcement at the intermediate support;

ΣP_{Rdi} is the sum of the resistance of all studs located between this intermediate support and the cross-section with the maximal positive bending moment.



By this simplification, the full connection in negative bending zone is automatically obtained.

- (3) For a beam with a usual composite section with partial **concrete** encasement, the degree of shear connection is calculated according to **(1)**, where Formula **(214)** is replaced by:

$$N_{cx} = \min\{N_{c,Rd}; N_{a,Rd} + N_{es,Rd}\} \quad (219)$$

where: $N_{es,Rd}$ is the resistance to tension of the steel reinforcement of the partial **concrete** encasement, given by:

$$N_{es,Rd} = \Sigma A_{esi} f_{sed} \quad (220)$$

8.2.3. Limitation of partial shear connection

- (1) Partial shear connection will be possible for the segment of the main span in positive bending moment. Additionally, partial connection will be possible only if the welded studs fulfil the following condition:

$$h_{sc} / d \geq 4 \quad (221)$$



This condition is required in order to obtain ductile connectors, as defined in EN 1994-1-1 § 6.6.1.1 (5).

- (2) Partial shear connection should fulfil the following criterion:

$$\eta \geq \eta_0 \geq \eta_{\min} \quad (222)$$

where η is the degree of shear connection, as calculated in § 8.2.2;

$$\eta_{\min} = 0,4 ;$$

η_0 is the minimum degree of shear connection, as given in **(3)** to **(5)**.

- (3) For steel sections with equal flanges, η_0 is given by:

$$\text{if } L_e \leq 25 \text{ m} \quad \eta_0 = 1 - \frac{355}{f_y} (0,75 - 0,03 L_e) \quad (223)$$

$$\text{if } L_e > 25 \text{ m} \quad \eta_0 = 1$$

where L_e is obtained according to § 5.1.2 **(2)**;

f_y is the minimum value of f_{yfs} , f_{yfi} and f_{yw} .

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- (4) For steel profiles having a lower flange with an area equal to three times the area of the upper flange, η_0 is given by:

$$\begin{aligned} \text{if } L_e \leq 20 \text{ m} \quad \eta_0 &= 1 - \frac{355}{f_y} (0,30 - 0,015 L_e) \\ \text{if } L_e > 20 \text{ m} \quad \eta_0 &= 1 \end{aligned} \quad (224)$$

- (5) For steel profiles having a bottom flange with an area exceeding the area of the upper flange but less than three times that area, η_0 will be obtained by a linear interpolation from Formulas (223) and (224).

- (6) For steel profiles outside this scope, partial shear connection will not be possible (i.e. $\eta_0 = 1$).



In normal mode, the steel profiles are always within this scope – see § 2.2.5.

- (7) If the option for the second generation of Eurocodes is selected, Formula (225) applies, instead of Formula (222):

$$\eta \geq \eta_0 \rho_m^2 k_{up} \geq \eta_{min} \quad (225)$$

where: $\eta_{min} = 0,4$;

η_0 as defined in (3) to (5);

k_{up} is a factor taking into account the method of construction, given by:

$k_{up} = 1$ if the beam is propped during construction (fully propped or with point propping);

$k_{up} = 1 - \rho_{up}$ if the beam is unpropped;

ρ_{up} is defined as follows:

$$\rho_{up} = \min\{0,15; M_{a,Ed} / M_{pl,Rd}\} \text{ if } M_{Ed} / M_{Rd}(\eta) \leq 0,95$$

$$\rho_{up} = 0 \text{ otherwise;}$$

M_{Ed} is the maximal positive bending moment along the beam;

$M_{a,Ed}$ is the part of M_{Ed} that is applied to the structural steel before the composite action is achieved, due to load case G_1 ;

$M_{pl,Rd}$ is the bending moment resistance for full shear connection, obtained according to § 5.5;

$M_{Rd}(\eta)$ is the design moment bending resistance, considering the applied shear connection at the point of maximal bending moment.

ρ_m is the utilisation ratio, defined by:

$$\rho_m = \frac{M_{Ed}}{0,95 M_{Rd}(\eta)} \quad \text{but } 0,8 \leq \rho_m \leq 1 \quad (226)$$



$M_{a,Ed} = 0$ in case of fully propped beams.

- (8) For beams where the degree of shear connection is lower than the condition required by Formulas (222) or (225), plastic design will not be possible and the verifications will be carried out according to elastic design procedure.

8.3. Elastic design of the connection

- (1) Elastic design of the shear connection between the slab and the steel profile is applied when:

- some cross-section along the beam are of Class 3;
- the minimum degree of shear connection is not met;
- or the elastic design option has been selected by the User.

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- (2) For each cross-section, the shear connection should fulfil the following criterion:

$$\Gamma_{cx} = \frac{v_{L,Ed}}{v_{L,Rd}} \leq 1 \quad (227)$$

where: $v_{L,Ed}$ is the design longitudinal shear force per unit of length (see **(3)**);

$v_{L,Rd}$ is the design longitudinal shear resistance per unit of length (see **(5)**).

- (3) $v_{L,Ed}$ is given by:

$$v_{L,Ed} = \sum v_{L,Ed,i} \quad (228)$$

where: $v_{L,Ed,i}$ is the design longitudinal shear force per unit of length for the load case i where the beam acts compositely (see **(4)**), taking into account the relevant combination factors.

- (4) In positive bending segments, $v_{L,Ed,i}$ is given by:

$$v_{L,Ed,i} = \frac{V_{Ed,i} S_{h,i}}{I_{h,i}} \quad (229)$$

where: $V_{Ed,i}$ is the design shear force for the load case i ;

$I_{h,i}$ is the second moment of area of the composite transformed cross-section, considering the modular ratio for the concrete of the slab n_i , according to **§ 5.3**;

$S_{h,i}$ is the first moment of area of the composite transformed cross-section constituted by the concrete slab, about the elastic neutral axis, given by:

$$S_{h,i} = \frac{b_{eff} t_c}{n_i} \left(t_h + t_p + \frac{t_c}{2} - z_{el,i} \right) \quad (230)$$

$z_{el,i}$ is the location of the neutral axis of transformed composite section.

- (5) In positive bending moment, if the reinforcements under compression are taken into account (see **§ 5.5.5 (1)**), Formula **(230)** in **(4)** will be replaced by:

$$S_{h,i} = \frac{b_{eff} t_c}{n_i} \left(t_h + t_p + \frac{t_c}{2} - z_{el,i} \right) + \sum \frac{A_{si}}{n_s n_i} (n_i - n_s) (z_i - z_{el,i}) \quad (231)$$

where: $n_s = E_a / E_s$.

- (6) In negative bending moments, **(3)** and **(4)** apply, where Formula **(230)** is replaced by:

$$S_{h,i} = \sum \frac{A_{si}}{n_s} (z_i - z_{el,i}) \quad (232)$$

- (7) $v_{L,Rd}$ is given by:

$$v_{L,Rd} = \frac{n_r P_{Rd}}{s_x} \quad (233)$$

where: P_{Rd} is the design shear resistance of a connector, according to **§ 5.7**;

n_r is the number of shear connectors per line;

s_x is the spacing between two lines of connectors (see **§ 2.6.2**).

8.4. Automatic design of the connection

- (1) An automatic design of the shear connection between the slab and the profile will be proposed. Its scope of application will be limited to beams where all applied loads fulfil the following conditions:

- no point loads;
- distributed loads on the full span length only, with constant load per unit of length.

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- (2) Both plastic and elastic design will be possible. For the plastic design, the User will specify if a full or partial connection is required. When a partial connection is chosen, the software provides the minimum number of connectors that fulfils all the requirements for the minimum degree of shear connection, the ultimate limit state requirements and the serviceability limit state criteria.
- (3) The provided solution will correspond to the connection with the lowest number of studs, considering a uniform distribution of studs along connection zones.

8.5. Design of transverse reinforcement for usual composite sections

8.5.1. General and assumptions

- (1) The software will automatically assess the section of transverse reinforcement that is required to resist the longitudinal shear in the connection.
- (2) For solid slabs, two layers will be assumed, one below and the other one above the head of the studs. For composite slabs, only one layer will be considered.
- (3) For solid slabs with haunches, a third bended layer is assumed to be placed in the haunch, when the following condition is met (**Figure 54 b**):

$$h_{sc} - 70 \text{ mm} \leq t_h \quad (234)$$



This criterion assumes:

- the depth of the stud head is 10 mm;
- the lower layer of reinforcement should be 30 mm below the head;
- a concrete coverage below the layer is 20 mm;
- the diameter of reinforcement is 10 mm.

In this case, the potential area of shear failure d-d is assumed.

- (4) The transverse reinforcement area is assumed to be uniformly distributed along each connection zone.

8.5.2. Minimum transverse reinforcement

- (1) The area of the transverse reinforcement should fulfil the following criterion, according to EN 1992-1-1, § 9.2.2 (5):

$$\rho_t \leq \rho_{t,min} = 0,08 \frac{\sqrt{f_{ck}}}{f_{ys}} \quad (235)$$

where: f_{ck} see § 2.10.5;

f_{ys} see § 2.10.6;

ρ_t is the ratio of transverse reinforcement, obtained by:

$$\rho_t = \frac{A_b + A_t}{s_f t_c} \quad \text{for solid slabs}$$

$$\rho_t = \frac{A_t}{s_f t_c} \quad \text{for composite slabs}$$

8.5.3. Configurations of slabs and potential areas of shear failure

- (1) The potential areas of shear failure considered in the calculation depend on the configuration of the slab, according to **Figure 54**.

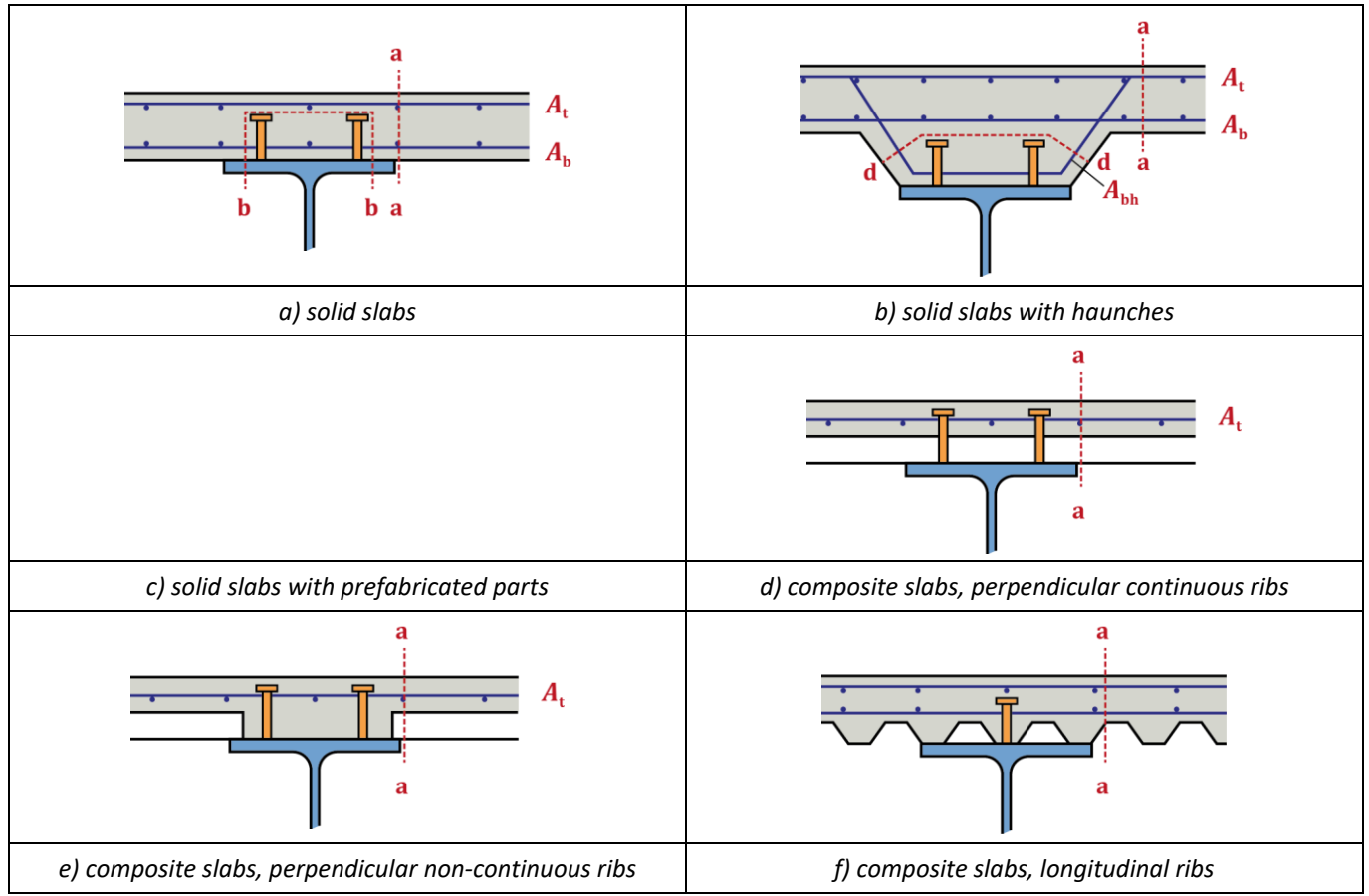


Figure 54 : Potential areas of shear failure

8.5.4. Longitudinal shear force

- (1) In each connection zone, the longitudinal shear force per unit length to be resisted by the transverse reinforcement is given by:

$$v_{cx,Ed} = \frac{n_r P_{Rd}}{s_x} \quad (236)$$

where: n_r is the number of connectors per file (perpendicularly to the beam axis);

s_x is the longitudinal spacing of studs files;

P_{Rd} is the shear resistance of a connector – see § 5.7.



This formula assumes a capacity design of the transverse reinforcement, in order to resist the maximal shear force in the connection. This formula is very close to the actual design force in the case of partial shear connection. It is also accurate in the case of full shear connection where the number of studs have been optimised.

- (2) The longitudinal shear force per unit of length in the potential area of shear failure is then obtained by:

$$v_{sf,Ed} = k_{sf} v_{cx,Ed} \quad (237)$$

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where: k_{sf} is the distribution factor, given by:

$$\begin{aligned} \text{for a-a shear surfaces: } k_{sf} &= \max \left\{ \frac{b_{ei} - b_0 / 2}{b_{eff,m}} \right\} && \text{in positive bending zone} \\ k_{sf} &= \max \left\{ \frac{\beta_i b_{ei} - b_0 / 2}{b_{eff,s}} \right\} && \text{in negative bending zone} \end{aligned}$$

$$\text{for b-b shear surfaces: } k_{sf} = 1$$

$$\text{for c-c shear surfaces: } k_{sf} = 1$$

$$\text{for d-d shear surfaces: } k_{sf} = 1$$

b_0 is the spacing between the two edge studs in a file, obtained by:

$$b_0 = (n_r - 1) b_{0,min}$$

$b_{0,min}$ is the minimum transverse spacing of studs:

$$b_{0,min} = 4 d \text{ for composite slabs;}$$

$$b_{0,min} = 2,5 d \text{ for solid slabs.}$$

$b_{eff,m}$ and $b_{eff,s}$ are the effective widths of the concrete slab, respectively at mid-span and at support, see § 5.1.2;

b_{ei} : see § 5.1.2.

- (3) The shear stress in the potential area of shear failure is then obtained by:

$$\tau_{Ed} = \frac{v_{sf,Ed}}{h_f} = k_{sf} \frac{n_r P_{Rd}}{h_f s_x} \quad (238)$$

where: h_f is the developed length of the potential area of shear failure, given by:

$$\text{for a-a shear surface: } h_f = 2 h_{sc} + b_0$$

$$\text{for b-b shear surface: } h_f = t_c$$

for c-c shear surface:

$$\text{for d-d shear surface: } h_f = b_0 + 2 \frac{b_f - b_0 + h_{cs} \tan \theta_h}{\sqrt{1 + (\tan \theta_h)^2}}$$

8.5.5. Verification of the concrete crushing

- (1) The inclination angle of the compression concrete strut is sought from the criterion for the resistance to compression of the concrete strut, according to 6.2.4 (4) of EN 1992-1-1. This criterion is given by:

$$\tau_{Ed} \leq v f_{cd} \sin \theta_f \cos \theta_f \quad (239)$$

where: f_{cd} : see § 2.10.5;

v is a reduction factor that takes into account the cracking of concrete in shear, defined by formula (6.6N) of EN 1992-1-1:

$$v = 0,6 (1 - f_{ck}/250)$$

For the second generation of Eurocodes, $v = 0,5$;

θ_f is the compression strut inclination angle.

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- (2) For the determination of the compression strut inclination angle θ_f , the following formula will be used, in order to minimise the angle:

$$\theta_f = \frac{1}{2} \arcsin \left(\frac{2 \tau_{Ed}}{\nu f_{cd}} \right) \quad (240)$$

with: $\theta_{f,min} \leq \theta_f \leq \theta_{f,max}$

where the smallest possible value, $\theta_{f,min}$, is given by:

- if the connection zone is fully in positive bending moment: $\theta_{f,min} = 27^\circ$;
- if the connection zone is partly in negative bending moment: $\theta_{f,min} = 36^\circ$;

The maximum value of this angle is: $\theta_{f,max} = 45^\circ$.

- (3) Once the inclination angle θ_f of the concrete compression struts is defined, the criterion for the resistance of these strut is calculated by:

$$\Gamma_{sf} = \frac{\tau_{Ed}}{\nu f_{cd} \sin \theta_f \cos \theta_f} \leq 1 \quad (241)$$

8.5.6. Design of transverse reinforcement

- (1) From the value of the inclination angle as defined in § 8.5.5, reinforcement is designed in order to fulfil the following criterion:

$$\frac{A_{sf}}{s_f} f_{sd} \geq \frac{\tau_{Ed} h_f}{\cotan \theta_f} \quad (242)$$

where: A_{sf} / s_f is the area of reinforcement by length unit;

f_{sd} : see § 2.10.6.

- (2) For configurations of slabs with at least two layers of transverse reinforcement, the area of top layer per length unit is denoted A_t and the area of bottom layer per length unit is denoted A_b . For haunches, a third layer may be required and is denoted A_{bh} . See **Table 51** for the configurations of layers of reinforcement and potential areas of shear failure.

	Potential areas of shear failure	Solid slab without prefabricated part	Solid slab with haunch	Other configurations
Intermediate beam	a-a	$A_t + A_b$	$A_t + A_b$	A_t
	b-b	$2 A_b$	-	-
	d-d	-	$2 A_{bh}$	-
Edge beam	a-a	$A_t + A_b$	$A_t + A_b$	A_t
	b-b	A_b	-	-
	d-d	-	$2 A_{bh}$	-

Table 51: Design of transverse reinforcement

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8.5.7. Design of transverse reinforcement with continuous profiled steel sheetings

- (1) For composite slabs with continuous steel profiled sheeting (configuration PE1 according to § 2.3.2), the contribution of the profiled deck to the resistance to longitudinal shear is taken into account as follows, according to 6.6.6.4 (4) of EN 1994-1-1. In this case, Formula (242) will be replaced by:

$$\frac{A_{sf}}{s_f} f_{sd} \geq \frac{\tau_{Ed} h_f}{\cotan \theta_f} - A_{pe} f_{pd} \quad \text{and} \quad \frac{A_{sf}}{s_f} f_{sd} \geq 0 \quad (243)$$

where: f_{pd} is the design value of the yield strength for the steel profiled deck, given by:

$$f_{pd} = f_p / \gamma_p$$

A_{pe} is the area of the profiled deck per length unit, given by:

$$A_{pe} = \frac{t_p}{e_p} \left(e_p + b_b - b_t + \sqrt{(b_b - b_t)^2 + 4 h_p^2} \right)$$

t_p, b_b, b_t, h_p and e_p : see § 2.5.1;

f_p : see § 2.5.2;

γ_p : see § 2.12.2.

8.6. Design of transverse reinforcement for slim floors

- (1) For slim floor composite sections with studs welded on the upper flange (Group B in Table 6), § 8.5.2 to 8.5.6 apply, considering one layer of reinforcement.
- (2) Transverse reinforcement for other types of connections (Groups A and C) is not calculated by the software.

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9. DESIGN OF THE CONNECTION FOR PARTIAL CONCRETE ENCASEMENT

9.1. Introduction

- (1) Chapter 9 applies to the connection between the steel profile and the partial concrete encasement, if any. It applies to beams with usual steel or composite section, with partial concrete encasement.
- (2) For the connection of partial concrete encasement, following types of design will be available in the software:
 - plastic design, according to § 9.2, applied when a plastic design of cross-sections is performed, i.e. all cross-sections are of Class 1 or 2;
 - elastic design, according to § 9.3, applied when an elastic design of cross-sections is carried out, either because some of the cross-sections are of Class 3 or because the elastic design option has been selected.
- (3) When plastic design is applied, full connection will always be considered, whatever the sign of the bending moment is.

9.2. Plastic design of the connection between the partial concrete encasement and the steel profile

- (1) Plastic design of the connection between the steel profile and the concrete encasement is performed only when all the cross-section of the beam are of Class 1 or 2 and the elastic design option is not selected.
- (2) For each critical zone, the software will provide the plastic shear flow to be resisted by the connection.

9.3. Elastic design of the connection between the partial concrete encasement and the steel profile

- (1) Elastic design of the connection between the steel profile and the concrete encasement is performed when some of the cross-section of the beam are of Class 3 or if the elastic design option is selected.
- (2) The software will provide the maximal elastic shear flow to be resisted by the connection.
- (3) The elastic design shear flow in the connection will be obtained from Formulas (228) and (229), where the first moment of area $S_{h,i}$ is calculated as follows.
- (4) If the partial concrete encasement is fully in compression and the longitudinal reinforcement is not considered, $S_{h,i}$ will be obtained by:

$$S_{h,i} = \frac{A_{ce}}{n_i} \left(-\frac{h_s}{2} - z_{el,i} \right) \quad (244)$$

- (5) If the partial concrete encasement is fully in tension, $S_{h,i}$ will be obtained by:

$$S_{h,i} = \sum \frac{n_{esi} d_{esi}^2}{4 n_s} (z_{esi} - z_{el,i}) \quad (245)$$

- (6) If the partial concrete encasement is partly in tension and in compression, $S_{h,i}$ will be obtained numerically according to the principles given in Annex B.

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10. FIRE DESIGN

10.1. General

- (1) The software will assess the standard fire resistance, assuming fire temperatures according to ISO 834 curve.
- (2) Fire calculations will be activated only if fire combinations are defined by the User.
- (3) For usual steel or composite section, it will be possible to define fire protection parameters, through generic protection products.

10.2. Principles for beams without partial concrete encasement

- (1) The fire calculations in the software follow the steps 1 to 5 (Figure 55):
 1. Design effect of actions in fire situation
 2. Cross-section classification
 - Steps 3 to 4 are iterative, starting from the initial time and considering a time increment:
 3. Section temperature at time t
 4. Design resistance criteria for bending moments, lateral torsional buckling and vertical shear, considering the calculated temperature at time t
 5. Steps 3 and 4 are repeated until the first design criterion reaches its limit, which gives the standard fire resistance.
- (2) The design resistance step (step no 4) will be carried for the following predefined times:
 - 30 min;
 - 60 min;
 - 90 min;
 - 120 min;
 - 180 min;
 - 240 min.

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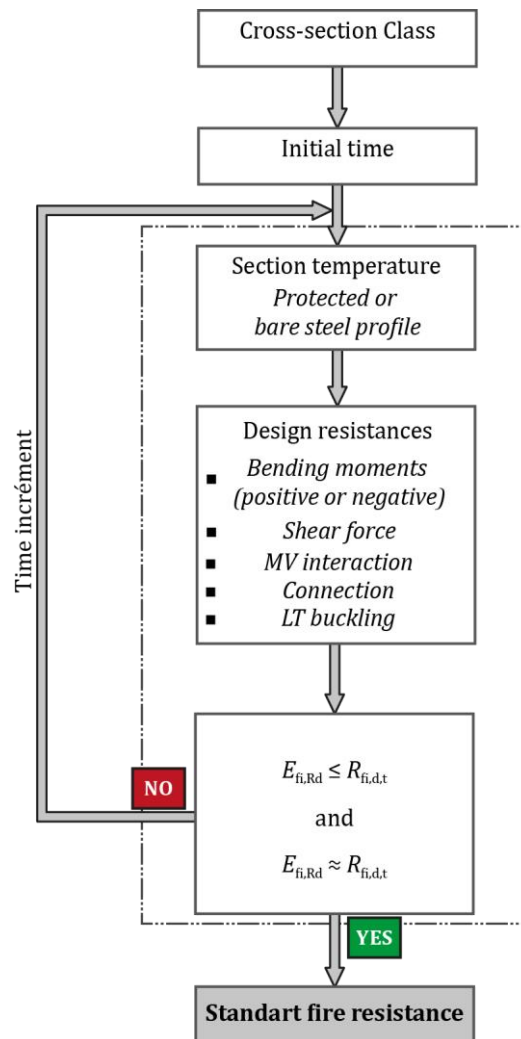


Figure 55: Calculation flowchart for the fire design in ABC-PMX – Beam without partial concrete encasement

10.3. Principles for beams with partial concrete encasement

- (1) For steel or composite beams with partial concrete encasement, the fire calculations in the software follow the steps 1 to 5 (Figure 56):
 1. Design effect of actions in fire situation
 2. Cross-section classification
 - Step 3 is incremental, starting from the initial time and considering a time increment:
 3. Design resistance criteria for bending moments, lateral torsional buckling and vertical shear, considering the resistance at time t
 4. Step 3 is repeated until the first design criterion reaches its limit, which gives the standard fire resistance.
- (2) The design resistance step (step no 3) will be carried for the following predefined times:
 - 30 min;
 - 60 min;
 - 90 min;
 - 120 min;
 - 180 min.

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- (3) The design resistance at time t is obtained according to the simplified method of the Annex F of EN 1994-1-2, as described in **Annex F**.

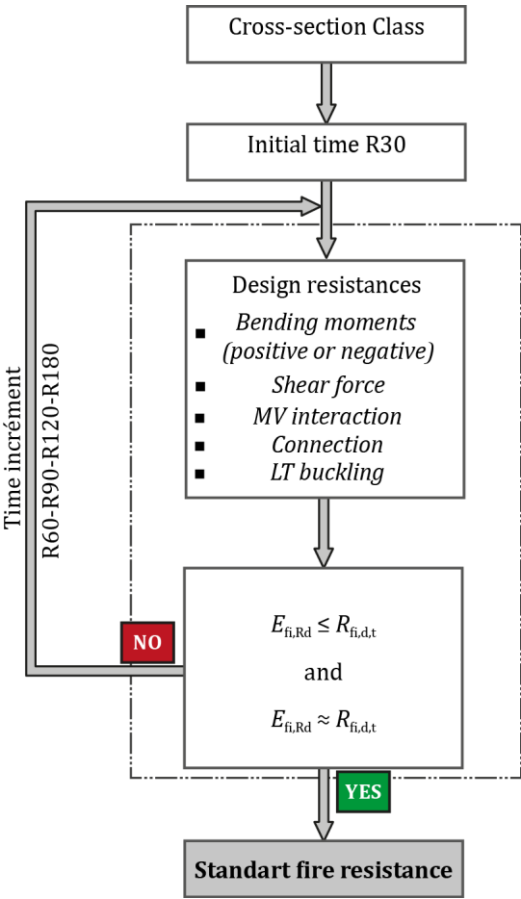


Figure 56: Calculation flowchart for the fire design in ABC-PMX – Beam with partial concrete encasement

10.4. Principles for slim-floor beams

10.5. Parameters for the fire design

- (1) See § 3.3 and § 3.4.

10.6. Section properties - Classification

- (1) Sections are classified according to § 5.4, where the steel parameter ε is replaced by:

$$\varepsilon = 0,85 \sqrt{235/f_y} \tag{246}$$

where f_y is the yield strength of steel.

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10.7. Heating of the section

10.7.1. Usual unprotected steel sections

- (1) A uniform temperature distribution is assumed over the cross-section.
- (2) The computation of temperature at time t_i (°C);
The process is started at time t_0 where $\theta(t_0) = 20^\circ\text{C}$.
- (3) For unprotected beams, the variation of steel temperature is obtained according to EN 1993-1-2 § 4.2.5.1, as follows:

$$\Delta\theta_a(t_i) = \frac{k_{sh} (A_m/V)}{c_a \rho_a} \dot{h}_{net,d} \Delta t \quad (247)$$

where: A_m/V is the section factor of the unprotected beam (m^{-1}), given by § 10.8.1;

k_{sh} is the shadow factor, given in (4);

c_a is the steel specific heat [in J/(kg.K)], given as a function of the temperature in (5);

ρ_a is the unit mass of steel [in kg/m³], which is assumed to be independent of the temperature (see § 2.10.4);

$\dot{h}_{net,d}$ is the increment of total received heat flux (W/m^2), which is given in (6);

Δt is the time increment [in s] – see § 10.3

In Expert mode, the User may define Δt (see § 3.4). The default value is 5 s, and according to EN 1994-1-2, 4.3.4.2.2 (5), Δt should not be taken greater than 5 s when assessing formula (247).

- (4) The shadow factor k_{sh} is obtained according to EN 1993-1-2 § 4.2.5.1, as follows:

$$k_{sh} = 0,9 \frac{(A_m/V)_b}{(A_m/V)} \quad (248)$$

where $(A_m/V)_b$ is the box value of the section factor - see § 10.8.1 (6).

- (5) The steel specific heat c_a is calculated according to EN 1993-1-2 § 3.4.1.2, as follows:

- for $20^\circ\text{C} \leq \theta_a < 600^\circ\text{C}$

$$c_a = 425 + 0,773 \theta_a - \frac{1,69}{10^3} \theta_a^2 + \frac{2,22}{10^6} \theta_a^3 \quad [\text{J}/(\text{kg} \cdot \text{K})]$$

- for $600^\circ\text{C} \leq \theta_a < 735^\circ\text{C}$

$$c_a = 666 + \frac{13\,002}{738 - \theta_a} \quad [\text{J}/(\text{kg} \cdot \text{K})]$$

- for $735^\circ\text{C} \leq \theta_a < 900^\circ\text{C}$

$$c_a = 545 + \frac{17\,820}{\theta_a - 731} \quad [\text{J}/(\text{kg} \cdot \text{K})]$$

- for $900^\circ\text{C} \leq \theta_a < 1200^\circ\text{C}$

$$c_a = 650 \quad [\text{J}/(\text{kg} \cdot \text{K})]$$

- (6) The increment of total received heat flux $\dot{h}_{net,d}$ is composed of the radiative flux $\dot{h}_{net,r}$ and the convective flux $\dot{h}_{net,c}$ (see § 3.1 of EN 1991-1-2):

$$\dot{h}_{net,d} = \dot{h}_{net,r} + \dot{h}_{net,c} \quad (249)$$

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where:

$$\dot{h}_{\text{net},r} = \Phi \varepsilon_m \varepsilon_f \sigma [(\theta_r + 273)^4 - (\theta_m + 273)^4]$$

$$\dot{h}_{\text{net},c} = \alpha_c (\theta_g - \theta_m)$$

σ is the Stefan-Boltzman constant ($\sigma = 5,67 \times 10^{-8} \text{ W/m}^2\text{K}^4$) – see § 3.3;

ε_f is the emissivity of fire; by default $\varepsilon_f = 1$ – see § 3.4;

ε_m is the emissivity of the steel, see (9);

θ_g is the gas temperature (°C) – see (7);

θ_m is the surface temperature of the member (°C), which is assumed equal to θ_a ;

θ_r is effective radiation temperature of the fire environment (°C), which is assumed equal to θ_g ;

Φ is the view factor (see (8));

α_c is the coefficient of heat transfer by convection ($\text{W/m}^2\text{K}$). The default value is $\alpha_c = 25 \text{ W/m}^2\text{K}$. – see § 3.4.

- (7) The temperature of hot gases at time t_i , θ_g , is given by the following formula (according to 3.2.1 (1) in EN 1991-1-2):

$$\theta_g = 20 + 345 \log_{10}(8 t_i + 1) \quad (250)$$

where t_i is the time expressed in minutes.

- (8) The view factor Φ is by default equal to 1. The User can specify another value in the range of [0; 1] – see § 3.4.
- (9) The emissivity of steel is taken from Table 52.

Type of steel	$\varepsilon_m (\theta \leq 500^\circ\text{C})$	$\varepsilon_m (\theta > 500^\circ\text{C})$
Carbon steel	0,7	0,7
Hot-dip galvanised steel	0,35	0,7

Table 52: Emissivity of steel



See Table 5.1 in prEN 1994-1-2:2024

10.7.2. Usual protected steel sections

- (1) A uniform temperature distribution is assumed over the cross-section.
- (2) The computation of the steel temperature θ_a at time t_i is carried out according to § 10.7.1 (2), where the variation of steel temperature is obtained according to EN 1993-1-2 § 4.2.5.2, as follows:

$$\Delta\theta_a(t_i) = \frac{\lambda_p (A_p/V)}{d_p c_a \rho_a} \frac{1}{1 + \phi/3} (\theta_g - \theta_a) \Delta t - (e^{\phi/10} - 1) \Delta\theta_g \quad (251)$$

with: c_a , ρ_a , θ_g , θ_a , Δt : see § 10.7.1;

λ_p is the thermal conductivity of the insulating element (W/m K) – see Table 14;

A_p/V is the section factor of the protected section (m^{-1}) – see § 10.8.2;

$\Delta\theta_g$ is the increase of the ambient gas temperature (°C) during the time interval Δt ;

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ϕ is defined by:

$$\phi = \frac{c_p \rho_p}{c_a \rho_a} d_p \frac{A_p}{V} \quad (252)$$

d_p is the thickness of the fire protection material (m);

c_p is the specific heat of the insulating material (J/kg K) – see **Table 14**;

ρ_p is the unit mass of the insulating material (kg/m³) – see **Table 14**.

Δt is the time increment [in s] – see § 10.3

In Expert mode, the User may define Δt (see § 3.4). The default value is 10 s, and according to EN 1994-1-2, 4.3.4.2.2 (8), Δt should not be taken greater than 30 s when assessing formula (251).

10.7.3. Usual unprotected composite sections

- (1) According to EN 1994-1-2, 4.3.4.2.2, usual unprotected composite sections are divided in the four following parts:
 - the lower flange;
 - the web (including root fillet for hot rolled profiles);
 - the upper flange;
 - the slab.

- (2) The temperature in each part is calculated independently, with the following assumptions:
 - the temperature in each steel part is uniform;
 - no heat transfer takes place between these different parts nor between the upper flange and the concrete slab;
 - the temperature of the slab varies within the thickness and is uniform along the width.

- (3) The temperature in a steel part is calculated according to § 10.7.1 (2) and (3), where, in Formula (247):
 - the shadow factor k_{sh} is calculated as follows (see § 2.2.2 for the parameters):

$$k_{sh} = 0,9 \frac{t_{fs} + t_{fi} + b_{fi}/2 + \sqrt{h_w^2 + (b_{fs} - b_{fi})^2}/4}{h_w + b_{fi} + b_{fs}/2 + t_{fs} + t_{fi} - t_w} \quad (253)$$

- the section factors are obtained from (4) to (6).

- (4) The section factor of the lower flange is calculated by:

$$\frac{A_m}{V} = 2 \frac{b_{fi} + t_{fi}}{b_{fi} t_{fi}} \quad (254)$$

- (5) The section factor of the upper flange is calculated by the following formula if its upper side is covered by the slab, according to the conditions in § 10.8.1 (5):

$$\frac{A_m}{V} = \frac{b_{fs} + 2 t_{fs}}{b_{fs} t_{fs}} \quad (255)$$

If the conditions for the coverage by the slab are not met, the section factor is obtained by:

$$\frac{A_m}{V} = 2 \frac{b_{fs} + t_{fs}}{b_{fs} t_{fs}} \quad (256)$$

- (6) The section factor for the web is obtained by:

$$\frac{A_m}{V} = \frac{2 d_w + \pi (r_{cs} + r_{ci})}{h_w t_w + (4 - \pi)(r_{cs}^2 + r_{ci}^2)/2} \quad (257)$$

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- (7) The temperature in the slab will be obtained with simplified tabulated methods as given in § 10.7.4 and in § 10.7.5, or using a numerical calculation, as described in § 10.7.6. The choice between these methods is according to the calculation options.

10.7.4. Slab – Simplified method – First generation of EN

- (1) The temperature in plain slabs, with or without prefabricated parts, assuming the following assumptions:
- the slab is divided in up to 14 layers of uniform thickness, with a thickness of 5 mm in the first 60 mm, an additional layer of 20 mm if needed, and a last one with the remaining thickness, if needed (**Figure 57**);
 - for normal concretes, the temperature in each of these layers is as given in **Table 53**, according to Table D.5 of EN 1994-1-2;
 - for light weight concrete, the temperature in each layer will be as given in **Table 53**, but reduced by a factor of 0,9.

- (2) The temperature in composite slabs will be assessed by applying (1), using the effective depth of the slab $h_{c,eff}$, obtained as follows:

- if $h_p/t_c \leq 1,5$:

$$h_{c,eff} = t_c + 0,5 h_p \frac{b_b + b_t}{e_p} \quad (258)$$

- if $h_p/t_c > 1,5$:

$$h_{c,eff} = t_c \left(1 + 0,75 h_p \frac{b_b + b_t}{e_p} \right) \quad (259)$$

where: b_b , b_t , h_p and e_p are the dimensions of the profiled steel sheeting, as defined in § 2.5.1;
 t_c see § 5.2.2.

	Thickness	zTop	Temperature in concrete layers (°C) after a fire duration of					
	(mm)	(mm)	30'	60'	90'	120'	180'	240'
1	5	5	535	705				
2	5	10	470	642	738			
3	5	15	415	581	681	754		
4	5	20	350	525	627	697		
5	5	25	300	469	571	642	738	
6	5	30	250	421	519	591	689	740
7	5	35	210	374	473	542	635	700
8	5	40	180	327	428	493	590	670
9	5	45	160	289	387	454	549	645
10	5	50	140	250	345	415	508	550
11	5	55	125	200	294	369	469	520
12	5	60	110	175	271	342	430	495
13	20	80	80	140	220	270	330	395
14	≥ 20	≥100	60	100	160	210	260	305

Table 53: Temperature distribution in solid slabs with normal concretes

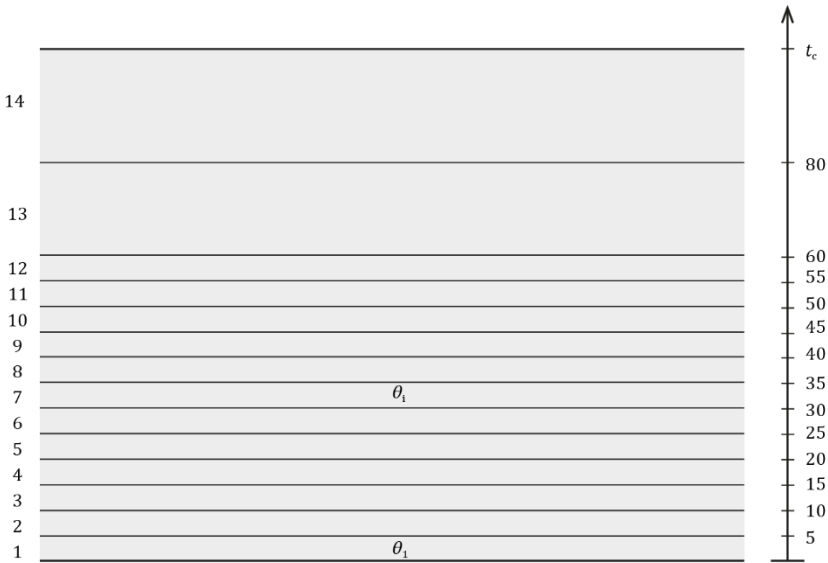


Figure 57: Temperature distribution in the slab

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10.7.5. Slab – Simplified method – Second generation of EN

- (1) The same principles as in § 10.7.4 are used, but with **Table 54** (according to Table B.6 in prEN 1994-1-2:2023). This Table is not applicable to R240 calculations.

	Thickness	zTop	Temperature in concrete layers (°C) after a fire duration of				
	(mm)	(mm)	30'	60'	90'	120'	180'
	2,5	2,5	675	831	912	967	1042
	7,5	10	513	684	777	842	932
	10	20	363	531	629	698	797
	10	30	260	418	514	583	685
	10	40	187	331	423	491	591
	10	50	135	263	349	415	514
	10	60	101	209	290	352	448
	10	70	76	166	241	300	392
	10	80	59	133	200	256	344
	10	90	46	108	166	218	303
	10	100	37	89	138	186	267
	10	110	31	73	117	159	236
	10	120	27	61	100	137	209
	10	130	24	51	86	119	186
	10	140	23	44	74	105	166
	10	150	22	38	65	94	149

Table 54: Temperature distribution in solid slabs with normal concretes – Second generation of EN

- (2) For any slab with a thickness greater than 150 mm, the temperature in the slab will be assumed constant and uniform in the upper layer above 150 mm.

10.7.6. Slab – Numerical method

- (1) The temperature in solid slabs, with or without prefabricated parts, will be assessed by a Finite Difference Analysis (with assumptions as given in **Annex G**).

10.7.7. Usual protected composite sections

- (1) For protected composite section, the same principles as in § 10.7.3 (1) and (2) apply. For sections with board thermal protection, the steel profile is represented by one common part.
- (2) The temperature in a steel part is calculated according to § 10.7.2 (2), applying the section factors in (3) to (4) below.
- (3) For section with contour protection, the section factors of the parts are obtained according to § 10.7.3 (4) to (6).

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- (4) For section with board thermal protection, the section factor of the steel profile is obtained according to **§ 10.8.2**.
- (5) The temperature in the slab is obtained with simplified tabulated methods as given in **§ 10.7.4** and in **§ 10.7.5**, or with a numerical calculation, as described in **§ 10.7.6**. The choice between these methods is according to the calculation options.

10.7.8. Slim floor sections

- (1) The temperature in slim floor steel or composite sections will be assessed by a Finite Difference Analysis.

10.8. Section factors

10.8.1. Usual unprotected steel or composite sections

- (1) The section factor of an unprotected steel section is given by:

$$\frac{A_m}{V} = \frac{p}{A_a} \quad (260)$$

where: p is the perimeter of the section exposed to fire

A_a is the area of the steel profile (see **§ 5.5**);

- (2) For a section exposed on all four sides, the perimeter p is given by:

$$p = 2 (h_s + b_{fs} + b_{fi} - t_w) + (\pi - 3) (r_{cs} + r_{ci}) \quad (261)$$

For hot rolled profiles, this equation may be simplified as follows:

$$p = 2 (h_s + 2 b_f - t_w + (\pi - 3) r)$$

- (3) For a section exposed on three sides only, the perimeter p is given by:

$$p = b_{fs} + 2 (h_s + b_{fi} - t_w) + (\pi - 3) (r_{cs} + r_{ci}) \quad (262)$$

- (4) An unprotected steel section will be considered as exposed on three sides if the upper side of the upper flange is covered by the slab as defined in **(5)**.

- (5) The upper flange is covered by the slab if one of the following condition is met:

- the slab is a solid slab, with or without prefabricated parts:
- the slab is composite where the ratio of coverage of the upper side of the upper flange is not less than 0,85.

For composite slabs with transverse ribs with configurations PE 1 or PE2 (see **Figure 14**), the ratio of coverage of the upper flange is given by:

$$c_r = b_b / e_p \quad (263)$$

where b_b and e_p are defined in **§ 2.5.1**.

For composite slabs with longitudinal ribs, the ratio of coverage of the upper flange is equal to:

$$c_r = b_b / b_{fs} \quad (264)$$

where b_{fs} is defined in **§ 2.2.2**.

For composite slabs with transverse ribs with configuration PE3, the upper flange is assumed to be covered.

- (6) The box value of the section factor of an unprotected steel section is given by:

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$$\left(\frac{A_m}{V}\right)_b = \frac{p_b}{A_a} \quad (265)$$

where: p_b is the box perimeter of the section exposed to fire

A_a is the area of the steel profile (see § 5.5);

- (7) For a section exposed on all four sides, the box perimeter p_b is given by:

$$p_b = 2 (h_s + \max\{b_{fs}; b_{fi}\}) \quad (266)$$

For hot-rolled profiles, this equation may be simplified as follows:

$$p_b = 2 (h_s + b_f)$$

- (8) For a section exposed on three sides only, the box perimeter p_b is given by:

$$p_b = 2 h_s + \max\{b_{fs}; b_{fi}\} \quad (267)$$

10.8.2. Usual protected steel or composite sections

- (1) The section factor of a protected steel section is given by:

$$\frac{A_p}{V} = \frac{p}{A_a} \quad (268)$$

where: p is the perimeter of the protected section exposed to fire

A_a is the area of the steel profile (see § 5.5);

- (2) For contour protection (i.e. spray or intumescent paint), the perimeter p is obtained according to § 10.8.1 (2) or (3).
- (2) For a section with board protection, the perimeter p is given by:

$$p = 2 h_s + \max\{b_{fs}; b_{fi}\} \quad (269)$$



In case of board protection, it is assumed that the upper side of the upper flange is always protected.

10.9. Reduction factors for mechanical properties

10.9.1. Construction steel

- (1) The steel properties vary when the temperature increases. These variations are taken into account through the following parameters:
- factor for the variation of the yield strength f_y : $k_{y,\theta}$
 - factor for the variation of the Young's modulus E_a : $k_{E,\theta}$
 - factor for the variation of the ultimate strength f_u : $k_{u,\theta}$
- (2) The values defined by Table 3.1 of EN 1993-1-2 are used by the software for factors $k_{y,\theta}$, $k_{u,\theta}$ and $k_{E,\theta}$ – see Table 55. For temperatures between the values of this table, linear interpolation is applied for the calculation of the temperature coefficients.

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Steel temperature θ (°C)	Reduction factor $k_{y,\theta}$	Reduction factor $k_{E,\theta}$	Reduction factor $k_{u,\theta}$
20	1,00	1,000	1,25
100	1,00	1,000	1,25
200	1,00	0,900	1,25
300	1,00	0,800	1,25
400	1,00	0,700	1,00
500	0,78	0,600	0,78
600	0,47	0,310	0,47
700	0,23	0,130	0,23
800	0,11	0,090	0,11
900	0,06	0,0675	0,06
1000	0,04	0,0450	0,04
1100	0,02	0,0225	0,02
1200	0,00	0,000	0,00

Table 55: Temperature reduction factors applied to the steel properties

10.9.2. Concrete

- (1) The concrete properties vary when the temperature increases. The resistance to compression is reduced by the factor $k_{c,\theta}$ as given in **Table 56** (according to Table 3.3 in EN 1994-1-2).

Temperature of concrete θ (°C)	Reduction factor $k_{c,\theta}$ for normal weight concrete	Reduction factor $k_{c,\theta}$ for light weight concrete
20	1,00	1,0
100	1,00	1,0
200	0,95	1,0
300	0,85	1,0
400	0,75	0,88
500	0,60	0,76
600	0,45	0,64
700	0,30	0,52
800	0,15	0,40
900	0,08	0,28
1000	0,04	0,16
1100	0,01	0,04
1200	0	0

Table 56: Temperature reduction factors applied to the concrete properties

10.9.3. Reinforcing steel

- (1) The properties of steel reinforcement vary when the temperature increases. The yield strength is reduced by the factor $k_{s,\theta}$ as given in **Table 57**, according to 3.2.3 (3) of EN 1994-1-2 for cold worked reinforcing steels.

Steel temperature θ (°C)	Reduction factor $k_{s,\theta}$ for normal concrete	Reduction factor $k_{s,\theta}$ see (2)
	cold worked	hot rolled
20	1,00	1,00
100	1,00	1,00
200	1,00	1,00
300	1,00	1,00
400	0,94	1,00
500	0,67	0,78
600	0,40	0,47
700	0,12	0,23
800	0,11	0,11
900	0,08	0,06
1000	0,05	0,04
1100	0,03	0,02
1200	0	0

Table 57: Temperature reduction factors applied to the reinforcement properties



The values for hot-rolled rebars are identical to the values used for structural steel – see **Table 55**. They are used with the first generation of Eurocodes, according to EN 1994-1-2, 3.2.3 (2). With the second generation, these values are according to Table 5.4 in prEN 1994-1-2:2024.

10.10. Fire resistance criteria

10.10.1. Principles

- (1) The following checks are performed in fire situation:
- resistance of the cross-section to bending moment;
 - resistance of the cross-section to shear force;
 - when relevant, resistance of the cross-section to MV interaction;
 - resistance of the beam to lateral torsional buckling.

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10.10.2. Resistance to bending moment

- (1) The resistance of a cross-section to bending moment is checked through the following criterion:

$$\Gamma_{M,fi} = \frac{M_{fi,Ed}}{M_{fi,\theta,Rd}} \quad (270)$$

where: $M_{fi,Ed}$ is the bending moment under the relevant combination of load in fire situation;
 $M_{fi,\theta,Rd}$ is the moment resistance of the cross-section at temperature θ .

10.10.3. Resistance to shear force

- (1) The resistance of a cross-section to shear force is checked through the following criterion:

$$\Gamma_{V,fi} = \frac{V_{fi,Ed}}{V_{fi,\theta,Rd}} \quad (271)$$

where: $V_{fi,Ed}$ is the shear force under the relevant fire combination of load;
 $V_{fi,\theta,Rd}$ is the resistant shear force of the cross-section at temperature θ .

- (2) The resistance to shear force in fire situation considering a uniform temperature θ of the cross-section is given by (according to 4.2.3.3 (1) of EN 1993-1-2):

$$V_{fi,\theta,Rd} = k_{y,\theta} \frac{\gamma_{M0}}{\gamma_{Mfi}} V_{Rd} \quad (272)$$

where: V_{Rd} is the resistant shear force at room temperature;
 $k_{y,\theta}$ is the reduction for the yield strength of steel at temperature θ as given in § 10.9.1;
 γ_{Mfi} : see § 2.12.2.

10.10.4. Resistance to MV interaction

- (1) When $\Gamma_{V,fi} > 0,5$, the resistance to both bending moment and shear force, taking account of the influence of the shear force on the bending resistance, is checked through the following criterion, for [Class 1, 2 or 3](#) sections:

$$\Gamma_{M,fi} = \frac{M_{fi,Ed}}{M_{V,fi,\theta,Rd}} \quad (273)$$

where: $M_{V,fi,Ed}$ is the resistant bending moment of the cross-section at temperature θ , taking into account the influence of the presence of V. It is obtained by reducing the yield strength of the web by $(1 - \rho_{fi})$, where ρ_{fi} is given by:

$$\rho_{fi} = (2 \Gamma_{V,fi} - 1)^2 \quad (274)$$



The yield resistance of the web is also reduced by $k_{y\theta}$.

10.10.5. Resistance to shear buckling

- (1) Resistance to shear buckling is checked when the following criterion is fulfilled (acc. to EN 1993-1-5, 5.1 (2)):

$$\frac{h_w}{t_w} > 72 \frac{\varepsilon_{fi,w}}{\eta} \quad (275)$$

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where: η see § 2.12.4 (1).

$\varepsilon_{fi,w}$ is the steel factor for the web, see § 10.6.

h_w is the height of the web and t_w is its thickness.

- (2) The resistance to shear buckling under the effect of a shear force V_{Ed} is checked through the following criterion:

$$\Gamma_{V,fi} = \frac{V_{fi,Ed}}{V_{b,fi,Rd}} \leq 1 \quad (276)$$

- (3) $V_{b,fi,Rd}$ is obtained by:

$$V_{b,fi,Rd} = \chi_w \frac{h_w t_w k_{y,\theta} f_{yw}}{\sqrt{3} \gamma_{M,fi}} \quad (277)$$

where: χ_w is the reduction factor for shear buckling, which is given in **Table 46**, using the reduced slenderness $\bar{\lambda}_{fi,w}$ – see (4);

- (4) The reduced slenderness $\bar{\lambda}_{fi,w}$ is calculated by:

$$\bar{\lambda}_{fi,w} = \frac{h_w}{t_w} \frac{1}{37,4 \varepsilon_{fi,w} \sqrt{k_\tau}} \sqrt{\frac{k_{y,\theta}}{k_{E,\theta}}} \quad (278)$$

where: $k_{y,\theta}$ and $k_{E,\theta}$ are temperature reduction factors for structural steel, according to **Table 55**;

$$k_\tau = 5,34.$$



§ 10.10.5 is applied to all type of sections, by an engineering judgement of § E.4 from EN 1994-1-2.

10.10.6. Resistance to lateral torsional buckling

- (1) Resistance to lateral torsional buckling will be checked in fire situation only for usual steel section, either with or without partial **concrete** encasement.
- (2) For beams without partial **concrete** encasement, resistance to lateral torsion resistance at temperature θ is checked by applying § 7.6.2, where:

- γ_{M1} is replaced by $\gamma_{M,fi}$;
- M_{Rk} is replaced by $k_{y,\theta} M_{Rk}$;
- $\bar{\lambda}_{LT}$ is replaced by $\bar{\lambda}_{LT,fi}$ obtained from

$$\bar{\lambda}_{LT,fi} = \sqrt{\frac{k_{y,\theta} M_{Rk}}{k_{E,\theta} M_{cr}}} \quad (279)$$

- $k_{y,\theta}$ and $k_{E,\theta}$ are the temperature reduction factors for structural steel at temperature θ given by **Table 55**.

- (3) For beams with partial **concrete** encasement, no lateral torsional buckling checking is carried out (see § 4.3.4.3.4 (7) in EN 1994-1-2).

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10.11. Verifications in fire situation

10.11.1. Usual steel sections

- (1) Beams with usual steel sections in fire situations are checked according to § 10.10.2 to 10.10.6.
- (2) The resistant bending moment in fire situation considering a uniform temperature θ of the cross-section is given by (according to 4.2.3.3 (1) of EN 1993-1-2):

$$M_{fi,\theta,Rd} = k_{y,\theta} \frac{\gamma_{M0}}{\gamma_{Mfi}} M_{Rd} \quad (280)$$

where: M_{Rd} is the resistant bending moment at room temperature - $M_{Rd} = M_{pl,Rd}$ for Class 1 or 2 sections, and $M_{Rd} = M_{el,Rd}$ for Class 3 sections, where the class of the section is defined according to § 10.6;
 $k_{y,\theta}$ is the reduction factor for the yield strength of steel at temperature θ as given in § 10.9.1;
 γ_{Mfi} : see § 2.12.2.

10.11.2. Usual composite sections

- (1) Beams with usual composite sections in fire situations are checked according to § 10.10.2 to 10.10.6.
- (2) When § 10.10.2 (1) is applied, the resistant bending moment in fire situation of a usual composite section is obtained according to the principle of the plastic calculation in § 5.5.5, where the following assumptions are considered:

- the area of the lower flange is stressed to the following yield strength:

$$k_{y,\theta} \frac{f_{y,fi}}{\gamma_{M,fi,a}}$$

where $k_{y,\theta}$ is obtained according to § 10.9.1 from the temperature in the lower flange;

- the area of the web, including the root fillet if any, is stressed to the following yield strength:

$$k_{y,\theta} (1 - \rho_{fi}) \frac{f_{y,w}}{\gamma_{M,fi,a}}$$

where $k_{y,\theta}$ is obtained according to § 10.9.1 from the temperature in the web;

- the area of the upper flange is stressed to the following yield strength:

$$k_{y,\theta} \frac{f_{y,fs}}{\gamma_{M,fi,a}}$$

where $k_{y,\theta}$ is obtained according to § 10.9.1 from the temperature in the upper flange;

- the area of each layer of the slab is stressed in compression to the following strength:

$$\alpha_{slab} k_{c,\theta} \frac{f_c}{\gamma_{M,fi,c}}$$

where $k_{c,\theta}$ is obtained according to § 10.9.2 from the temperature in the layer;

$$\alpha_{slab} = 1;$$

Starting from the top of the slab, only the layers which fulfil the following condition are considered in the calculation of the plastic moment:

$$\alpha_{slab} \sum_j A_j k_{c,\theta} f_{ck} / \gamma_{M,fi,c} \leq \eta_{fi} N_{cx,fi} \quad (281)$$

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where: A_j is the area of the layer;

η_{fi} is the degree of shear connection, as given in § 10.12 (3);

$N_{cx,fi}$ is the longitudinal shear force in the connection for a full connection, as given in § 10.12 (2).

- the area of a reinforcement layer is stressed in tension to the following strength:

$$k_{s,\theta} \frac{f_{sk}}{\gamma_{M,fi,s}}$$

where $k_{s,\theta}$ is obtained according to § 10.9 from the temperature in the layer, assuming that the latter is equal to the temperature of the slab layer where the reinforcement is located.

- (3) An option will be proposed for usual section composite without partial concrete encasement to assess the bending resistance according to Annex E of EN 1994-1-2 (Annex C of prEN 1994-1-2:2023). In this case, the plastic moment will be obtained applying (2) with the assumption that $k_{c,\theta} = 1$ as long as $\theta_c \leq 250^\circ\text{C}$.

10.11.3. Profiles with partial concrete encasement

- The resistance of steel or composite beams with partial concrete encasement under fire situation is checked at the following time steps: 30 min; 60 min; 90 min; 120 min; 180 min.
- The bending moment resistance under positive or negative moments is checked through the method given in Annex F.
- The resistance to shear force is checked according to F.6.
- The resistance of cross-section is limited by the depth of slab according to Table 58

Standart fire resistance	Minimum slab thickness $t_d \geq t_{d,min}$ [mm]
R30	60
R60	80
R90	100
R120	120
R180	150

Table 58: Minimum slab thickness

10.11.4. Slim floors

- § 10.11.1 is applied for steel sections.
- § 10.11.2 is applied for composite sections.

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10.12. Connection

- (1) The shear resistance of a welded stud in fire situation is obtained according to § 5.7.2, where (see § 4.3.4.2.5 in EN 1994-1-2):

- the partial factor γ_v is replaced by $\gamma_{v,fi}$;
- Formula (109) is replaced by:

$$P_{Rd,s} = 0,8^2 k_{u,\theta} \frac{f_{u,sc} \pi d^2}{4 \gamma_{M,fi,V}} \quad (282)$$

- Formula (110) is replaced by:

$$P_{Rd,c} = 0,29 k_{c,\theta} \frac{\alpha d^2 \sqrt{f_{ck} E_{cm}}}{\gamma_{M,fi,V}} \quad (283)$$

where, in Formulas (282) and (283):

$k_{u,\theta}$ is the temperature reduction factor for steel, obtained from **Table 55**, considering a temperature of the connector θ_v equal to 80 % of the temperature of the upper flange;

$k_{c,\theta}$ is the temperature reduction factor for concrete, obtained from **Table 56**, considering a temperature of concrete θ_c equal to 40 % of the temperature of the upper flange.



The temperatures θ_v and θ_c are always taken greater or equal than the initial room temperature θ_{ref} .

- (2) The longitudinal shear force in the connection along a critical segment of a usual composite beam without partial concrete encasement, for a full connection, is obtained by:

$$N_{cx,fi} = \min\{N_{c,fi,Rd}; N_{a,fi,Rd}\} \quad (284)$$

where: $N_{a,fi,Rd}$ is the axial resistance of the steel profile in fire situation, given by:

$$N_{a,fi,Rd} = \sum_j A_j k_{y,\theta} f_{yj} / \gamma_{M,fi} \quad (285)$$

the summation is carried out on the different parts j considered for the calculation of the temperature in the steel profile;

A_j is the area of part no. j ;

f_{yj} is the yield strength of part no. j ;

$k_{y,\theta}$ is temperature reduction factor, according to **Table 55**, considering the temperature in part no j ;

$N_{c,fi,Rd}$ is the axial resistance to compression of the slab in fire situation, given by:

$$N_{c,fi,Rd} = \alpha_{slab} \sum_j A_j k_{c,\theta} f_{ck} / \gamma_{c,fi} \quad (286)$$

the summation is carried out the different layers of slab j considered for the calculation of the temperature in the slab;

A_j is the area of part no j ;

f_{ck} is the resistance to compression of slab part no j ;

$k_{c,\theta}$ is the temperature reduction factor, according to **Table 56**, considering the temperature in the slab layer no j ;

$\alpha_{slab} = 1$.

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- (3) The degree of shear connection in fire situation is obtained through the following equation:

$$\eta_{fi} = \frac{\sum P_{Rd,fi,i}}{N_{cx,fi}} \quad (287)$$

where: $N_{cx,fi}$ is the maximal longitudinal shear force in the connection, given by **(2)**:

$\sum P_{Rd,fi,i}$ is the sum of the resistance of all the studs $P_{Rd,fi}$ located between the point of 0 moment in the bending moment diagram and the cross-section with the maximal positive bending moment, calculated according to **(1)**.

11. VERIFICATIONS AT SERVICEABILITY LIMIT STATES

11.1. Deflections

- (1) The deflections of the beam will be calculated through the Finite Element numerical analysis (see **Annex C**).
- (2) The calculation sheet will report, for each span:
- the maximal deflection for each elementary load case;
 - the maximal deflection for each SLS combination of loads.

For multi-spans beams, the report will give the deflection in the main span and at the end of the cantilever spans.

- (3) For composite beams with partial degree of connection, the effect of lack of interaction is taken into account, according to the method provided by the second generation of EN, by replacing the second moment of area of composite cross-sections in positive bending segments by the following one:

$$I_{yh,eff} = I_{y,ch} + I_{y,a} + \frac{A_{c,eff} A_a}{A_{c,eff} + A_a} z_{ch}^2 \quad (288)$$

where: $I_{y,ch}$ is the second moment of area of the transformed area of the slab, assuming uncracked concrete;

$I_{y,a}$ is the second moment of area of the steel profile (including the partial concrete encasement if any);

$A_{c,eff}$ is obtained by:

$$A_{c,eff} = A_c / n_{eff}$$

A_c is the area of the slab, given by: $A_c = b_{eff} t_c$

n_{eff} is the effective modular ratio, given by:

$$n_{eff} = n \left(1 + \frac{\pi^2 E_a A_c / n}{L_e^2 c_s} \right)$$

n is the modular ratio according to § 5.3;

L_e is the distance between points of zero moment, as given in § 5.1.2 (2);

c_s is the stiffness of the shear connection, given by:

$$c_s = n_r k_{sc} / s_x$$

n_r is the number of studs per file, see § 2.6.2;

s_x is the longitudinal spacing between files of studs, idem;

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k_{sc} is the stiffness of an individual stud. In absence of any formula in prEN 1994-1-1, it will be assumed that the deformation of the stud is equal to δ_d when the shear resistance P_{Rd} is reached, where $\delta_d = 0,5$ mm. k_{sc} is then obtained by: $k_{sc} = P_{Rd} / \delta_d$;

z_{ch} is the distance between the centroids of the concrete slab and of the steel section.



Formula (288) will be applied specifically for each connection zone.

δ_d is a calculation option of the software. See § 3.2.3 (2).

Reference: prEN 1994-1-1, 9.3.1 (5).

11.2. Eigen frequencies

- (1) Eigen frequencies of the beam will be assessed by Finite Element analysis, using modal analysis/Rayleigh analysis. The adopted method will be decided later.
- (2) Eigen frequencies of the beam will be assessed for the following combinations of loads:
 - G
 - $G + 0,1 Q$
 - $G + 0,2 Q$
 - $G + 0,3 Q$
 - $G + 0,4 Q$
 - $G + 0,5 Q$
 where Q means Q_1 , Q_2 or $Q_1 + Q_2$.
- (3) Each force of any load case will be transferred in the model as an equivalent mass by dividing the force by g .

11.3. Comfort to vibrations

11.3.1. General

- (1) The assessment of the comfort to vibrations is carried out according to the Hivoss method [13]. It implies the calculations of the dynamic properties of the beam (eigen-frequency and modal mass) and of the damping of the floor.

11.3.2. Dynamic properties of the floor

- (1) The eigen-frequency of the beam is assessed by the software according to § 11.2, considering the combination of masses as defined by the User, see § 2.16.1.
- (2) Considering Table 7-2 of [13], the modal mass of a single span beam is given by: $m_{mod} = m_{beam}$, where m_{beam} is the full mass of the beam considering the combination of masses.
- (3) The damping is obtained by summing up the appropriate values D_1 to D_3 , defined as follows (Table 7-4 of [13]):
 - Structural damping D_1
According to the type of the ACB+ beam:
 - Steel beam: $D_1 = 1\%$
 - Composite beam: $D_1 = 1\%$
 - Damping for furniture D_2
According to the type of use defined by the user in the following list:

- Traditional office with partition walls: $D_2 = 2 \%$
- Paperless office: $D_2 = 0 \%$
- Open plan office: $D_2 = 1 \%$
- Library: $D_2 = 1 \%$
- Houses: $D_2 = 1 \%$
- Schools: $D_2 = 2 \%$
- Gymnastic: $D_2 = 0 \%$
- User defined: D_2 between 0 and 5 %
- Damping due to the finishing D_3

According to the type of finishing defined by the User in the following list:

- Ceiling under the floor: $D_3 = 1 \%$
- Free floating floor: $D_3 = 0 \%$
- Swimming screed: $D_3 = 1 \%$

Total damping: $D = D_1 + D_2 + D_3$

Limits: D is an integer value between $1\% \leq D \leq 9\%$

11.3.3. Determination of the $OS-RMS_{90}$ class

- (1) Considering the values of the eigen-frequency f , the modal mass m_{mod} and the damping D , the class of the $OS - RMS_{90}$ will be obtained from the numerical transcription of the Figure 7-4 to 7-12 of [13].

The $OS - RMS_{90}$ classes are defined according to the Table 59.

Class	$OS-RMS_{90}$ limits
A	$OS-RMS_{90} \leq 0,1$
B	$0,1 < OS-RMS_{90} \leq 0,2$
C	$0,2 < OS-RMS_{90} \leq 0,8$
D	$0,8 < OS-RMS_{90} \leq 3,2$
E	$3,2 < OS-RMS_{90} \leq 12,8$
F	$12,8 < OS-RMS_{90} \leq 51,2$
G	$51,2 < OS-RMS_{90}$

Table 59: $OS-RMS_{90}$ Classes

11.3.4. Vibration performance assessment

- (1) According to $OS - RMS_{90}$ class and to the use of the floor defined by the User, a vibration assessment will be given to the floor, between: recommended, critical, not recommended.

This assessment will follow the Table 7-5 of [13].

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11.4. Cracking of concrete for beams with usual composite sections

11.4.1. General

- (1) § 11.4 applies to beams with usual composite sections, with or without partial concrete encasement.
- (2) At the support of beam with no adjacent span, if no limitation of concrete cracking is imposed (see § 2.3.6) and if there is a continuity of the slab at the support (which is not accounted for in the design, see § 2.1 (5)), § 11.4.2 applies.
- (3) At the support of beam with no adjacent span, if a limitation of concrete cracking is imposed (see § 2.3.6) and if there is a continuity of the slab at the support (which is not accounted for in the design, see § 2.1 (5)), § 11.4.3 applies.
- (4) At the support of beam with an adjacent span, if a limitation of concrete cracking is imposed (see § 2.3.6), § 11.4.4 applies.

11.4.2. Minimum reinforcement for simply supported beams

- (1) The minimum ratio of longitudinal reinforcement at a simple support of the beam with slab continuity should fulfil the following criteria (according to EN 1994-1-1, 7.4.1 (4)):

$$\begin{aligned}\rho_s &\geq 0,4 \% \text{ for propped construction;} \\ \rho_s &\geq 0,2 \% \text{ for un-propped construction}\end{aligned}\tag{289}$$

where ρ_s is the ratio of longitudinal reinforcement – see § 2.3.3 (5)



Fully propped beams and point propped beams are considered as propped construction.

11.4.3. Minimum reinforcement for crack width limitation without direct loading

- (1) For a slab at the support of the beam with no adjacent span and with slab continuity, the crack width limitation is obtained by fulfilling the following condition (according to EN 1994-1-1, 7.4.2):

$$A_s \geq A_{s,min} = k_s k_c k f_{ct,eff} A_{ct} / \sigma_s\tag{290}$$

where: k is a factor which allows for the effect of non-uniform self-equilibrating stresses: $k = 0,8$;

k_s is a factor which allows for the effect of the reduction of the normal force of the concrete slab due to initial cracking and local slip of the shear connection: $k_s = 0,9$;

k_c is given by Formula (198) in § 7.7;

A_{ct} is the area of the tensile zone of concrete immediately prior to cracking, given by: $A_{ct} = b_{eff} t_c$;

t_c is the thickness of the active part of the concrete flange, see § 5.2;

b_{eff} : see § 5.1;

σ_s is the maximal stress permitted in the reinforcement immediately after cracking. It is obtained as the minimum value of f_{sk} the characteristic yield strength of the reinforcement (see § 2.10.6) and the value σ_{s1} obtained from clause (2).

- (2) σ_{s1} is obtained from the maximal thickness of concrete crack $w_{k,max}$ and from the diameter d_s of rebars, according to Table 60.

Stress in rebars σ_{s1} [MPa]	Diameter of rebars d_s [mm]		
	$w_{k,max} = 0,4$ mm	$w_{k,max} = 0,3$ mm	$w_{k,max} = 0,2$ mm
160	40	32	25
200	32	25	16
240	20	16	12
280	16	12	8
320	12	10	6
360	10	8	5
400	8	6	4
450	6	5	-

Table 60: Relation between stress in rebars and diameter of rebars

11.4.4. Control of cracking due to direct loading

- (1) For beams with a continuity at support, i.e. where negative bending moments appear at support, the stresses in the reinforcements of the slab will be assessed according to § 5.9, including the effect of tension stiffening, and considering quasi-permanent SLS combinations of loads (see § 2.15.2 (4)).
- (2) From these calculations, the maximal value σ_s is obtained.
- (3) The maximal diameter of rebars is calculated from σ_s according to Table 60. The maximal spacing of rebars is calculated from σ_s according to Table 61.

Stress in rebars σ_{s1} [MPa]	Spacing of longitudinal rebars e_s [mm]		
	$w_{k,max} = 0,4$ mm	$w_{k,max} = 0,3$ mm	$w_{k,max} = 0,2$ mm
160	300	300	200
200	300	250	150
240	250	200	100
280	200	150	50
320	150	100	-
360	100	50	-

Table 61: Relation between stress in rebars and spacing of rebars

- (4) The calculation report will display these two values for rebars.

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11.4.5. Minimum reinforcement for crack width limitation in partial concrete encasement

- (1) For beams with partial concrete encasement, the longitudinal reinforcement of the encasement (see § 2.9) should fulfil condition of Formula (290), according to EN 1994-1-1, 7.4.2 (6), where:

A_s is the area of the lower layer of reinforcement, given by: $A_s = A_{se,inf}$;

A_c is the area of the concrete of partial encasement, given by Formula (31);

$k_c = 0,6$;

$k = 0,8$;

f_{sk} is replaced by f_{sek} , the characteristic yield strength of the reinforcing steel used in partial encasement.

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Annex A Specific configuration of composite slabs with Cofraplus 220

- (1) This Annex provides details on the specific configuration of composite slabs using Cofraplus 220 profiled steel sheeting.
- (2) This Annex applies to usual steel or composite beams.
- (3) When a composite slab uses Cofraplus 220 profiled steel sheeting, the following specific details apply (**Figure 58**):
 - the ribs are perpendicular to the beam axis;
 - the steel sheeting is supported by wings that are either bolted or welded to the profile's web;
 - the steel sheeting upper side and the upper fibre of the profile are at the same level.
- (4) When composite slabs with Cofraplus 220 are used with usual composite beams, the presence of the ribs has no effect on the shear connection and the shear resistance of the connectors is calculated considering plain slabs.
- (5) The software does not cover the design of the support wings nor their link with the web.

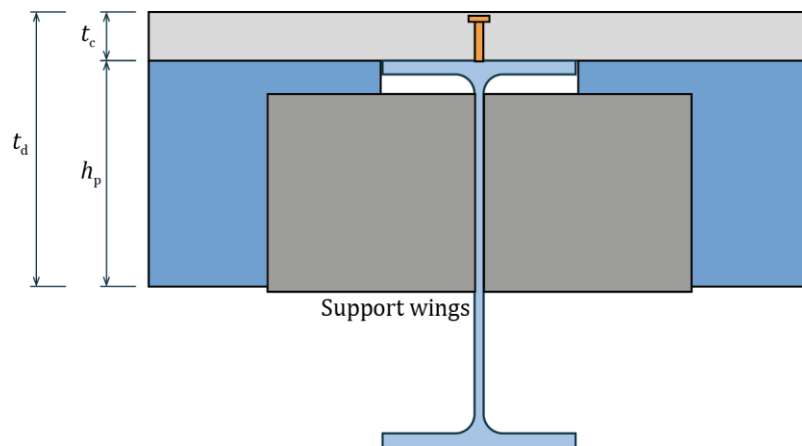


Figure 58 : Specific configuration of a composite slabs with Cofraplus 220 sheeting

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Annex B Procedure for the calculation of section bending properties

B.1 General

- (1) A general procedure will be developed for the calculation of the bending properties of all types of cross-sections. This procedure will give the values of:
 - the location of the elastic and plastic neutral axes;
 - the second moment of area of the transformed section for positive or negative bending moments;
 - the plastic resistance moment for positive or negative bending moments.
- (2) The procedure is based on a representation of the cross-section by section elements, with four types of elements, as described in this Annex and in **Figure 59**.

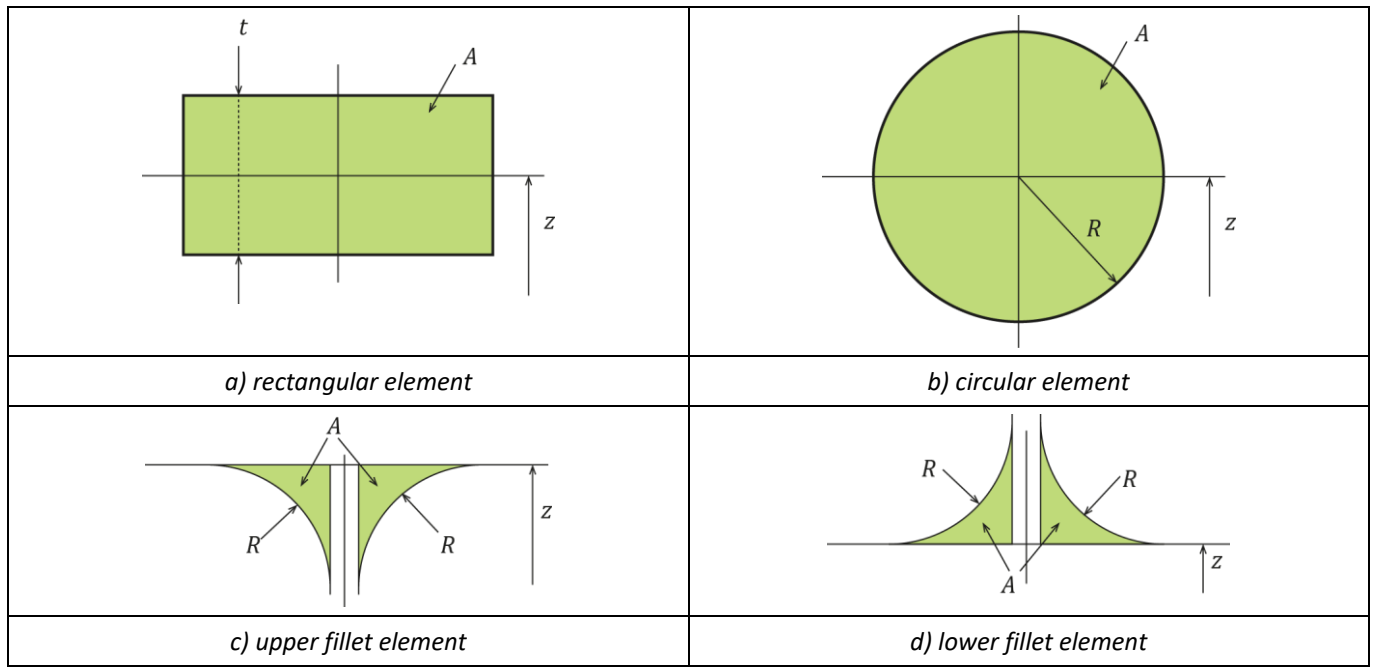


Figure 59: Type of elements

B.2 Section properties

Elastic properties

- (1) The location of the elastic neutral axis z_{el} is obtained by an iterative process, until the first moment of area of the section is equal to 0.
- (2) The first moment of area of the section about the yy axis at location z_{axe} is given by:

$$S_{axe} = \sum_i S_i(z_{axe}) \quad (291)$$

where: $S_i(z_{axe})$ is the first moment of area of the element no. i about this same axis.

- (3) Once the location of the elastic neutral axis has been obtained, the second moment of area of the transformed section about the y-y axis is given by:

$$I_{h,y} = \sum_i I_{y,h,i}(z_{el}) \quad (292)$$

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where: $I_{y,h,i}(z_{axe})$ is the second moment of area of the element no. i about the neutral axis.

- (4) The area of the transformed section, $A_{h,y}$, is obtained by the following equation:

$$A_{h,y} = \sum_i \frac{A_i}{n_i} \quad (293)$$

- (4) The elastic moment of the section is obtained as follows:

$$M_{el} = I_y \times \min_i k_{\sigma,i} \quad (294)$$

where: $k_{\sigma,i}$ is the constraint ratio of the element no i .

Plastic properties

- (1) The location of the plastic neutral axis z_{pl} is obtained by an iterative process, until the axial resultant of the plastic stress diagram is equal to N_{Ed} , where N_{Ed} is the internal axial force in the cross-section. In the absence of MN interaction, $N_{Ed} = 0$.
- (2) The axial resultant of the plastic stress diagram, considering a plastic neutral axis at the location z_{axe} , is given by:

$$N_{pl,axe} = \sum_i N_{pl,i}(z_{axe}) \quad (295)$$

where: $N_{pl,i}(z_{axe})$ is the contribution of the element no i .

- (3) Once the location of the plastic neutral axis is known, the associated plastic moment is calculated with the following formula:

$$M_{pl} = \sum_i M_{pl,i}(z_{pl}) \quad (296)$$

where: $M_{pl,i}(z_{axe})$ is the contribution of the element no i .

B.3 Formulation of elements

Principles

- (1) For each type of element, the procedure first assesses the proportion p or area located above the reference axis.
- (2) The properties of each part of the element located above and below the reference axis are then calculated, taking into account the indicators δ_{Ap} for the part above the axis and δ_{Am} for the part below the axis. These indicators are given by **Table 62**.

Sign of the bending moment	δ_{Ap}	δ_{Am}
$M_{Ed} > 0$	δ_{ci}	δ_{ti}
$M_{Ed} < 0$	δ_{ti}	δ_{ci}

Table 62: Indicators for upper and lower parts

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Rectangular elements

(1) A rectangular elements is described by the following parameters:

- A_i area;
- n_i number of units;
- z_i location of the element centroid;
- t_i thickness (in the direction perpendicular to the reference axis);
- $t_{i,eff}$ effective thickness of the element for the plastic properties (by default, $t_{i,eff} = t_i$)
- $n_{eq,i}$ modular ratio;
- f_{ki} nominal or characteristic yield strength;
- γ_i partial factor;
- k_{pli} coefficient applied to the yield strength for the plastic analysis;
- δ_{ti} tension indicator, = 1 if the element is active in tension, 0 otherwise;
- δ_{ci} compression indicator, = 1 if the element is active in compression, 0 otherwise;
- Type*: rectangular.

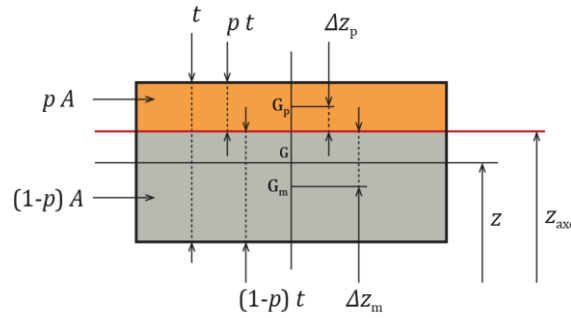


Figure 60: Parts of the element delimited by the reference axis

(2) Let us consider the reference axis with location z_{axe} . The proportion p of the element located above the axis is given by:

$$p = \frac{1}{2} + \frac{\Delta z}{t_i} \quad \text{but } 0 \leq p \leq 1$$

where: $\Delta z = z_i - z_{axe}$

(3) For the plastic analysis, it is necessary to deal with effective thickness, e.g. in order to deal with partial connection of slabs. The latter formulae is thus replaced by:

$$p = \frac{1}{2} + \frac{\Delta z}{t_{eff,i}} \quad \text{but } 0 \leq p \leq 1$$

where: $\Delta z = z_{eff,i} - z_{axe}$

$$z_{eff,i} = z_i + (t_i - t_{eff,i}) / 2$$

(4) The first moment of area of a rectangular element about the reference axis (location z_{axe}) is obtained by:

$$S_i(z_{axe}) = n_i \frac{A_i}{n_{eq,i}} \left[\Delta z (p \delta_{Ap} + (1-p) \delta_{Am}) + p(1-p) \frac{t_i}{2} (\delta_{Ap} - \delta_{Am}) \right] \quad (297)$$

where: Δz and p : see (3);

δ_{Ap} and δ_{Am} : see Table 62.

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- (5) The second moment of area of the rectangular element, $I_{h,y}$, about the reference axis is given by:

$$I_{h,y,i} = n_i \frac{A_i}{n_{eq,i}} \left[p \delta_{Ap} \left(\Delta z_p^2 + \frac{p^2 t_i^2}{12} \right) + (1-p) \delta_{Am} \left(\Delta z_m^2 + \frac{(1-p)^2 t_i^2}{12} \right) \right] \quad (298)$$

where: Δz and p : see **(3)**;

$$\Delta z_p = \Delta z + (1-p) t_i / 2 \quad \textbf{(Figure 60)}$$

$$\Delta z_m = \Delta z - p t_i / 2$$

δ_{Ap} and δ_{Am} : see **Table 62**.

- (6) Under the effect of a bending moment M_o applied around the elastic neutral axis, the stress ratio $k_{\sigma i}$ of the element is calculated for both distance v_j as follows:

$$v_j = \Delta z \pm t_i / 2 \quad (299)$$

if $Sign(M_o \times v_j) > 0$ and $\delta_t > 0$

or if $Sign(M_o \times v_j) < 0$ and $\delta_c > 0$ then :

$$k_{\sigma i} = n_{eq,i} \frac{f_{ki}}{v_j}$$

If design values are required, $k_{\sigma i}$ is also divided by γ_i

- (7) The contribution of a rectangular element to the stress axial resultant in the plastic stress diagram whose neutral axis location is denoted z_{axe} is given by:

$$N_{pl,i}(z_{axe}) = n_i A_i k_{pli} f_{ki} (p \delta_{Ap} - (1-p) \delta_{Am}) / \gamma_i \quad (300)$$

where: p : see **(3)**;

δ_{Ap} and δ_{Am} : see **Table 62**.

- (8) The contribution of a rectangular element to the stress bending resultant in the plastic stress diagram whose neutral axis location is denoted z_{axe} is given by:

$$M_{pl,i}(z_{axe}) = n_i \frac{A_i k_{pli} f_{ki}}{\gamma_i} \left[\Delta z (p \delta_{Ap} - (1-p) \delta_{Am}) + p (1-p) \frac{t_i}{2} (\delta_{Ap} + \delta_{Am}) \right] \quad (301)$$

where: Δz et p : see **(3)**;

δ_{Ap} et δ_{Am} : see **Table 62**.

- (9) For the assessment of the characteristic values of the plastic properties, (7) and (8) apply where γ_i is replaced by 1.

Circular elements

- (1) The same parameters as for a rectangular element are used, except the thickness that is not considered and with the additional following parameters:

r_i radius;

Type: circular

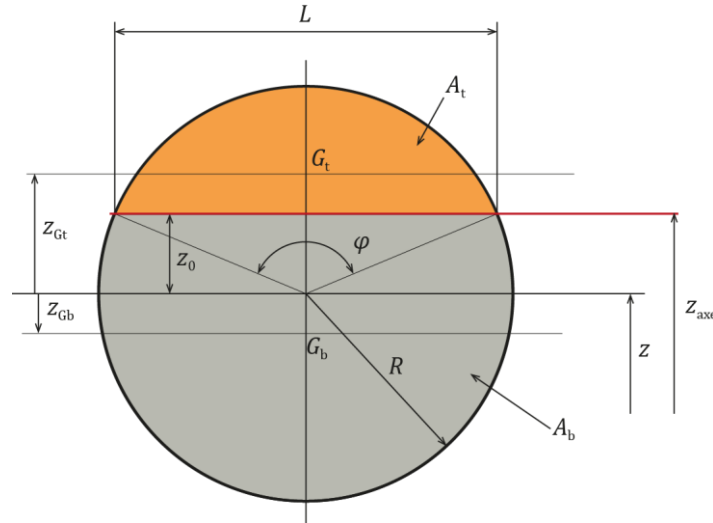


Figure 61: Parts of the element delimited by the reference axis

- (2) Let us consider the reference axis with location z_{axe} . The proportion p of the element located above the axis is given by:

$$p = \frac{1}{2} + \frac{\Delta z}{2 R_i} \quad \text{avec } 0 \leq p \leq 1$$

where: $\Delta z = z_i - z_{axe}$

- (3) The angle φ for the location of the reference axis within the element (**Figure 61**) is given by:

$$\varphi = 2 \arccos(1 - 2p)$$

- (4) The area of the parts located respectively above and below the reference axis are denoted A_t and A_b . They are calculated by:

$$A_t = \frac{R_i^2}{2} (\varphi - \sin \varphi)$$

$$A_b = \pi R_i^2 - A_t$$

- (5) The location of the centroids of areas A_t and A_b , respectively denoted z_{Gt} and z_{Gb} , is calculated about the centre of the circle as follows:

$$z_{Gt} = \frac{4 R_i (\sin(\varphi/2))^3}{3 (\varphi - \sin \varphi)}$$

$$z_{Gb} = -z_{Gs} A_t / A_b$$

- (6) I_{ypt} is the second moment of area of the upper area about its centroid:

$$I_{ypt} = \frac{R_i^4}{8} \left(\varphi - \frac{1}{2} \sin(2\varphi) \right) - A_t (z_{Gt})^2$$

I_{ypb} is the second moment of area of the lower area about its centroid:

$$I_{ypb} = \pi \frac{R_i^4}{4} - \frac{R_i^4}{8} \left(\varphi - \frac{1}{2} \sin(2\varphi) \right) - A_b (z_{Gb})^2$$

- (7) The first moment of area of a circular element about the reference axis (location z_{axe}) is obtained by:

$$S_i(z_{axe}) = \frac{n_i}{n_{eq,i}} [A_t (z_{Gt} + z_i - z_{axe}) \delta_{Ap} + A_b (z_{Gb} + z_i - z_{axe}) \delta_{Am}] \quad (302)$$

where: δ_{Ap} and δ_{Am} : see **Table 62**;

A_t , A_b , z_{Gt} and z_{Gb} : see **(4)** and **(5)**.

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- (8) The second moment of area of a circular element about the reference axis (location z_{axe}) is obtained by:

$$I_{h,y,i} = \frac{n_i}{n_{eq,i}} \left[(I_{ypt} + A_t (\Delta z + z_{Gt})^2) \delta_{Ap} + (I_{ypb} + A_b (\Delta z + z_{Gb})^2) \delta_{Am} \right] \quad (303)$$

where: δ_{Ap} et δ_{Am} : see **Table 62**;

Δz : see **(2)**;

A_t , A_b , z_{Gt} and z_{Gb} : see **(4)** and **(5)**;

I_{ypt} et I_{ypb} : see **(6)**.

- (9) Under the effect of a bending moment M_0 applied around the elastic neutral axis, the stress ratio $k_{\sigma i}$ of the element is calculated for both distance v_j as follows:

$$v_j = \Delta z \pm r_i / 2 \quad (304)$$

if $Sign(M_0 \times v_j) > 0$ and $\delta_t > 0$

or if $Sign(M_0 \times v_j) < 0$ and $\delta_c > 0$ then :

$$k_{\sigma i} = n_{eq,i} \frac{f_{ki}}{v_j}$$

If design values are required, $k_{\sigma i}$ is also divided by γ_i

- (10) The contribution of a circular element to the stress axial resultant in the plastic stress diagram whose neutral axis location is denoted z_{axe} is given by:

$$N_{pl,i}(z_{axe}) = n_i \frac{k_{pli} f_{ki}}{\gamma_i} (A_t \delta_{Ap} - A_b \delta_{Am}) \quad (305)$$

where: A_t et A_b : cf. **(4)**;

δ_{Ap} et δ_{Am} : see **Table 62**

- (11) The contribution of a circular element to the stress bending resultant in the plastic stress diagram whose neutral axis location is denoted z_{axe} is given by:

$$M_{pl,i}(z_{axe}) = n_i \frac{k_{pli} f_{ki}}{\gamma_i} \left[\delta_{Ap} A_t (z_{Gt} + z_i - z_{axe}) - \delta_{Am} A_b (z_{Gb} + z_i - z_{axe}) \right] \quad (306)$$

with: A_t , A_b , z_{Gt} et z_{Gb} : see **(4)** and **(5)**;

δ_{Ap} et δ_{Am} : see **Table 62**.

- (12) For the assessment of the characteristic values of the plastic properties, **(9)** and **(10)** apply where γ_i is replaced by 1.

Upper root elements

- (1) The parameters for an upper fillet element are the same than for a circular element. The location z_i of the element corresponds to the interface between the fillet and the upper flange.

- (2) The area of the element is given by:

$$A_i = (4 - \pi) r_i^2 / 2$$

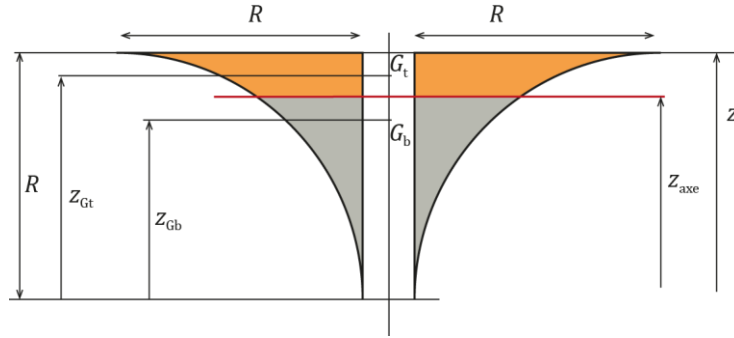


Figure 62: Parts of the upper root element delimited by the reference axis

- (3) Let us consider the reference axis with location z_{axe} . The proportion p of the element located above the axis is given by:

$$p = \frac{\Delta z}{R_i} \quad \text{with } 0 \leq p \leq 1$$

where: $\Delta z = z_i - z_{axe}$

- (4) The angle φ associated to the limit between the parts above and below the reference axis is given by:

$$\varphi = 2 \arccos(1 - p)$$

- (5) The areas of the parts located respectively above and below the reference axis are denoted A_t and A_b . They are calculated by:

$$A_t = \frac{R_i^2}{2} (4p - \varphi + \sin \varphi)$$

$$A_b = A_i - A_t = \frac{R_i^2}{2} (4(1 - p) - \pi + \varphi - \sin \varphi)$$

- (6) The location of the centroids of areas A_t and A_b , respectively denoted z_{Gt} and z_{Gb} , is calculated about the centre of the circle, i.e. about to lower fibres of the element, as follows:

$$z_{Gt} = 2 R_i \frac{p(2 - p) - \frac{2}{3} \sin^3 \left(\frac{\varphi}{2} \right)}{4p - \varphi + \sin \varphi}$$

$$z_{Gb} = 2 R_i \frac{(1 - p)^2 - \frac{2}{3} \left(1 - \sin^3 \left(\frac{\varphi}{2} \right) \right)}{4(1 - p) - \pi + \varphi - \sin \varphi}$$

- (7) When $p = 0$, $A_t = 0$ and $z_{Gt} = R_i$.

When $p = 1$, $A_b = 0$ and $z_{Gb} = 0$.

- (8) I_{ypt} is the second moment of area of the upper area about its centroid:

$$I_{ypt} = \frac{R_i^4}{8} \left(\frac{16}{3} (1 - p)^3 - \pi + \varphi - \frac{1}{2} \sin(2\varphi) \right) - A_t (z_{Gt})^2$$

I_{ypb} is the second moment of area of the lower area about its centroid:

$$I_{ypb} = \frac{R_i^4}{8} \left(\frac{16}{3} p (p^2 - 3p + 3) - \varphi + \frac{1}{2} \sin(2\varphi) \right) - A_b (z_{Gb})^2$$

- (9) The first moment of area of a circular element about the reference axis (location z_{axe}) is obtained by:

$$S_i(z_{axe}) = \frac{n_i}{n_{eq,i}} [A_t (\Delta z + z_{Gt} - R_i) \delta_{Ap} + A_b (\Delta z + z_{Gb} - R_i) \delta_{Am}] \quad (307)$$

with: δ_{Ap} and δ_{Am} : see **Table 62**;

Δz , A_t , A_b , z_{Gt} and z_{Gb} : see **(3)**, **(5)** and **(6)**.

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- (10) The second moment of area of a circular element about the reference axis (location z_{axe}) is obtained by:

$$I_{h,y,i} = \frac{n_i}{n_{eq,i}} \left[(I_{ypt} + A_t (\Delta z + z_{Gt} - R_i)^2) \delta_{Ap} + (I_{ypb} + A_b (\Delta z + z_{Gb} - R_i)^2) \delta_{Am} \right] \quad (308)$$

where: δ_{Ap} and δ_{Am} : see **Table 62**;

Δz : see

A_t, A_b : see **(5)**;

z_{Gt} et z_{Gb} : see **(6)**;

I_{ypt} et I_{ypb} : see **(8)**.

- (11) The contribution of a upper root element to the stress axial resultant in the plastic stress diagram whose neutral axis location is denoted z_{axe} is given by:

$$N_{pl,i}(z_{axe}) = n_i \frac{k_{pli} f_{ki}}{\gamma_i} (A_t \delta_{Ap} - A_b \delta_{Am}) \quad (309)$$

with: A_t and A_b : cf. **(5)** ;

δ_{Ap} and δ_{Am} : see **Table 62**.

- (12) The contribution of an upper root element to the stress bending resultant in the plastic stress diagram whose neutral axis location is denoted z_{axe} is given by:

$$M_{pl,i}(z_{axe}) = n_i \frac{k_{pli} f_{ki}}{\gamma_i} \left[\delta_{Ap} A_t (\Delta z + z_{Gt} - R_i) - \delta_{Am} A_b (\Delta z + z_{Gb} - R_i) \right] \quad (310)$$

where: A_t, A_b, z_{Gt} et z_{Gb} : cf. **(5)** and **(6)**;

δ_{Ap} et δ_{Am} : see **Table 62**.

- (13) For the assessment of the characteristic values of the plastic properties, **(11)** and **(12)** apply where γ_i is replaced by 1.

Lower root elements

- (1) The parameters for an lower fillet element are the same than for a circular element. The location z_i of the element corresponds to the interface between the fillet and the lower flange.
- (2) The area of the element is given by:

$$A_i = (4 - \pi) r_i^2 / 2$$

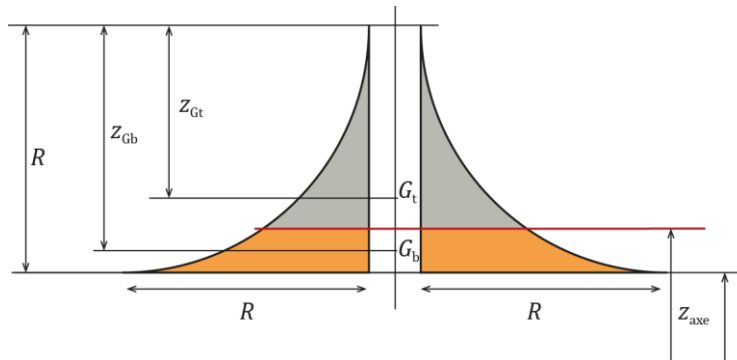


Figure 63 :Découpage d'un élément congés supérieurs par un axe

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- (3) Let us consider the reference axis with location z_{axe} . The proportion p of the element located above the axis is given by:

$$p = 1 - \frac{\Delta z}{R_i} \quad \text{with } 0 \leq p \leq 1$$

where: $\Delta z = z_i - z_{\text{axe}}$

- (4) The angle φ associated to the limit between the parts above and below the reference axis is given by:

$$\varphi = 2 \arccos(p)$$

- (5) The areas of the parts located respectively above and below the reference axis are denoted A_t and A_b . They are calculated by:

$$A_t = \frac{R_i^2}{2} (4p - \pi + \varphi - \sin \varphi)$$

$$A_b = A_i - A_t = \frac{R_i^2}{2} (4(1 - p) - \varphi + \sin \varphi)$$

- (6) The location of the centroids of areas A_t and A_b , respectively denoted z_{Gt} and z_{Gb} , is calculated about the centre of the circle, i.e. about to lower fibres of the element, as follows:

$$z_{Gt} = \frac{2}{3} R_i \frac{3p^2 - 2(1 - \sin^3(\varphi/2))}{4p - \pi + \varphi - \sin \varphi}$$

$$z_{Gb} = 2 R_i \frac{1 - p^2 - \frac{2}{3} \sin^3(\varphi/2)}{4(1 - p) - \varphi + \sin \varphi}$$

- (7) When $p = 0$, $A_t = 0$ and $z_{Gt} = R_i$.

When $p = 1$, $A_b = 0$ and $z_{Gb} = 0$.

- (8) I_{ypt} is the second moment of area of the upper area about its centroid:

$$I_{ypt} = \frac{R_i^4}{8} \left(\frac{16}{3} p^3 - \pi + \varphi - \frac{1}{2} \sin(2\varphi) \right) - A_t (z_{Gt})^2$$

I_{ypb} is the second moment of area of the lower area about its centroid:

$$I_{ypb} = \frac{R_i^4}{8} \left(\frac{16}{3} (1 - p) (p^2 + p + 1) - \varphi + \frac{1}{2} \sin(2\varphi) \right) - A_b (z_{Gb})^2$$

- (9) The first moment of area of a circular element about the reference axis (location z_{axe}) is obtained by:

$$S_i(z_{\text{axe}}) = \frac{n_i}{n_{\text{eq},i}} [A_t (\Delta z - z_{Gt} + R_i) \delta_{Ap} + A_b (\Delta z - z_{Gb} + R_i) \delta_{Am}] \quad (311)$$

with: δ_{Ap} and δ_{Am} : see **Table 62**;

Δz , A_t , A_b , z_{Gt} and z_{Gb} : see **(3)**, **(5)** and **(6)**.

- (10) The second moment of area of a circular element about the reference axis (location z_{axe}) is obtained by:

$$I_{h,y,i} = \frac{n_i}{n_{\text{eq},i}} [(I_{ypt} + A_t (\Delta z - z_{Gt} + R_i)^2) \delta_{Ap} + (I_{ypb} + A_b (\Delta z - z_{Gb} + R_i)^2) \delta_{Am}] \quad (312)$$

where: δ_{Ap} and δ_{Am} : see **Table 62**;

Δz , A_t , A_b , z_{Gt} and z_{Gb} : see **(3)**, **(5)** and **(6)**;

I_{ypt} and I_{ypb} : see **(8)**.

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- (11) The contribution of a lower root element to the stress axial resultant in the plastic stress diagram whose neutral axis location is denoted z_{axe} is given by:

$$N_{pl,i}(z_{axe}) = n_i \frac{k_{pli} f_{ki}}{\gamma_i} (A_t \delta_{Ap} - A_b \delta_{Am}) \quad (313)$$

where: A_t and A_b : see **(5)**;

δ_{Ap} and δ_{Am} : see **Table 62**.

- (12) The contribution of a lower root element to the stress bending resultant in the plastic stress diagram whose neutral axis location is denoted z_{axe} is given by:

$$M_{pl,i}(z_{axe}) = n_i \frac{k_{pli} f_{ki}}{\gamma_i} [\delta_{Ap} A_t (\Delta z - z_{Gt} + R_i) - \delta_{Am} A_b (\Delta z - z_{Gb} + R_i)] \quad (314)$$

where: A_t , A_b , z_{Gt} and z_{Gb} : see **(5)** and **(6)** ;

δ_{Ap} and δ_{Am} : see **Table 62**.

- (13) For the assessment of the characteristic values of the plastic properties, **(11)** and **(12)** apply where γ_i is replaced by 1.

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Annex C Procedure for the Finite Element calculations

C.1 Principles

- (1) All global analyses of the beam are carried out with Finite Elements numerical analysis, assuming an elastic behaviour.
- (2) The present Annex gives the general method for applying the Finite Element analyses in the software.

C.2 Numerical modeling

- (1) Nodes are imposed at each following section of the beam:
 - beam ends;
 - beam supports;
 - at point propping;
 - at point with point lateral restraints;
 - at support with beam continuity, point at 0,15 times the span length (which is the simplified threshold between positive and negative bending).

- (2) Nodes are added between the imposed nodes, so as to fulfil the following requirements:
 - the spacing between two adjacent nodes (i.e. the element length) should not be greater than $d_{el,max}$;
 - the minimum number of nodes per cantilever span should not be lower than $n_{can,min}$;
 - the minimum number of nodes per intermediate span should not be lower than $n_{int,min}$;

where $d_{el,max}$, $n_{can,min}$, $n_{int,min}$ are calculation options of the software that may be changed by the User. Their default values are:

$$d_{el,max} = 0,5 \text{ m}$$

$$n_{can,min} = 5$$

$$n_{int,min} = 10$$

- (3) The number of nodes is denoted N_{nodes} .
All nodes are ranked by increasing order of their position x , measured from the left end of the beam. Node no 1 is on the left end and node no N_{nodes} is on the right end of the beam.
- (4) One beam element is placed between two adjacent nodes. The number of elements is denoted N_{elts} , where:

$$N_{elts} = N_{nodes} - 1$$

Between the nodes no i and no $i + 1$ is located the element no i .

- (5) Elements are assumed to be continuous all along the beam. There is no relaxation at nodes.
- (6) Elements are all aligned horizontally on the same line representing the beam axis. For composite beams, the variation of the vertical location of the centroid is not taken into account.
- (7) The main directions are defined as follows:
 - x is the horizontal direction of the beam axis, from the left to the right end;
 - z is the vertical upwards direction;
 - y is the horizontal direction that forms an orthonormal basis.
- (8) There are two degrees of freedom for each node: the vertical displacement uz and the rotation ry .
- (9) At each support of the beam, the degree of freedom uz is fixed.

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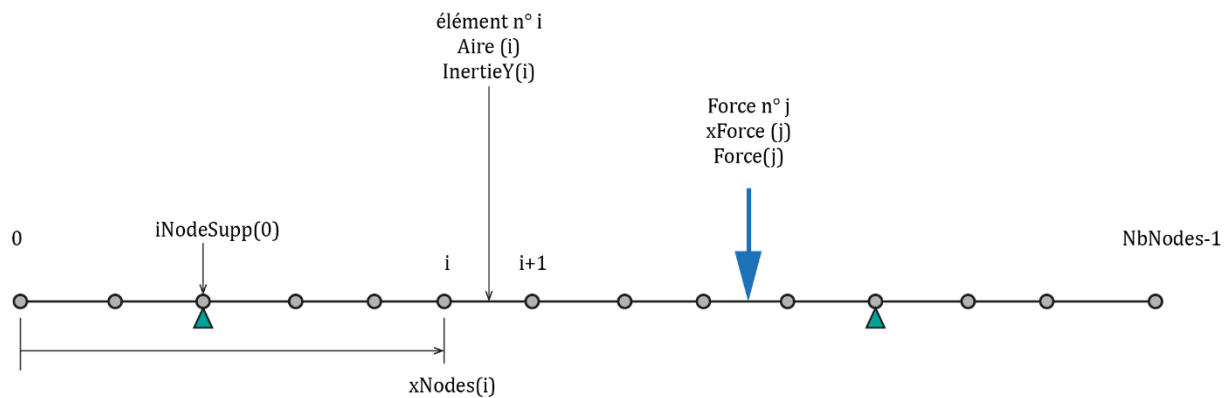


Figure 64 : Nodes and elements along the beam

- (10) For each load case, the Finite Element numerical analysis will provide the following results, for each node:
 - the vertical displacement u_z ;
 - the rotation u_y ;
 - the shear force, at LHS and RHS;
 - the bending moment, at LHS and RHS;
- (11) Additionally, the analysis will provide, for each support node, the vertical reaction R_Z .

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Annex D Critical moments for LT Buckling

D.1 Scope of application

- (1) This Annex describes the calculation of the critical moment for lateral torsional buckling of segments of beams which are located between two cross-sections, where both flanges are laterally restrained.

D.2 Critical moments

- (1) The critical moment to lateral torsional buckling of the beam segment will be assessed through the LTBeamN module.
- (2) The LTBeamN module is based on a separate DLL.
- (3) Each segment of beam will be represented in LTBeamN. At each end of the segment, the internal moment associated to the relevant combination of loads will be applied, as well as all loads applied on the segment, with the relevant combination factors.

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Annex E Critical moment for LT Buckling taking account of the steel deck

E.1 Scope of application

- (1) This Annex applies to beams with usual composite cross-section (with or without partial concrete encasement), with a composite slab where the ribs are perpendicular to the beam.
- (2) The thickness of the profiled steel sheeting should be greater than 0,75 mm: $t_p \geq 0,75$ mm.

E.2 Calculation of the critical moment

- (1) The critical moment for LT Buckling, M_{cr} , is calculated by using the LTBeamN module, considering the shear and the flexural stiffnesses (respectively S_{act} –see § E.3 – and k_θ – see § E.4) provided by the steel profiled sheeting.

E.3 Shear rigidity of the steel profiled sheeting

General

- (1) The distributed shear stiffness of the steel profiled sheeting is obtained by:

$$S_{act} = \frac{L}{c} \quad (315)$$

where: L is the length of the span;

c is the total flexibility of the steel deck.

- (2) c is obtained by:

$$c = c_{1,1} + c_{1,2} + c_{2,1} + c_{2,2} \quad (316)$$

where: $c_{1,1}$ is the contribution of the steel profiled sheets distortion;

$c_{1,2}$ is the contribution of shear strain in the profiled deck;

$c_{2,1}$ is the contribution of the sheet-to-beam fasteners deformation;

$c_{2,2}$ is the contribution of the seam fasteners deformation.

Flexibilities due to the sheet deformation

- (1) The components $c_{1,j}$, associated to the sheet deformation, either due to the distortion or to the shear strain, may be obtained with the following formulas:

$$c_{1,1} = \frac{L \alpha_5 K}{E_a d^2} \left(\frac{e_p}{t_p} \right)^{2,5} \quad (317)$$

$$c_{1,2} = \frac{2 L (1 + \nu_a) (1 + 2h_p/e_p)}{E_a t_p d} \quad (318)$$

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where: L is the span length of the beam;

E_a and ν_a are respectively the Young's modulus and the Poisson's ratio for steel –see § 2.10.4;

h_p , t_p and e_p are respectively the height, the thickness and the rib spacing of the steel sheets (cf. § 2.5.1) ;

K is a sheeting constant, which depends on the geometry and the connection of the deck. K is equal to K_1 when a fastener is used in every rib and to K_2 when a fastener is used every two ribs.

K_1 and K_2 predefined for profiled steel sheeting from the database? To be discussed with ArcelorMittal.

Table 66 to **Table 69** give the values of K_1 and K_2 coefficients for any geometry of steel profiled sheeting with open rib, according to ECCS Recommendations, respectively for one fastener at every rib and at every two ribs.

Values K_1 and K_2 from ECCS Recommendations are valid only for open ribs, with $\theta_p \geq 0^\circ$. For profiled steel sheeting with re-entrant ribs, where $\theta_p < 0^\circ$, ECCS recommendations provides no values. It could be decided to limit the method to sheeting with open ribs only. Or potentially to sheetings with re-entrant ribs, where K_1 and K_2 are already known, by numerical simulations. See for example Table 4 in [16].

α_5 is a factor which accounts for the influence of the overlapping and the continuity of profiled sheets. It is obtained according to **Table 63**.

n_t Number of sheets in the transverse direction	Continuity of sheets at beam supports	
	With overlapping	Without overlapping
1 or 2	1,0 / m	1,0 / m
3	0,9 / m	1,0 / m
4	0,8 / m	1,0 / m
≥ 5	0,7 / m	1,0 / m
Coefficients m and n_t : see § 3.1		

Table 63: α_5 coefficient

Flexibilities due to the fasteners deformation

- (1) The components $c_{2,j}$, associated to the deformations of the beam-to-deck fasteners ($c_{2,1}$) or of the seam fasteners ($c_{2,2}$), may be obtained with the following formulas:

$$c_{2,1} = 2 \frac{L s_p e_L}{d^2} \quad (319)$$

$$c_{2,2} = \frac{s_s s_p (n_L - 1)}{n_c s_p + \beta_1 s_s} \quad (320)$$

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where: s_p is the slip of the beam-to-deck fasteners (see Table 12);

s_s is the slip of the seam fasteners (see Table 13);

e_L is the longitudinal spacing between beam-to-deck fasteners (see Figure 26);

n_L is the number of sheets in the direction of the beams;

n_c is the number of seam fasteners along the deck span, excluding those which are fixed to the secondary beam;

β_1 is a coefficient whose value is given by Table 64, as a function of the number of fasteners on the width of a profiled sheet.

n_f	β_1
2	1,0
3	1,0
4	1,04
5	1,13
6	1,22
7	1,33
8	1,45
9	1,56
10	1,68

Table 64: β_1 coefficient

Slip of fasteners

- (1) According to Table 5.1 of the ECCS recommendations [14], the slip to be considered for the beam-to-sheeting fasteners is given in Table 12 (see § 2.11.3), depending on the type of fastener. For seam fasteners, the values to be taken into account are given in Table 13 (idem).

E.4 Stiffness of the torsional restraint

- (1) The stiffness of the torsional restraint, k_θ , opposing the torsional rotation $\Delta\theta_x$ of the beam, is calculated as follows:

$$k_\theta = \frac{1}{\frac{1}{k_{\theta,A}} + \frac{1}{k_{\theta,C}}} \quad (321)$$

where: $k_{\theta,A}$ is the rotational stiffness due to the beam-to-deck connection;

$k_{\theta,C}$ is the rotational stiffness corresponding to the flexural stiffness of the deck.

- (2) $k_{\theta,C}$ stiffness will be obtained with formula (322):

$$k_{\theta,C} = \frac{k E I_{\text{eff}}}{d} \quad (322)$$

where: I_{eff} is the effective second moment of area per unit width of the deck;

d is the span of the deck, i.e. the spacing between beams;

k is a factor which takes into account the sheeting support conditions:

$k = 6$ for an intermediate beam;

$k = 3$ for an edge beam.

(3) $k_{\theta,A}$ stiffness may be obtained with formula (323):

$$k_{\theta,A} = k_{100} k_{ba} k_t k_{bR} k_g k_{bb} \quad (323)$$

where: k_{100} is the torsional stiffness for a flange width equal to 100 mm, which is given in Table 65.

Geometry of the deck	Fastener spacing	k_{100} (kNm/m)	b_{bmax} (mm)
$b_r \leq b_b$	$e_L = e_p^{(1)}$	5,2	40
	$e_L = 2 e_p^{(2)}$	3,1	40
$b_r > b_b$	$e_L = e_p^{(1)}$	3,1	120
	$e_L = 2 e_p^{(2)}$	2,0	120
(1) : One fastener in every rib			
(2) : One fastener in every two ribs			

Table 65: Values of k_{100} and b_{bmax}

k_{ba} is a factor which accounts for the actual flange width b_{fs} , as given by the following formulas:

$$k_{ba} = \left(\frac{b_{fs}}{100} \right)^2 \quad \text{when } 100 \text{ mm} \leq b_{fs} < 125 \text{ mm}$$

$$k_{ba} = 1,25 \left(\frac{b_{fs}}{100} \right) \quad \text{when } 125 \text{ mm} \leq b_{fs} \leq 200 \text{ mm}$$

Formula (323) has been calibrated for flange widths up to 200 mm. For flange widths above 200 mm, it is possible, from the safety point of view, to perform the calculations considering a width of 200 mm.

k_t is a coefficient taking into account the influence of the thickness of the deck, when this is different from 0,75 mm. k_t may be obtained by :

$$k_t = \left(\frac{t_{nom}}{0,75} \right)^{1,1} \quad \text{when } t_{nom} \geq 0,75 \text{ mm and } b_r \leq b_b$$

$$k_t = \left(\frac{t_{nom}}{0,75} \right)^{1,5} \quad \text{when } t_{nom} \geq 0,75 \text{ mm and } b_r > b_b$$

$$k_t = \left(\frac{t_{nom}}{0,75} \right)^{1,5} \quad \text{when } t_{nom} < 0,75 \text{ mm}$$

t_{nom} is the nominal thickness of the deck, in mm ;

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k_{bR} is a coefficient associated to the spacing e_p between the ribs (in mm), obtained as follows:

$$k_{bR} = 1 \quad \text{when } e_p \leq 185 \text{ mm}$$

$$k_{bR} = \frac{185}{e_p} \quad \text{when } e_p > 185 \text{ mm}$$

k_g is a coefficient to take into account the stabilising effect of gravity loads applied to the deck and transmitted to the beam. k_g may be obtained as follows:

$$k_g = 1 + 0,08 (q_g - 1) \quad \text{when } t_{nom} = 0,75 \text{ mm and } b_r \leq b_b$$

$$k_g = 1 + 0,16 (q_g - 1) \quad \text{when } t_{nom} = 0,75 \text{ mm and } b_r > b_b$$

$$k_g = 1 + 0,095 (q_g - 1) \quad \text{when } t_{nom} \geq 1 \text{ mm}$$

q_g is the distributed load transmitted by the sheeting to the beam, expressed in kN/m, not exceeding 12 kN/m in the calculations. For composite slabs in the construction phase, this maximum value can be considered conservatively in the calculations with regards to the self-weight and construction loads;

For nominal thickness values between 0,75 and 1 mm, a linear interpolation can be used to calculate k_g ;

k_{bb} is a coefficient taking into account the width of the rib bottom, obtained by the following formula:

$$k_{bb} = \min\{1; \sqrt{b_{bmax} / b_b}\}$$

b_{bmax} is defined in **Table 65**.

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<i>K₁ coefficient for fasteners in every trough (acc. to [14])</i>										
θ_p	b_r/e_p h_p/e_p	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9
0°	0,1	0,013	0,030	0,041	0,041	0,046	0,050	0,066	0,103	0,193
	0,2	0,042	0,096	0,131	0,142	0,142	0,153	0,199	0,311	0,602
	0,3	0,086	0,194	0,264	0,285	0,283	0,302	0,388	0,601	1,188
	0,4	0,144	0,323	0,438	0,473	0,468	0,494	0,629	0,972	1,935
	0,5	0,216	0,438	0,654	0,703	0,695	0,729	0,922	1,420	2,837
	0,6	0,302	0,674	0,911	0,980	0,965	1,008	1,266	1,938	3,892
	0,7	0,402	0,895	1,208	1,300	1,277	1,329	1,661	2,536	5,098
	0,8	0,516	1,146	1,546	1,662	1,631	1,692	2,107	3,208	6,453
5°	0,1	0,014	0,031	0,041	0,044	0,044	0,049	0,066	0,107	0,205
	0,2	0,050	0,099	0,128	0,134	0,132	0,146	0,198	0,336	0,652
	0,3	0,107	0,202	0,253	0,260	0,254	0,280	0,386	0,681	1,548
	0,4	0,188	0,338	0,413	0,417	0,404	0,448	0,629	1,158	2,639
	0,5	0,295	0,507	0,604	0,601	0,578	0,648	0,934	1,783	
	0,6	0,429	0,706	0,823	0,826	0,772	0,877	1,306	2,586	
	0,7	0,591	0,935	1,066	1,028	0,983	1,385	1,756	3,605	
	0,8	0,780	1,191	1,328	1,264	1,208	1,423	2,990	4,838	
10°	0,1	0,016	0,031	0,040	0,042	0,042	0,048	0,065	0,111	0,221
	0,2	0,056	0,101	0,123	0,125	0,123	0,139	0,200	0,366	0,873
	0,3	0,125	0,204	0,238	0,233	0,260	0,264	0,402	0,786	
	0,4	0,222	0,338	0,375	0,356	0,345	0,418	0,689	1,445	
	0,5	0,349	0,494	0,526	0,486	0,473	0,605	1,082	2,428	
	0,6	0,502	0,668	0,682	0,615	0,608	0,837	1,607		
	0,7	0,677	0,851	0,834	0,736	0,752	1,128	2,308		
	0,8	0,869	1,035	0,975	0,844	0,907	1,494	3,200		
15°	0,1	0,017	0,031	0,040	0,041	0,041	0,047	0,066	0,115	0,241
	0,2	0,062	0,102	0,118	0,115	0,113	0,134	0,209	0,403	
	0,3	0,139	0,202	0,218	0,204	0,200	0,254	0,440	0,945	
	0,4	0,244	0,321	0,325	0,293	0,294	0,414	0,796		
	0,5	0,370	0,448	0,426	0,371	0,396	0,636	1,329		
	0,6	0,508	0,568	0,508	0,434	0,513	0,941			
	0,7	0,646	0,668	0,561	0,483	0,664	1,349			
	0,8	0,768	0,735	0,578	0,527	0,861				
20°	0,1	0,018	0,032	0,039	0,039	0,039	0,046	0,066	0,111	0,276
	0,2	0,068	0,101	0,111	0,106	0,104	0,131	0,221	0,452	
	0,3	0,148	0,193	0,194	0,174	0,177	0,255	0,492		
	0,4	0,249	0,289	0,267	0,230	0,259	0,444	0,931		
	0,5	0,356	0,372	0,315	0,270	0,364	0,725			
	0,6	0,448	0,420	0,326	0,303	0,512				
	0,7	0,509	0,423	0,301	0,346					
	0,8	0,521	0,372	0,259	0,413					

Table 66: Values of K_1 for θ_p between 0° and 20°

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<i>K₁ coefficient for fasteners in every trough (acc. to [14])</i>										
θ_p	$\frac{b_r}{e_p}$ $\frac{h_p}{e_p}$	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9
25°	0,1	0,019	0,032	0,038	0,038	0,038	0,045	0,068	0,126	0,313
	0,2	0,072	0,099	0,103	0,095	0,095	0,129	0,236	0,513	
	0,3	0,151	0,178	0,166	0,144	0,160	0,268	0,557		
	0,4	0,238	0,244	0,204	0,176	0,247	0,494			
	0,5	0,306	0,272	0,203	0,204	0,376				
	0,6	0,333	0,248	0,172	0,241					
	0,7	0,300	0,174	0,142						
	0,8	0,204	0,081							
30°	0,1	0,020	0,032	0,037	0,036	0,036	0,044	0,070	0,133	
	0,2	0,075	0,095	0,094	0,084	0,087	0,132	0,256		
	0,3	0,148	0,157	0,135	0,116	0,152	0,291			
	0,4	0,208	0,186	0,139	0,139	0,253				
	0,5	0,226	0,161	0,112	0,176					
	0,6	0,180	0,089	0,093						
	0,7	0,077								
	0,8									
35 °	0,1	0,021	0,032	0,036	0,034	0,034	0,043	0,072	0,142	
	0,2	0,076	0,089	0,083	0,072	0,082	0,137	0,281		
	0,3	0,137	0,130	0,102	0,093	0,151				
	0,4	0,162	0,119	0,082	0,120					
	0,5	0,123	0,059							
	0,6	0,032								
	0,7									
	0,8									
40°	0,1	0,023	0,032	0,034	0,032	0,032	0,043	0,075		
	0,2	0,075	0,081	0,070	0,060	0,077	0,146			
	0,3	0,116	0,096	0,068	0,078					
	0,4	0,100	0,053	0,048						
	0,5	0,024								
	0,6									
	0,7									
	0,8									
45°	0,1	0,024	0,031	0,032	0,029	0,030	0,043	0,079		
	0,2	0,071	0,069	0,056	0,050	0,073				
	0,3	0,086	0,057	0,041						
	0,4	0,032								
	0,5									
	0,6									
	0,7									
	0,8									

Table 67: Values of K_1 for θ_p between 25° and 45°

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<i>K₂ coefficient for fasteners in every two troughs (acc. to [14])</i>										
θ_p	b_r/e_p h_p/e_p	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9
0°	0,1	0,014	0,025	0,036	0,046	0,054	0,061	0,070	0,108	0,211
	0,2	0,031	0,065	0,099	0,129	0,151	0,169	0,206	0,318	0,649
	0,3	0,054	0,123	0,192	0,252	0,294	0,328	0,402	0,608	1,269
	0,4	0,084	0,202	0,316	0,414	0,482	0,535	0,653	0,968	2,056
	0,5	0,123	0,299	0,468	0,614	0,712	0,790	0,958	1,410	3,006
	0,6	0,169	0,415	0,649	0,846	0,982	1,090	1,318	1,928	4,113
	0,7	0,222	0,549	0,855	1,108	1,286	1,433	1,730	2,525	5,383
	0,8	0,284	0,699	1,086	1,398	1,398	1,818	2,196	3,198	6,811
5°	0,1	0,089	0,138	0,184	0,288	0,269	0,311	0,359	0,432	0,590
	0,2	0,300	0,433	0,564	0,690	0,810	0,934	1,091	1,358	2,046
	0,3	0,627	0,872	1,113	1,345	1,569	1,806	2,215	2,710	4,441
	0,4	1,076	1,453	1,826	2,187	2,535	2,910	3,446	4,498	8,057
	0,5	1,644	2,171	2,694	3,205	3,703	4,244	5,058	6,671	12,940
	0,6	2,280	2,961	3,639	4,313	4,999	5,797	6,971	9,571	
	0,7	2,961	3,803	4,620	5,443	6,347	7,479	9,206	13,010	
	0,8	3,802	4,838	5,788	6,612	7,701	9,257	11,760	17,200	
10°	0,1	0,091	0,140	0,186	0,229	0,270	0,312	0,362	0,440	0,627
	0,2	0,312	0,446	0,575	0,699	0,817	0,943	1,112	1,425	2,472
	0,3	0,665	0,907	1,144	1,370	1,589	1,835	2,204	2,979	
	0,4	1,156	1,529	1,891	2,239	2,578	2,984	3,655	5,251	
	0,5	1,793	2,313	2,819	3,305	3,782	4,397	5,519	7,872	
	0,6	2,533	3,206	3,858	4,509	5,192	6,096	7,875		
	0,7	3,334	4,148	4,949	5,780	6,737	8,112	10,820		
	0,8	4,236	5,170	6,051	7,066	8,404	10,470	12,590		
15°	0,1	0,093	0,142	0,198	0,231	0,271	0,313	0,364	0,448	0,682
	0,2	0,325	0,458	0,586	0,707	0,824	0,953	1,140	1,523	
	0,3	0,703	0,942	1,174	1,393	1,610	1,874	2,316	3,411	
	0,4	1,237	1,602	1,953	2,285	2,624	3,089	3,981		
	0,5	1,937	2,443	2,926	3,379	3,869	4,640	6,256		
	0,6	2,778	3,428	4,058	4,664	5,366	6,581			
	0,7	3,692	4,488	5,273	6,081	7,138	8,902			
	0,8	4,648	5,570	6,516	7,628	9,910				
20°	0,1	0,096	0,144	0,190	0,232	0,273	0,315	0,368	0,459	0,680
	0,2	0,339	0,472	0,597	0,716	0,832	0,966	1,177	1,659	
	0,3	0,743	0,976	1,204	1,416	1,633	1,927	2,461		
	0,4	1,317	1,673	2,009	2,325	2,679	3,246	3,840		
	0,5	2,075	2,559	3,011	3,436	3,993	4,969			
	0,6	3,006	3,625	4,194	4,752	5,588				
	0,7	4,042	4,789	5,494	6,272					
	0,8	5,122	6,013	6,883	7,861					

Table 68: Values of K_2 for θ_p between 0° and 20°

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<i>K₂ coefficient for fasteners in every two troughs (acc. to [14])</i>										
θ_p	b_r/e_p h_p/e_p	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9
25°	0,1	0,098	0,147	0,192	0,234	0,274	0,317	0,373	0,475	0,665
	0,2	0,355	0,485	0,609	0,725	0,840	0,983	1,226	1,566	
	0,3	0,784	1,015	1,233	1,437	1,660	2,000	2,589		
	0,4	1,398	1,740	2,057	2,359	2,753	3,427			
	0,5	2,205	2,659	3,064	3,490	4,114				
	0,6	2,000	3,752	4,218	4,797					
	0,7	4,318	4,941	5,480						
	0,8	5,487	6,132							
30°	0,1	0,101	0,150	0,194	0,236	0,276	0,319	0,378	0,495	
	0,2	0,372	0,500	0,621	0,734	0,850	1,005	1,298		
	0,3	0,827	1,051	1,260	1,456	1,697	2,098			
	0,4	1,477	1,801	2,092	2,393	2,830				
	0,5	2,319	2,727	3,075	3,499					
	0,6	3,320	3,728	4,041						
	0,7	4,378								
	0,8									
35 °	0,1	0,105	0,153	0,197	0,238	0,278	0,322	0,385	0,525	
	0,2	0,390	0,516	0,634	0,744	0,862	1,035	1,329		
	0,3	0,872	1,088	1,284	1,548	1,741				
	0,4	1,553	1,849	2,105	2,412					
	0,5	2,400	2,713							
	0,6	3,278								
	0,7									
	0,8									
40°	0,1	0,109	0,156	0,200	0,241	0,280	0,325	0,394	0,569	
	0,2	0,411	0,538	0,647	0,753	0,878	1,077			
	0,3	0,919	1,122	1,301	1,496					
	0,4	1,614	1,859	2,085						
	0,5	2,376								
	0,6									
	0,7									
	0,8									
45°	0,1	0,114	0,160	0,203	0,243	0,282	0,329	0,409		
	0,2	0,434	0,553	0,661	0,764	0,899				
	0,3	0,965	1,148	1,306						
	0,4	1,634								
	0,5									
	0,6									
	0,7									
	0,8									

Table 69: Values of K_2 for θ_p between 25° and 45°

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Annex F Bending moment resistance in fire situation of partially encased profiles

F.1 Scope of application

- (1) This Annex describes the simplified method used in ACB-PMX to assess the fire design resistance at time t of cross-sections for steel or composite beams with partially encased profiles.
- (2) The simplified method of the Annex F of EN 1994-1-2:2005 are applied (Annex D in prEN 1994-1-2:2023).
- (3) The design resistance are calculated by iteration at the following time steps:
 - 30 min;
 - 60 min;
 - 90 min;
 - 120 min;
 - 180 min.

F.2 Bending resistance under positive moment

- (1) The positive bending resistant moment at a time step under fire situation is obtained using the plastic calculation according to the principles shown in **Figure 65** and as detailed below.

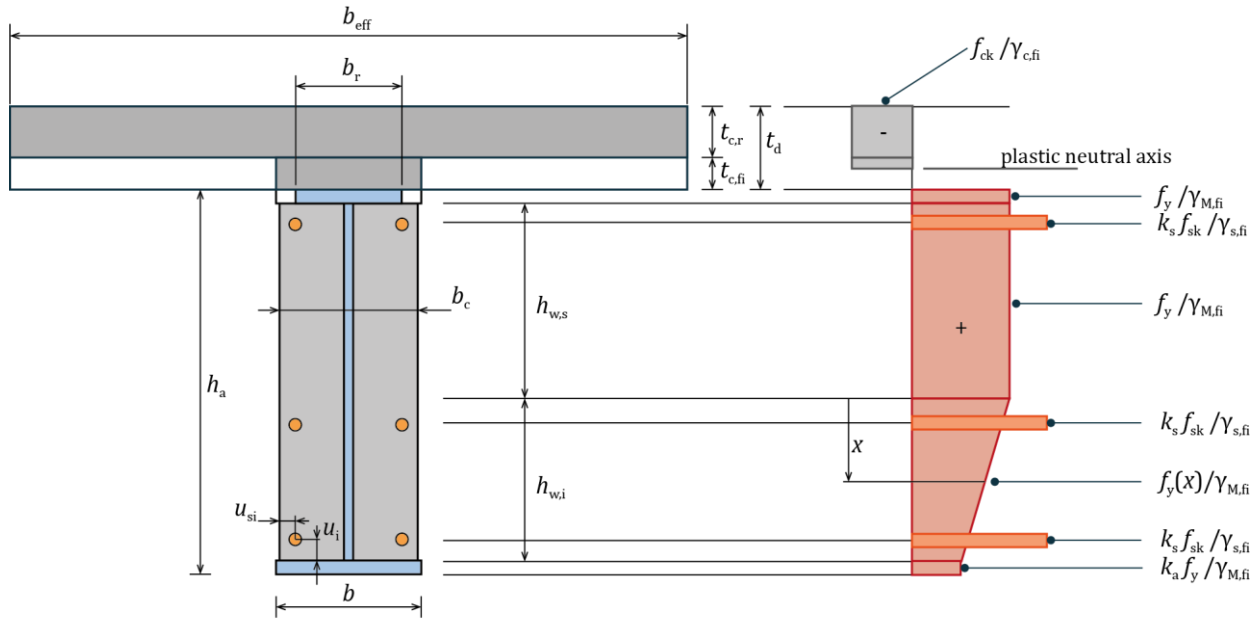


Figure 65 : Stress distribution for the calculation of the positive bending resistance

- (2) The lower flange is fully yielded to its reduced strength, given by:

$$k_a f_y / \gamma_{M,fi} \quad (324)$$

where: f_y is the yield strength of steel at room temperature, as defined in § 2.10;

$\gamma_{M,fi}$ is the partial factor, defined in § 2.12.2;

k_a is a reduction factor given in **Table 70**, where:

h_a is the height of the steel profile;

t_f is a thickness of the flanges;

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b_c is the width of the concrete encasement.

Time step	Reduction factor k_a	$k_{a,min}$	$k_{a,max}$
R30	$\left[1,12 - \frac{84}{b_c} + \frac{h_a}{22 b_c}\right] a_0$	0,5	0,8
R60	$\left[0,21 - \frac{26}{b_c} + \frac{h_a}{24 b_c}\right] a_0$	0,12	0,4
R90	$\left[0,12 - \frac{17}{b_c} + \frac{h_a}{38 b_c}\right] a_0$	0,06	0,12
R120	$\left[0,1 - \frac{15}{b_c} + \frac{h_a}{40 b_c}\right] a_0$	0,05	0,1
R180	$\left[0,03 - \frac{3}{b_c} + \frac{h_a}{50 b_c}\right] a_0$	0,03	0,06
where:		$a_0 = 0,018 t_f + 0,7$	

Table 70: Reduction factor k_a

- (3) The web is divided in two parts. The upper part has a height $h_{w,s}$ and the lower part a height $h_{w,i}$. When $1 < h_a/b_c < 2$, $h_{w,i}$ is defined by **Table 71**. For other cases, it is defined by Formula (325), where a_1 , a_2 and $h_{w,i,min}$ are given in **Table 72**.

$$h_{w,i} = \frac{a_1}{b_c} + a_2 \frac{t_w}{b_c h_a} \quad \text{and} \quad h_{w,i} \leq h_{w,i,min} \quad (325)$$

$h_{w,s}$ is given by: $h_{w,s} = h_w - h_{w,i}$

Time step	$h_{w,i}$	$h_{w,i,min}$
R30	$h_{w,i} = 3600/b_c$	20
R60	$h_{w,i} = \frac{9\,500}{b_c} + 20\,000 \frac{t_w}{b_c h_a} \left(2 - \frac{h_a}{b_c}\right)$	30
R90	$h_{w,i} = \frac{14\,000}{b_c} + 75\,000 \frac{t_w}{b_c h_a} \left(\frac{86}{75} + 2 - \frac{h_a}{b_c}\right)$	40
R120	$h_{w,i} = \frac{23\,000}{b_c} + 110\,000 \frac{t_w}{b_c h_a} \left(\frac{7}{11} + 2 - \frac{h_a}{b_c}\right)$	45
R180	$h_{w,i} = \frac{35\,000}{b_c} + 250\,000 \frac{t_w}{b_c h_a} \left(\frac{15}{25} + 2 - \frac{h_a}{b_c}\right)$	55

Table 71: Height of the lower part of the web $h_{w,i}$ – case $1 < h_a/b_c < 2$

Range	Time step	a_1 (mm ²)	a_2 (mm ²)	$h_{w,i,min}$ (mm)
$h_a/b_c \leq 1$	R30	3 600	0	20
	R60	9 500	20 000	30
	R90	14 000	160 000	40
	R120	23 000	180 000	45
	R180	35 000	400 000	55
$h_a/b_c \geq 2$	R30	3 600	0	20
	R60	9 500	0	30
	R90	14 000	75 000	40
	R120	23 000	110 000	45
	R180	35 000	250 000	55

Table 72: Height of the lower part of the web $h_{w,i}$ (all parameters in mm)

- (4) The upper part of the web is yielded to its design strength, given by:

$$f_y / \gamma_{M,fi} \quad (326)$$

- (5) The lower part of the web is yielded to a trapezoidal strength, whose shape is given by:

$$f_y(x) / \gamma_{M,fi} \quad (327)$$

$$f_y(x) = f_y \left(1 - \frac{x}{h_{w,i}} (1 - k_a) \right)$$

where: k_a is as defined in (2);

x is the distance of the fibre to frontier between the upper and lower parts of the web.

- (6) The effective width of the upper flange is yielded to its design strength, given by:

$$f_y / \gamma_{M,fi} \quad (328)$$

The effective width of the upper flange is given by:

$$b_r = b - 2b_{fi} \quad (329)$$

where b_{fi} is the width reduction, as given in Table 73.

Time step	b_{fi} (mm)
R30	$t_f/2 + (b - b_c)/2$
R60	$t_f/2 + 10 + (b - b_c)/2$
R90	$t_f/2 + 30 + (b - b_c)/2$
R120	$t_f/2 + 40 + (b - b_c)/2$
R180	$t_f/2 + 60 + (b - b_c)/2$

Table 73: Width reduction $b_{f,i}$ (in mm)

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- (7) The reinforcements in the concrete of the encasement are yielded to their reduced design strength, given by:

$$k_s f_{sk} / \gamma_{s,fi} \quad (330)$$

where: f_{sk} is the characteristic yield strenght of the reinforcement, see § 2.10.6;

$\gamma_{s,fi}$ is the partial factor, defined in § 2.12.2;

k_s is a reduction factor, defined by:

$$k_s = \frac{(u a_3 + a_4) a_5}{\sqrt{A_m/V}} \quad \text{but} \quad k_{s,min} \leq k_s \leq k_{s,max} \quad (331)$$

$$k_{s,min} = 0,1$$

$$k_{s,max} = 1$$

$$A_m = 2 h_a + b_c$$

$$V = h_a b_c$$

$$u = \frac{1}{\frac{1}{u_i} + \frac{1}{u_{si}} + \frac{1}{b_c - t_w - u_{si}}}$$

u_i is the distance (in mm) between the rebar axis and the inner face of the lower flange;

u_{si} is the distance (in mm) between the rebar axis and the outer face of the encasement;

a_3, a_4, a_5 are given in **Table 74**.

Time step	a_3	a_4	a_5
R30	0,062	0,16	0,126
R60	0,034	-0,04	0,101
R90	0,026	-0,154	0,090
R120	0,026	-0,284	0,082
R180	0,024	-0,562	0,076

Table 74: a_i factors for the calculation of the reduction factor k_s

- (7) A solid slab is divided in two parts:

- the upper part, whose dimensions are $b_{eff} \times t_{d,r}$;
- the lower part, whose dimensions are $b \times t_{d,fi}$;

where : b_{eff} is the effective width of the slab, as given in § 5.1;

b is the width of the upper flange;

$t_{c,fi}$ is the thickness reduction of the solid slab, given in **Table 75**;

$t_{c,r}$ is the remaining thickness: $t_{c,r} = t_d - t_{c,fi}$.

- (8) Both parts of the slab are stressed in compression to their strength, given by:

$$f_{ck} / \gamma_{M,c,fi} \quad (332)$$

where: f_{ck} is the characteristic resistance to compression of concrete, see § 2.10.5;

$\gamma_{M,c,fi}$ is the partial factor for concrete in fire situations, defined in § 2.12.2.

Time step	$t_{c,fi}$ (mm)
R30	10
R60	20
R90	30
R120	40
R180	55

Table 75: Thickness reduction of a solid slab $t_{c,fi}$

- (9) For composite slabs with open profiles, disposed perpendicularly to the beam axis (configurations PE1 and PE2, see § 2.3.2), the thickness reduction $t_{c,fi}$ is applied from the upper face of the steel deck (Figure 66) (for a steel deck with or without an upper stiffener). The thickness of the remaining fully effective part is given by: $t_{c,r} = t_c - t_{c,fi} = t_d - h_p - t_{c,fi}$.

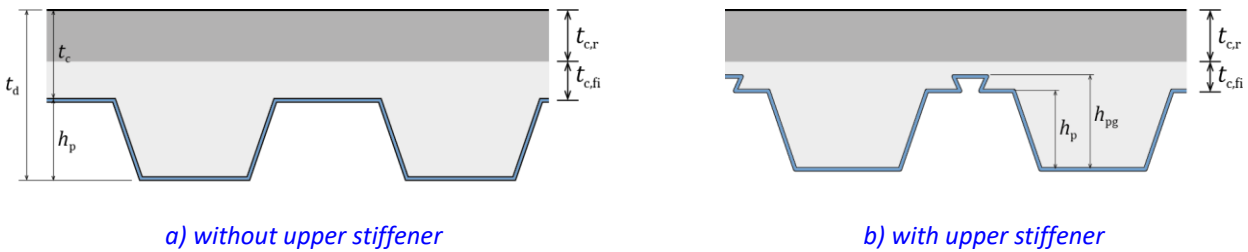


Figure 66 : Thickness reduction for composite slabs with perpendicular open profiles decks

- (10) For composite slabs with reintrant profiles (i.e. with $b_t < b_b$, see § 2.5.1 and Figure 20 b), disposed perpendicularly to the beam axis (configurations PE1 and PE2, see § 2.3.2), the thickness reduction $t_{c,fi}$ is applied from the lower face of the steel deck, but without being lower than the deck depth (Figure 67).

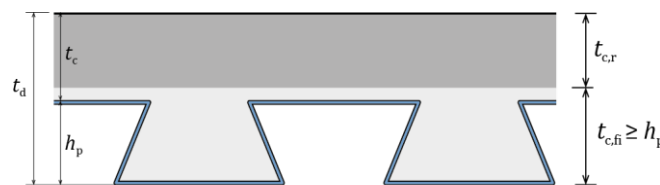


Figure 67 : Thickness reduction for composite slabs with perpendicular reintrant profile decks

- (11) For composite slabs with reintrant profiles, disposed parallel to the beam, the thickness reduction $t_{c,fi}$ is applied from the lower face of the steel deck. This also applies to composite slabs with perpendicular ribs, with configuration PE3 according to § 2.3.2.
- (12) For composite slabs with open profiles, disposed parallel to the beam, the thickness reduction $t_{c,fi}$ is applied considering the effective depth of the slab $h_{c,eff}$ defined in § 10.7.4 (2) (Figure 68).

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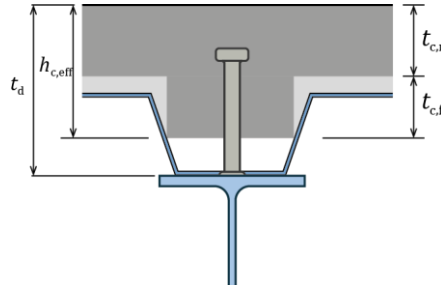


Figure 68 : Thickness reduction for composite slabs with parallel open profile decks

- (13) For solid slabs with prefabricated parts, the thickness reduction is applied from the lower face of the precast elements, but not lower that the joints filled with concrete (Figure 69).



Figure 69 : Thickness reduction for solid slabs with precast elements

- (14) For solid slabs with haunches, the thickness reduction is applied from the lower face of the haunch (Figure 70).

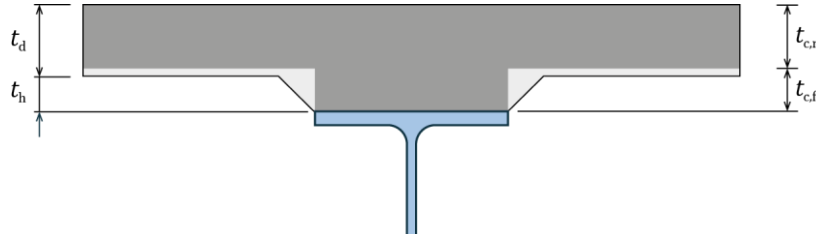


Figure 70 : Thickness reduction for solid slabs with precast elements

- (15) Clauses (8) to (14) apply only to composite beams. For non-composite beams with concrete encasement, the slab is ignored.

F.3 Longitudinal shear in the connection under positive moment

- (1) Under positive moment, the longitudinal shear force in the connection between the slab and the profile along a critical segment of a usual composite beam with partial concrete encasement, for a full connection, is obtained by:

$$N_{cx,fi} = \min\{N_{c,fi,Rd}; N_{a,fi,Rd} + N_{s,fi,Rd}\} \quad (333)$$

where: $N_{a,fi,Rd}$ is the axial resistance of the steel profile in fire situation, given by:

$$N_{a,fi,Rd} = [b_r t_{fs} + b_{fi} t_{fi} + t_w (h_{ws} + h_{wi} (1 + k_a))] f_{yj} / \gamma_{M,fi} \quad (334)$$

$N_{c,fi,Rd}$ is the axial resistance to compression of the slab in fire situation, given by:

$$N_{c,fi,Rd} = [b_{eff} t_{cr} + b_{fs} t_{c,fi}] f_{ck} / \gamma_{c,fi} \quad (335)$$

$N_{s,fi,Rd}$ is the axial resistance to tension of the reinforcement in the encasement in fire situation, given by:

$$N_{s,fi,Rd} = \sum A_{si} k_s f_{sk} / \gamma_{s,fi} \quad (336)$$

Time step	$k_{s,d}$	$k_{s,d,min}$	$k_{s,d,max}$
R30	1	0	1
R60	$0,022 u + 0,34$		
R90	$0,0275 u - 0,1$		
R120	$0,022 u - 0,2$		
R180	$0,018 u - 0,26$		

Table 76: Reduction factor $k_{s,d}$

- (5) The effective part of the encasement concrete in compression is taken into account, assuming that it is stressed in compression to its strength, given by:

$$f_{cek} / \gamma_{c,fi} \quad (338)$$

where: f_{cek} is the characteristic resistance to compression of the encasement concrete, see § 2.10.5;

$\gamma_{c,fi}$ is the partial factor for concrete in fire situations, defined in § 2.12.2.

- (6) On each side of the concrete encasement, the ineffective width of concrete $b_{c,fi}$ is neglected. $b_{c,fi}$ is given by Table 77.

Time step	$b_{c,fi}$ (mm)	$b_{c,fi,min}$ (mm)
R30	25	25
R60	$60 - 0,15 b_c$	30
R90	$70 - 0,1 b_c$	35
R120	$75 - 0,1 b_c$	45
R180	$85 - 0,1 b_c$	55

Table 77: Ineffective width $b_{c,fi}$

- (7) The lower height of the concrete encasement $h_{c,fi}$ is ineffective and is neglected in the calculations. $h_{c,fi}$ is given by Table 78.

Time step	$h_{c,fi}$ (mm)	$h_{c,fi,min}$ (mm)
R30	25	25
R60	$165 - 0,4 b_c - 8 h_a/b_c$	30
R90	$220 - 0,5 b_c - 8 h_a/b_c$	45
R120	$290 - 0,6 b_c - 10 h_a/b_c$	55
R180	$360 - 0,7 b_c - 10 h_a/b_c$	65

Table 78: Ineffective height $h_{c,fi}$

- (8) The reinforcement of the encasement is taken into account either in tension or in compression, according to § F.2 (7).
- (9) The concrete of the slab is neglected.

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- (10) For steel sections with partial concrete encasement, the contribution of the slab reinforcement is neglected.

F.5 Longitudinal shear in the connection under negative moment

- (1) Under positive moment, the longitudinal shear force in the connection between the slab and the profile along a critical segment of an usual composite beam with partial concrete encasement, for a full connection, is obtained by:

$$N_{cx,fi} = \min\{N_{sc,fi,Rd}; N_{ce,fi,Rd} + N_{se,fi,Rd} + N_{f,fi,Rd}\} \quad (339)$$

where: $N_{f,fi,Rd}$ is the axial resistance of the upper flange in fire situation, given by:

$$N_{f,fi,Rd} = b_r t_{fs} f_{yj} / \gamma_{M,fi} \quad (340)$$

$N_{ce,fi,Rd}$ is the axial resistance to compression of the encasement in fire situation, given by:

$$N_{ce,fi,Rd} = \left[(b_c - 2 b_{ci} - t_w)(h_w - h_{c,fi}) - (4 - \pi) \frac{r^2}{2} \right] f_{ck} / \gamma_{c,fi} \quad (341)$$

$N_{sc,fi,Rd}$ is the axial resistance to tension of the reinforcement in the slab in fire situation, given by:

$$N_{sc,fi,Rd} = \sum A_{si} k_{s,d} f_{sk} / \gamma_{s,fi} \quad (342)$$

$N_{se,fi,Rd}$ is the axial resistance to compression of the reinforcement in the encasement in fire situation, given by (only if the relevant option is activated):

$$N_{se,fi,Rd} = \sum A_{si} k_s f_{sk} / \gamma_{s,fi} \quad (343)$$

see F.4 for the other parameters.



The critical length is accounted between the point of zero moment and the maximal negative bending moment.

F.6 Resistance to shear force

- (1) The shear resistance of the section is calculated from the following equation:

$$V_{pl,Rd,fi} = \frac{A_v f_y}{\sqrt{3} \gamma_{M,fi}} \frac{h_{ws} + h_{wi} (1 + k_a) / 2}{h_w} \quad (344)$$

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Annex G Numerical procedure for the assessment of the temperature in a solid slab in fire situation

G.1 Scope of application

- (1) This Annex describes the numerical that may be used in the software to assess the temperature within a solid slab under the effect of an ISO fire based underneath the slab

G.2 Assumptions

- (1) The slab of thickness t_c will be modelled by a set of n layers, each of ones of thickness t_{ci} (the thickness of layers may vary along the depth of the slab).
- (2) At time t , the temperature of gazes below the slab is denoted $\theta_g(t)$. It is obtained by Formula (250) in § 10.7.1 (7). The temperature of gazes above the slab is $\theta_0 = 20^\circ\text{C}$.
- (3) The coefficients of heat transfer by convection are as follows:
- α_c on the exposed side of the slab (below);
 - α_{cc} on the non-exposed side of the slab (above);
 - see § 10.5 for α_c and α_{cc} .
- (4) The calculation will take into account the following parameters (see § 10.5):
- the emissivity of fire ε_f ;
 - the Boltzmann constant σ ;
- (5) The density of concrete will be considered either as constant or as varying with its temperature (according to the option activated). If the density is assumed as constant, the value ρ_c defined by the User will be considered (see § 2.10.5 (3)). If it is assumed to vary with the temperature, following equations will be used. With the first generation of EN, according to EN 1994-1-2: 2005, Formula (3.11):

$$\rho_c(\theta_c) = 2354 - 23,47 \frac{\theta_c}{100} \quad (345)$$

where ρ_c is in $[\text{kg}/\text{m}^3]$ and θ_c in $[\text{°C}]$.

With the second generation of EN, according to prEN 1994-1-2 : 2024, Formulas (5.21) to (5.24):

$$\begin{aligned} \rho_c(\theta_c) &= \rho_{c0} & \text{for } 20^\circ\text{C} \leq \theta_c \leq 115^\circ\text{C} \\ \rho_c(\theta_c) &= \rho_{c0} \left(1 - 0,02 \frac{\theta_c - 115}{85} \right) & \text{for } 115^\circ\text{C} < \theta_c \leq 200^\circ\text{C} \\ \rho_c(\theta_c) &= \rho_{c0} \left(0,98 - 0,03 \frac{\theta_c - 200}{200} \right) & \text{for } 200^\circ\text{C} < \theta_c \leq 400^\circ\text{C} \\ \rho_c(\theta_c) &= \rho_{c0} \left(0,95 - 0,07 \frac{\theta_c - 400}{800} \right) & \text{for } 400^\circ\text{C} < \theta_c \leq 1200^\circ\text{C} \end{aligned} \quad (346)$$

where $\rho_{c0} = \rho_c$ as defined in 2.10.5 (3) for the normal situation.

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- (6) The specific heat of concrete is a function of the temperature, which is obtained as follows (according to EN 1994-1-2 :2005 3.3.2 (4) (ou prEN 1994-1-2 :2024, 5.2.2.3 (1))):

$$\begin{aligned}
 c_c(\theta_c) &= 900 & \text{for } 20^\circ\text{C} \leq \theta_c \leq 100^\circ\text{C} \\
 c_c(\theta_c) &= 900 + (\theta_c - 100) & \text{for } 100^\circ\text{C} < \theta_c \leq 200^\circ\text{C} \\
 c_c(\theta_c) &= 1000 + (\theta_c - 100)/2 & \text{for } 200^\circ\text{C} < \theta_c \leq 400^\circ\text{C} \\
 c_c(\theta_c) &= 1100 & \text{for } 400^\circ\text{C} < \theta_c \leq 1200^\circ\text{C}
 \end{aligned} \tag{347}$$

where ρ_c is in [J/kg.K] and θ_c in [°C].

- (7) In the case of normal concrete, the content of moisture U will be taken into account as follows.

With the first generation of EN, according to EN 1994-1-2 : 2005, 3.3.2 (8):

Maximal value of c_c^* at temperature 115°C, according to the value of U :

$$\begin{aligned}
 c_c^* &= 900 \text{ J/kg.K} & \text{for } U = 0\% ; \\
 c_c^* &= 2020 \text{ J/kg.K} & \text{for } U = 3\% ; \\
 c_c^* &= 5600 \text{ J/kg.K} & \text{for } U = 10\% .
 \end{aligned}$$

For other values of U , the value will be obtained by linear interpolation.

For temperatures between 100°C and 115 °C: linear interpolation of c_c between 900 J/kg.K and c_c^* ;

For temperatures between 115°C and 200°C: linear interpolation of c_c between c_c^* and $c_c(200^\circ\text{C})$.

With the second generation of EN, according to EN 1994-1-2 : 2024, 5.2.2.3 (3):

Constant value of c_c^* between 100°C and 115°C, according to the value of U :

$$\begin{aligned}
 c_c^* &= 900 \text{ J/kg.K} & \text{for } U = 0\% ; \\
 c_c^* &= 1470 \text{ J/kg.K} & \text{for } U = 1,5\% ; \\
 c_c^* &= 2020 \text{ J/kg.K} & \text{for } U = 3\% ; \\
 c_c^* &= 5600 \text{ J/kg.K} & \text{for } U = 10\% .
 \end{aligned}$$

For other values of U , the value will be obtained by linear interpolation.

For temperatures between 115°C and 200°C: linear interpolation of c_c between c_c^* and $c_c(200^\circ\text{C})$.

- (8) With the first generation of EN, the thermal conductivity of concrete will be obtained as follows.

For normal concretes, outside the French NA, according to the upper limit in EN 1994-1-2 :2005 3.3.2 (10):

$$\lambda_c(\theta_c) = 2 - 0,2451 \frac{\theta_c}{100} + 0,0107 \left(\frac{\theta_c}{100} \right)^2 \tag{348}$$

where λ_c is in [W/m.K] and θ_c in [°C].

For normal concretes, according to the French National Annex of EN 1994-1-2:

$$\begin{aligned}
 \lambda_c(\theta_c) &= 2 - 0,2451 \frac{\theta_c}{100} + 0,0107 \left(\frac{\theta_c}{100} \right)^2 & \text{for } 20^\circ\text{C} \leq \theta_c \leq 140^\circ\text{C} \\
 \lambda_c(\theta_c) &= -0,02604 \theta_c + 5,324 & \text{for } 140^\circ\text{C} < \theta_c \leq 160^\circ\text{C} \\
 \lambda_c(\theta_c) &= 1,36 - 0,136 \frac{\theta_c}{100} + 0,0057 \left(\frac{\theta_c}{100} \right)^2 & \text{for } 160^\circ\text{C} < \theta_c \leq 1200^\circ\text{C}
 \end{aligned} \tag{349}$$

For light weight concretes, according to EN 1994-1-2 :2005 3.3.2 (12), a constant value will be used for all temperatures: $\lambda_c(\theta_c) = 1,6 \text{ W/(m.K)}$.

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- (8) With the second generation of EN, the thermal conductivity of concrete will be obtained as follows.
For normal concretes, according to prEN 1994-1-2 :2024 5.2.2.2 (1), $\lambda_c(\theta_c)$ will be obtained with Formulas **(349)**.

For light concretes, according to prEN 1994-1-2 :2024 5.2.2.2 (2):

$$\lambda_c(\theta_c) = 1 - \frac{\theta_c}{1600} \quad \text{for } 20^\circ\text{C} \leq \theta_c \leq 800^\circ\text{C} \quad (350)$$

$$\lambda_c(\theta_c) = 0,5 \quad \text{for } 800^\circ\text{C} < \theta_c \leq 1200^\circ\text{C}$$